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CBSE Class 10

MATHEMATICS

KHATAM IN 99 PAGES

Session 2026-27

FREE e-book

Algebra | Geometry | Trigonometry | Mensuration & Stats

By Shivam Bhaiya

One Shot handwritten notes | All Formulas + Tricks | Most Important Questions in every chapter

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Theory 80 marks
+ 20 internal assessment

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Is book ko kaise padhein

- ✓ Purple **Formula Box** = chapter ke saare formulas.
- ✓ Blue box = solved example, steps ke saath.
- ✓ Orange "Yaad karo" = trick / mnemonic; red dashed = common mistakes.
- ✓ Answers green mein: pehle khud solve karo, phir milao.

Shivam Bhaiya says

Maths padhne se nahi, **likh ke practice** karne se aati hai. Har chapter: pehle notes + Formula Box, phir solved examples khud karo, phir Most Important Questions bina dekhe solve karke green answers se milao.

Board paper pattern (80 marks)

Section	Type	Marks
A	18 MCQ + 2 A-R	20 x 1
B	5 very short	5 x 2
C	6 short	6 x 3
D	4 long	4 x 5
E	3 case based	3 x 4

Unit-wise weightage

Unit	Marks
Number Systems	6
Algebra	20
Coordinate Geometry	6
Geometry	15
Trigonometry	12
Mensuration	10
Statistics & Probability	11

Last 7 days plan (is book se)

Day	Kya padhna hai
1	Real Numbers, Polynomials, Linear Equations
2	Quadratic Equations, AP
3	Triangles, Coordinate Geometry, Circles
4	Trigonometry + Applications
5	Areas Related to Circles, Surface Areas and Volumes
6	Statistics, Probability + Formula Sheet
7	Mock Papers 1 + 2 (3 hours each) + Last-night revision

Is book mein kya kya hai

14 chapters, NCERT 2026-27	Notes + Formula Box
500+ questions	sabke answers green mein
180+ solved examples	step by step
60+ Yaad karo tricks	formulas yaad rakhne ke liye
2 full mock papers	80 marks, board pattern
3-page formula sheet	+ 56 rapid-fire checks

Most asked: HCF/LCM word problems, prove √2, √3, √5 irrational

REAL NUMBERS

1. Primes, Composites and FTA

A **prime number** has exactly two factors, 1 and itself (2, 3, 5, 7, 11, 13 ...). A **composite number** has more than two factors (4, 6, 8, 9 ...).

→ **1** is neither prime nor composite.
→ **2** is the only even prime.
→ Two numbers are **coprime** if their HCF is 1 (e.g. 8 and 15).

Theorem: Fundamental Theorem of Arithmetic

Every composite number can be expressed (factorised) as a product of primes, and this factorisation is **unique**, apart from the order in which the prime factors occur.

Standard form: primes in ascending order with powers, e.g. $1260 = 2^2 \times 3^2 \times 5 \times 7$.

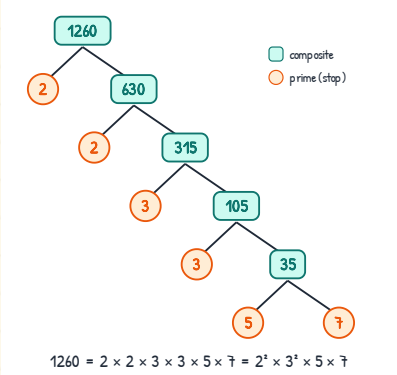


Fig: factor tree of 1260 (split till every leaf is prime)

Meme samjho

Factor tree kaise bhi shuru karo (2 x 630 ya 4 x 315), end mein wahi primes aate hain? **ACHHAA!!**

Asli baat: FTA says the prime factorisation is unique: $1260 = 2^2 \times 3^2 \times 5 \times 7$ every time. Only the order can change.

Solved Example 1

Express 3780 as a product of its prime factors.

1. $3780 \div 2 = 1890, 1890 \div 2 = 945$
2. $945 \div 3 = 315, 315 \div 3 = 105, 105 \div 3 = 35$
3. $35 \div 5 = 7, 7 \div 7 = 1$

✔ $3780 = 2^2 \times 3^3 \times 5 \times 7$

2. HCF and LCM by Prime Factorisation

Yaad karo: "HCF = Hamesha Chhoti, LCM = Lambi"

HCF = Hamesha Chhoti power, sirf **common** primes ki. LCM = Lambi (badi) power, **saare** primes ki. Check: HCF hamesha LCM ko divide karega.

Formula Box: Real Numbers

→ HCF = product of the smallest power of each **common** prime factor
→ LCM = product of the greatest power of **every** prime factor
→ For two positive integers:

$$\text{HCF}(a,b) \times \text{LCM}(a,b) = a \times b$$

→ So $\text{LCM} = \dfrac{a \times b}{\text{HCF}}$ (NOT true for 3 numbers)
→ HCF always divides LCM; HCF of coprimes = 1, LCM of coprimes = product
→ a^n ends in 0 only if a has both 2 and 5 as prime factors
→ If prime p divides a^2, then p divides a

2	3780	
2	1890	divide by the smallest prime each time
3	945	
3	315	
3	105	
5	35	7 (prime)

Fig: division ladder for 3780 = 2 x 2 x 3 x 3 x 3 x 5 x 7

Solved Example 2

Find the HCF and LCM of 120 and 144, and verify HCF x LCM = product of the numbers.

1. $120 = 2^3 \times 3 \times 5, 144 = 2^4 \times 3^2$
2. $\text{HCF} = 2^3 \times 3 = 24$ (smallest powers of common primes)
3. $\text{LCM} = 2^4 \times 3^2 \times 5 = 720$
4. Check: $24 \times 720 = 17280 = 120 \times 144$ ✔

✔ HCF = 24, LCM = 720

Solved Example 3

Find the HCF and LCM of 6, 72 and 120.

1. $6 = 2 \times 3, 72 = 2^3 \times 3^2, 120 = 2^3 \times 3 \times 5$
2. $\text{HCF} = 2 \times 3 = 6; \text{LCM} = 2^3 \times 3^2 \times 5 = 360$
3. Note: $6 \times 360 = 2160$ but $6 \times 72 \times 120 = 51840$, so the product rule fails for 3 numbers.

✔ HCF = 6, LCM = 360

Can a^n end with digit 0?

A number ends with 0 only if it is divisible by $10 = 2 \times 5$, so its prime factorisation must contain **both 2 and 5**.

→ $6^n = 2^n \times 3^n$: no 5, so it **never** ends with 0.
→ $15^n = 3^n \times 5^n$: no 2, so never 0 (it ends with 5).
→ $20^n = 2^{2n} \times 5^n$: ends with 0 always.

Solved Example 4

Show that $7 \times 11 \times 13 + 13$ is a composite number.

1. Take 13 common: $13\,(7 \times 11 + 1) = 13 \times 78$
2. It has factors other than 1 and itself (13, 78, ...)
3. Indeed it equals $1014 = 2 \times 3 \times 13^2$

✔ Composite (not prime)

Shivam Bhaiya says

Board mein sirf answer $2^2 \times 3^3 \times 5 \times 7$ likh ke mat chhodna. Division ladder ya factor tree **dikhana compulsory** hai, working ke marks wahi se aate hain.

3. HCF or LCM? Word Problems

Words in question	Use
greatest / largest / maximum size, tile, container, rows, groups	HCF
least / smallest / next time together, bells, lights, laps, meet again	LCM

Solved Example 5

A room is 8.25 m long and 6.75 m wide. Find the side of the largest square tile that can pave it exactly, and the number of tiles.

1. In cm: 825 and 675
2. $825 = 3 \times 5^2 \times 11, 675 = 3^3 \times 5^2$
3. HCF $= 3 \times 5^2 = 75$ cm
4. Tiles $= \frac{825}{75} \times \frac{675}{75} = 11 \times 9 = 99$

✔ Side 75 cm, 99 tiles

Solved Example 6

Three bells toll at intervals of 18, 24 and 32 minutes. They toll together at 9:00 a.m. When will they next toll together?

1. $18 = 2 \times 3^2, 24 = 2^3 \times 3, 32 = 2^5$
2. LCM $= 2^5 \times 3^2 = 288$ min
3. 288 min = 4 h 48 min

✔ At 1:48 p.m.

Remainder problems

→ **Greatest number** that divides x, y leaving remainders r_1, r_2 : HCF of $(x - r_1)$ and $(y - r_2)$.

→ **Smallest number** which when divided by a, b, c leaves remainder r in each case: LCM$(a, b, c) + r$.

Yaad karo: "Greatest = Ghatao, Smallest = Sum"

Greatest wale sawal mein pehle remainder Ghatao, phir HCF. Smallest wale mein pehle LCM, phir remainder Sum (jodo). Ulta kiya to answer gaya!

Recall $\sqrt{p}$ is irrational for every prime p.

Solved Example 7

Find the greatest number that divides 398, 436 and 542 leaving remainders 7, 11 and 15 respectively.

1. Subtract remainders: $398 - 7 = 391, 436 - 11 = 425, 542 - 15 = 527$
2. $391 = 17 \times 23, 425 = 5^2 \times 17, 527 = 17 \times 31$
3. HCF $= 17$

✔ 17

4. Irrational Numbers

Definition

A number s is **irrational** if it cannot be written as $\frac{p}{q}$, where p, q are integers and $q \neq 0$. Examples: $\sqrt{2}, \sqrt{3}, \sqrt{5}, \pi, 3 + \sqrt{2}$, 0.1011011101...

Its decimal expansion is **non-terminating, non-repeating**.

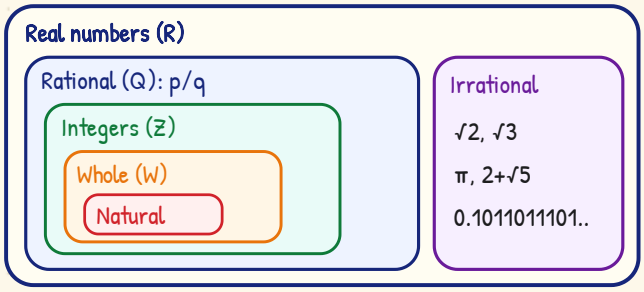


Fig: number family. Every real number is either rational or irrational.

Operation	Result	Example
rational $\pm$ irrational	irrational	$5 - \sqrt{3}$
non-zero rational $\times$ irrational	irrational	$2\sqrt{5}$
irrational + irrational	may be rational	$\sqrt{2} + (-\sqrt{2}) = 0$
irrational $\times$ irrational	may be rational	$\sqrt{2} \times \sqrt{8} = 4$

Theorem used in proofs

Let p be a prime. If p divides a^2 (a a positive integer), then p divides a.

Yaad karo: "MSD + Captain" (proof steps)

Maano rational ($\frac{a}{b}$, coprime), Square karo, Divide dikhao (prime divides a, phir b), Contradiction likho. MSD = Dhoni, aur proof ka Captain = Contradiction!

Quick fact Smallest prime = 2, smallest composite = 4, smallest odd prime = 3.

Yaad rakho HCF of two distinct primes = 1, LCM = their product.

Solved Example 8 (proof)

Prove that $\sqrt{2}$ is irrational.

1. Assume, to the contrary, $\sqrt{2} = \frac{a}{b}$, where a, b are **coprime** integers, $b \neq 0$.
2. Squaring: $2b^2 = a^2$, so 2 divides a^2, hence 2 divides a. Write $a = 2c$.
3. Then $2b^2 = 4c^2 \Rightarrow b^2 = 2c^2$, so 2 divides b^2, hence 2 divides b.
4. So 2 divides both a and b: this **contradicts** that a, b are coprime.

✔ Hence $\sqrt{2}$ is irrational.

Solved Example 9

Given that $\sqrt{3}$ is irrational, prove that $5 - 2\sqrt{3}$ is irrational.

1. Assume $5 - 2\sqrt{3} = \frac{a}{b}$, a, b integers, $b \neq 0$.
2. Then $2\sqrt{3} = 5 - \frac{a}{b}$, so $\sqrt{3} = \frac{5b - a}{2b}$.
3. RHS is rational (integers, $2b \neq 0$), so $\sqrt{3}$ would be rational.
4. This contradicts the fact that $\sqrt{3}$ is irrational.

✔ So $5 - 2\sqrt{3}$ is irrational.

Meme samjho

$\sqrt{2}$ ko kabhi $\frac{p}{q}$ mein likh paoge, koi **guarantee deta hai??**

Asli baat: Koi nahi! $\sqrt{2}$ is irrational: it can never be written as $\frac{p}{q}$ with integers p, q ($q \neq 0$).

Common mistakes

- → Using HCF x LCM = product for **three** numbers (works only for two).
- → Forgetting to write "a, b are **coprime**" in the proof: then there is no contradiction.
- → Remainder sums: taking HCF of the numbers **before** subtracting the remainders.
- → Calling 1 a prime, or 51 a prime ($51 = 3 \times 17$).
- → Saying "sum of two irrationals is always irrational" ($\sqrt{2} + (-\sqrt{2}) = 0$).

Shivam Bhaiya says

Given that $\sqrt{3}$ is irrational likha ho, to $\sqrt{3}$ ka poora proof dobara mat likho. Bas rearrange karo aur dikhao $\sqrt{3}$ rational ban jayega. Time bachao, marks pakke.

Recall π is irrational; $\frac{22}{7}$ is only its rational approximation.

5. Extra Practice: Board + Competency

Primes up to 100 (25 primes)

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

↳ Divisibility check: kisi number n ke liye sirf $\sqrt{n}$ tak ke primes se divide karke dekho.

Solved Example 10

Three runners take 36 s, 48 s and 60 s for one round of a track. They start together. After how many minutes will they meet again at the start? How many rounds will the fastest have done?

1. $36 = 2^2 \times 3^2, 48 = 2^4 \times 3, 60 = 2^2 \times 3 \times 5$
2. LCM $= 2^4 \times 3^2 \times 5 = 720$ s $= 12$ min
3. Fastest (36 s): $\frac{720}{36} = 20$ rounds

✔ After 12 min; 20 rounds

Solved Example 11

Find the largest number which divides 70 and 125, leaving remainders 5 and 8 respectively.

1. $70 - 5 = 65 = 5 \times 13$
2. $125 - 8 = 117 = 3^2 \times 13$
3. HCF $= 13$ (check: $70 = 13 \times 5 + 5, 125 = 13 \times 9 + 8$ ✔)

✔ 13

Solved Example 12

Prove that $\sqrt{2} + \sqrt{3}$ is irrational. (HOTS)

1. Assume $\sqrt{2} + \sqrt{3} = r$, rational. Then $\sqrt{3} = r - \sqrt{2}$.
2. Square: $3 = r^2 - 2\sqrt{2}\,r + 2 \Rightarrow \sqrt{2} = \frac{r^2 - 1}{2r}$, which is rational.
3. Contradiction, since $\sqrt{2}$ is irrational.

✔ $\sqrt{2} + \sqrt{3}$ is irrational

MOST IMPORTANT QUESTIONS

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. The HCF of the smallest prime number and the smallest composite number is 1M

(a) 1 (b) 2 (c) 4 (d) 8

Ans. (b) 2. Smallest prime = 2, smallest composite = 4, HCF(2, 4) = 2.

Q2. If $a = p^3q^2$ and $b = p^2q^4$ (p, q primes), then LCM(a, b) is 1M Board fav

(a) p^2q^2 (b) p^5q^6 (c) p^3q^4 (d) p^2q^4

Ans. (c) p^3q^4 (greatest powers).

Q3. The LCM of two numbers is 2400. Which of these can NOT be their HCF? 1M

(a) 400 (b) 500 (c) 600 (d) 800

> **Ans.** (b) 500, since HCF must divide LCM and $2400 \div 500 = 4.8$.

Q4. If HCF(26, 91) = 13, then LCM(26, 91) is 1M

(a) 91 (b) 182 (c) 364 (d) 2366

> **Ans.** (b) $\dfrac{26 \times 91}{13} = 182$.

Q5. $(2+\sqrt{3})(2-\sqrt{3})$ is 1M

(a) irrational (b) rational (c) zero (d) negative

> **Ans.** (b) $= 4 - 3 = 1$, a rational number.

Q6. For a natural number n, 12^n can never end with the digit 1M

(a) 2 (b) 4 (c) 0 (d) 6

> **Ans.** (c) 0, because $12^n = 2^{2n} \times 3^n$ has no factor 5.

Q7. **A:** HCF of two consecutive natural numbers is 1. **R:** Two consecutive natural numbers are always coprime. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

> **Ans.** (a) Both true and R explains A.

Q8. **A:** 5^n ends with 0 for some natural number n. **R:** A number ending in 0 has both 2 and 5 as prime factors. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

> **Ans.** (d) A is false (5^n has no factor 2), R is true.

Q9. The ratio of LCM and HCF of the least composite number and the least prime number is 1M

(a) 1 : 2 (b) 1 : 1 (c) 2 : 1 (d) 4 : 1

> **Ans.** (c) LCM$(4, 2) = 4$, HCF $= 2$, ratio $2 : 1$.

Q10. $\dfrac{\sqrt{75}}{\sqrt{3}}$ is equal to 1M

(a) $\sqrt{5}$ (b) 5 (c) 25 (d) $5\sqrt{3}$

> **Ans.** (b) $\sqrt{25} = 5$, a rational number.

Very short (2 marks)

Q11. Express 2520 as a product of its prime factors. 2M

> **Ans.** $2520 = 2^3 \times 3^2 \times 5 \times 7$ (ladder: 2520, 1260, 630, 315, 105, 35, 7, 1).

Q12. Given HCF(144, 198) = 18, find LCM(144, 198). 2M Board fav

> **Ans.** LCM $= \dfrac{144 \times 198}{18} = \dfrac{28512}{18} = 1584$.

Q13. Check whether 8^n can end with the digit 0 for any natural number n. 2M

> **Ans.** $8^n = 2^{3n}$. To end with 0 it needs prime factors 2 and 5; 5 is absent (FTA: factorisation is unique). So 8^n never ends with 0.

Q14. Find the HCF and LCM of 30, 72 and 432. 2M

> **Ans.** $30 = 2 \times 3 \times 5, 72 = 2^3 \times 3^2, 432 = 2^4 \times 3^3$. HCF $= 2 \times 3 = 6$, LCM $= 2^4 \times 3^3 \times 5 = 2160$.

Short answer (3 marks)

Q15. Prove that $\sqrt{5}$ is irrational. 3M Board fav

> (contd.) Assume $\sqrt{5} = \frac{a}{b}$, a, b coprime, $b \neq 0$. Then $5b^2 = a^2$, so 5 divides a^2, hence 5 divides a; let $a = 5c$. Then $5b^2 = 25c^2$, $b^2 = 5c^2$, so 5 divides b. 5 divides both a and b: contradiction (coprime). Hence $\sqrt{5}$ is irrational.

Q16. Given that $\sqrt{3}$ is irrational, prove that $2\sqrt{3} - 1$ is irrational. 3M

> **Ans.** Let $2\sqrt{3} - 1 = \frac{a}{b}$ (a, b integers, $b \neq 0$). Then $\sqrt{3} = \dfrac{a+b}{2b}$, which is rational. Contradiction, since $\sqrt{3}$ is irrational. So $2\sqrt{3} - 1$ is irrational.

Q17. Two tankers contain 850 litres and 680 litres of petrol. Find the maximum capacity of a container which can measure the petrol of either tanker an exact number of times. 3M

> **Ans.** $850 = 2 \times 5^2 \times 17, 680 = 2^3 \times 5 \times 17$. HCF $= 2 \times 5 \times 17 = 170$. Container = 170 litres (5 times and 4 times).

Q18. Find the smallest number which when divided by 35, 56 and 91 leaves remainder 7 in each case. 3M

> **Ans.** $35 = 5 \times 7, 56 = 2^3 \times 7, 91 = 7 \times 13$. LCM $= 2^3 \times 5 \times 7 \times 13 = 3640$. Required number $= 3640 + 7 = 3647$.

Long answer (5 marks)

Q19. Prove that $\sqrt{3}$ is irrational. Hence show that $7 + 2\sqrt{3}$ is irrational. 5M Board fav

> **Ans.** Assume $\sqrt{3} = \frac{a}{b}$, a, b coprime, $b \neq 0$. $3b^2 = a^2$, so 3 divides a; $a = 3c$ gives $b^2 = 3c^2$, so 3 divides b. Contradiction, so $\sqrt{3}$ is irrational. Now if $7 + 2\sqrt{3} = \frac{p}{q}$, then $\sqrt{3} = \dfrac{p - 7q}{2q}$ would be rational: contradiction. So $7 + 2\sqrt{3}$ is irrational.

Q20. A sweet seller has 420 kaju barfis and 130 badam barfis. He stacks them so that each stack has the same number and only one kind of barfi. (a) Find the maximum number of barfis per stack. (b) How many stacks will there be? (c) Find LCM(420, 130) using (a). 5M

> **Ans.** (a) $420 = 2^2 \times 3 \times 5 \times 7, 130 = 2 \times 5 \times 13$, HCF $= 2 \times 5 = 10$ barfis per stack. (b) $\frac{420}{10} + \frac{130}{10} = 42 + 13 = 55$ stacks. (c) LCM $= \dfrac{420 \times 130}{10} = 5460$.

More practice (board pattern)

Q21. The exponent of 2 in the prime factorisation of 144 is 1M

(a) 4 (b) 5 (c) 6 (d) 3

> **Ans.** (a) 4, since $144 = 2^4 \times 3^2$.

Q22. Which of these is irrational? 1M

(a) $\sqrt{0.25}$ (b) $\sqrt{196}$ (c) $\sqrt{12}$ (d) $\frac{\sqrt{50}}{\sqrt{2}}$

> **Ans.** (c) $\sqrt{12} = 2\sqrt{3}$ is irrational; $\sqrt{0.25} = 0.5$, $\sqrt{196} = 14$ and $\frac{\sqrt{50}}{\sqrt{2}} = 5$ are rational.

Q23. Two numbers are in the ratio 3 : 4 and their HCF is 6. Their LCM is 1M

(a) 36 (b) 72 (c) 144 (d) 24

> **Ans.** (b) Numbers are 18 and 24; LCM $= 72$.

Do you know? π was proved irrational by Lambert in 1761.

Q24. Find the largest number that divides 245 and 1029 leaving remainder 5 in each case. 2M

> **Ans.** $245-5=240=2^4\times3\times5, 1029-5=1024=2^{10}$. HCF $=2^4=16$.

Q25. Ravi and Seema run around a circular track; Ravi takes 12 min and Seema 18 min for one round. Starting together at the same point and direction, after how many minutes will they meet again at the start? 2M

> **Ans.** LCM$(12,18)$: $12=2^2\times3, 18=2\times3^2$, LCM $=2^2\times3^2=36$ min.

Q26. An army contingent of 616 members marches behind a band of 32 members. Both groups march in the same number of columns. Find the maximum number of columns. 3M Board fav

> **Ans.** $616=2^3\times7\times11, 32=2^5$. HCF $=2^3=8$ columns.

Q27. If n is a natural number, then $9^{2n}-4^{2n}$ is always divisible by 1M

(a) 5 only (b) 13 only (c) both 5 and 13 (d) neither

> **Ans.** (c) a^2-b^2 form: $(9^n-4^n)(9^n+4^n)$, and $9^{2n}-4^{2n}=81^n-16^n$ is divisible by $81-16=65=5\times13$, so by both 5 and 13.

Q28. The product of a non-zero rational number and an irrational number is 1M

(a) always rational (b) always irrational (c) rational or irrational (d) one

> **Ans.** (b) always irrational (e.g. $3\sqrt{2}$).

Q29. Write the HCF of $2^3\times3^2\times5$ and $2^2\times3^3\times7$. 1M

> **Ans.** $2^2\times3^2=36$ (smallest powers of common primes 2 and 3).

Q30. Find the least number of square tiles needed to pave a floor 15 m 17 cm long and 9 m 2 cm broad. 3M

> **Ans.** In cm: 1517 and 902. $1517=37\times41, 902=2\times11\times41$. HCF $=41$ cm (largest tile). Number $=\frac{1517\times902}{41\times41}=37\times22=814$ tiles.

Q31. In a school, 390 boys and 312 girls are to be seated in rows so that each row has only boys or only girls, with the same number in every row. Find the minimum number of rows. 3M

> **Ans.** $390=2\times3\times5\times13, 312=2^3\times3\times13$. HCF $=78$ per row. Rows $=\frac{390}{78}+\frac{312}{78}=5+4=9$.

Q32. **Competency.** A shopkeeper packs 1260 pens and 1080 pencils into identical gift packs, each pack having the same number of pens and the same number of pencils, with nothing left over. (i) Maximum number of packs? (ii) Pens and pencils per pack? 3M

> **Ans.** (i) $1260=2^2\times3^2\times5\times7, 1080=2^3\times3^3\times5$; HCF $=2^2\times3^2\times5=180$ packs. (ii) $\frac{1260}{180}=7$ pens and $\frac{1080}{180}=6$ pencils per pack.

Q33. Explain why $3\times5\times7+7$ is a composite number. 2M

> **Ans.** $7(3\times5+1)=7\times16=112$; it has factors other than 1 and itself, so it is composite.

Q34. If LCM$(x,18)=36$ and HCF$(x,18)=2$, find x. 2M

> **Ans.** $x\times18=36\times2\Rightarrow x=\frac{72}{18}=4$.

Q35. **Competency.** Two neon signs flash every 15 s and 20 s, and a third every 25 s. They flash together at 8:00 p.m. (i) After how many seconds will they flash together again? (ii) How many times will they flash together between 8:00 p.m. and 8:30 p.m. (excluding 8:00)? 3M

> **Ans.** (i) LCM$(15,20,25)$: $15=3\times5, 20=2^2\times5, 25=5^2$; LCM $=2^2\times3\times5^2=300$ s (5 min). (ii) 30 min $=1800$ s; $\frac{1800}{300}=6$ times.

Q36. Find the least perfect square number divisible by each of 18, 24 and 30. (HOTS) 2M

> **Ans.** LCM $=2^3\times3^2\times5=360$. Make every power even: $2^4\times3^2\times5^2=3600$.

Case based (4 marks)

Q37. **Traffic lights.** At three crossings on a road, the lights change after every 48 s, 72 s and 108 s. They all change together at 7:00:00 a.m. (i) Write 108 as a product of primes. (1) (ii) After how many seconds will they next change together? At what time? (2) (iii) Find HCF(48, 72, 108). (1) 4M

> **Ans.** (i) $108=2^2\times3^3$. (ii) $48=2^4\times3, 72=2^3\times3^2$; LCM $=2^4\times3^3=432$ s = 7 min 12 s, so at 7:07:12 a.m. (iii) HCF $=2^2\times3=12$.

Ek nazar mein: Real Numbers

Type of sawal	Method	Marks
Prime factorise	ladder / tree, powers	1-2
HCF, LCM	smallest / greatest powers	2-3
One of HCF/LCM given	HCF x LCM = a x b	1-2
Word problem	greatest = HCF, next together = LCM	3
Prove irrational	contradiction, coprime	3-5

Last minute: 8 one-liners

1. FTA: factorisation into primes is unique (order aside).
2. HCF: smallest powers of common primes; LCM: greatest powers of all primes.
3. HCF x LCM = product: only for 2 numbers.
4. HCF of coprimes = 1; LCM of coprimes = product.
5. Ends in 0 needs both 2 and 5.
6. Irrational: not of the form $\frac{p}{q}$; non-terminating non-repeating.
7. Proof: assume, coprime, square, divide, contradiction.
8. Greatest + remainder: subtract then HCF; smallest + remainder: LCM then add.

Do you know? If a divides b, then HCF$(a,b)=a$ and LCM$(a,b)=b$.

Check trick LCM is always a multiple of HCF; if not, recheck!

Most asked: zeroes + verify relation, form quadratic, $\alpha^2+\beta^2$ type

POLYNOMIALS

1. Polynomial Basics

A **polynomial** in x has only whole-number powers of x, e.g. $3x^2 - 5x + 2$. The highest power is its **degree**.

Degree	Name	General form	Max zeroes
1	Linear	$ax + b$	1
2	Quadratic	$ax^2 + bx + c$	2
3	Cubic	$ax^3 + bx^2 + cx + d$	3

↳ $a \neq 0$ hamesha! Warna degree hi badal jayegi.

Key terms

Value of $p(x)$ at $x = k$: put k in place of x, i.e. $p(k)$.

Zero of $p(x)$: a number k such that $p(k) = 0$. A polynomial of degree n has **at most** n zeroes.

$\sqrt{x}$, $\frac{1}{x}$, x^{-2} terms make an expression **NOT** a polynomial.

2. Geometrical Meaning of Zeroes

Zeroes of $p(x)$ are the **x-coordinates of the points where the graph of $y = p(x)$ meets the x-axis**.

→ Linear $y = ax + b$: a straight line, cuts x-axis at exactly **one** point $\left(-\frac{b}{a}, 0\right)$.

→ Quadratic: a **parabola**; it may cut the x-axis at 2 points, touch at 1 point, or not meet it (0 zeroes).

→ Cubic: meets the x-axis at most 3 times.

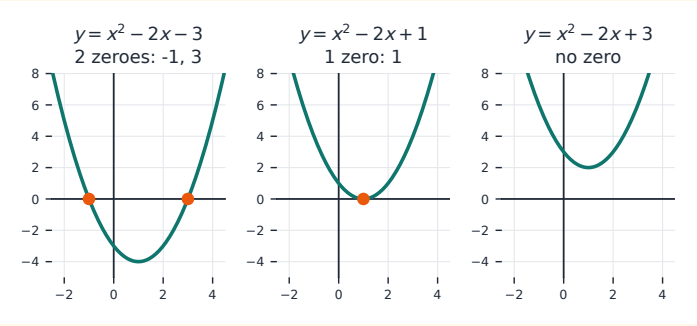


Fig: 2 zeroes, 1 zero, no zero (count the dots on the x-axis)

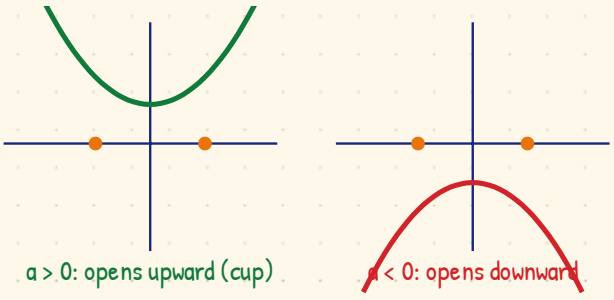


Fig: shape of $y = ax^2 + bx + c$ depends on the sign of a; orange dots = zeroes

Yaad karo: "Graph se zero = cut ginno"

Graph ka sawal aaye to sirf dekho: curve x-axis ko **kitni baar kaata/chhua**. Utne hi zeroes. y-axis wale cuts ginna galti hai!

3. Zeroes and Coefficients

Formula Box: Polynomials

→ Quadratic $ax^2 + bx + c$, zeroes α, β:

$$\alpha + \beta = -\frac{b}{a}, \qquad \alpha\beta = \frac{c}{a}$$

→ Quadratic from zeroes: $k\left[x^2 - (\alpha + \beta)x + \alpha\beta\right]$

→ Cubic $ax^3 + bx^2 + cx + d$, zeroes α, β, γ: $\alpha + \beta + \gamma = -\frac{b}{a}$

→ $\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$, $\alpha\beta\gamma = -\frac{d}{a}$

→ $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

→ $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha+\beta}{\alpha\beta}$, $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

→ $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

Yaad karo: "Sum mein Minus, Product mein Plus"

Quadratic: Sum $= -\frac{b}{a}$ (minus wala), Product $= +\frac{c}{a}$. Cubic mein signs alternate: $-\frac{b}{a}$, $+\frac{c}{a}$, $-\frac{d}{a}$ ("minus, plus, minus" = MPM).

Solved Example 1

Find the zeroes of $6x^2 - 7x - 3$ and verify the relationship between zeroes and coefficients.

1. Split middle term: $6x^2 - 9x + 2x - 3 = 3x(2x - 3) + 1(2x - 3) = (2x - 3)(3x + 1)$
2. Zeroes: $x = \frac{3}{2}$ and $x = -\frac{1}{3}$
3. Sum $= \frac{3}{2} - \frac{1}{3} = \frac{7}{6} = -\frac{(-7)}{6} = -\frac{b}{a}$ ✔
4. Product $= \frac{3}{2} \times \left(-\frac{1}{3}\right) = -\frac{1}{2} = \frac{-3}{6} = \frac{c}{a}$ ✔

✔ Zeroes $\frac{3}{2}, -\frac{1}{3}$; relation verified

Quick fact A constant polynomial like 7 has degree 0; the zero polynomial has no defined degree.

Yaad rakho If $c = 0$, one zero of $ax^2 + bx + c$ is always 0.

Solved Example 2

Find the zeroes of $4u^2 + 8u$ and of $t^2 - 15$.

1. $4u^2 + 8u = 4u(u + 2)$, zeroes $0, -2$ (sum $-2 = -\frac{8}{4}$, product $0 = \frac{0}{4}$)
2. $t^2 - 15 = (t - \sqrt{15})(t + \sqrt{15})$, zeroes $\pm\sqrt{15}$ (sum 0, product -15)

✔ $0, -2$ and $\sqrt{15}, -\sqrt{15}$

Solved Example 3

Find a quadratic polynomial whose sum and product of zeroes are $\frac{1}{4}$ and -1.

1. $p(x) = k\left[x^2 - \frac{1}{4}x - 1\right]$
2. Take $k = 4$ to clear the fraction

✔ $4x^2 - x - 4$

Meme samjho

Zeroes nikalne ki zarurat hi nahi agar $\alpha^2 + \beta^2$ poocha hai, seedha $S^2 - 2P$ laga do!

Asli baat: For $x^2 - 5x + 6$: $\alpha^2 + \beta^2 = (5)^2 - 2(6) = 13$, no need to find $\alpha = 2, \beta = 3$ first.

Solved Example 4

If α, β are the zeroes of $x^2 - 5x + 6$, find $\alpha^2 + \beta^2$ and $\frac{1}{\alpha} + \frac{1}{\beta}$.

1. $\alpha + \beta = 5, \alpha\beta = 6$
2. $\alpha^2 + \beta^2 = 25 - 12 = 13$
3. $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{5}{6}$

✔ 13 and $\frac{5}{6}$

Solved Example 5

If -3 is a zero of $p(x) = (k-1)x^2 + kx + 1$, find k.

1. $p(-3) = 9(k-1) - 3k + 1 = 0$
2. $9k - 9 - 3k + 1 = 0 \Rightarrow 6k = 8$

✔ $k = \frac{4}{3}$

Shivam Bhaiya says

Verify wale sawal mein 3 kaam: zeroes nikalo, **sum check**, **product check**. Teeno likhoge tabhi poore 3 marks. Aur sign ka dhyan: $-\frac{b}{a}$ mein b ka apna sign bhi lagta hai.

Recall If a and c have opposite signs, the zeroes have opposite signs (product $\frac{c}{a} < 0$).

Solved Example 6

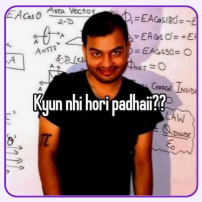
Verify that $1, -1, 3$ are the zeroes of $x^3 - 3x^2 - x + 3$ and check the relations.

1. $p(1) = 1 - 3 - 1 + 3 = 0, p(-1) = -1 - 3 + 1 + 3 = 0, p(3) = 27 - 27 - 3 + 3 = 0$
2. Sum $= 3 = -\frac{-3}{1}$; pairs $= -1 - 3 + 3 = -1 = \frac{c}{a}$
3. Product $= -3 = -\frac{d}{a} = -\frac{3}{1}$

✔ All three relations hold

Common mistakes

- → Sum of zeroes written as $\frac{b}{a}$ instead of $-\frac{b}{a}$.
- → For $4u^2 + 8u$, forgetting the zero $u = 0$ (here $c = 0$).
- → Counting y-axis cuts as zeroes on a graph.
- → Writing the polynomial as $x^2 + Sx + P$ (correct: $x^2 - Sx + P$).
- → Calling $x^2 + \frac{1}{x}$ a polynomial.



Meme samjho

$-\frac{b}{a}$ ka minus bhool ke poochte ho: **Kyun nahi ho rahi padhai??**

Asli baat: Sum of zeroes of $ax^2 + bx + c$ is $-\frac{b}{a}$. For $2x^2 - 5x + 3$ it is $+\frac{5}{2}$, not $-\frac{5}{2}$.

Yaad karo: "Naye zeroes? S aur P naye banao"

Agar naye zeroes $\frac{1}{\alpha}, \frac{1}{\beta}$ ya $2\alpha + 1, 2\beta + 1$ type hon: pehle **naya S** aur **naya P** nikalo (purane S, P se), phir $x^2 - Sx + P$. Zeroes nikalna zaroori nahi.

Yaad karo: "Polynomial banana = x² minus S x plus P"

Zeroes diye hain? Bas yaad rakho **"x square, minus Sum x, plus Product"**. Fraction aaye to poore ko LCM (k) se multiply kar do.

4. Reading Zeroes from a Graph

For a graph question, mark every point where the curve **cuts or touches** the x-axis. Their x-coordinates are the zeroes; their count is the number of zeroes.

Check trick Zeroes found? Put them back: $p(\alpha)$ must be 0.

Do you know? The graph of a quadratic polynomial is called a parabola.

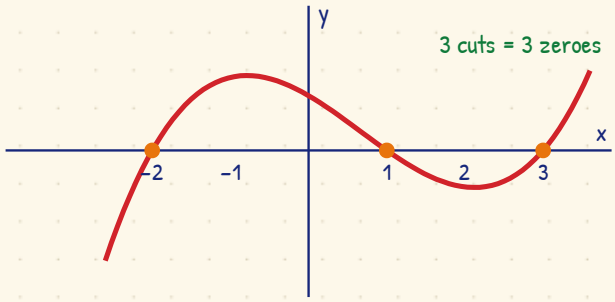


Fig: a cubic meeting the x-axis at $x = -2, 1, 3$

Here $p(x)$ has zeroes $-2, 1, 3$, so $p(x) = k(x+2)(x-1)(x-3)$ for some $k \neq 0$.

Solved Example 7

Find a quadratic polynomial, the sum and product of whose zeroes are $\sqrt{2}$ and $\frac{1}{3}$.

1. $p(x) = k\left[x^2 - \sqrt{2}\,x + \frac{1}{3}\right]$
2. Multiply by $k = 3$

✔ $3x^2 - 3\sqrt{2}\,x + 1$

Solved Example 8

If α, β are zeroes of $x^2 - 7x + 10$, find $\alpha - \beta$ (given $\alpha > \beta$) and $\alpha^3 + \beta^3$.

1. $S = 7, P = 10$; $(\alpha - \beta)^2 = 49 - 40 = 9 \Rightarrow \alpha - \beta = 3$
2. $\alpha^3 + \beta^3 = S^3 - 3PS = 343 - 210 = 133$
3. Check: zeroes 5, 2: $5 - 2 = 3$, $125 + 8 = 133$ ✔

✔ 3 and 133

Ek nazar mein: Polynomials

Asked	Do this
zeroes of quadratic	split middle term / $a^2 - b^2$
verify relation	check sum $-\frac{b}{a}$ and product $\frac{c}{a}$
polynomial from S, P	$k[x^2 - Sx + P]$
$\alpha^2 + \beta^2$, $\frac{1}{\alpha} + \frac{1}{\beta}$	use identities, not the zeroes
one zero given, find k	put it in: $p(\text{zero}) = 0$
graph given	count x-axis meeting points

5. Extra Practice: Board + Competency

Solved Example 9

Find the zeroes of $x^2 + 7x + 10$ and verify the relationship with coefficients.

1. $x^2 + 5x + 2x + 10 = (x+5)(x+2)$, zeroes $-5, -2$
2. Sum $= -7 = -\frac{7}{1}$ ✔; product $= 10 = \frac{10}{1}$ ✔

✔ Zeroes $-5, -2$

Solved Example 10

If α, β are zeroes of $3x^2 + 5x - 2$, find $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$.

1. $S = -\frac{5}{3}, P = -\frac{2}{3}$
2. $\alpha^2 + \beta^2 = S^2 - 2P = \frac{25}{9} + \frac{4}{3} = \frac{37}{9}$
3. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{37}{9} \div \left(-\frac{2}{3}\right) = -\frac{37}{6}$

✔ $-\frac{37}{6}$

Solved Example 11

Find a cubic polynomial whose zeroes are 1, 2 and 3.

1. Sum $= 6$, sum of products in pairs $= 2 + 6 + 3 = 11$, product $= 6$
2. $p(x) = x^3 - 6x^2 + 11x - 6$ (any non-zero multiple also works)

✔ $x^3 - 6x^2 + 11x - 6$

Shivam Bhaiya says

Polynomials ke 6-8 marks pakke hain: zeroes + verify, polynomial banana, aur $\alpha^2 + \beta^2$ type. Ye teen type 10-10 baar practice kar lo, chapter khatam!

Yaad karo: "Cubic ke 3 relations = S, P-pairs, P"

Cubic mein yaad rakho **"Sum, Pair-sum, Product"** = $-\frac{b}{a}, +\frac{c}{a}, -\frac{d}{a}$. Coefficients b, c, d order mein, signs minus-plus-minus.

MOST IMPORTANT QUESTIONS

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. The graph of $y = p(x)$ cuts the x-axis at 3 points and touches it nowhere else. The number of zeroes of $p(x)$ is [1M]

(a) 1 (b) 2 (c) 3 (d) 0

Ans. (c) 3

Q2. The zeroes of $x^2 - 2x - 8$ are [1M]

(a) $4, -2$ (b) $-4, 2$ (c) $8, -1$ (d) $2, 4$

Ans. (a) $(x-4)(x+2)$ gives 4 and -2.

Q3. If α, β are zeroes of $x^2 + 5x + 6$, then $\alpha + \beta + \alpha\beta =$ [1M] [Board fav]

(a) -1 (b) 1 (c) 11 (d) -11

Ans. (b) $-5 + 6 = 1$

Q4. A quadratic polynomial whose zeroes are -3 and 4 is [1M]

(a) $x^2 + x - 12$ (b) $x^2 - x + 12$ (c) $x^2 + x + 12$ (d) $x^2 - x - 12$

(contd.) (d) Sum 1, product -12: $x^2 - x - 12$.

Q5. If one zero of $x^2 - 3x + k$ is 1, the other zero is 1M
(a) 1 (b) 2 (c) 3 (d) -2

Ans. (b) Sum $= 3$, so other $= 2$ (and $k = 2$).

Q6. The polynomial with sum of zeroes 0 and product -5 is 1M
(a) $x^2 - 5$ (b) $x^2 + 5$ (c) $x^2 - 5x$ (d) $x^2 + 5x$

Ans. (a) $x^2 - 0x - 5$

Q7. A: $x^2 + 2$ has no real zero. R: Zeroes of $p(x)$ are the x-coordinates of the points where $y = p(x)$ meets the x-axis. 1M
(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true

Ans. (a) Both true; the parabola $y = x^2 + 2$ lies above the x-axis, so no meeting point.

Q8. A: If the sum of zeroes of $x^2 - kx + 6$ is 3, then $k = 3$. R: Sum of zeroes of $ax^2 + bx + c$ is $-\frac{b}{a}$. 1M
(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true

Ans. (a) Sum $= -\frac{-k}{1} = k = 3$; R explains A.

Q9. If one zero of $x^2 - 3x + k$ is 2, then $k =$ 1M
(a) 2 (b) -2 (c) 10 (d) -10

Ans. (a) $4 - 6 + k = 0 \Rightarrow k = 2$.

Q10. If $p(x) = x^2 + 5x + 6$, then $p(-2) + p(-3) =$ 1M
(a) 0 (b) 6 (c) 12 (d) -5

Ans. (a) Both are zeroes: $0 + 0 = 0$.

Very short (2 marks)

Q11. Find the zeroes of $4x^2 - 4x + 1$ and verify the relation with coefficients. 2M

Ans. $(2x - 1)^2$: zeroes $\frac{1}{2}, \frac{1}{2}$. Sum $1 = -\frac{-4}{4}$, product $\frac{1}{4} = \frac{1}{4}$ ✔

Q12. Find a quadratic polynomial whose zeroes are $2 + \sqrt{3}$ and $2 - \sqrt{3}$. 2M Board fav

Ans. Sum $= 4$, product $= 4 - 3 = 1$. Polynomial: $x^2 - 4x + 1$.

Q13. If α, β are zeroes of $2x^2 - 5x + 3$, find $\alpha^2\beta + \alpha\beta^2$. 2M

Ans. $\alpha\beta(\alpha + \beta) = \frac{3}{2} \times \frac{5}{2} = \frac{15}{4}$.

Q14. If -2 is a zero of $x^2 + kx - 6$, find k. 2M

Ans. $4 - 2k - 6 = 0 \Rightarrow k = -1$.

Short answer (3 marks)

Q15. Find the zeroes of $3x^2 - x - 4$ and verify the relationship between zeroes and coefficients. 3M Board fav

Ans. $3x^2 - 4x + 3x - 4 = (3x - 4)(x + 1)$: zeroes $\frac{4}{3}, -1$. Sum $= \frac{1}{3} = -\frac{(-1)}{3}$, product $= -\frac{4}{3} = \frac{c}{a}$ ✔

Q16. If α, β are zeroes of $x^2 - 6x + 4$, find (i) $\alpha^2 + \beta^2$ (ii) $\frac{1}{\alpha} + \frac{1}{\beta}$ (iii) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$. 3M

Ans. $S = 6, P = 4$. (i) $36 - 8 = 28$ (ii) $\frac{6}{4} = \frac{3}{2}$ (iii) $\frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{28}{4} = 7$.

Q17. The zeroes of $x^2 + (a + 1)x + b$ are 2 and -3. Find a and b. 3M

Ans. Sum $= -1 = -(a + 1) \Rightarrow a = 0$. Product $= -6 = b$. So $a = 0, b = -6$.

Q18. If α, β are zeroes of $x^2 - x - 2$, find a quadratic polynomial whose zeroes are $2\alpha + 1$ and $2\beta + 1$. 3M

Ans. $x^2 - x - 2 = (x - 2)(x + 1)$: $\alpha = 2, \beta = -1$. New zeroes 5 and -1: sum 4, product -5. Polynomial $x^2 - 4x - 5$.

Long answer (5 marks)

Q19. Verify that $4, -2$ and $\frac{1}{2}$ are the zeroes of $2x^3 - 5x^2 - 14x + 8$. Also verify all three relationships between zeroes and coefficients. 5M

Ans. $p(4) = 128 - 80 - 56 + 8 = 0, p(-2) = -16 - 20 + 28 + 8 = 0, p(\frac{1}{2}) = \frac{1}{4} - \frac{5}{4} - 7 + 8 = 0$. Sum $= \frac{5}{2} = -\frac{-5}{2}$; pairs $= -8 - 1 + 2 = -7 = \frac{-14}{2}$; product $= -4 = -\frac{8}{2}$ ✔

Q20. If the sum of the squares of the zeroes of $x^2 - 8x + k$ is 40, find k. Hence find the zeroes. 5M

Ans. $\alpha^2 + \beta^2 = S^2 - 2P = 64 - 2k = 40 \Rightarrow k = 12$. $x^2 - 8x + 12 = (x - 2)(x - 6)$, zeroes 2 and 6 (check $4 + 36 = 40$ ✔).

More practice (board pattern)

Q21. The number of polynomials having zeroes -2 and 5 is 1M
(a) 1 (b) 2 (c) 3 (d) more than 3

Ans. (d) infinitely many: $k(x^2 - 3x - 10)$ for every $k \neq 0$.

Q22. If the product of zeroes of $ax^2 - 6x - 6$ is 4, then $a =$ 1M
(a) $-\frac{3}{2}$ (b) $\frac{3}{2}$ (c) $-\frac{2}{3}$ (d) $\frac{2}{3}$

Ans. (a) $\frac{-6}{a} = 4 \Rightarrow a = -\frac{3}{2}$.

Q23. If α, β are zeroes of $x^2 - p(x + 1) - c$, then $(\alpha + 1)(\beta + 1) =$ 1M
(a) $c - 1$ (b) $1 + c$ (c) $1 - c$ (d) $-c$

Ans. (c) $x^2 - px - (p + c)$: $\alpha\beta + \alpha + \beta + 1 = -(p + c) + p + 1 = 1 - c$.

Q24. Find the zeroes of $x^2 + \frac{1}{6}x - 2$. 2M

Ans. Multiply by 6: $6x^2 + x - 12 = (3x - 4)(2x + 3)$, zeroes $\frac{4}{3}, -\frac{3}{2}$ (sum $-\frac{1}{6}$ ✔).

Q25. If one zero of $2x^2 - 11x + k$ is 5, find k and the other zero. 2M

Ans. $50 - 55 + k = 0 \Rightarrow k = 5$. Sum $= \frac{11}{2}$, other zero $= \frac{11}{2} - 5 = \frac{1}{2}$.

Q26. If α, β are zeroes of $x^2 - 3x + 2$, find a quadratic polynomial whose zeroes are $\frac{1}{\alpha}$ and $\frac{1}{\beta}$. 3M

> **Ans.** $S = 3, P = 2$. New sum $= \frac{S}{P} = \frac{3}{2}$, new product $= \frac{1}{P} = \frac{1}{2}$. Polynomial $k[x^2 - \frac{3}{2}x + \frac{1}{2}]$, i.e. $2x^2 - 3x + 1$.

Q27. Which of the following is NOT a polynomial? 1M
(a) $x^2 - \sqrt{2}x$ (b) $\sqrt{x} + 3$ (c) 5 (d) $\frac{x^3}{2}$

> **Ans.** (b) $\sqrt{x}$ has power $\frac{1}{2}$, not a whole number.

Q28. If α, β are the zeroes of $x^2 + x - 12$, find $\frac{1}{\alpha} + \frac{1}{\beta}$ without finding the zeroes. Then check by finding them. 2M

> **Ans.** $S = -1, P = -12$: $\frac{S}{P} = \frac{-1}{-12} = \frac{1}{12}$. Zeroes $3, -4$: $\frac{1}{3} - \frac{1}{4} = \frac{1}{12}$ ✔

Q29. Find the zeroes of $x^2 - 3$ and verify the relationship with coefficients. 2M

> **Ans.** $(x - \sqrt{3})(x + \sqrt{3})$: zeroes $\pm\sqrt{3}$. Sum $0 = -\frac{0}{1}$, product $-3 = \frac{-3}{1}$ ✔

Q30. If the zeroes of $x^2 - kx + 16$ are equal, find the possible values of k and the zeroes. 3M

> **Ans.** Let both zeroes be α: $\alpha^2 = 16 \Rightarrow \alpha = \pm 4$; $k = 2\alpha = \pm 8$. $k = 8$: zeroes $4, 4$; $k = -8$: zeroes $-4, -4$.

Q31. If the graph of $y = p(x)$ is a parabola that just touches the x-axis at one point, the number of zeroes of $p(x)$ is 1M
(a) 0 (b) 1 (c) 2 (d) 3

> **Ans.** (b) 1 (two equal zeroes, counted as one point).

Q32. If one zero of $4x^2 - 8kx - 9$ is the negative of the other, then $k =$ 1M
(a) 0 (b) 1 (c) -1 (d) 2

> **Ans.** (a) Sum $= 0 = \frac{8k}{4} \Rightarrow k = 0$.

Q33. Find a quadratic polynomial whose zeroes are $3 + \sqrt{2}$ and $3 - \sqrt{2}$. 2M

> **Ans.** Sum $= 6$, product $= 9 - 2 = 7$: $x^2 - 6x + 7$.

Q34. If α, β are zeroes of $x^2 - 4x + 3$, find $\alpha^3 + \beta^3$ without finding the zeroes. 2M

> **Ans.** $S^3 - 3PS = 64 - 36 = 28$ (check: zeroes 1, 3: $1 + 27 = 28$ ✔).

Q35. If α, β are zeroes of $2x^2 + 5x + k$ such that $\alpha^2 + \beta^2 + \alpha\beta = \frac{21}{4}$, find k. 3M Board fav

> **Ans.** $\alpha^2 + \beta^2 + \alpha\beta = S^2 - P = \frac{25}{4} - \frac{k}{2} = \frac{21}{4} \Rightarrow \frac{k}{2} = 1 \Rightarrow k = 2$.

Q36. The zeroes of $x^2 + px + q$ are double the zeroes of $2x^2 - 5x - 3$. Find p and q. 3M

> **Ans.** $2x^2 - 5x - 3 = (2x + 1)(x - 3)$: zeroes $3, -\frac{1}{2}$. Doubled: $6, -1$. Polynomial $x^2 - 5x - 6$, so $p = -5, q = -6$.

Q38. Find the zeroes of $2x^2 - 7x + 3$ and verify the relation between zeroes and coefficients. 2M

> **Ans.** $(2x - 1)(x - 3)$: zeroes $\frac{1}{2}, 3$. Sum $\frac{7}{2} = -\frac{-7}{2}$, product $\frac{3}{2} = \frac{3}{2}$ ✔

Q37. Competency: water rocket. In a science fair, the height of a water rocket is modelled by $h(t) = -4t^2 + 16t$ (scaled units). (i) Find the zeroes of $h(t)$ and say what they mean. (ii) Find the sum and product of the zeroes from the coefficients. 3M

> **Ans.** (i) $-4t(t - 4) = 0$: $t = 0$ (launch) and $t = 4$ (lands back). (ii) Sum $= -\frac{16}{-4} = 4$, product $= \frac{0}{-4} = 0$ ✔

Q39. If the sum of the zeroes of $kx^2 + 2x + 3k$ equals their product, find k. 2M

> **Ans.** $-\frac{2}{k} = \frac{3k}{k} = 3 \Rightarrow k = -\frac{2}{3}$.

Q40. If α, β are zeroes of $x^2 - 5x + k$ and $\alpha - \beta = 1$, find k. 3M Board fav

> **Ans.** $(\alpha - \beta)^2 = S^2 - 4P \Rightarrow 1 = 25 - 4k \Rightarrow k = 6$ (zeroes 3 and 2 ✔).

Case based (4 marks)

Q41. Cricket shot. The height of a ball hit by Rohit follows $p(x) = -x^2 + 2x + 8$ (on a scaled graph). (i) Find the zeroes of $p(x)$. (2) (ii) Find the sum and product of the zeroes. (1) (iii) Does the graph open upward or downward? Why? (1) 4M

> **Ans.** (i) $-x^2 + 2x + 8 = 0 \Rightarrow x^2 - 2x - 8 = 0 \Rightarrow (x - 4)(x + 2) = 0$: $x = 4, -2$. (ii) Sum $= -\frac{2}{-1} = 2$, product $= \frac{8}{-1} = -8$. (iii) Downward, since $a = -1 < 0$.

Q42. Arch gate. The shape of an arch is given by $p(x) = -x^2 + 6x - 5$ (x in metres along the ground). (i) Find the points where the arch meets the ground. (2) (ii) Find the width of the arch at the ground. (1) (iii) Find the sum of the zeroes using coefficients. (1) 4M

> **Ans.** (i) $x^2 - 6x + 5 = 0 \Rightarrow (x - 1)(x - 5) = 0$: $x = 1$ and $x = 5$. (ii) Width $= 5 - 1 = 4$ m. (iii) Sum $= -\frac{6}{-1} = 6 = 1 + 5$ ✔

Last minute: 6 one-liners

1. Zero of $p(x)$: value of x where $p(x) = 0$.
2. Degree n: at most n zeroes.
3. Zeroes = x-coordinates where graph meets x-axis.
4. $\alpha + \beta = -\frac{b}{a}$, $\alpha\beta = \frac{c}{a}$.
5. Quadratic from zeroes: $x^2 - Sx + P$.
6. Parabola opens up if $a > 0$, down if $a < 0$.

Yaad rakho Zeroes of $(x - a)(x - b)$ are a and b: just change the signs.

Recall A linear polynomial $ax + b$ has exactly one zero, $-\frac{b}{a}$.

Apne notes / galtiyan yahan likho

Most asked: ratio test for k, elimination, word problems (age, digits, fare)

Pair of Linear Equations

Chapter at a glance

- ✓ Pair of lines: meet once, never (parallel) or everywhere (coincident).
- ✓ Ratio test decides it without solving.
- ✓ Methods in syllabus: graphical, substitution, elimination.
- ✓ Exam favourites: value of k, elimination and word problems.

Formula Box

- → General pair: $a_1x + b_1y + c_1 = 0,\ a_2x + b_2y + c_2 = 0$
- → Unique solution: $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ (intersecting)
- → Infinitely many: $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ (coincident)
- → No solution: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$ (parallel)
- → Consistent = unique or infinitely many; inconsistent = none
- → Two-digit number $10x + y$; reversed $10y + x$
- → Relative speed: same direction $u - v$, opposite $u + v$
- → Total cost = fixed + rate $\times$ quantity

1. Linear Equation in Two Variables

An equation of the form $ax + by + c = 0$, where a, b, c are real numbers and a, b **are not both zero**, is a linear equation in two variables x and y.

- → Its graph is always a **straight line**. Every point on the line is a solution, so one equation has infinitely many solutions.
- → Two such equations in the same two variables = a **pair of linear equations**:

$$a_1x + b_1y + c_1 = 0, \qquad a_2x + b_2y + c_2 = 0$$

Key terms

Solution of the pair: values of x and y that satisfy both equations together. On the graph it is the point common to both lines.

Consistent: has at least one solution. **Inconsistent:** has no solution. **Dependent:** consistent with infinitely many solutions (lines coincide).

↳ Check a solution: put x, y in BOTH equations. Ek mein bhi fail = solution nahi.

2. Graphical Method

1. For each equation make a table of **2-3 points** (put $x = 0$, then $y = 0$, then one more value).
2. Plot both lines on the **same axes** with the same scale.
3. Read the picture: lines **intersect** = unique solution (the meeting point); **parallel** = no solution; **coincident** = infinitely many.

Solved Example 1

Solve graphically: $2x + y = 8$ and $x - y = 1$.

$2x + y = 8$	$(0, 8)$	$(4, 0)$	$(2, 4)$
$x - y = 1$	$(0, -1)$	$(1, 0)$	$(4, 3)$

1. Plot both lines (figure). They meet at $(3, 2)$.
2. Check: $2(3) + 2 = 8$ and $3 - 2 = 1$.

✓ $x = 3,\ y = 2$ **(unique solution, consistent)**

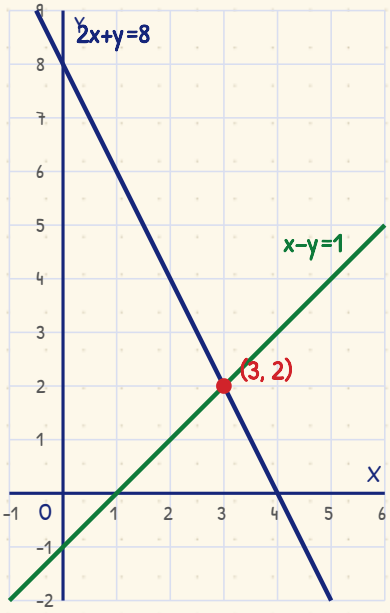


Fig 3.1: $2x + y = 8$ and $x - y = 1$ meet at $(3, 2)$

Quick fact Graph of $ax + by + c = 0$ is always a straight line.

Yaad rakho Consistent = at least one solution; inconsistent = none.

Do you know? $y = 0$ is the x-axis and $x = 0$ is the y-axis.

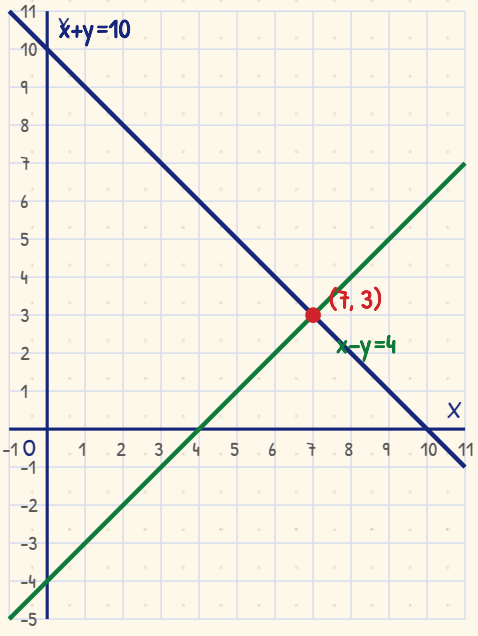


Fig 3.2: quiz problem, lines meet at $(7, 3)$

Solved Example 2

10 students took part in a maths quiz. The number of girls is 4 more than the number of boys. Find the numbers graphically.

1. Girls x, boys y: $x + y = 10$ and $x - y = 4$.
2. $x + y = 10$: $(0, 10), (10, 0), (5, 5)$. $x - y = 4$: $(4, 0), (0, -4), (6, 2)$.
3. The lines meet at $(7, 3)$ (Fig 3.2).

✔ 7 girls and 3 boys

Solved Example 3

Show graphically that $x + 2y = 4$ and $2x + 4y = 12$ have no solution.

1. $x + 2y = 4$: $(0, 2), (4, 0), (2, 1)$.
2. $2x + 4y = 12$: $(0, 3), (6, 0), (2, 2)$.
3. Lines are parallel (Fig 3.3): no common point. Ratio check: $\frac{1}{2} = \frac{2}{4} \neq \frac{4}{12}$.

✔ No solution: the pair is inconsistent

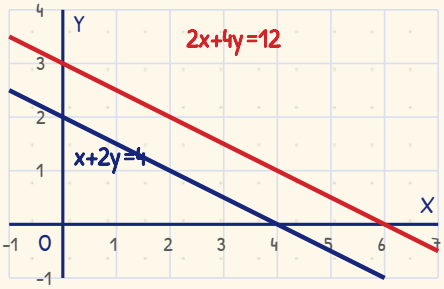


Fig 3.3: parallel lines, same slope, never meet

Exam tip Always verify (x, y) in both equations.

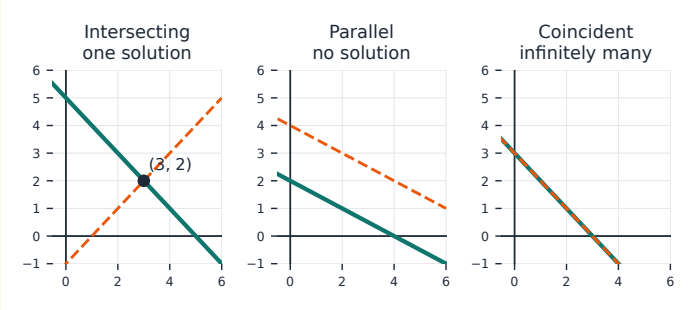


Fig 3.4: intersecting, parallel and coincident lines

3. Ratio Test (no graph needed)

Write both equations in the **same form**, then compare the ratios of coefficients:

Ratios	Lines	Solutions	System
$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$	Intersecting	Exactly one	Consistent
$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$	Coincident	Infinitely many	Consistent (dependent)
$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$	Parallel	No solution	Inconsistent

Yaad karo: "Alag, Sab Same, Bas c alag"

Alag (a ratio $\neq b$ ratio) = lines alag direction, ek point pe milti hain (1 solution). **Sab same** (all 3 ratios equal) = ek hi line, infinite. **Bas c alag** = parallel, kabhi nahi milti (0).

✓ $x = a$ is a line parallel to the **y-axis**; $y = b$ is parallel to the **x-axis**; they always meet at (a, b).

✓ $y = mx$ always passes through the origin. $x = 0$ is the y-axis itself; $y = 0$ is the x-axis.

Solved Example 4

Find k so that $kx + 4y = k - 4$ and $16x + ky = k$ have infinitely many solutions.

1. Need $\frac{k}{16} = \frac{4}{k} = \frac{k-4}{k}$.
2. First two: $k^2 = 64 \Rightarrow k = 8$ or $k = -8$.
3. Last two: $4 = k - 4 \Rightarrow k = 8$.
4. Check $k = 8$: $\frac{8}{16} = \frac{4}{8} = \frac{4}{8} = \frac{1}{2}$. ($k = -8$ fails: $-\frac{1}{2} \neq \frac{3}{2}$.)

✔ $k = 8$

k-questions in one line

Unique: use only $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$. **No solution:** $\frac{a_1}{a_2} = \frac{b_1}{b_2}$ gives k, then check $\neq \frac{c_1}{c_2}$. **Infinitely many:** all three equal; keep only the k that satisfies every ratio.

Solved Example 5

Find a and b for which $3x + 4y = 12$ and $(a+b)x + 2(a-b)y = 5a - 1$ have infinitely many solutions.

1. Need $\frac{3}{a+b} = \frac{4}{2(a-b)} = \frac{12}{5a-1}$.
2. First two: $6(a-b) = 4(a+b) \Rightarrow a = 5b$.
3. Last two: $4(5a-1) = 24(a-b) \Rightarrow -4a + 24b = 4$.
4. Put $a = 5b$: $4b = 4 \Rightarrow b = 1,\ a = 5$. Check: $\frac{3}{6} = \frac{4}{8} = \frac{12}{24}$ ✓

✔ $a = 5,\ b = 1$

Meme samjho

Do parallel lines ek doosre ko: "Hum kabhi milenge?" Ratio test: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$. "Acha!? Kabhi nahi."

Asli baat: parallel lines have the same slope but different intercepts, so there is no common point: the pair is inconsistent.

4. Substitution Method

1. From one equation, express one variable in terms of the other (pick the one with coefficient 1 or -1).
2. Substitute it in the **other** equation: you get an equation in one variable. Solve it.
3. Put this value back in step 1 to get the second variable. **Verify** in both.

Solved Example 6

Solve by substitution: $3x - y = 7$ and $2x + 3y = 1$.

1. From the first: $y = 3x - 7$.
2. Put in the second: $2x + 3(3x-7) = 1 \Rightarrow 11x - 21 = 1 \Rightarrow x = 2$.
3. $y = 3(2) - 7 = -1$. Check: $2(2) + 3(-1) = 1$ ✓

✔ $x = 2,\ y = -1$

↳ Substitution ke baad agar $0 = 0$ aaye to infinitely many solutions; agar $0 = 5$ jaisa false statement aaye to no solution.

5. Elimination Method

1. Multiply the equations by suitable numbers so that the coefficients of one variable become **equal** (or opposite).
2. **Subtract** (same signs) or **add** (opposite signs) to eliminate that variable.
3. Solve for the other variable, substitute back, verify.

Solved Example 7

Solve by elimination: $5x + 2y = 16$ and $3x - 4y = 20$.

1. Multiply the first by 2: $10x + 4y = 32$.
2. Add to $3x - 4y = 20$: $13x = 52 \Rightarrow x = 4$.
3. $5(4) + 2y = 16 \Rightarrow y = -2$. Check: $12 + 8 = 20$ ✓

✔ $x = 4,\ y = -2$

Equations with fractions or decimals

First **clear the fractions** (multiply by the LCM) or decimals (multiply by 10 or 100), then use any method.

Solved Example 8

Solve: $\frac{x}{3} + \frac{y}{4} = 4$ and $\frac{x}{2} - \frac{y}{4} = 1$.

1. Multiply by 12 and by 4: $4x + 3y = 48$ and $2x - y = 4$.
2. From the second, $y = 2x - 4$. Then $4x + 6x - 12 = 48 \Rightarrow x = 6$.
3. $y = 2(6) - 4 = 8$. Check: $2 + 2 = 4$ ✓, $3 - 2 = 1$ ✓

✔ $x = 6,\ y = 8$

Solved Example 9

Solve: $0.3x + 0.2y = 1.3$ and $0.5x + 0.4y = 2.3$.

1. Multiply by 10: $3x + 2y = 13$ and $5x + 4y = 23$.
2. Double the first: $6x + 4y = 26$. Subtract the second: $x = 3$.
3. $9 + 2y = 13 \Rightarrow y = 2$.

✔ $x = 3,\ y = 2$

Clear fractions, same form → Ratio test: 1, 0 or infinite? → Solve + verify

Yaad karo: "Same Sign Subtract"

SSS: coefficients ka sign **same** hai to **subtract**, opposite hai to **add**. Aur hamesha us variable ko udaao jiska LCM chhota ho: kam multiplication, kam galti.

Shivam Bhaiya says

Question mein method likha hai ("by elimination", "by substitution") to **wahi method** use karo, warna marks kat sakte hain. Method nahi likha to elimination sabse fast hai. Last step mein dono equations mein **verify** zaroor karo.

6. Word Problems: Turn Words into 2 Equations

Let the two unknowns be x and y, write **two equations** from the two conditions, solve, and answer in words with units.

Type	How to write it
Two-digit number	Number $= 10x + y$; reversed $= 10y + x$ (tens digit x, units y)
Fraction	$\frac{x}{y}$; "1 added to both" gives $\frac{x+1}{y+1}$
Ages	n years ago: $x - n$; after n years: $x + n$ (for BOTH persons)
Fixed + per unit	Total $=$ fixed charge $+$ rate $\times$ quantity
Two speeds	Same direction: gap closes at $u - v$; opposite: at $u + v$
Angles	Supplementary: $x + y = 180^\circ$; triangle: sum 180°

Solved Example 10

3 pencils and 5 pens cost Rs 62; 5 pencils and 3 pens cost Rs 50. Find the cost of each.

1. $3x + 5y = 62$, $5x + 3y = 50$.
2. Add: $8x + 8y = 112 \Rightarrow x + y = 14$. Subtract: $2y - 2x = 12 \Rightarrow y - x = 6$.
3. So $y = 10$, $x = 4$.

✔ Pencil Rs 4, pen Rs 10

Yaad karo: "Ulte coefficients = jodo-ghatao"

Jab coefficients ulte-palte hon ($3x + 5y$, $5x + 3y$), seedha **add** aur **subtract** karo: $x + y$ aur $x - y$ turant mil jaate hain. Multiply karne ki zaroorat nahi.

Solved Example 11

A hostel charges a fixed amount plus a daily food cost. A student pays Rs 2560 for 22 days and Rs 3040 for 28 days. Find the fixed charge and the cost per day.

1. Let fixed $= x$, per day $= y$: $x + 22y = 2560$, $x + 28y = 3040$.
2. Subtract: $6y = 480 \Rightarrow y = 80$.
3. $x = 2560 - 22(80) = 800$.

✔ Fixed charge Rs 800; food Rs 80 per day

Yaad karo: "Pehle bhi, Baad mein bhi"

Age ke sawal mein "n saal pehle" ya "n saal baad" **dono logon** ki age pe lagta hai: $(x - 5)$ AND $(y - 5)$. Sirf ek pe lagaya to answer pakka galat.

Solved Example 12

Five years ago, A was 3 times as old as B. Ten years later, A will be twice as old as B. Find their present ages.

1. $x - 5 = 3(y - 5) \Rightarrow x - 3y = -10$.
2. $x + 10 = 2(y + 10) \Rightarrow x - 2y = 10$.
3. Subtract: $y = 20$, then $x = 50$. Check: $45 = 3(15)$, $60 = 2(30)$ ✓

✔ A = 50 years, B = 20 years

Solved Example 13

A two-digit number exceeds the number formed by reversing its digits by 18. The sum of its digits is 10. Find the number.

1. $(10x + y) - (10y + x) = 18 \Rightarrow x - y = 2$.
2. $x + y = 10$. Add: $2x = 12 \Rightarrow x = 6$, $y = 4$.

✔ The number is 64 (check: $64 - 46 = 18$)

Solved Example 14

In $\triangle ABC$, $\angle C = 2\angle B$ and $\angle A$ exceeds $\angle B$ by 20°. Find all three angles.

1. Let $\angle A = x$, $\angle B = y$, so $\angle C = 2y$.
2. Angle sum: $x + 3y = 180$; given: $x - y = 20$.
3. Subtract: $4y = 160 \Rightarrow y = 40$, $x = 60$, $\angle C = 80$.

✔ $\angle A = 60^\circ$, $\angle B = 40^\circ$, $\angle C = 80^\circ$

Solved Example 15

A boat goes 30 km downstream in 2 hours and 18 km upstream in 2 hours. Find the speed of the boat in still water and of the stream.

1. Boat u, stream v (km/h). Downstream: $2(u + v) = 30 \Rightarrow u + v = 15$.
2. Upstream: $2(u - v) = 18 \Rightarrow u - v = 9$.
3. Add: $u = 12$, then $v = 3$.

✔ Boat 12 km/h, stream 3 km/h

Yaad karo: "Down = Dost, Up = Ulta"

Downstream mein stream dost hai, speeds jud jaati hain: $u + v$. **Up**stream mein stream ulta dhakka deta hai: $u - v$. Same rule cars ke liye: aamne-saamne $u + v$, ek hi direction $u - v$.

Recall Cross-multiplication method is NOT in the 2026-27 syllabus.

Common mistakes

- → Taking c_1/c_2 with one equation in $ax+by+c=0$ form and the other in $ax+by=c$ form: the sign flips.
- → For "infinitely many" or "no solution" checking only $a_1/a_2=b_1/b_2$: the third ratio decides which k is valid.
- → In ages, applying "5 years ago" to only one person.
- → Writing the reversed number as yx instead of $10y+x$.
- → Graph: fewer than 2 points per line, unlabelled axes, no scale.

Meme samjho

Solve karte karte aaya $0=0$. "Hain? Ye kaise ho gaya?"

Asli baat: a true statement like $0=0$ means the two equations are the same line: **infinitely many solutions** (coincident). A false one like $0=7$ means parallel lines: no solution.

7. Board Exam Strategy

Question type	Marks	What examiners want
Nature of pair / value of k	1-2	Ratios written, condition stated, final k
Solve by a given method	2-3	Same method, each step, verification
Graphical solution	3-5	Tables of 3 points, scale, labelled lines, point named
Word problem	3-5	"Let ... be ...", two equations, solve, answer in words + units
Case study	4	Form equations first; later parts use earlier answers

Shivam Bhaiya says

Word problem mein pehli line **"Let the ... be x and ... be y"** likhna mat bhoolna, iske bhi marks hain. Answer ke saath unit (km/h, Rs, years) likho. Ye chapter **full marks** wala hai, bas practice karo!

Last-minute checklist

- ✓ Both equations in the same form before taking ratios.
- ✓ Two variables = two conditions = two equations.
- ✓ Solution checked in BOTH equations.
- ✓ Graph: 3 points each, scale written, intersection labelled.
- ✓ Negative length, speed or age? Recheck the setup.

Practice yourself (answers at the end)

1. Solve: $3x+4y=10$, $2x-2y=2$.
2. Find k if $3x+y=1$ and $(2k-1)x+(k-1)y=2k+1$ have no solution.
3. Sum of two numbers is 35 and their difference is 13. Find them.
4. Is the pair $x-2y=5$, $3x-6y=15$ consistent?
5. 5 chairs and 2 tables cost Rs 6500; 3 chairs and 1 table cost Rs 3500. Find each cost.

↳Ans: (1) $x=2,\ y=1$ (2) $k=2$ (3) 24, 11 (4) yes, infinitely many (5) chair Rs 500, table Rs 2000

Extra practice: board pattern (answers below)

1. Solve: $2x+y=7$, $x-y=2$.
2. Solve: $3x+2y=11$, $2x+3y=4$.
3. Solve: $x+y=5$, $2x-3y=0$.
4. Nature: $6x-3y+10=0$, $2x-y+9=0$.
5. Nature: $5x-8y+1=0$, $3x-\frac{24}{5}y+\frac{3}{5}=0$.
6. Find k for a unique solution: $x+2y=3$, $5x+ky=7$.
7. Two numbers differ by 26 and one is 3 times the other. Find them.
8. Mother + daughter ages $=50$; 5 years ago mother was 7 times the daughter. Present ages?
9. A rectangle's length exceeds its breadth by 4 m; perimeter 40 m. Find the sides.
10. A library charges a fixed amount plus a daily charge: Rs 27 for 7 days, Rs 21 for 5 days. Find both.

↳Ans: (1) $(3,1)$ (2) $(5,-2)$ (3) $(3,2)$ (4) parallel, no solution (5) coincident, infinitely many (6) $k\neq 10$ (7) 39, 13 (8) 40, 10 (9) 12 m, 8 m (10) fixed Rs 6, Rs 3 per day

MOST IMPORTANT QUESTIONS

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. The pair $3x-5y=7$ and $-6x+10y=3$ has 1M Board fav

(a) a unique solution (b) no solution (c) infinitely many solutions (d) exactly two solutions

Ans. (b) no solution. $\frac{a_1}{a_2}=\frac{3}{-6}=-\frac{1}{2}$, $\frac{b_1}{b_2}=\frac{-5}{10}=-\frac{1}{2}$, $\frac{c_1}{c_2}=\frac{7}{3}$: parallel lines.

Q2. For what value of k do $2x+ky=5$ and $6x+12y=7$ have no solution? 1M

(a) 2 (b) 3 (c) 4 (d) 6

(contd.) (c) $k = 4$: $\frac{2}{6} = \frac{k}{12} \Rightarrow k = 4$, and $\frac{5}{7} \neq \frac{1}{3}$.

Q3. If the lines $3x + 2ky = 2$ and $2x + 5y + 1 = 0$ are parallel, then $k =$ 1M

(a) $\frac{15}{2}$ (b) $\frac{4}{15}$ (c) $\frac{15}{4}$ (d) $\frac{3}{2}$

Ans. (c) $\frac{3}{2} = \frac{2k}{5} \Rightarrow k = \frac{15}{4}$ (and $\frac{-2}{1} \neq \frac{3}{2}$).

Q4. If $x = 2,\ y = 3$ is a solution of $5x - py = 1$, then $p =$ 1M

(a) 2 (b) 3 (c) -3 (d) 9

Ans. (b) 3: $10 - 3p = 1 \Rightarrow p = 3$.

Q5. The lines $2x + y = 6$ and $2x - y + 2 = 0$ intersect at 1M

(a) $(1, 4)$ (b) $(4, 1)$ (c) $(2, 2)$ (d) $(-1, 8)$

Ans. (a) Add: $4x + 2 = 6 \Rightarrow x = 1, y = 4$.

Q6. The lines $x = 3$ and $y = -2$ intersect at 1M

(a) $(-2, 3)$ (b) $(2, -3)$ (c) $(-3, 2)$ (d) $(3, -2)$

Ans. (d) $(3, -2)$.

Q7. The pair $2x - 3y = 8$ and $4x - 6y = 9$ is 1M

(a) consistent with a unique solution (b) consistent with infinitely many solutions (c) inconsistent (d) none of these

Ans. (c) inconsistent: $\frac{2}{4} = \frac{-3}{-6} = \frac{1}{2}$ but $\frac{8}{9} \neq \frac{1}{2}$.

Q8. The pair $x = 0$ and $x = 5$ represents 1M

(a) intersecting lines (b) parallel lines (c) coincident lines (d) the two axes

Ans. (b) parallel lines: both are parallel to the y-axis, so no solution.

Q9. Which pair has the unique solution $x = 2,\ y = -3$? 1M Board fav

(a) $x + y = -1,\ 2x - 3y = 13$ (b) $x - y = 5,\ 2x + y = 3$ (c) $2x + y = 1,\ x - y = 4$ (d) $x + 2y = 4,\ x - y = 5$

Ans. (a): $2 - 3 = -1$ and $4 + 9 = 13$ ✓

Q10. **A:** The pair $x + 2y = 3$ and $2x + 4y = 6$ has infinitely many solutions. **R:** If $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$, the lines are coincident. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (a) All ratios are $\frac{1}{2}$, so the lines coincide.

Q11. **A:** The pair $2x + 3y = 5$ and $4x + 6y = 8$ is inconsistent. **R:** Two parallel lines have exactly one common point. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (c) Ratios $\frac{1}{2} = \frac{1}{2} \neq \frac{5}{8}$, so A is true; parallel lines have NO common point, so R is false.

Very short (2 marks)

Q12. For what value of k does the pair $kx - y = 2$ and $6x - 2y = 3$ have a unique solution? 2M

Ans. Unique: $\frac{k}{6} \neq \frac{-1}{-2} = \frac{1}{2} \Rightarrow k \neq 3$. So every real k except 3.

Q13. For what k do $2x + 3y = 4$ and $(k + 2)x + 6y = 3k + 2$ have infinitely many solutions? 2M

Ans. $\frac{2}{k+2} = \frac{3}{6} \Rightarrow k = 2$; $\frac{3}{6} = \frac{4}{3k+2} \Rightarrow 3k + 2 = 8 \Rightarrow k = 2$. So $k = 2$.

Q14. Is $(2, -1)$ a solution of the pair $3x + y = 5$ and $x - 2y = 4$? Justify. 2M

Ans. $3(2) + (-1) = 5$ ✓ and $2 - 2(-1) = 4$ ✓. Yes, it satisfies both equations.

Q15. Solve by elimination: $x + y = 14,\ x - y = 4$. 2M

Ans. Add: $2x = 18 \Rightarrow x = 9$; then $y = 5$.

Q16. Half the perimeter of a rectangular garden is 36 m and its length is 4 m more than its width. Find its dimensions. 2M

Ans. $l + w = 36,\ l - w = 4 \Rightarrow l = 20$ m, $w = 16$ m.

Q17. Write the pair for: "the sum of two numbers is 50 and one exceeds the other by 14", and solve it. 2M

Ans. $x + y = 50,\ x - y = 14 \Rightarrow x = 32,\ y = 18$.

Short answer (3 marks)

Q18. Solve $2x + 3y = 11$ and $2x - 4y = -24$, and hence find m for which $y = mx + 3$. 3M

Ans. Subtract: $7y = 35 \Rightarrow y = 5$; $2x = 11 - 15 \Rightarrow x = -2$. Then $5 = -2m + 3 \Rightarrow m = -1$.

Q19. Solve by substitution: $3x - 4y = 10$ and $x + 2y = 10$. 3M

Ans. $x = 10 - 2y$. Then $3(10 - 2y) - 4y = 10 \Rightarrow 30 - 10y = 10 \Rightarrow y = 2, x = 6$. Check: $18 - 8 = 10$ ✓

Q20. Check whether $x + 3y = 6$ and $2x - 3y = 12$ is consistent. If so, solve it. 3M

Ans. $\frac{1}{2} \neq \frac{3}{-3}$: unique solution, so consistent. Add: $3x = 18 \Rightarrow x = 6$; then $3y = 0 \Rightarrow y = 0$. Solution $(6, 0)$.

Q21. A fraction becomes $\frac{2}{3}$ if 1 is added to both numerator and denominator. If 1 is subtracted from both, it becomes $\frac{1}{2}$. Find the fraction. 3M

Ans. $3(x + 1) = 2(y + 1) \Rightarrow 3x - 2y = -1$; $2(x - 1) = y - 1 \Rightarrow y = 2x - 1$. So $3x - 4x + 2 = -1 \Rightarrow x = 3,\ y = 5$. Fraction $= \frac{3}{5}$.

Q22. 3 pens and 2 notebooks cost Rs 110, while 4 pens and 3 notebooks cost Rs 155. Find the cost of one pen and one notebook. 3M

Ans. $3p + 2n = 110, 4p + 3n = 155$. Multiply by 3 and 2: $9p + 6n = 330, 8p + 6n = 310$. Subtract: $p = 20$, then $n = 25$. Pen Rs 20, notebook Rs 25.

Q23. The larger of two supplementary angles exceeds the smaller by $18°$. Find the angles. 3M

Ans. $x + y = 180, x - y = 18 \Rightarrow 2x = 198$, so $x = 99°, y = 81°$.

Q24. Two places P and Q are 120 km apart. Two cars start from P and Q at the same time. Going in the same direction they meet in 6 hours; going towards each other they meet in 1.5 hours. Find their speeds. 3M

> **Ans.** Same direction: $6(u-v)=120 \Rightarrow u-v=20$. Opposite: $1.5(u+v)=120 \Rightarrow u+v=80$. So $u=50$ km/h, $v=30$ km/h.

Q25. The cost of 2 kg apples and 1 kg grapes is Rs 160; after a month, 4 kg apples and 2 kg grapes cost Rs 300. Represent algebraically and check the nature of the pair. 3M

> **Ans.** $2x+y=160$ and $4x+2y=300$. Ratios $\frac{2}{4}=\frac{1}{2}=\frac{1}{2}$ but $\frac{160}{300}=\frac{8}{15}$, not $\frac{1}{2}$. Lines are parallel: no solution (prices cannot be the same as before, the situation is inconsistent).

Q26. Five years hence, a man's age will be three times his son's age. Five years ago, he was seven times as old as his son. Find their present ages. 3M

> **Ans.** $x+5=3(y+5) \Rightarrow x-3y=10$. $x-5=7(y-5) \Rightarrow x-7y=-30$. Subtract: $4y=40 \Rightarrow y=10, x=40$. Man 40 years, son 10 years.

Q27. Seven times a two-digit number equals four times the number obtained by reversing its digits. The digits differ by 3. Find the number. 3M

> **Ans.** $7(10x+y)=4(10y+x) \Rightarrow 66x=33y \Rightarrow y=2x$. $y-x=3 \Rightarrow x=3,\ y=6$. Number 36 (check: $7\times 36=252=4\times 63$).

Long answer (5 marks)

Q28. Solve graphically: $x-y+2=0$ and $2x+y-8=0$. Also find the vertices of the triangle formed by these lines and the x-axis, and its area. 5M

> **Ans.** Points: $x-y+2=0$: $(0,2),(-2,0),(2,4)$; $2x+y-8=0$: $(0,8),(4,0),(2,4)$. Lines meet at $A(2,4)$, so $x=2,\ y=4$. They cut the x-axis at $B(-2,0)$ and $C(4,0)$. Base $BC=6$, height $=4$; area $=\frac{1}{2}\times 6\times 4=12$ sq units.

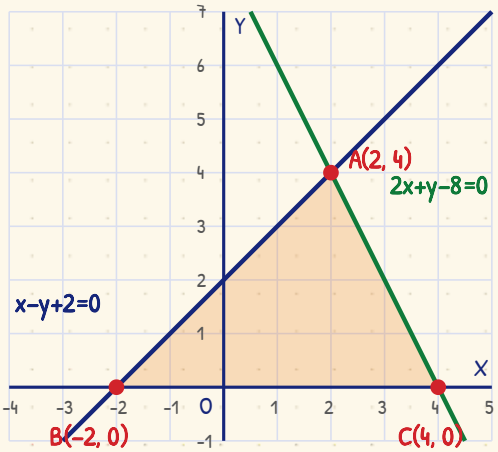


Fig 3.5: Q28, triangle ABC with the x-axis (shaded)

Q29. Two years ago a father was 5 times as old as his son. Two years later, the father's age will be 8 more than 3 times the son's age. Find their present ages. 5M

> **Ans.** $x-2=5(y-2) \Rightarrow x-5y=-8$. $x+2=3(y+2)+8 \Rightarrow x-3y=12$. Subtract: $2y=20 \Rightarrow y=10, x=42$. Father 42 years, son 10 years. Check: $40=5(8), 44=3(12)+8$ ✓

Q30. The sum of a two-digit number and the number obtained by reversing its digits is 110. The digits differ by 4. Find the number. 5M

> **Ans.** $(10x+y)+(10y+x)=110 \Rightarrow 11(x+y)=110 \Rightarrow x+y=10$. Digits differ by 4: $x-y=4$ or $y-x=4$. Case 1: $x=7,\ y=3$, number 73. Case 2: $x=3,\ y=7$, number 37. Both are valid: 73 or 37.

Q31. A train covers a certain distance at a uniform speed. If it were 10 km/h faster, it would take 2 hours less; if it were 10 km/h slower, it would take 4 hours more. Find the distance. 5M

> **Ans.** Speed x, time t, distance xt. $(x+10)(t-2)=xt \Rightarrow -2x+10t=20$. $(x-10)(t+4)=xt \Rightarrow 4x-10t=40$. Add: $2x=60 \Rightarrow x=30$; then $10t=80 \Rightarrow t=8$. Distance $=30\times 8=240$ km. Check: $40\times 6=240, 20\times 12=240$ ✓

Q32. The area of a rectangle decreases by 8 sq units if its length is reduced by 4 units and breadth increased by 2 units. It increases by 48 sq units if length is increased by 3 and breadth by 2. Find its length and breadth. 5M

> **Ans.** $(x-4)(y+2)=xy-8 \Rightarrow 2x-4y=0 \Rightarrow x=2y$. $(x+3)(y+2)=xy+48 \Rightarrow 2x+3y=42$. So $7y=42, y=6, x=12$. Length 12, breadth 6.

Case based (4 marks)

Q33. **City taxi:** A taxi charges a fixed amount plus a rate per km. For a 10 km ride the fare is Rs 105 and for a 15 km ride it is Rs 155. (i) Write the pair of equations. (1) (ii) Find the fixed charge and the rate per km. (2) (iii) How much will a 25 km ride cost? (1) 4M

> **Ans.** (i) $x+10y=105, x+15y=155$ (x = fixed, y = rate per km). (ii) Subtract: $5y=50 \Rightarrow y=10; x=5$. Fixed Rs 5, Rs 10 per km. (iii) $5+25(10)=$ Rs 255.

Q34. **Cinema tickets:** 1 adult and 2 child tickets cost Rs 400; 2 adult and 3 child tickets cost Rs 700. (i) Write the equations. (1) (ii) Find the price of each ticket. (2) (iii) What do 3 adult and 4 child tickets cost? (1) 4M

> **Ans.** (i) $a+2c=400, 2a+3c=700$. (ii) $a=400-2c$; $800-4c+3c=700 \Rightarrow c=100, a=200$. Adult Rs 200, child Rs 100. (iii) $600+400=$ Rs 1000.

Apne notes / galtiyan yahan likho

Most asked: discriminant + nature of roots, value of k, word problems

QUADRATIC EQUATIONS

Chapter at a glance

- ✓ $ax^2+bx+c=0$, $a\neq 0$: at most two roots.
- ✓ Solve by factorisation or the quadratic formula.
- ✓ $D=b^2-4ac$ tells the nature of the roots.
- ✓ Exam favourites: nature of roots, value of k and word problems.

Formula Box

- → Standard form: $ax^2+bx+c=0,\ a\neq 0$
- → Splitting: two numbers with product ac and sum b
- → Quadratic formula: $x=\dfrac{-b\pm\sqrt{b^2-4ac}}{2a}$
- → Discriminant: $D=b^2-4ac$
- → $D>0$: two distinct real roots; $D=0$: equal roots $-\frac{b}{2a}$; $D<0$: no real roots
- → Real roots: $D\geq 0$
- → Check (from polynomials): sum of roots $=-\frac{b}{a}$, product $=\frac{c}{a}$

1. What is a Quadratic Equation?

A quadratic equation in x is an equation of the form

$$ax^2+bx+c=0,\qquad a\neq 0$$

where a, b, c are real numbers. This is the **standard form**. Highest power of x = 2 (degree 2).

- → If $a=0$, the x^2 term disappears and it becomes linear, so the condition $a\neq 0$ is a must.
- → To check whether an equation is quadratic, **expand both sides first**, bring everything to one side, then look at the highest power. The x^2 terms may cancel!

Solved Example 1

Check whether these are quadratic: (i) $(x-2)^2+1=2x-3$ (ii) $x(x+1)+8=(x+2)(x-2)$ (iii) $(x+2)^3=x^3-4$

1. (i) $x^2-4x+5=2x-3\Rightarrow x^2-6x+8=0$: **quadratic**.
2. (ii) $x^2+x+8=x^2-4\Rightarrow x+12=0$: degree 1, **not quadratic**.
3. (iii) $x^3+6x^2+12x+8=x^3-4\Rightarrow 6x^2+12x+12=0$: **quadratic** ($x^3$ cancels).

✓ (i) yes (ii) no (iii) yes

Key terms

Root (zero): a real number α is a root of $ax^2+bx+c=0$ if $a\alpha^2+b\alpha+c=0$. The roots of the equation are the zeroes of the polynomial ax^2+bx+c.

A quadratic equation has at most two roots. Graphically, the roots are where the parabola $y=ax^2+bx+c$ cuts the x-axis.

Solved Example 2

If $x=3$ is a root of $2x^2+kx-15=0$, find k and the other root.

1. Put $x=3$: $18+3k-15=0\Rightarrow k=-1$.
2. Equation: $2x^2-x-15=0\Rightarrow (x-3)(2x+5)=0$.

✓ $k=-1$; other root $=-\frac{5}{2}$

2. Solving by Factorisation (Splitting the Middle Term)

1. Write in standard form $ax^2+bx+c=0$ and note a, b, c with signs.
2. Find two numbers whose **product** $=ac$ and **sum** $=b$.
3. Split bx into these two terms, group in pairs and take common factors.
4. Put each factor $=0$ to get the two roots.

Yaad karo: "PS: Product ac, Sum b"

Middle term todne ka mantra Product $=ac$, Sum $=b$. Product negative hai to dono numbers ke sign alag; product positive aur sum negative hai to dono negative.

Solved Example 3

Solve: $6x^2-x-2=0$.

1. $ac=-12$, $b=-1$: numbers -4 and 3.
2. $6x^2-4x+3x-2=2x(3x-2)+1(3x-2)$
3. $=(3x-2)(2x+1)=0$

✓ $x=\frac{2}{3}$ or $x=-\frac{1}{2}$

Quick fact A quadratic equation has at most 2 roots.

Yaad rakho Equal roots: each root $=-\frac{b}{2a}$.

Do you know? Completing the square is NOT in the 2026-27 syllabus.

Solved Example 4

Solve: $\sqrt{3}\,x^2 + 10x + 7\sqrt{3} = 0$.

1. $ac = \sqrt{3} \times 7\sqrt{3} = 21$, $b = 10$: numbers 7 and 3.
2. $\sqrt{3}x^2 + 3x + 7x + 7\sqrt{3} = \sqrt{3}x(x + \sqrt{3}) + 7(x + \sqrt{3})$
3. $= (x + \sqrt{3})(\sqrt{3}x + 7) = 0$

✔ $x = -\sqrt{3}$ or $x = -\frac{7}{\sqrt{3}} = -\frac{7\sqrt{3}}{3}$

Solved Example 5

Solve: $x^2 - (\sqrt{3}+1)x + \sqrt{3} = 0$.

1. $ac = \sqrt{3}$, $b = -(\sqrt{3}+1)$: numbers $-\sqrt{3}$ and -1.
2. $x^2 - \sqrt{3}x - x + \sqrt{3} = x(x - \sqrt{3}) - 1(x - \sqrt{3}) = (x-1)(x - \sqrt{3})$.

✔ $x = 1$ or $x = \sqrt{3}$

Solved Example 6

Solve: $49x^2 - 28x + 4 = 0$.

1. $ac = 196$, $b = -28$: numbers -14 and -14.
2. $49x^2 - 14x - 14x + 4 = 7x(7x-2) - 2(7x-2) = (7x-2)^2$

✔ $x = \frac{2}{7}, \frac{2}{7}$ (two equal roots)

Solved Example 7

Solve: $x - \dfrac{3}{x} = 2,\ x \neq 0$.

1. Multiply by x: $x^2 - 2x - 3 = 0$.
2. $(x-3)(x+1) = 0$.

✔ $x = 3$ or $x = -1$

Meme samjho

Student: "Ye middle term toot kyun nahi raha?" Bhaiya: "Do numbers dhundho jinka product ac aur sum b ho."

Asli baat: if no such pair of integers exists, don't waste time: use the quadratic formula. It works for every quadratic.

Use factorisation when	Use the formula when
ac has an easy factor pair adding to b	no integer pair works, or roots are irrational
question says "by factorisation"	question says "by quadratic formula"
coefficients are small integers	coefficients are big, or contain surds/decimals

Solved Example 8

Solve $2x^2 - 7x + 3 = 0$ by factorisation AND by the formula, and compare.

1. Factorisation: $ac = 6$, $b = -7$: split $-6x - x$. $2x(x-3) - 1(x-3) = (2x-1)(x-3)$.
2. Formula: $D = 49 - 24 = 25$, $x = \dfrac{7 \pm 5}{4} = 3$ or $\dfrac{1}{2}$.

✔ $x = 3,\ \frac{1}{2}$ by both methods

Solved Example 9

Solve: $\dfrac{1}{x+3} - \dfrac{1}{x-5} = \dfrac{2}{3},\ x \neq -3, 5$.

1. LHS $= \dfrac{(x-5)-(x+3)}{(x+3)(x-5)} = \dfrac{-8}{(x+3)(x-5)}$.
2. So $(x+3)(x-5) = -12 \Rightarrow x^2 - 2x - 3 = 0$.
3. $(x-3)(x+1) = 0$.

✔ $x = 3$ or $x = -1$

Find $D = b^2 - 4ac$ → $D < 0$: stop, no real roots → $D \geq 0$: factorise or formula

3. Quadratic Formula

For $ax^2 + bx + c = 0$ with $b^2 - 4ac \geq 0$, the roots are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

↳ Ye formula har quadratic pe chalta hai. Factor na bane to seedha formula lagao.

Yaad karo: formula ka gaana

"Minus b, plus-minus root of b square minus 4ac, sab upon 2a". Pehle D alag se nikaalo, phir formula mein daalo: kam galti.

Solved Example 10

Solve by the quadratic formula: $3x^2 - 5x + 1 = 0$.

1. $a = 3,\ b = -5,\ c = 1$; $D = 25 - 12 = 13$.
2. $x = \dfrac{5 \pm \sqrt{13}}{6}$

✔ $x = \dfrac{5+\sqrt{13}}{6}$ or $\dfrac{5-\sqrt{13}}{6}$

Recall Sum of roots $= -\frac{b}{a}$, product $= \frac{c}{a}$.

Exam tip Always reject negative length, speed or age.

Solved Example 11

Solve: $x^2 - 2x - 4 = 0$.

1. $D = 4 + 16 = 20$, $\sqrt{20} = 2\sqrt{5}$.
2. $x = \dfrac{2 \pm 2\sqrt{5}}{2} = 1 \pm \sqrt{5}$

✔ $x = 1 + \sqrt{5}$ or $1 - \sqrt{5}$

4. Discriminant and Nature of Roots

$D = b^2 - 4ac$ is the discriminant: it decides the nature of the roots without solving.

Value of D	Nature of roots	Graph
$D > 0$	Two distinct real roots $\frac{-b \pm \sqrt{D}}{2a}$	Cuts x-axis at 2 points
$D = 0$	Two equal real roots, each $-\frac{b}{2a}$	Touches x-axis at 1 point
$D < 0$	No real roots	Does not meet x-axis

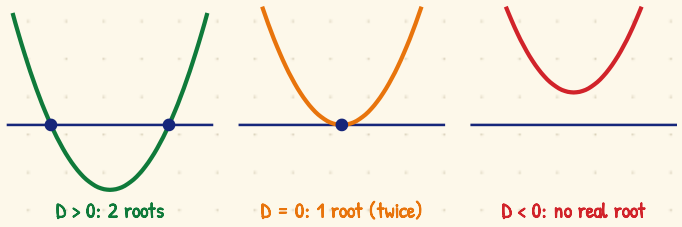


Fig 4.1: $y = x^2 - 2x + c$ for $c = -3,\ 1,\ 3$

Yaad karo: "D ka mood"

D positive = do alag dost (2 distinct). **D zero** = judwa (2 equal). **D negative** = koi real root nahi. "Real roots" bole to $D \geq 0$; "equal roots" bole to $D = 0$.

Solved Example 12

Find the nature of the roots of $2x^2 - 6x + 3 = 0$.

1. $D = 36 - 24 = 12 > 0$.

✔ Two distinct real roots

Solved Example 13

Find k so that $kx(x - 3) + 9 = 0$ has two equal roots.

1. $kx^2 - 3kx + 9 = 0$; $D = 9k^2 - 36k = 0$.
2. $9k(k - 4) = 0 \Rightarrow k = 0$ or $k = 4$.
3. $k = 0$ is rejected (then $a = 0$, not quadratic).

✔ $k = 4$ (equation $4x^2 - 12x + 9 = (2x - 3)^2$, root $\frac{3}{2}$)

Solved Example 14

For what values of k does $2x^2 + kx + 2 = 0$ have real roots?

1. Real roots: $D = k^2 - 16 \geq 0$.
2. $k^2 \geq 16 \Rightarrow k \leq -4$ or $k \geq 4$.

✔ $k \leq -4$ or $k \geq 4$

Solved Example 15

Is it possible to design a rectangular park of perimeter 60 m and area 225 m²? What about area 250 m²?

1. $l + b = 30$; let $l = x$, $b = 30 - x$: $x(30 - x) = 225 \Rightarrow x^2 - 30x + 225 = 0$.
2. $D = 900 - 900 = 0$: equal roots $x = 15$. Possible: a 15 m square.
3. For 250 m²: $x^2 - 30x + 250 = 0$, $D = 900 - 1000 = -100 < 0$: not possible.

✔ 225 m²: yes (15 m by 15 m); 250 m²: no

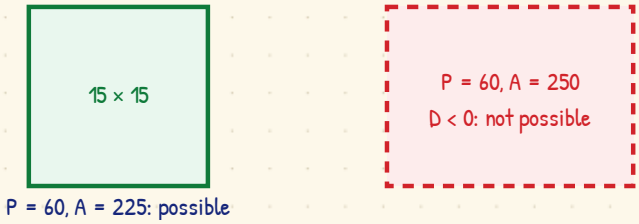


Fig 4.2: same perimeter, different areas

Solved Example 16

Is this possible? The sum of the ages of two friends is 20 years. Four years ago, the product of their ages was 48.

1. Ages now x and $20 - x$: $(x - 4)(16 - x) = 48$.
2. $-x^2 + 20x - 64 = 48 \Rightarrow x^2 - 20x + 112 = 0$.
3. $D = 400 - 448 = -48 < 0$: no real roots.

✔ Not possible

Solved Example 17

A cottage industry makes some toys in a day. The cost of making each toy (in Rs) is 60 minus the number of toys made. On a day the total cost was Rs 800. How many toys were made?

1. $x(60 - x) = 800 \Rightarrow x^2 - 60x + 800 = 0$.
2. $(x - 20)(x - 40) = 0$.

✔ 20 toys or 40 toys (both give total Rs 800)

↳Yahan dono roots valid hain: dono positive whole numbers hain. Har word problem mein dono roots ko situation se check karo, phir hi reject karo.

Solved Example 18

Find k for which $kx^2 + 2x + 1 = 0$ has two distinct real roots.

1. $D = 4 - 4k > 0 \Rightarrow k < 1$.
2. Also $k \neq 0$ (else not quadratic).

✔ $k < 1$ and $k \neq 0$

Shivam Bhaiya says

"Value of k" wale sawal mein bas teen kaam: equation ko standard form mein likho, D ki condition lagao ($= 0$, > 0 ya ≥ 0), aur jo k se $a = 0$ ho jaye usko **reject** karo. Ye 2 marks almost pakke hain.

Meme samjho

Paper mein dikha "Find the nature of roots". Me, $D = b^2 - 4ac$ yaad karke: "Aaj to maza aa jayega!"

Asli baat: nature of roots needs only D, no solving. $D > 0$: distinct, $D = 0$: equal, $D < 0$: no real roots.

5. Word Problems

Let the unknown be x, form the equation, bring to standard form, solve, then **reject** roots that make no sense (negative length, speed, age, time, number of items).

Type	Equation idea
Consecutive integers	$x,\ x+1$ (odd/even: $x,\ x+2$)
Rectangle / right triangle	Area $= lb$; Pythagoras: $h^2 = p^2 + b^2$
Speed, distance	time $= \frac{\text{distance}}{\text{speed}}$; difference of times given
Stream	upstream $u - v$, downstream $u + v$
Work / taps	work in 1 hour $= \frac{1}{x}$; add rates
Cost per item	cost per item $= \frac{\text{total}}{\text{number}}$

Solved Example 19

The altitude of a right triangle is 7 cm less than its base. The hypotenuse is 17 cm. Find the other two sides.

1. Base x, altitude $x - 7$: $x^2 + (x-7)^2 = 17^2$.
2. $2x^2 - 14x - 240 = 0 \Rightarrow x^2 - 7x - 120 = 0$.
3. $(x-15)(x+8) = 0 \Rightarrow x = 15$ ($x = -8$ rejected).

✔ Base 15 cm, altitude 8 cm

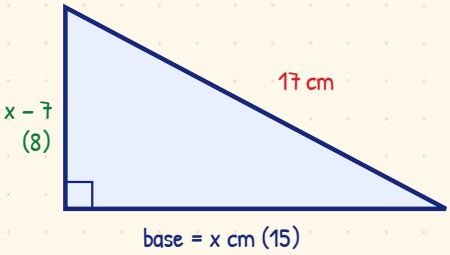


Fig 4.3: $8^2 + 15^2 = 17^2$

Solved Example 20

A train travels 480 km at a uniform speed. If the speed had been 8 km/h more, it would have taken 2 hours less. Find the speed.

1. $\frac{480}{x} - \frac{480}{x+8} = 2$
2. $480 \times 8 = 2x(x+8) \Rightarrow x^2 + 8x - 1920 = 0$.
3. $D = 64 + 7680 = 7744 = 88^2$; $x = \frac{-8 + 88}{2} = 40$.

✔ 40 km/h (the negative root -48 is rejected)

Solved Example 21

The sum of the squares of two consecutive positive integers is 313. Find them.

1. $x^2 + (x+1)^2 = 313 \Rightarrow 2x^2 + 2x - 312 = 0$.
2. $x^2 + x - 156 = 0 \Rightarrow (x+13)(x-12) = 0$.

✔ 12 and 13 ($144 + 169 = 313$)

Solved Example 22

Find two consecutive odd positive integers, the sum of whose squares is 394.

1. $x^2 + (x+2)^2 = 394 \Rightarrow 2x^2 + 4x - 390 = 0$.
2. $x^2 + 2x - 195 = 0 \Rightarrow (x+15)(x-13) = 0 \Rightarrow x = 13$.

✔ 13 and 15 ($169 + 225 = 394$)

Solved Example 23

A rectangular mango grove is twice as long as it is broad, and its area is 800 m². Find its length and breadth.

1. $b(2b) = 800 \Rightarrow b^2 = 400 \Rightarrow b = 20$ (-20 rejected).

✔ Breadth 20 m, length 40 m

Forming a quadratic from its roots

If the roots are α and β, the equation is $x^2 - (\alpha + \beta)x + \alpha\beta = 0$. Example: roots 2 and -5 give $x^2 + 3x - 10 = 0$.

Yaad karo: "x square minus S x plus P"

Roots diye hain to equation: $x^2 - Sx + P = 0$, jahan **S** = sum, **P** = product. Ulta bhi: $ax^2 + bx + c = 0$ mein $S = -\frac{b}{a}$, $P = \frac{c}{a}$: answer check karne ka shortcut.

Yaad karo: "Negative ko Na"

Length, speed, age, time, number of books: ye kabhi **negative** nahi hote. Do roots aaye to dono check karo, galat wale ko likh ke reject karo: "$x = -8$ is rejected as length cannot be negative."

Common mistakes

- → Dividing both sides by x: the root $x = 0$ is lost. Factorise instead: $x^2 = 3x \Rightarrow x(x-3) = 0$.
- → Reading a, b, c before writing the standard form, so signs go wrong.
- → Calling an equation quadratic without expanding: x^2 may cancel.
- → Writing $\frac{-b \pm \sqrt{D}}{2}$ instead of dividing by $2a$.
- → Keeping k that makes $a = 0$, or not rejecting impossible roots in word problems.

Pattern pehchano: one-line approach

Question says	Do this
"is it quadratic?"	expand both sides, check degree 2
"is $x = p$ a root?"	substitute, LHS $= 0$?
"equal roots / real roots"	$D = 0$ / $D \geq 0$, solve for k
"by factorisation"	product ac, sum b, split
"by quadratic formula"	write a, b, c, find D, then x
"is it possible ...?"	form equation, check sign of D
speed / time / work	difference of two fractions $=$ given value

6. Board Exam Strategy

Question type	Marks	What to write
Is it quadratic? / is it a root?	1	Expand, simplify, conclude
Nature of roots / value of k	1-2	D formula, condition, answer
Solve (factorise or formula)	2-3	Standard form, split or D, both roots
Word problem	3-5	Let x, equation, solve, reject, answer in words

Shivam Bhaiya says

Word problem ka equation banne ke baad **half marks** wahin mil jaate hain. Isliye "Let speed $= x$ km/h" aur equation saaf likho. Answer ke baad ek line ka **check** likho: examiner khush!

Practice yourself (answers below)

1. Solve: $x^2 - 5x - 14 = 0$.
2. Find the nature of roots: $3x^2 + 4x + 2 = 0$.
3. Find k for equal roots: $x^2 + kx + 25 = 0$.
4. Two numbers differ by 3 and their product is 108. Find them (positive).

↳Ans: (1) $7, -2$ (2) $D = -8$, no real roots (3) $k = \pm 10$ (4) 9 and 12

Extra practice A (answers below)

1. Solve: $x^2 + 7x + 12 = 0$.
2. Solve: $2x^2 - 5x + 3 = 0$.
3. Solve: $4x^2 - 4x + 1 = 0$.
4. Solve by formula: $x^2 - 6x + 4 = 0$.
5. Nature of roots: $2x^2 - 3x + 5 = 0$.

↳Ans: (1) $-3, -4$ (2) $1, \frac{3}{2}$ (3) $\frac{1}{2}, \frac{1}{2}$ (4) $3 \pm \sqrt{5}$ (5) $D = -31$, no real roots

Extra practice B (answers below)

6. Nature of roots: $4x^2 - 12x + 9 = 0$.
7. Find k for equal roots: $2x^2 + kx + 3 = 0$.
8. Find k for real roots: $kx^2 + 6x + 1 = 0$.
9. Sum of squares of two consecutive even positive numbers is 340. Find them.
10. A rectangle's length exceeds its breadth by 7 cm and its area is 144 cm². Find its sides.

↳Ans: (6) $D = 0$, equal roots (7) $\pm 2\sqrt{6}$ (8) $k \leq 9,\ k \neq 0$ (9) 12, 14 (10) 9 cm, 16 cm

MOST IMPORTANT QUESTIONS

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. Which of the following is a quadratic equation? 1M Board fav

(a) $x^2 + 2x + 1 = (4 - x)^2 + 3$ (b) $x(2x + 3) = x^2 + 1$ (c) $(x + 2)^2 = x^2 + 5$ (d) $(x - 3)(x + 3) = x^2 + 2x$

Ans. (b): $x(2x + 3) = x^2 + 1 \Rightarrow x^2 + 3x - 1 = 0$. The others reduce to degree 1.

Q2. The roots of $2x^2 - 4x + 3 = 0$ are 1M
(a) real and distinct (b) real and equal (c) rational (d) not real
Ans. (d) $D = 16 - 24 = -8 < 0$: no real roots.

Q3. If $x = 2$ is a root of $x^2 + kx - 6 = 0$, then $k =$ 1M
(a) 1 (b) -1 (c) 2 (d) 3
Ans. (a) $4 + 2k - 6 = 0 \Rightarrow k = 1$.

Q4. The sum of the roots of $3x^2 - 7x + 2 = 0$ is 1M
(a) $\frac{7}{3}$ (b) $-\frac{7}{3}$ (c) $\frac{2}{3}$ (d) $-\frac{2}{3}$
Ans. (a) $-\frac{b}{a} = \frac{7}{3}$ (roots are 2 and $\frac{1}{3}$).

Q5. The values of k for which $9x^2 + kx + 4 = 0$ has equal roots are 1M Board fav
(a) ± 6 (b) ± 8 (c) ± 12 (d) ± 36
Ans. (c) $k^2 - 144 = 0 \Rightarrow k = \pm 12$.

Q6. The discriminant of $\sqrt{3}x^2 + 2\sqrt{2}x - 2\sqrt{3} = 0$ is 1M
(a) 32 (b) 16 (c) -16 (d) 8
Ans. (a) $D = (2\sqrt{2})^2 - 4(\sqrt{3})(-2\sqrt{3}) = 8 + 24 = 32$.

Q7. If one root of $5x^2 + 13x + k = 0$ is the reciprocal of the other, then $k =$ 1M
(a) 0 (b) 5 (c) $\frac{1}{6}$ (d) 6
Ans. (b) product of roots $= \frac{k}{5} = 1 \Rightarrow k = 5$.

Q8. The roots of $x^2 - 3x = 0$ are 1M
(a) 3 only (b) 0 and 3 (c) -3 and 3 (d) 0 and -3
Ans. (b) $x(x - 3) = 0 \Rightarrow x = 0, 3$.

Q9. A quadratic equation whose roots are 3 and -4 is 1M
(a) $x^2 - x - 12 = 0$ (b) $x^2 + 7x + 12 = 0$ (c) $x^2 + x - 12 = 0$ (d) $x^2 - x + 12 = 0$
Ans. (c) $x^2 - (3 - 4)x + (3)(-4) = x^2 + x - 12 = 0$.

Q10. **A:** $3x^2 - 6x + 3 = 0$ has two equal roots. **R:** If $D = 0$, a quadratic equation has two equal real roots. 1M
(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true
Ans. (a) $D = 36 - 36 = 0$, and R is the reason.

Q11. **A:** $x^2 + x + 1 = 0$ has two distinct real roots. **R:** A quadratic equation has at most two roots. 1M
(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true
Ans. (d) $D = 1 - 4 = -3 < 0$, so A is false; R is true.

Very short (2 marks)

Q12. Solve by factorisation: $2x^2 + x - 6 = 0$. 2M
Ans. $ac = -12$: split as $4x - 3x$. $2x(x + 2) - 3(x + 2) = (2x - 3)(x + 2) = 0 \Rightarrow x = \frac{3}{2}, -2$.

Q13. Find k if $x^2 - 2kx + 16 = 0$ has equal roots. 2M
Ans. $D = 4k^2 - 64 = 0 \Rightarrow k^2 = 16 \Rightarrow k = \pm 4$.

Q14. The product of two consecutive positive integers is 342. Find them. 2M
(contd.) $x(x + 1) = 342 \Rightarrow x^2 + x - 342 = 0 \Rightarrow (x + 19)(x - 18) = 0$. Integers 18 and 19.

Q15. Find the nature of the roots of $3x^2 - 2\sqrt{6}x + 2 = 0$ and find them. 2M
Ans. $D = 24 - 24 = 0$: two equal roots, each $\frac{2\sqrt{6}}{6} = \frac{\sqrt{6}}{3}$.

Q16. Is $x = -\frac{1}{2}$ a root of $2x^2 + 3x + 1 = 0$? Find the other root. 2M
Ans. $2(\frac{1}{4}) - \frac{3}{2} + 1 = 0$ ✓ Yes. $(2x + 1)(x + 1) = 0$, other root $x = -1$.

Q17. For what value of k is 3 a root of $x^2 - 2kx - 6 = 0$? 2M
Ans. $9 - 6k - 6 = 0 \Rightarrow k = \frac{1}{2}$.

Q18. Find the roots of $x^2 + 2\sqrt{2}x - 6 = 0$. 2M
Ans. $D = 8 + 24 = 32$, $\sqrt{D} = 4\sqrt{2}$. $x = \frac{-2\sqrt{2} \pm 4\sqrt{2}}{2} = \sqrt{2}$ or $-3\sqrt{2}$.

Q19. Write the quadratic equation whose roots are $\frac{1}{2}$ and -3 . 2M
Ans. Sum $= -\frac{5}{2}$, product $= -\frac{3}{2}$: $x^2 + \frac{5}{2}x - \frac{3}{2} = 0$, i.e. $2x^2 + 5x - 3 = 0$.

Short answer (3 marks)

Q20. If -5 is a root of $2x^2 + px - 15 = 0$ and $p(x^2 + x) + k = 0$ has equal roots, find k. 3M
Ans. $2(25) - 5p - 15 = 0 \Rightarrow p = 7$. Then $7x^2 + 7x + k = 0$: $D = 49 - 28k = 0 \Rightarrow k = \frac{7}{4}$.

Q21. Solve by the quadratic formula: $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$. 3M
Ans. $D = 49 - 4(\sqrt{2})(5\sqrt{2}) = 49 - 40 = 9$. $x = \frac{-7 \pm 3}{2\sqrt{2}}$: $x = -\sqrt{2}$ or $x = -\frac{5}{\sqrt{2}} = -\frac{5\sqrt{2}}{2}$.

Q22. Solve using the quadratic formula: $2x^2 + x - 4 = 0$. 3M
Ans. $a = 2,\ b = 1,\ c = -4$; $D = 1 + 32 = 33$. $x = \frac{-1 \pm \sqrt{33}}{4}$.

Q23. Find the values of k for which $(k + 4)x^2 + (k + 1)x + 1 = 0$ has equal roots. 3M
Ans. $D = (k + 1)^2 - 4(k + 4) = k^2 - 2k - 15 = 0 \Rightarrow (k - 5)(k + 3) = 0$. $k = 5$ or $k = -3$ (neither makes $k + 4 = 0$).

Q24. The sum of a number and its reciprocal is $\frac{10}{3}$. Find the number. 3M
Ans. $x + \frac{1}{x} = \frac{10}{3} \Rightarrow 3x^2 - 10x + 3 = 0 \Rightarrow (3x - 1)(x - 3) = 0$. Number $= 3$ or $\frac{1}{3}$.

Q25. Find two numbers whose sum is 27 and product is 182. 3M
Ans. $x(27 - x) = 182 \Rightarrow x^2 - 27x + 182 = 0 \Rightarrow (x - 13)(x - 14) = 0$. Numbers 13 and 14.

Q26. The diagonal of a rectangular field is 16 m more than the shorter side, and the longer side is 14 m more than the shorter side. Find the sides. 3M

Ans. $x^2 + (x+14)^2 = (x+16)^2 \Rightarrow x^2 - 4x - 60 = 0 \Rightarrow (x-10)(x+6) = 0$. $x = 10$. Sides 10 m and 24 m (diagonal 26 m).

Q27. The product of a girl's age 4 years ago and her age 6 years from now is 119. Find her present age. 3M

Ans. $(x-4)(x+6) = 119 \Rightarrow x^2 + 2x - 143 = 0 \Rightarrow (x+13)(x-11) = 0$. Age 11 years ($7 \times 17 = 119$).

Q28. Solve for x: $\dfrac{1}{x} - \dfrac{1}{x-2} = 3$, $x \neq 0, 2$. 3M

Ans. $\frac{(x-2)-x}{x(x-2)} = 3 \Rightarrow -2 = 3x^2 - 6x \Rightarrow 3x^2 - 6x + 2 = 0$. $D = 36 - 24 = 12$; $x = \frac{6 \pm 2\sqrt{3}}{6} = 1 \pm \frac{\sqrt{3}}{3}$.

Q29. The sum of the areas of two squares is 356 m² and the difference of their perimeters is 24 m. Find their sides. 3M

Ans. Sides x and $x+6$ ($4 \times 6 = 24$). $x^2 + (x+6)^2 = 356 \Rightarrow x^2 + 6x - 160 = 0 \Rightarrow (x+16)(x-10) = 0$. Sides 10 m and 16 m.

Q30. Find two consecutive odd positive integers whose product is 143. 3M

Ans. $x(x+2) = 143 \Rightarrow x^2 + 2x - 143 = 0 \Rightarrow (x+13)(x-11) = 0$. Integers 11 and 13.

Q31. Solve for x: $4x^2 - 4a^2x + (a^4 - b^4) = 0$. 3M

Ans. $a' = 4$, $b' = -4a^2$, $c' = a^4 - b^4$. $D = 16a^4 - 16(a^4 - b^4) = 16b^4$, $\sqrt{D} = 4b^2$. $x = \dfrac{4a^2 \pm 4b^2}{8} = \dfrac{a^2 + b^2}{2}$ or $\dfrac{a^2 - b^2}{2}$.

Long answer (5 marks)

Q32. Two taps together fill a tank in 6 hours. The larger tap alone takes 5 hours less than the smaller one. Find the time each takes alone. 5M

Ans. Smaller x h, larger $x-5$ h. $\frac{1}{x} + \frac{1}{x-5} = \frac{1}{6} \Rightarrow 6(2x-5) = x(x-5) \Rightarrow x^2 - 17x + 30 = 0 \Rightarrow (x-15)(x-2) = 0$. $x = 2$ gives $x - 5 < 0$, rejected. Smaller 15 h, larger 10 h.

Q33. A motorboat whose speed in still water is 15 km/h takes 1 hour more to go 36 km upstream than to return downstream. Find the speed of the stream. 5M

Ans. $\frac{36}{15-x} - \frac{36}{15+x} = 1 \Rightarrow 72x = 225 - x^2 \Rightarrow x^2 + 72x - 225 = 0$. $D = 5184 + 900 = 6084 = 78^2$; $x = \frac{-72+78}{2} = 3$. Stream 3 km/h.

Apne notes / galtiyan yahan likho

Q34. A shopkeeper buys some books for Rs 80. If he had bought 4 more books for the same amount, each book would cost Rs 1 less. Find the number of books. 5M

Ans. $\frac{80}{x} - \frac{80}{x+4} = 1 \Rightarrow 320 = x^2 + 4x \Rightarrow x^2 + 4x - 320 = 0 \Rightarrow (x+20)(x-16) = 0$. He bought 16 books.

Q35. A plane left 30 minutes late. To reach its destination 600 km away on time, it increased its usual speed by 100 km/h. Find its usual speed. 5M

Ans. $\frac{600}{x} - \frac{600}{x+100} = \frac{1}{2} \Rightarrow 120000 = x^2 + 100x \Rightarrow x^2 + 100x - 120000 = 0$. $D = 490000 = 700^2$, $x = \frac{-100+700}{2} = 300$. Usual speed 300 km/h.

Q36. In a test, a student's Maths and English marks add up to 40. Had she got 3 more in Maths and 4 less in English, the product of the marks would be 360. Find her marks. 5M

Ans. Maths x, English $40 - x$. $(x+3)(36-x) = 360 \Rightarrow x^2 - 33x + 252 = 0$. $D = 1089 - 1008 = 81$; $x = \frac{33 \pm 9}{2} = 21$ or 12. Maths 21, English 19; or Maths 12, English 28.

Q37. An express train takes 1 hour less than a passenger train to travel 180 km. The express is 15 km/h faster. Find the speeds of both trains. 5M

Ans. $\frac{180}{x} - \frac{180}{x+15} = 1 \Rightarrow 2700 = x^2 + 15x \Rightarrow x^2 + 15x - 2700 = 0$. $D = 225 + 10800 = 11025 = 105^2$, $x = \frac{-15+105}{2} = 45$. Passenger 45 km/h, express 60 km/h.

Case based (4 marks)

Q38. Prayer hall: A rectangular hall has area 528 m². Its length is 1 m more than twice its breadth. (i) Form the quadratic equation. (1) (ii) Find its discriminant. (1) (iii) Find the length, breadth and perimeter. (2) 4M

Ans. (i) $b(2b+1) = 528 \Rightarrow 2b^2 + b - 528 = 0$. (ii) $D = 1 + 4224 = 4225 = 65^2$. (iii) $b = \frac{-1+65}{4} = 16$ m, length $= 33$ m, perimeter $= 2(33+16) = 98$ m.

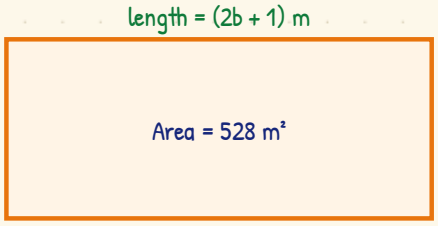


Fig 4.4: Q38, the prayer hall

Q39. Ball throw: A ball's height after t seconds is $h = 30t - 5t^2$ metres. (i) When is it at 40 m? (2) (ii) When does it come back to the ground? (1) (iii) Can it reach 50 m? (1) 4M

Ans. (i) $5t^2 - 30t + 40 = 0 \Rightarrow t^2 - 6t + 8 = 0 \Rightarrow t = 2$ s (going up) and $t = 4$ s (coming down). (ii) $30t - 5t^2 = 0 \Rightarrow t = 6$ s. (iii) $t^2 - 6t + 10 = 0$: $D = 36 - 40 < 0$, so no.

Apne notes / galtiyan yahan likho

Most asked: a_n, S_n, which term, word problems (savings, logs)

ARITHMETIC PROGRESSIONS

Chapter at a glance

- ✓ AP = add the same d every time.
- ✓ Any term: $a_n = a + (n-1)d$. Last term from the end: $l - (n-1)d$.
- ✓ Total: $S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a + l)$.
- ✓ Exam favourites: which term, sum of n terms, savings and rows problems.

Formula Box

- → Common difference: $d = a_2 - a_1 = a_{k+1} - a_k$
- → nth term: $a_n = a + (n-1)d$
- → nth term from the end: $l - (n-1)d$
- → Number of terms: $n = \dfrac{l-a}{d} + 1$
- → Sum: $S_n = \dfrac{n}{2}[2a + (n-1)d] = \dfrac{n}{2}(a + l)$
- → $a_n = S_n - S_{n-1}$, $a_1 = S_1$
- → a, b, c in AP $\Leftrightarrow 2b = a + c$
- → $1 + 2 + \cdots + n = \dfrac{n(n+1)}{2}$; sum of first n odd numbers $= n^2$

1. What is an AP?

A list of numbers in a definite order is a **sequence**; its members are **terms** $a_1, a_2, a_3, \ldots$

An **arithmetic progression (AP)** is a list of numbers in which each term (except the first) is obtained by **adding a fixed number** to the previous term. This fixed number is the **common difference** d.

$$d = a_2 - a_1 = a_3 - a_2 = \cdots = a_{k+1} - a_k$$

- → General form: $a,\ a+d,\ a+2d,\ a+3d,\ \ldots$ (a = first term).
- → d can be **positive** (increasing AP), **negative** (decreasing) or **zero** (all terms equal: 5, 5, 5, ...).
- → **Finite AP:** has a last term (e.g. 3, 7, 11, ..., 99). **Infinite AP:** no last term (e.g. 1, 3, 5, 7, ...).
- → Examples: roll numbers 1, 2, 3, ...; taxi fare Rs 15, 23, 31, 39, ... for 1, 2, 3, 4 km (d = 8).

Yaad karo: "Teen term, do difference"

AP check karne ke liye kam se kam **teen terms** lo aur **do differences** nikaalo. Dono same = AP. Sirf ek difference dekh ke "AP hai" mat bolo. Root wale terms ko pehle simplify karo ($\sqrt{8} = 2\sqrt{2}$).

Solved Example 1

Which of these are APs? Give d if yes. (i) $2, 6, 10, 14, \ldots$ (ii) $1, 4, 9, 16, \ldots$ (iii) $\sqrt{2}, \sqrt{8}, \sqrt{18}, \sqrt{32}, \ldots$ (iv) $3, 3, 3, 3, \ldots$

1. (i) differences $4, 4, 4$: AP, $d = 4$.
2. (ii) differences $3, 5, 7$: not an AP.
3. (iii) $\sqrt{2}, 2\sqrt{2}, 3\sqrt{2}, 4\sqrt{2}$: AP, $d = \sqrt{2}$.
4. (iv) AP with $d = 0$.

✓ (i), (iii), (iv) are APs

2. nth Term (General Term)

Term	a_1	a_2	a_3	a_n
Value	a	$a+d$	$a+2d$	$a+(n-1)d$

$$a_n = a + (n-1)d$$

Also written $l = a + (n-1)d$ when a_n is the **last term** l.

The nth term from the end $= l - (n-1)d$.

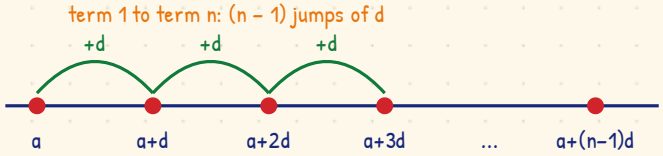
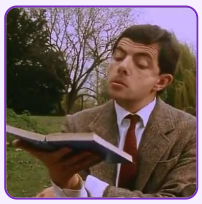


Fig 5.1: to reach the nth term you take $(n-1)$ jumps of d



Meme samjho

Me reading "$a_n = a + nd$" in my own notes: wait... ye galat kaise likh diya? **Asli baat:** $a_n = a + (n-1)d$. The first term already sits at a, so the nth term is only $(n-1)$ jumps away.

Solved Example 2

Which term of the AP $3, 8, 13, \ldots$ is 248?

1. $a = 3,\ d = 5$: $3 + (n-1)5 = 248$.
2. $(n-1)5 = 245 \Rightarrow n - 1 = 49$.

✓ $n = 50$: the 50th term

Solved Example 3

Is 400 a term of the AP $7, 13, 19, \ldots$?

1. $7 + (n-1)6 = 400 \Rightarrow (n-1)6 = 393$.
2. $n - 1 = 65.5$: not a whole number.

✓ No, 400 is not a term

Yaad karo: "n natural hi hona chahiye"

"Is it a term?" wale sawal mein n nikaalo. n **positive whole number** aaya to haan, fraction ya negative aaya to **na**. Bas itna hi.

Solved Example 4

The 5th term of an AP is 20 and the 12th term is 48. Find the AP.

1. $a+4d=20,\ a+11d=48.$
2. Subtract: $7d=28\Rightarrow d=4;\ a=20-16=4.$

✔ AP: $4,8,12,16,\ldots$

Solved Example 5

Check whether -145 is a term of the AP $11,8,5,2,\ldots$

1. $11+(n-1)(-3)=-145\Rightarrow 3(n-1)=156$.
2. $n-1=52\Rightarrow n=53$, a natural number.

✔ Yes, it is the 53rd term

Solved Example 6

Find the number of terms in the AP $7,13,19,\ldots,205$.

1. $n=\dfrac{205-7}{6}+1=33+1.$

✔ 34 terms

Solved Example 7

The 4th term of an AP is 19 and the 9th term exceeds the 6th term by 15. Find the AP.

1. $a_9-a_6=3d=15\Rightarrow d=5.$
2. $a+3d=19\Rightarrow a=4.$

✔ AP: $4,9,14,19,\ldots$

Solved Example 8

Find the 10th term from the end of the AP $4,9,14,\ldots,254.$

1. $l=254,\ d=5.$
2. 10th from end $=l-(10-1)d=254-45.$

✔ 209

Yaad karo: count karne ka formula

Kitne terms hain? $n=\dfrac{l-a}{d}+1$. "Last minus first, divide by d, **plus one**". Plus one bhoole to answer 1 kam aayega.

Solved Example 9

How many three-digit numbers are divisible by 6?

1. AP: $102,108,\ldots,996$ $(a=102,\ d=6,\ l=996)$.
2. $996=102+(n-1)6\Rightarrow n-1=149.$

✔ 150 numbers

Solved Example 10

For what value of n are the nth terms of the APs $57,60,63,\ldots$ and $7,12,17,\ldots$ equal?

1. $57+3(n-1)=7+5(n-1).$
2. $50=2(n-1)\Rightarrow n=26.$
3. Both 26th terms $=57+75=132.$

✔ $n=26$

Solved Example 11

Two APs have the same common difference. The difference between their 100th terms is 100. What is the difference between their 1000th terms?

1. $(a+99d)-(b+99d)=a-b=100.$
2. $(a+999d)-(b+999d)=a-b.$

✔ Still 100 (same d, so the gap never changes)

3. Three Terms in AP

If a,b,c are in AP then $b-a=c-b$, i.e. $\boxed{2b=a+c}$: b is the **arithmetic mean** of a and c.

When the sum of three terms is given, take them as $a-d,\ a,\ a+d$ (four terms: $a-3d,\ a-d,\ a+d,\ a+3d$).

Solved Example 12

For what k are $2k+1,\ 3k+3,\ 5k-1$ in AP?

1. $2(3k+3)=(2k+1)+(5k-1)$
2. $6k+6=7k\Rightarrow k=6.$

✔ $k=6$; the terms are $13,21,29$

Yaad karo: "Beech wala double"

Teen terms AP mein hain to **beech wala double = baaki dono ka sum**: $2b=a+c$. Aur sum diya ho to $a-d,a,a+d$ lo, d apne aap cut jaata hai.

4. Sum of First n Terms

Young **Gauss** added $1+2+\cdots+100$ in seconds: pair the first and last terms, each pair $=101$, 50 pairs, total 5050. Same idea for any AP:

$$S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a + l)$$

→ Use $\frac{n}{2}(a+l)$ when the last term l is known.

→ Sum of first n natural numbers: $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$.

→ Sum of first n odd numbers $= n^2$; first n even numbers $= n(n+1)$.

→ If S_n is given, $a_n = S_n - S_{n-1}$ and $a_1 = S_1$.

Meme samjho

Teacher: "1 se 100 tak jodo." Gauss, 10 second mein: "5050." Class: "Hain? Ye kaise ho gaya?"

Asli baat: $1 + 100 = 2 + 99 = \cdots = 101$, and there are 50 such pairs: $50 \times 101 = 5050 = \frac{100}{2}(1 + 100)$.

Solved Example 13

Find the sum of the first 25 terms of $7, 3, -1, \ldots$

1. $a = 7,\ d = -4,\ n = 25$.
2. $S_{25} = \frac{25}{2}[14 + 24(-4)] = \frac{25}{2}(-82)$.

✔ -1025

Solved Example 14

How many terms of the AP $27, 24, 21, \ldots$ must be taken so that their sum is 126? Explain the double answer.

1. $\frac{n}{2}[54 + (n-1)(-3)] = 126 \Rightarrow n(57 - 3n) = 252$.
2. $3n^2 - 57n + 252 = 0 \Rightarrow n^2 - 19n + 84 = 0$.
3. $(n-7)(n-12) = 0 \Rightarrow n = 7$ or 12.

✔ 7 or 12 terms. Both are right: terms 8th to 12th are $6, 3, 0, -3, -6$, whose sum is 0.

Solved Example 15

The sum of the first n terms of an AP is $S_n = 3n^2 + 5n$. Find a_n and the 20th term.

1. $a_n = S_n - S_{n-1} = 3n^2 + 5n - [3(n-1)^2 + 5(n-1)]$
2. $= 6n + 2$. Check: $a_1 = 8 = S_1$ ✓

✔ $a_n = 6n + 2$; $a_{20} = 122$

Solved Example 16

Find the sum of all multiples of 7 between 100 and 300.

1. AP: $105, 112, \ldots, 294$; $n = \frac{294-105}{7} + 1 = 28$.
2. $S = \frac{28}{2}(105 + 294) = 14 \times 399$.

✔ 5586

Solved Example 17

In an AP, $a_2 = 13$ and $a_3 = 17$. Find the sum of the first 40 terms.

1. $d = 17 - 13 = 4$, $a = 13 - 4 = 9$.
2. $S_{40} = \frac{40}{2}[18 + 39(4)] = 20 \times 174$.

✔ 3480

Solved Example 18

Given $a = 7$, $d = 4$, $a_n = 83$, find n and S_n.

1. $7 + (n-1)4 = 83 \Rightarrow n - 1 = 19 \Rightarrow n = 20$.
2. $S_{20} = \frac{20}{2}(7 + 83) = 10 \times 90$.

✔ $n = 20$, $S_{20} = 900$

Solved Example 19

Find the sum of the first 40 positive integers divisible by 6.

1. $6 + 12 + \cdots + 240 = 6(1 + 2 + \cdots + 40)$.
2. $= 6 \times \frac{40 \times 41}{2} = 6 \times 820$.

✔ 4920

Solved Example 20

The sum of the first n terms of an AP is $5n - n^2$. Find the first term, the second term, d and the 10th term.

1. $a_1 = S_1 = 4$; $S_2 = 10 - 4 = 6 \Rightarrow a_2 = 6 - 4 = 2$.
2. $d = 2 - 4 = -2$; $a_n = 4 + (n-1)(-2) = 6 - 2n$.

✔ $a_1 = 4,\ a_2 = 2,\ d = -2,\ a_{10} = -14$

Series	Sum of first n terms
$1 + 2 + 3 + \ldots$	$\frac{n(n+1)}{2}$
$1 + 3 + 5 + \ldots$ (odd)	n^2
$2 + 4 + 6 + \ldots$ (even)	$n(n+1)$
$k + 2k + 3k + \ldots$ (multiples of k)	$k \cdot \frac{n(n+1)}{2}$

Yaad karo: "Sum = gin lo x average"

$S_n = n \times \frac{\text{first} + \text{last}}{2}$: kitne terms × pehle aur aakhri ka average. Gauss wala jodi trick, har AP pe chalta hai.

Quick fact d can be positive, negative or zero.

Yaad rakho $1 + 2 + \cdots + 100 = 5050$.

Do you know? Sum of first n odd numbers $= n^2$.

Shivam Bhaiya says
Pehle decide karo sawal **term** maang raha hai ya **sum**. "Kitna hoga 10th week mein" = a_n. "Total kitna" = S_n. Isi confusion se sabse zyada marks jaate hain.

Which formula? Given, then use

Given	Use
a, d; want one term	$a_n = a + (n-1)d$
two terms, e.g. a_p, a_q	subtract: $(p-q)d = a_p - a_q$
a, d, n; want total	$S_n = \frac{n}{2}[2a + (n-1)d]$
first and last term	$S_n = \frac{n}{2}(a+l), n = \frac{l-a}{d} + 1$
S_n as an expression in n	$a_n = S_n - S_{n-1}$
sum given, want n	quadratic in n; keep natural n

AP in daily life

- ✓ Monthly instalments that rise by a fixed amount.
- ✓ Seats in rows of an auditorium, logs or bricks in stacked rows.
- ✓ Simple interest: amount after 1, 2, 3, ... years (same interest added each year).
- ✓ Taxi fare: fixed start plus the same charge for each extra km.

5. Word Problems

Spot a and d from the story, decide a_n or S_n, solve, and answer in words.

Solved Example 21

Rs 1600 is to be given as 8 cash prizes, each prize Rs 20 less than the one before. Find each prize.

1. $d = -20,\ n = 8$: $\frac{8}{2}[2a + 7(-20)] = 1600$.
2. $4(2a - 140) = 1600 \Rightarrow 2a = 540 \Rightarrow a = 270$.

✓ Rs 270, 250, 230, 210, 190, 170, 150, 130

Solved Example 22

165 logs are stacked with 20 in the bottom row, 19 in the next, 18 in the next, and so on. How many rows are there, and how many logs in the top row?

1. $a = 20,\ d = -1$: $\frac{n}{2}[40 - (n-1)] = 165 \Rightarrow n(41 - n) = 330$.
2. $n^2 - 41n + 330 = 0 \Rightarrow (n-11)(n-30) = 0$.
3. $n = 30$ gives top row $20 - 29 = -9$ logs: impossible. So $n = 11$.

✓ 11 rows; top row has $20 - 10 = 10$ logs

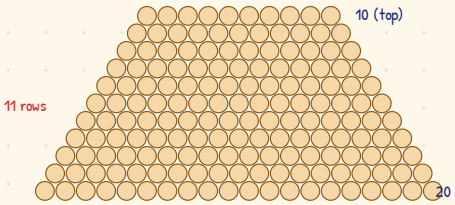


Fig 5.2: 20, 19, ..., 10 logs: 11 rows, 165 logs

Solved Example 23

A spiral is made of semicircles of radii $0.5, 1.0, 1.5, \ldots$ cm. Find the total length of 14 such semicircles. (Take $\pi = \frac{22}{7}$.)

1. Length of a semicircle $= \pi r$: $0.5\pi, 1.0\pi, 1.5\pi, \ldots$ is an AP with $a = 0.5\pi,\ d = 0.5\pi$.
2. $S_{14} = \frac{14}{2}[\pi + 13(0.5\pi)] = 7 \times 7.5\pi = 52.5\pi$.

✓ $52.5 \times \frac{22}{7} = 165$ cm

Solved Example 24

Potato race: a bucket is at the start line. The first potato is 4 m from the bucket and the others are 3 m apart in a straight line; there are 8 potatoes. A runner picks up one potato at a time, runs back to the bucket, and drops it. Find the total distance run.

1. One-way distances: $4, 7, 10, \ldots, 25$ (8 terms).
2. Each is run twice: total $= 2 \times \frac{8}{2}(4 + 25) = 2 \times 116$.

✓ 232 m

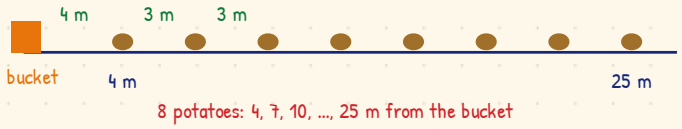


Fig 5.3: potato race, distances form an AP

Solved Example 25

A contractor is fined Rs 200 for the first day of delay, Rs 250 for the second, Rs 300 for the third, and so on. How much does a 30-day delay cost him?

1. $a = 200,\ d = 50,\ n = 30$.
2. $S_{30} = 15\,[400 + 29(50)] = 15 \times 1850$.

✓ Rs 27750

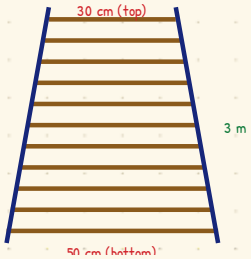


Fig 5.4: 11 rungs from 50 cm to 30 cm

Recall $a_n = S_n - S_{n-1}$.

Solved Example 26

A ladder has rungs 30 cm apart. The rungs shorten uniformly from 50 cm at the bottom to 30 cm at the top, and the top and bottom rungs are 3 m apart. How much wood is needed for the rungs?

1. Number of rungs $= \frac{300}{30} + 1 = 11$.
2. Lengths $50, 48, \ldots, 30$ form an AP: $S_{11} = \frac{11}{2}(50 + 30) = 440$.

✔ 440 cm $= 4.4$ m of wood

Solved Example 27

Houses in a row are numbered 1 to 288. Show that there is a house x such that the sum of the numbers before it equals the sum of the numbers after it, and find x.

1. Before: $\frac{(x-1)x}{2}$. After: $\frac{288\times289}{2} - \frac{x(x+1)}{2}$.
2. Equate: $x^2 = \frac{288\times289}{2} = 41616 \Rightarrow x = 204$.

✔ House number 204 (both sums are 20706)

Solved Example 28

Find the sum of the first 15 terms of the AP whose nth term is $3 + 4n$.

1. $a_1 = 7$, $a_{15} = 63$.
2. $S_{15} = \frac{15}{2}(7 + 63)$.

✔ 525

Common mistakes

- → $a_n = a + nd$ instead of $a + (n-1)d$.
- → Taking $d = a_1 - a_2$ (wrong sign). Always $d = a_2 - a_1$.
- → Forgetting "+1" while counting terms: $n = \frac{l-a}{d} + 1$.
- → Mixing a_n (one term) with S_n (total) in word problems.
- → Not rejecting the impossible value of n (negative, fraction, or giving negative logs).

6. Board Exam Strategy

Question type	Marks	Key step
Find d / next term / is it AP	1	Two differences
Which term / is it a term	2	n must be a natural number
Two terms given, find AP	3	Two equations in a, d
Sum, number of terms	3-5	S_n formula, quadratic in n
Case study (savings, rows)	4	Identify a, d; a_n vs S_n

Shivam Bhaiya says

Har sawal ki pehli line: "Here $a = \ldots$, $d = \ldots$, $n = \ldots$". Ye likhte hi aadha sawal solve ho jaata hai aur step marks bhi pakke. AP ke sawal seedhe **formula-based** hote hain: practice karo, free marks hain!

Practice yourself (answers below)

1. Find the 15th term of $2, 7, 12, \ldots$
2. Find the sum of the first 30 even natural numbers.
3. How many terms are in $10, 14, 18, \ldots, 126$?
4. If $a_n = 5n - 3$, find S_{20}.
5. Find the 8th term from the end of $5, 9, 13, \ldots, 185$.
6. Find k if $k + 2$, $4k - 6$, $3k - 2$ are in AP.
7. Sum of all natural numbers from 1 to 150 divisible by 5?
8. Which term of $5, 2, -1, \ldots$ is -82?

↳Ans: (1) 72 (2) 930 (3) 30 (4) 990 (5) 157 (6) $k = 3$ (7) 2325 (8) 30th

Extra practice A (answers below)

1. Find the 30th term of $10, 7, 4, \ldots$
2. Find d if $a = -18$ and $a_{10} = 0$.
3. Which term of $3, 15, 27, 39, \ldots$ is 132 more than its 54th term?
4. How many multiples of 4 lie between 10 and 250?
5. Find S_{15} if $a_n = 9 - 5n$.
6. An AP has $a = 2, l = 74, n = 25$. Find its sum.
7. Find S_{12} of $\frac{1}{15}, \frac{1}{12}, \frac{1}{10}, \ldots$

↳Ans: (1) -77 (2) 2 (3) 65th (4) 60 (5) -465 (6) 950 (7) $\frac{19}{10}$

Extra practice B (answers below)

8. A salary starts at Rs 8000 in 2015 with an increment of Rs 500 every year. In which year does it reach Rs 12000?
9. A girl saves Rs 10 in week 1 and Rs 2.50 more every week. In which week does she save Rs 45?
10. $S_{10} = 155$ and $S_{15} = 345$. Find a and d.

↳Ans: (8) 2023 (9) 15th week (10) $a = 2$, $d = 3$

Exam tip Write a, d, n first in every AP sum.

Last-minute checklist

- ✓ $d = a_2 - a_1$ (later minus earlier).
- ✓ n must be a natural number; reject fractions and negatives.
- ✓ Term or total? a_n vs S_n.
- ✓ Count terms with $\frac{l-a}{d} + 1$.
- ✓ Write "$a = \ldots,\ d = \ldots,\ n = \ldots$" first.

MOST IMPORTANT QUESTIONS

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. Which of these is an AP? 1M
(a) $1, 2, 4, 8, \ldots$ (b) $1, 4, 9, 16, \ldots$ (c) $10, 6, 2, -2, \ldots$ (d) $1, 3, 6, 10, \ldots$

Ans. (c) differences are all -4.

Q2. The common difference of the AP $\frac{1}{p}, \frac{1-p}{p}, \frac{1-2p}{p}, \ldots$ is 1M Board fav
(a) 1 (b) -1 (c) p (d) $-p$

Ans. (b) $\frac{1-p}{p} - \frac{1}{p} = \frac{-p}{p} = -1$.

Q3. The 11th term of the AP $-3, -\frac{1}{2}, 2, \ldots$ is 1M
(a) 28 (b) -22 (c) 22 (d) -38

Ans. (c) $d = \frac{5}{2}$: $-3 + 10 \times \frac{5}{2} = 22$.

Q4. The next term of the AP $\sqrt{7}, \sqrt{28}, \sqrt{63}, \ldots$ is 1M
(a) $\sqrt{70}$ (b) $\sqrt{84}$ (c) $\sqrt{97}$ (d) $\sqrt{112}$

Ans. (d) terms are $\sqrt{7}, 2\sqrt{7}, 3\sqrt{7}$, so next $= 4\sqrt{7} = \sqrt{112}$.

Q5. If $2x,\ x+10,\ 3x+2$ are in AP, then $x =$ 1M
(a) 6 (b) 4 (c) 3 (d) 2

Ans. (a) $2(x+10) = 2x + 3x + 2 \Rightarrow x = 6$.

Q6. The sum of the first 20 odd natural numbers is 1M
(a) 100 (b) 210 (c) 300 (d) 400

Ans. (d) $n^2 = 400$.

Q7. The nth term of an AP is $7 - 4n$. Its common difference is 1M
(a) 4 (b) 7 (c) -4 (d) 3

Ans. (c) $a_{n+1} - a_n = -4$.

Q8. If the first term of an AP is -5 and $d = 2$, the sum of its first 6 terms is 1M
(a) -30 (b) 0 (c) 5 (d) 15

Ans. (b) $S_6 = 3[-10 + 5(2)] = 0$.

Q9. If $k, 2k-1, 2k+1$ are three consecutive terms of an AP, then $k =$ 1M
(a) 2 (b) -3 (c) 3 (d) 6

Ans. (c) $2(2k-1) = k + 2k + 1 \Rightarrow k = 3$.

Q10. The number of two-digit numbers divisible by 3 is 1M Board fav
(a) 31 (b) 30 (c) 33 (d) 29

Ans. (b) $12, 15, \ldots, 99$: $\frac{99-12}{3} + 1 = 30$.

Q11. **A:** $3, 6, 12, 24, \ldots$ is not an AP. **R:** In an AP the difference of consecutive terms is constant. 1M
(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true

Ans. (a) Differences $3, 6, 12$ are not equal; R explains A.

Q12. **A:** The 20th term of the AP $10, 7, 4, \ldots$ is -47. **R:** The nth term of an AP is $a + nd$. 1M
(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true

Ans. (c) $10 + 19(-3) = -47$, A true; formula is $a + (n-1)d$, so R false.

Very short (2 marks)

Q13. The 11th term of an AP is 41 and the 16th term is 66. Find the 31st term. 2M

Ans. $5d = 25 \Rightarrow d = 5; a = 41 - 50 = -9; a_{31} = -9 + 30(5) = 141$.

Q14. Find the sum of the first 15 multiples of 8. 2M

Ans. $8(1 + 2 + \cdots + 15) = 8 \times 120 = 960$.

Q15. Which term of the AP $21, 18, 15, \ldots$ is zero? 2M

Ans. $21 + (n-1)(-3) = 0 \Rightarrow n - 1 = 7 \Rightarrow n = 8$. The 8th term.

Q16. Find the middle term of the AP $213, 205, 197, \ldots, 37$. 2M

Ans. $n = \frac{37-213}{-8} + 1 = 23$; middle $=$ 12th term $= 213 + 11(-8) = 125$.

Q17. Find the sum of the first 30 terms of the AP whose n th term is $3 + 2n$. 2M

Ans. $a_1 = 5,\ a_{30} = 63$. $S_{30} = 15(5 + 63) = 1020$.

Q18. The 2nd term of an AP is 13 and the 5th term is 25. Find the 7th term. 2M

Ans. $3d = 12 \Rightarrow d = 4, a = 9; a_7 = 9 + 24 = 33$.

Short answer (3 marks)

Q19. The sum of the 4th and 8th terms of an AP is 32 and the sum of the 6th and 10th terms is 52. Find the first three terms. 3M

Ans. $2a + 10d = 32, 2a + 14d = 52 \Rightarrow 4d = 20, d = 5, a = -9$. Terms: $-9, -4, 1$.

Q20. How many terms of the AP $5, 9, 13, \ldots$ must be taken to give a sum of 275? 3M

Ans. $\frac{n}{2}[10 + 4(n-1)] = n(2n+3) = 275 \Rightarrow 2n^2 + 3n - 275 = 0$. $D = 9 + 2200 = 2209 = 47^2, n = \frac{-3+47}{4} = 11$.

Q21. Find the sum of all two-digit odd numbers. 3M

Ans. $11, 13, \ldots, 99$: $n = \frac{99-11}{2} + 1 = 45$. $S = \frac{45}{2}(11 + 99) = 2475$.

Q22. Three numbers in AP have sum 27 and product 648. Find them. 3M

> **Ans.** Take $9-d, 9, 9+d$ (sum $27 \Rightarrow$ middle 9). $9(81-d^2)=648 \Rightarrow d^2=9, d=\pm 3$. Numbers 6, 9, 12.

Q23. If the pth term of an AP is q and the qth term is p ($p \neq q$), show that the $(p+q)$th term is 0. 3M

> **Ans.** $a+(p-1)d=q, a+(q-1)d=p$. Subtract: $(p-q)d=q-p \Rightarrow d=-1, a=p+q-1$. $a_{p+q}=a+(p+q-1)d=(p+q-1)-(p+q-1)=0$.

Q24. Find the sum $(-5)+(-8)+(-11)+\cdots+(-230)$. 3M

> **Ans.** $a=-5,\ d=-3$: $n=\frac{-230+5}{-3}+1=76$. $S=\frac{76}{2}(-5-230)=38\times(-235)=-8930$.

Q25. Find the sum of the first 22 terms of an AP in which $d=7$ and the 22nd term is 149. 3M

> **Ans.** $149=a+21(7) \Rightarrow a=2$. $S_{22}=\frac{22}{2}(2+149)=11\times 151=1661$.

Q26. The first and last terms of an AP are 17 and 350, and the common difference is 9. How many terms are there and what is their sum? 3M

> **Ans.** $350=17+(n-1)9 \Rightarrow n=38$. $S=\frac{38}{2}(17+350)=19\times 367=6973$.

Q27. Show that $a_1, a_2, \ldots$ form an AP if $a_n=3+4n$, and find its common difference. 3M

> **Ans.** $a_{n+1}-a_n=(3+4n+4)-(3+4n)=4$, a constant, so it is an AP with $d=4$ (terms $7, 11, 15, \ldots$).

Q28. Find the AP whose first term is 5 and the sum of whose first four terms is half the sum of the next four terms. 3M

> **Ans.** $S_4=2(10+3d)=20+6d$. $S_8-S_4=4(10+7d)-(20+6d)=20+22d$. $20+6d=\frac{20+22d}{2} \Rightarrow d=2$. AP: $5, 7, 9, 11, \ldots$

Q29. How many terms of the AP $63, 60, 57, \ldots$ must be taken for the sum to be 693? Explain the double answer. 3M

> **Ans.** $\frac{n}{2}[126-3(n-1)]=693 \Rightarrow n^2-43n+462=0 \Rightarrow (n-21)(n-22)=0$. $n=21$ or 22: the 22nd term is $63-63=0$, so adding it does not change the sum.

Long answer (5 marks)

Q30. The sum of the first 7 terms of an AP is 63 and the sum of its first 17 terms is 323. Find the sum of its first n terms. 5M

> **Ans.** $S_7=7(a+3d)=63 \Rightarrow a+3d=9$. $S_{17}=17(a+8d)=323 \Rightarrow a+8d=19$. So $d=2,\ a=3$. $S_n=\frac{n}{2}[6+2(n-1)]=n^2+2n$.

Apne notes / galtiyan yahan likho

Q31. A sum of Rs 1050 is to be given as 6 prizes, each Rs 30 less than the previous one. Find the value of each prize. 5M

> **Ans.** $\frac{6}{2}[2a+5(-30)]=1050 \Rightarrow 2a-150=350 \Rightarrow a=250$. Prizes: Rs 250, 220, 190, 160, 130, 100 (sum $=1050$ ✓).

Q32. In an AP, the first term is 8, the last term is 350 and the common difference is 9. Find the number of terms and their sum. 5M

> **Ans.** $350=8+(n-1)9 \Rightarrow n-1=38 \Rightarrow n=39$. $S=\frac{39}{2}(8+350)=39\times 179=6981$.

Q33. If the sum of the first 14 terms of an AP is 1050 and its first term is 10, find the 20th term and the sum of the first 20 terms. 5M

> **Ans.** $\frac{14}{2}[20+13d]=1050 \Rightarrow 20+13d=150 \Rightarrow d=10$. $a_{20}=10+19(10)=200$; $S_{20}=10(10+200)=2100$.

Q34. The 17th term of an AP exceeds its 10th term by 7. Its first term is 4. Find the sum of its first 20 terms and the 30th term. 5M

> **Ans.** $7d=7 \Rightarrow d=1$. $a_{30}=4+29=33$; $S_{20}=10(8+19)=270$.

Q35. If the sum of the first m terms of an AP equals the sum of its first n terms ($m \neq n$), show that the sum of its first $(m+n)$ terms is zero. 5M

> **Ans.** $\frac{m}{2}[2a+(m-1)d]=\frac{n}{2}[2a+(n-1)d] \Rightarrow 2a(m-n)+d(m^2-m-n^2+n)=0$. Divide by $(m-n)$: $2a+(m+n-1)d=0$. So $S_{m+n}=\frac{m+n}{2}[2a+(m+n-1)d]=0$.

Q36. A factory produced 720 TV sets in its 3rd year and 840 in its 7th year. Production rises by the same number every year. Find the production in the 1st year and the 10th year, and the total in the first 7 years. 5M

> **Ans.** $4d=120 \Rightarrow d=30; a=720-60=660$. $a_{10}=660+270=930$. $S_7=\frac{7}{2}(1320+180)=7\times 750=5250$.

Case based (4 marks)

Q37. Weekly savings: Rohan saves Rs 50 in the first week and increases his saving by Rs 20 every week. (i) How much does he save in the 10th week? (1) (ii) How much has he saved in total in 10 weeks? (2) (iii) In which week will his weekly saving be Rs 450? (1) 4M

> **Ans.** $a=50,\ d=20$. (i) $a_{10}=50+9(20)=$ Rs 230. (ii) $S_{10}=5(100+180)=$ Rs 1400. (iii) $50+20(n-1)=450 \Rightarrow n=21$: the 21st week.

Q38. Auditorium: The first row of a hall has 18 seats and each next row has 2 more seats. There are 25 rows. (i) Seats in the last row? (1) (ii) Total seats? (2) (iii) Which row has 40 seats? (1) 4M

> **Ans.** (i) $18+24(2)=66$. (ii) $\frac{25}{2}(18+66)=1050$. (iii) $18+2(n-1)=40 \Rightarrow n=12$: the 12th row.

Most asked: BPT proof, AA similarity, shadow + ratio sums

TRIANGLES

1. Similar Figures

Two figures are **similar** if they have the same shape but not necessarily the same size. **Congruent** figures have same shape AND same size.

- ✓ Every congruent pair is similar, but every similar pair is not congruent.
- ✓ **Always similar:** all circles, all squares, all equilateral triangles, all line segments.
- ✓ Two polygons with the same number of sides are similar if **(i)** corresponding angles are equal AND **(ii)** corresponding sides are in the same ratio. Both are needed for polygons.
- ✓ Square vs rhombus (same side): sides proportional, angles not equal, so not similar. Square vs rectangle: angles equal, sides not proportional, so not similar.

Definition: Similar triangles

$\triangle ABC \sim \triangle DEF$ means $\angle A = \angle D$, $\angle B = \angle E$, $\angle C = \angle F$ and

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}$$

The ratio is called the **scale factor**. For triangles, if one condition holds the other holds automatically.

Yaad karo: "Naam ka order = partner ka order"

$\triangle ABC \sim \triangle QPR$ likha hai to A ka partner Q, B ka P, C ka R. Ratio bhi isi order mein: AB/QP = BC/PR = CA/RQ.

2. Basic Proportionality Theorem (BPT)

Also called **Thales Theorem**. **Statement:** If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the **same ratio**.

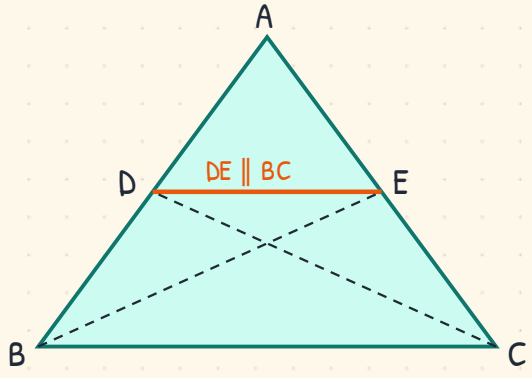


Fig: In triangle ABC, DE is parallel to BC

Quick fact Thales (Greek mathematician) gave the BPT, so it is also called Thales Theorem.

Theorem: Proof of BPT (3 to 5 marks)

Given: In $\triangle ABC$, $DE \parallel BC$, D on AB, E on AC. **To prove:** $\frac{AD}{DB} = \frac{AE}{EC}$.

Construction: Join BE and CD; draw $EN \perp AB$ and $DM \perp AC$.

1. $\text{ar}(ADE) = \frac{1}{2} AD \cdot EN$, $\text{ar}(BDE) = \frac{1}{2} DB \cdot EN$ so $\frac{\text{ar}(ADE)}{\text{ar}(BDE)} = \frac{AD}{DB}$ ...(1)
2. Similarly $\frac{\text{ar}(ADE)}{\text{ar}(DEC)} = \frac{AE}{EC}$ ...(2)
3. $\triangle BDE$, $\triangle DEC$ are on the same base DE and between the same parallels DE, BC, so $\text{ar}(BDE) = \text{ar}(DEC)$ ... (3)
4. From (1), (2), (3): $\frac{AD}{DB} = \frac{AE}{EC}$. **Hence proved.**

Converse of BPT

If a line divides any two sides of a triangle in the **same ratio**, then the line is parallel to the third side. Use it when the question says "show that DE ∥ BC".

Theorem: Converse of BPT (proof idea)

Given: $\frac{AD}{DB} = \frac{AE}{EC}$. **To prove:** $DE \parallel BC$.

1. Suppose DE is not parallel to BC. Draw $DE' \parallel BC$ with E' on AC.
2. By BPT, $\frac{AD}{DB} = \frac{AE'}{E'C}$, so $\frac{AE}{EC} = \frac{AE'}{E'C}$.
3. Add 1 to both sides: $\frac{AC}{EC} = \frac{AC}{E'C} \Rightarrow EC = E'C$.
4. So E and E' coincide, i.e. $DE \parallel BC$.

Solved Example 1

In $\triangle ABC$, $DE \parallel BC$. If $AD = x$, $DB = x - 2$, $AE = x + 2$ and $EC = x - 1$, find x.

1. BPT: $\frac{x}{x-2} = \frac{x+2}{x-1}$
2. $x(x-1) = (x-2)(x+2)$
3. $x^2 - x = x^2 - 4$
4. $-x = -4$

✔ $x = 4$ (check: $\frac{4}{2} = \frac{6}{3} = 2$)

Quick fact If $\triangle ABC \sim \triangle DEF$ with $k = \frac{AB}{DE}$, then perimeter ratio $= k$ too.

Formula Box

- BPT ($DE \parallel BC$): $\frac{AD}{DB} = \frac{AE}{EC}$
- Other form: $\frac{AD}{AB} = \frac{AE}{AC}$
- Other form: $\frac{DB}{AB} = \frac{EC}{AC}$
- $\triangle ADE \sim \triangle ABC$: $\frac{DE}{BC} = \frac{AD}{AB}$
- Similar triangles: $\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD} = k$
- Perimeters, altitudes, medians, angle bisectors are all in the same ratio k
- Trapezium $AB \parallel DC$, diagonals meet at O: $\frac{AO}{OC} = \frac{BO}{OD}$
- Trapezium, $EF \parallel AB$: $\frac{AE}{ED} = \frac{BF}{FC}$

Solved Example 2

E and F are points on sides PQ and PR of $\triangle PQR$. $PE = 4$ cm, $EQ = 4.5$ cm, $PF = 8$ cm, $FR = 9$ cm. Is $EF \parallel QR$?

1. $\frac{PE}{EQ} = \frac{4}{4.5} = \frac{8}{9}$
2. $\frac{PF}{FR} = \frac{8}{9}$
3. Ratios equal, so by converse of BPT, $EF \parallel QR$.

✔ Yes, EF is parallel to QR.

Yaad karo: "x-square cancel ho jaata hai"

BPT algebra mein cross multiply karte hi x^2 dono side cancel. Agar cancel nahi hua to check karo: kahin AD/AB ko AD/DB ke saath mix to nahi kiya?

Trapezium results (from BPT)

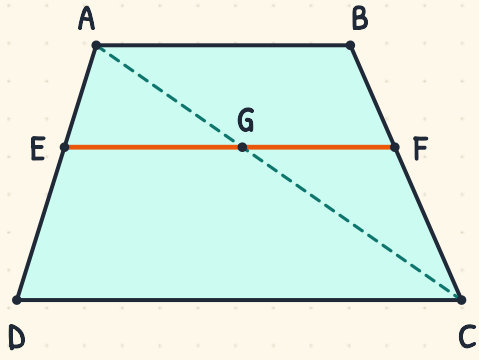


Fig: Trapezium ABCD, AB parallel to DC, EF parallel to AB

In trapezium ABCD with $AB \parallel DC$: (i) a line $EF \parallel AB$ meeting AD at E and BC at F gives $AE : ED = BF : FC$ (join AC, apply BPT in $\triangle ADC$ and $\triangle CAB$). (ii) Diagonals divide each other in the same ratio: $AO : OC = BO : OD$ (because $\triangle AOB \sim \triangle COD$ by AA).

Meme samjho

Tum: $\triangle ABC \sim \triangle PQR$ ka ratio likhte ho AB/QR. Examiner: **"Kya tapleek hai aapko?"**

Asli baat: Correspondence letter order se aati hai: A-P, B-Q, C-R, so AB/PQ = BC/QR = CA/RP. Galat pairing = poora sum galat.

3. Criteria for Similarity of Triangles

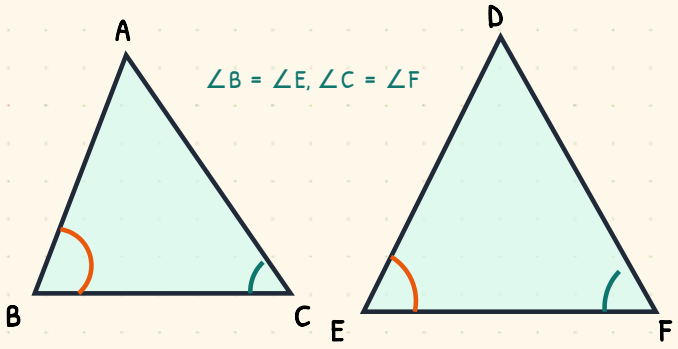


Fig: Equal angles (AA) and proportional sides (SSS, SAS)

Criterion	What you must show
AAA / AA	Corresponding angles equal. Two angles are enough (3rd is automatic, angle sum 180 degrees).
SSS	All three pairs of corresponding sides in the same ratio.
SAS	One angle equal AND the sides including it in the same ratio.

↳ Similarity ke sirf 3 criteria: AA, SSS, SAS. Congruence wale ASA, AAS, RHS yahan alag se nahi chahiye.

Yaad karo: "AA-SSS-SAS = 1 + 3 + 2"

AA mein 2 angle, SSS mein 3 side-ratio, SAS mein 2 side-ratio + beech ka 1 angle. Jo maanga hai utna hi dikhao, extra mat ghumao.

Common mistakes

- **SSA is NOT a criterion.** In SAS the equal angle must be between the two proportional sides.
- Writing ratios in wrong order: follow the vertex order of the similarity statement.
- Using AD/AB on one side and AE/EC on the other (mixed forms of BPT).
- Using BPT without first stating $DE \parallel BC$ (lose the reason mark).
- Square and rhombus "similar"? No: angles differ. Both conditions needed for polygons.

4. Using Similarity: Shadows and Heights

At the same time of day, sun rays make **equal angles**

with the ground. Object and its shadow form a right triangle, so the two triangles are similar by **AA**:

$$\frac{\text{height}_1}{\text{shadow}_1} = \frac{\text{height}_2}{\text{shadow}_2}$$

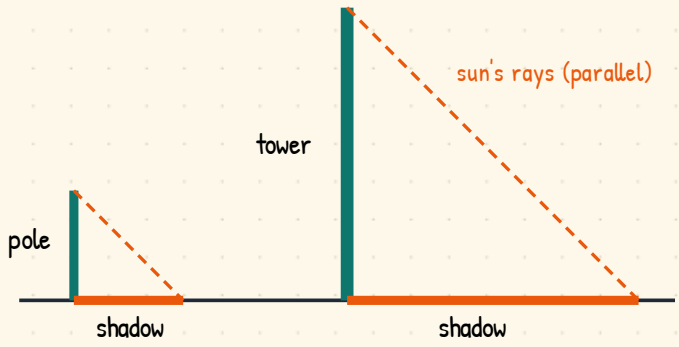


Fig: Pole and tower with their shadows at the same time

Solved Example 3

A vertical pole 6 m high casts a shadow 4 m long. At the same time a tower casts a shadow 28 m long. Find the height of the tower.

1. Both right triangles have equal sun angle, so they are similar (AA).
2. $\frac{h}{28} = \frac{6}{4}$
3. $h = \frac{6 \times 28}{4}$

✔ **Height of tower** $= 42$ m

Solved Example 4

A boy 1.5 m tall walks away from the base of a lamp-post at 1 m/s. The lamp is 4.5 m above the ground. Find the length of his shadow after 6 seconds.

1. Distance walked $= 1 \times 6 = 6$ m. Let shadow $= x$ m.
2. Boy and lamp-post are both vertical, light ray common: triangles similar (AA).
3. $\frac{x}{x+6} = \frac{1.5}{4.5} = \frac{1}{3} \Rightarrow 3x = x + 6$

✔ $x = 3$ m

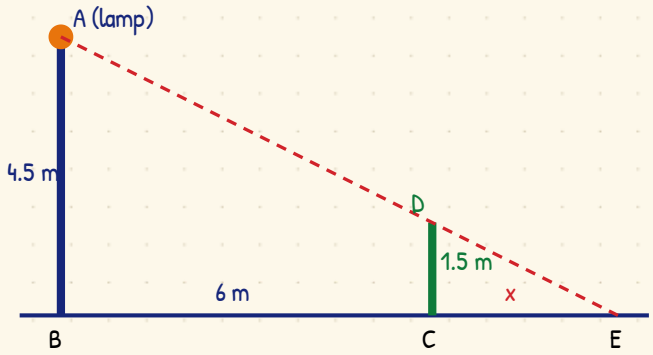


Fig: Solved Example 4. $\triangle ECD \sim \triangle EBA$ (AA); shadow CE = x

Solved Example 5

In $\triangle ABC$, $\angle A = 90°$ and $AD \perp BC$. Prove that $AB^2 = BC \cdot BD$.

1. In $\triangle ADB$ and $\triangle CAB$: $\angle ADB = \angle CAB = 90°$, $\angle B$ common.
2. So $\triangle ADB \sim \triangle CAB$ (AA).
3. $\frac{AB}{CB} = \frac{BD}{BA}$

✔ $AB^2 = BC \cdot BD$. **Hence proved.**

Shivam Bhaiya says

Triangles mein 5 marks ka proof pakka aata hai: **BPT proof** figure + Given + To prove + Construction ke saath likho. Figure ke bina proof = adhe marks. Aur similarity likhte waqt **reason bracket mein**: (AA), (SAS), (BPT).

NCERT exercise results to remember

- → Line through mid-point of one side, parallel to another side, bisects the third side.
- → Line joining mid-points of two sides is parallel to the third side.
- → In trapezium, a line parallel to the parallel sides cuts the legs proportionally.
- → If $AO : BO = CO : DO$ for diagonals of ABCD, then ABCD is a trapezium.
- → Ratio of medians, altitudes, angle bisectors and perimeters of similar triangles = ratio of sides.

Proof writing template

1. Name the two triangles in the correct order.
2. Write the equal angles/ratios with reasons (vertically opposite, alternate, common, given).
3. State the criterion: $\triangle__ \sim \triangle__$ (AA / SAS / SSS).
4. Write the ratio of corresponding sides and finish.

Meme samjho

SSA se similarity prove karke bhi itna confidence: **"Sir, perfect hai na?"**
Asli baat: SSA koi criterion nahi hai. SAS mein angle hamesha dono proportional sides ke **beech** wala hona chahiye.

5. Important Similarity Results (proof type)

These results come straight from AA similarity. Board mein proof ya numerical dono aate hain.

Quick fact Two triangles with two pairs of equal angles are always similar (AA).

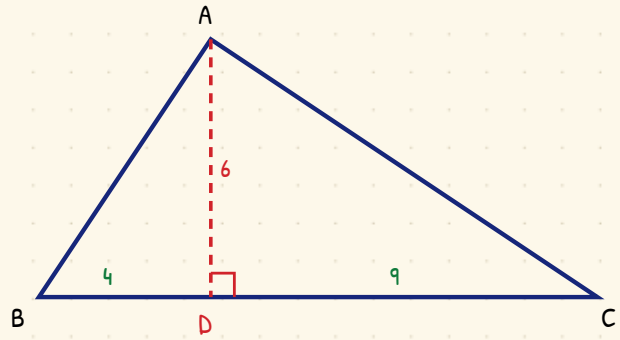


Fig: Solved Example 6. Angle A = 90 degrees, AD perpendicular to BC; BD = 4, DC = 9, AD = 6 (to scale)

Result: Perpendicular from the right angle

In $\triangle ABC$ with $\angle A = 90^\circ$ and $AD \perp BC$:

→ $\triangle ADB \sim \triangle CAB$ and $\triangle ADC \sim \triangle BAC$ (AA)

→ $\triangle ADB \sim \triangle CDA$ (AA), so $\frac{AD}{CD} = \frac{BD}{AD}$

$$AD^2 = BD \cdot DC$$

$$AB^2 = BD \cdot BC$$

$$AC^2 = CD \cdot CB$$

Solved Example 6

In $\triangle ABC$, $\angle A = 90^\circ$, $AD \perp BC$, $BD = 4$ cm and $DC = 9$ cm. Find AD, AB and AC.

1. $AD^2 = BD \cdot DC = 4 \times 9 = 36 \Rightarrow AD = 6$ cm
2. $AB^2 = BD \cdot BC = 4 \times 13 = 52 \Rightarrow AB = 2\sqrt{13}$ cm
3. $AC^2 = CD \cdot CB = 9 \times 13 = 117 \Rightarrow AC = 3\sqrt{13}$ cm

✔ $AD = 6$ cm, $AB = 2\sqrt{13}$ cm, $AC = 3\sqrt{13}$ cm

Result: Double BPT (NCERT favourite)

In $\triangle ABC$, D on AB, E on BC, F on BE with $DE \parallel AC$ and $DF \parallel AE$. Then $\frac{BF}{FE} = \frac{BE}{EC}$.

Proof: In $\triangle ABC$, $DE \parallel AC$: $\frac{BD}{DA} = \frac{BE}{EC}$ ...(1). In $\triangle ABE$, $DF \parallel AE$: $\frac{BD}{DA} = \frac{BF}{FE}$ ...(2). From (1), (2): $\frac{BF}{FE} = \frac{BE}{EC}$.

Parallel line in triangle → BPT ratio $\frac{AD}{DB} = \frac{AE}{EC}$ → Similar (AA) $\frac{DE}{BC} = \frac{AD}{AB}$

Quick fact Mid-point theorem is BPT with ratio $1:1$: the line joining mid-points is parallel to the third side and half of it.

HOTS: two poles and crossing wires

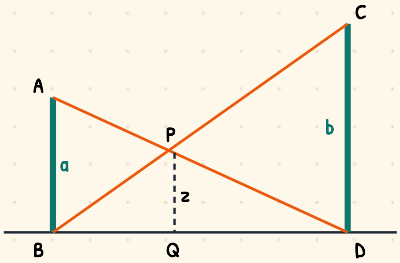


Fig: Poles AB = a and CD = b; wires AD and CB cross at P; PQ = z

$\triangle BQP \sim \triangle BDC$ (AA) gives $\frac{z}{b} = \frac{BQ}{BD}$; $\triangle DQP \sim \triangle DBA$ (AA) gives $\frac{z}{a} = \frac{DQ}{DB}$. Add: $\frac{z}{a} + \frac{z}{b} = \frac{BQ + DQ}{BD} = 1$.

$$\frac{1}{z} = \frac{1}{a} + \frac{1}{b}$$

↳Example: poles 6 m and 3 m: $\frac{1}{z} = \frac{1}{6} + \frac{1}{3} = \frac{1}{2}$, so wires cross 2 m above ground, poles kitni bhi door hon!

Ek nazar mein: kaunsa criterion kab?

Question mein dikhe	Use karo
Line parallel to a side	BPT, or AA (corresponding angles)
Vertically opposite / alternate / common angle	AA
All three sides given as numbers	SSS (check 3 ratios)
Two sides around a known angle	SAS
Shadow, lamp-post, mirror, ladder	AA, then ratio

Congruent (≅)

Same shape + same size. Criteria: SSS, SAS, ASA, AAS, RHS. Ratio $k = 1$.

Similar (∼)

Same shape, any size. Criteria: AA, SSS, SAS. Ratio k any positive number.

Yaad karo: "Shadow sum = H by S same"

Ek hi time pe sabka (height divided by shadow) same. Lamp-post sum mein shadow ka tip common vertex hai: boy / shadow = lamp / (shadow + distance).

Shivam Bhaiya says

Ratio sums mein pehle **figure banao aur parallel lines mark karo.** Phir decide karo: BPT (AD : DB wala form) ya small-big similar triangle (DE : BC = AD : AB wala form). Third side DE chahiye to BPT nahi, similarity lagao!

Chapter at a glance

Topic	Key line	Marks
Similar figures	Same shape; angles equal + sides proportional	1
BPT + converse	Parallel $\Leftrightarrow$ same ratio	1 to 5
AA, SSS, SAS	Name triangles in order, give reason	2 to 5
Applications	Shadow, lamp-post, mirror, trapezium	3 to 4

True or False? (quick self-check)

1. All isosceles triangles are similar. False (angles can differ)
2. All equilateral triangles are similar. True
3. Two congruent triangles are always similar. True ($k = 1$)
4. If two angles of one triangle equal two angles of another, they are similar. True (AA)
5. SSA proves similarity. False
6. Line joining mid-points of two sides is parallel to the third. True

MOST IMPORTANT QUESTIONS

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. If $\triangle ABC \sim \triangle PQR$, $\angle A = 50^\circ$ and $\angle Q = 60^\circ$, then $\angle R =$ 1M

(a) 50° (b) 60° (c) 70° (d) 80°

Ans. (c) $\angle R = \angle C = 180^\circ - 50^\circ - 60^\circ = 70^\circ$

Q2. In $\triangle ABC$, $DE \parallel BC$, $AD = 2$ cm, $AB = 6$ cm, $AE = 3$ cm. Then $AC =$ 1M Board fav

(a) 4.5 cm (b) 6 cm (c) 9 cm (d) 12 cm

Ans. (c) $\frac{AD}{AB} = \frac{AE}{AC} \Rightarrow \frac{2}{6} = \frac{3}{AC} \Rightarrow AC = 9$ cm

Q3. Which of these is NOT a criterion for similarity of triangles? 1M

(a) AA (b) SSS (c) SAS (d) SSA

Ans. (d) SSA

Q4. A 12 m pole casts an 8 m shadow. At the same time a tree casts a 20 m shadow. Height of the tree is 1M

(a) 24 m (b) 30 m (c) 32 m (d) 40 m

Ans. (b) $\frac{h}{20} = \frac{12}{8} \Rightarrow h = 30$ m

Q5. $\triangle ABC \sim \triangle DEF$, perimeters 30 cm and 18 cm, $BC = 9$ cm. Then $EF =$ 1M

(a) 5.4 cm (b) 6 cm (c) 7.5 cm (d) 15 cm

Ans. (a) ratio $= \frac{30}{18} = \frac{5}{3}$, so $EF = 9 \times \frac{3}{5} = 5.4$ cm

Q6. In $\triangle ABC$ and $\triangle PQR$, $\frac{AB}{QR} = \frac{BC}{PR} = \frac{CA}{PQ}$. Then 1M

(a) $\triangle PQR \sim \triangle CAB$ (b) $\triangle PQR \sim \triangle ABC$ (c) $\triangle CBA \sim \triangle PQR$ (d) $\triangle BCA \sim \triangle PQR$

Ans. (a) Pairing: A-Q, B-R, C-P, so $\triangle ABC \sim \triangle QRP$, i.e. $\triangle PQR \sim \triangle CAB$.

Q7. $\triangle ABC \sim \triangle DEF$, $AB = 4$ cm, $DE = 6$ cm, $EF = 9$ cm. Then $BC =$ 1M

(a) 4.5 cm (b) 6 cm (c) 7.5 cm (d) 13.5 cm

Ans. (b) $\frac{BC}{9} = \frac{4}{6} \Rightarrow BC = 6$ cm

Q8. **A:** In $\triangle ABC$, $DE \parallel BC$ and $\frac{AD}{DB} = \frac{3}{5}$, so $\frac{AE}{AC} = \frac{3}{8}$. **R:** A line parallel to one side of a triangle divides the other two sides in the same ratio. 1M

(a) Both true, R explains A (b) Both true, R does not explain A

(c) A true, R false (d) A false, R true

Ans. (a) $\frac{AE}{EC} = \frac{3}{5} \Rightarrow \frac{AE}{AC} = \frac{3}{3+5} = \frac{3}{8}$; R is BPT and explains A.

Q9. **A:** A square and a rhombus of the same side are similar. **R:** For similar polygons, corresponding angles must be equal and corresponding sides proportional. 1M

(a) Both true, R explains A (b) Both true, R does not explain A

(c) A true, R false (d) A false, R true

Ans. (d) A is false (angles of rhombus are not all 90 degrees); R is true.

Q10. **A:** If $\triangle ABC \sim \triangle DEF$ and $AB = 2DE$, then perimeter of $\triangle ABC$ is twice that of $\triangle DEF$. **R:** Ratio of perimeters of similar triangles equals the ratio of corresponding sides. 1M

(a) Both true, R explains A (b) Both true, R does not explain A

(c) A true, R false (d) A false, R true

Ans. (a) Both true and R explains A.

Q11. **A:** Triangles with sides 2, 3, 4 cm and 4, 6, 7 cm are similar. **R:** For SSS similarity all three pairs of sides must be in the same ratio. 1M

(a) Both true, R explains A (b) Both true, R does not explain A

(c) A true, R false (d) A false, R true

Ans. (d) $\frac{2}{4} = \frac{3}{6} \neq \frac{4}{7}$, so A is false; R is true.

Very short (2 marks)

Q12. In $\triangle ABC$, $DE \parallel BC$, $AD = x + 1$, $DB = 3x - 1$, $AE = x + 3$, $EC = 3x + 4$. Find x. 2M Board fav

Ans. $(x+1)(3x+4) = (x+3)(3x-1) \Rightarrow 3x^2 + 7x + 4 = 3x^2 + 8x - 3 \Rightarrow x = 7$. (AD = 8, DB = 20, AE = 10, EC = 25; both ratios $= 0.4$)

Q13. E, F lie on PQ, PR of $\triangle PQR$ with $PE = 4.2$ cm, $EQ = 3$ cm, $PF = 5.6$ cm, $FR = 4$ cm. Is $EF \parallel QR$? 2M

Ans. $\frac{PE}{EQ} = \frac{4.2}{3} = 1.4$, $\frac{PF}{FR} = \frac{5.6}{4} = 1.4$. Equal, so $EF \parallel QR$ (converse of BPT).

Q14. Are triangles with sides 3, 4, 5 cm and 4.5, 6, 7.5 cm similar? Give the scale factor. 2M

Ans. $\frac{3}{4.5} = \frac{4}{6} = \frac{5}{7.5} = \frac{2}{3}$. Yes, similar by SSS; ratio 2 : 3.

Q15. Diagonals of trapezium ABCD ($AB \parallel DC$) meet at O. $OA = 3x - 1$, $OC = 5x - 3$, $OB = 2x + 1$, $OD = 6x - 5$. Find x. 2M

(contd.) $\frac{OA}{OC} = \frac{OB}{OD} \Rightarrow (3x-1)(6x-5) = (5x-3)(2x+1) \Rightarrow 8x^2 - 20x + 8 = 0 \Rightarrow 2x^2 - 5x + 2 = 0$, $x = 2$ or $\frac{1}{2}$. $x = \frac{1}{2}$ makes OC negative, so $x = 2$.

Q16. If $\triangle ABC \sim \triangle DEF$, $\angle A = 47°$ and $\angle E = 63°$, find $\angle C$. 2M

Ans. $\angle B = \angle E = 63°$, so $\angle C = 180° - 47° - 63° = 70°$.

Q17. Perimeters of similar triangles ABC and PQR are 36 cm and 24 cm. If $PQ = 10$ cm, find AB. 2M

Ans. $\frac{AB}{PQ} = \frac{36}{24} = \frac{3}{2} \Rightarrow AB = 15$ cm.

Q18. In $\triangle ABC$, $DE \parallel BC$ with $AD = 4$ cm, $AB = 12$ cm and $BC = 9$ cm. Find DE. 2M

Ans. $\triangle ADE \sim \triangle ABC$ (AA): $\frac{DE}{9} = \frac{4}{12} \Rightarrow DE = 3$ cm.

Q19. $\triangle ABC \sim \triangle PQR$, $AB = 3.6$ cm, $PQ = 6$ cm and altitude $AL = 2.7$ cm. Find altitude PM. 2M

Ans. Altitudes are in the ratio of sides: $\frac{PM}{2.7} = \frac{6}{3.6} \Rightarrow PM = 4.5$ cm.

Q20. In the figure of Ex 6 ($\angle A = 90°$, $AD \perp BC$), $BD = 8$ cm and $AD = 4$ cm. Find DC and BC. 2M

Ans. $AD^2 = BD \cdot DC \Rightarrow 16 = 8 \cdot DC \Rightarrow DC = 2$ cm; $BC = 8 + 2 = 10$ cm.

Q21. In $\triangle ABC$, $DE \parallel BC$, $\frac{AD}{DB} = \frac{2}{3}$ and $AE = 4$ cm. Find EC and $\frac{DE}{BC}$. 2M

Ans. $\frac{4}{EC} = \frac{2}{3} \Rightarrow EC = 6$ cm. $\frac{DE}{BC} = \frac{AD}{AB} = \frac{2}{5}$.

Short answer (3 marks)

Q22. In $\triangle ABC$, P and Q lie on AB and AC with $PQ \parallel BC$. $AP = 2.4$ cm, $PB = 3.6$ cm, $AQ = 2$ cm, $BC = 7.5$ cm. Find QC and PQ. 3M

Ans. BPT: $\frac{2.4}{3.6} = \frac{2}{QC} \Rightarrow QC = 3$ cm. $\triangle APQ \sim \triangle ABC$ (AA): $\frac{PQ}{BC} = \frac{AP}{AB} = \frac{2.4}{6}$, so $PQ = 0.4 \times 7.5 = 3$ cm.

Q23. A girl 1.2 m tall walks away from a lamp-post at 1.5 m/s. The bulb is 4.8 m above the ground. Find the length of her shadow after 4 seconds. 3M Board fav

Ans. Distance $= 6$ m. $\frac{x}{x+6} = \frac{1.2}{4.8} = \frac{1}{4} \Rightarrow 4x = x + 6 \Rightarrow x = 2$ m.

Q24. ABCD is a trapezium with $AB \parallel DC$; diagonals meet at O. Prove that $\frac{AO}{BO} = \frac{CO}{DO}$. 3M

Ans. In $\triangle AOB$, $\triangle COD$: $\angle OAB = \angle OCD$ (alternate, $AB \parallel DC$), $\angle AOB = \angle COD$ (vertically opposite). So $\triangle AOB \sim \triangle COD$ (AA) $\Rightarrow \frac{AO}{CO} = \frac{BO}{DO} \Rightarrow \frac{AO}{BO} = \frac{CO}{DO}$.

Q25. $\triangle ABC \sim \triangle PQR$; CM and RN are medians. Prove $\triangle AMC \sim \triangle PNR$. 3M

Ans. $\frac{AB}{PQ} = \frac{AC}{PR}$ and $\angle A = \angle P$. $AM = \frac{AB}{2}$, $PN = \frac{PQ}{2}$, so $\frac{AM}{PN} = \frac{AB}{PQ} = \frac{AC}{PR}$. With $\angle A = \angle P$: $\triangle AMC \sim \triangle PNR$ (SAS).

Q26. In $\triangle PQR$, $\angle PQR = \angle PRQ$. S lies on QR and T on PQ such that $\frac{QR}{QS} = \frac{QT}{PR}$. Show that $\triangle PQS \sim \triangle TQR$. 3M

Ans. $\angle PQR = \angle PRQ \Rightarrow PQ = PR$. So $\frac{QR}{QS} = \frac{QT}{PQ} \Rightarrow \frac{PQ}{QT} = \frac{QS}{QR}$. $\angle Q$ is common, so $\triangle PQS \sim \triangle TQR$ (SAS).

Q27. E is a point on side AD produced of parallelogram ABCD, and BE meets CD at F. Show that $\triangle ABE \sim \triangle CFB$. 3M Board fav

Ans. $\angle A = \angle C$ (opposite angles of a parallelogram). $AE \parallel BC \Rightarrow \angle AEB = \angle CBF$ (alternate angles). So $\triangle ABE \sim \triangle CFB$ (AA).

Q28. ABCD is a parallelogram and P is a point on BC. DP produced meets AB produced at L. Prove that $\frac{DP}{PL} = \frac{DC}{BL}$. 3M

Ans. In $\triangle DPC$ and $\triangle LPB$: $\angle DPC = \angle LPB$ (vertically opposite), $\angle DCP = \angle LBP$ (alternate, $DC \parallel AL$). So $\triangle DPC \sim \triangle LPB$ (AA) $\Rightarrow \frac{DP}{LP} = \frac{DC}{LB}$.

Q29. M is the mid-point of side CD of parallelogram ABCD. BM meets AC at L and AD produced at E. Prove that $EL = 2BL$. 3M

Ans. $\triangle BMC \cong \triangle EMD$ (ASA: $MC = MD$, vertically opposite angles, alternate angles) $\Rightarrow DE = BC$, so $AE = AD + DE = 2BC$. $\triangle AEL \sim \triangle CBL$ (AA, $AE \parallel BC$) $\Rightarrow \frac{EL}{BL} = \frac{AE}{CB} = 2$. So $EL = 2BL$.

Q30. In $\triangle ABC$, D on AB and E on AC with $\frac{AD}{DB} = \frac{AE}{EC}$ and $\angle ADE = \angle ACB$. Prove that $\triangle ABC$ is isosceles. 3M Board fav

Ans. Equal ratios $\Rightarrow DE \parallel BC$ (converse of BPT) $\Rightarrow \angle ADE = \angle ABC$ (corresponding). Given $\angle ADE = \angle ACB$, so $\angle ABC = \angle ACB \Rightarrow AC = AB$. Isosceles.

Q31. A, B, C lie on OP, OQ, OR respectively with $AB \parallel PQ$ and $AC \parallel PR$. Show that $BC \parallel QR$. 3M

Ans. In $\triangle OPQ$: $AB \parallel PQ \Rightarrow \frac{OA}{AP} = \frac{OB}{BQ}$. In $\triangle OPR$: $AC \parallel PR \Rightarrow \frac{OA}{AP} = \frac{OC}{CR}$. So $\frac{OB}{BQ} = \frac{OC}{CR}$; by converse of BPT in $\triangle OQR$, $BC \parallel QR$.

Q32. Using BPT, prove that a line through the mid-point of one side of a triangle, parallel to another side, bisects the third side. 3M

Ans. In $\triangle ABC$, D mid-point of AB, $DE \parallel BC$. BPT: $\frac{AD}{DB} = \frac{AE}{EC}$. $AD = DB \Rightarrow \frac{AE}{EC} = 1 \Rightarrow AE = EC$. So E is the mid-point of AC.

Q33. Diagonals of quadrilateral ABCD meet at O such that $\frac{AO}{BO} = \frac{CO}{DO}$. Show that ABCD is a trapezium. 3M

Ans. Draw $OE \parallel AB$ meeting AD at E. In $\triangle ABD$: $\frac{AE}{ED} = \frac{BO}{OD}$. Given $\frac{AO}{OC} = \frac{BO}{OD}$, so $\frac{AE}{ED} = \frac{AO}{OC}$; by converse of BPT in $\triangle ADC$, $EO \parallel DC$. Hence $AB \parallel DC$: trapezium.

Q34. In the figure, $DE \parallel OQ$ and $DF \parallel OR$, where D lies on PO, E on PQ and F on PR. Show that $EF \parallel QR$. 3M

Ans. In $\triangle POQ$: $DE \parallel OQ \Rightarrow \frac{PE}{EQ} = \frac{PD}{DO}$. In $\triangle POR$:

(contd.) $DF \parallel OR \Rightarrow \frac{PF}{FR} = \frac{PD}{DO}$. So $\frac{PE}{EQ} = \frac{PF}{FR}$; by converse of BPT in $\triangle PQR$, $EF \parallel QR$.

Long answer (5 marks)

Q35. State and prove BPT. Using it, in $\triangle ABC$ with $DE \parallel BC$, $AD = 3$ cm, $DB = 5$ cm, $AC = 12$ cm, find AE. 5M Board fav

Ans. Statement and proof as in the theorem box above (Given, To prove, Construction, areas (1) (2) (3)). Then $\frac{AE}{AC} = \frac{AD}{AB} = \frac{3}{8} \Rightarrow AE = \frac{3}{8} \times 12 = 4.5$ cm.

Q36. Sides AB, BC and median AD of $\triangle ABC$ are proportional to sides PQ, QR and median PM of $\triangle PQR$. Show $\triangle ABC \sim \triangle PQR$. 5M

Ans. $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AD}{PM}$. Since $BD = \frac{BC}{2}$, $QM = \frac{QR}{2}$: $\frac{AB}{PQ} = \frac{BD}{QM} = \frac{AD}{PM}$, so $\triangle ABD \sim \triangle PQM$ (SSS) $\Rightarrow \angle B = \angle Q$. Now $\frac{AB}{PQ} = \frac{BC}{QR}$ and $\angle B = \angle Q$, so $\triangle ABC \sim \triangle PQR$ (SAS).

Q37. In $\triangle PQR$, $ST \parallel QR$ with S on PQ, T on PR. $PS = 4$ cm, $SQ = 6$ cm, $ST = 5$ cm, $PT = 3.2$ cm. Find QR, TR and the ratio of perimeters of $\triangle PST$ and $\triangle PQR$. 5M

Ans. $\triangle PST \sim \triangle PQR$ (AA, corresponding angles). $\frac{PS}{PQ} = \frac{4}{10} = \frac{2}{5}$. $QR = 5 \times \frac{5}{2} = 12.5$ cm. BPT: $\frac{PS}{SQ} = \frac{PT}{TR} \Rightarrow \frac{4}{6} = \frac{3.2}{TR} \Rightarrow TR = 4.8$ cm. Ratio of perimeters $= \frac{2}{5}$, i.e. $2 : 5$.

Q38. CD and GH bisect $\angle ACB$ and $\angle EGF$, with D on AB and H on FE, of $\triangle ABC \sim \triangle FEG$. Prove (i) $\frac{CD}{GH} = \frac{AC}{FG}$ (ii) $\triangle DCB \sim \triangle HGE$. 5M

Ans. $\angle A = \angle F$, $\angle C = \angle G \Rightarrow \angle ACD = \frac{\angle C}{2} = \frac{\angle G}{2} = \angle FGH$. (i) $\triangle ACD \sim \triangle FGH$ (AA) $\Rightarrow \frac{CD}{GH} = \frac{AC}{FG}$. (ii) $\angle B = \angle E$ and $\angle DCB = \angle HGE$ (half of equal angles) $\Rightarrow \triangle DCB \sim \triangle HGE$ (AA).

Q39. A vertical stick 1.8 m long casts a shadow 1.2 m long, and at the same time a tower casts a shadow 36 m long. (i) Find the height of the tower. (ii) Later the stick's shadow becomes 2.7 m. Find the tower's shadow then. (iii) Which criterion did you use? 5M

Ans. (i) $\frac{h}{36} = \frac{1.8}{1.2} \Rightarrow h = 54$ m. (ii) $\frac{54}{s} = \frac{1.8}{2.7} \Rightarrow s = 81$ m. (iii) AA: both are right triangles and the sun's angle of elevation is the same for both at one time.

Quick fact Congruent triangles are similar with scale factor $k = 1$.

Apne notes / galtiyan yahan likho

Case based (4 marks)

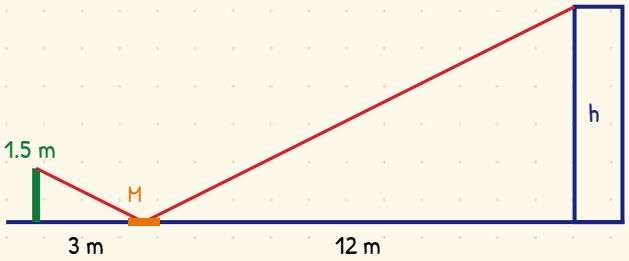


Fig: Q40. Mirror M on the ground; equal angles at M

Q40. Riya (eye level 1.5 m) keeps a small mirror on level ground 3 m in front of her and sees the top of a building in it. The mirror is 12 m from the building. (Angle of incidence = angle of reflection.) (i) Which criterion makes the two triangles similar? (ii) Find the height of the building. (iii) If she stands 2 m from the mirror instead, how far from the building must the mirror be? 4M

Ans. (i) AA (right angles at the ground, equal reflection angles). (ii) $\frac{h}{12} = \frac{1.5}{3} \Rightarrow h = 6$ m. (iii) $\frac{6}{d} = \frac{1.5}{2} \Rightarrow d = 8$ m.

Q41. Two triangular parks are similar. Sides of the smaller park are 30 m, 40 m and 50 m; the longest side of the bigger park is 75 m. (i) Find the scale factor (big : small). (ii) Find the other two sides of the bigger park. (iii) Find the perimeter of the bigger park. 4M

Ans. (i) $\frac{75}{50} = \frac{3}{2}$. (ii) $30 \times \frac{3}{2} = 45$ m, $40 \times \frac{3}{2} = 60$ m. (iii) $45 + 60 + 75 = 180$ m (check: $120 \times \frac{3}{2} = 180$).

Practice yourself (answers in brackets)

1. $DE \parallel BC$, $AD = 3$, $DB = 4.5$, $AE = 4$. Find EC. [6]
2. $\triangle ABC \sim \triangle PQR$, $BC = 8$, $QR = 12$, $PR = 9$. Find AC. [6]
3. A 2 m stick casts a 1.6 m shadow; a building casts 20 m. Height? [25 m]
4. Poles 10 m and 15 m, wires cross at height z. Find z. [6 m]
5. Sides 6, 8, 10 and 9, 12, 15: similar? Ratio? [Yes, SSS, 2 : 3]

Last minute: 8 one-liners

1. Similar = same shape; congruent = same shape + size.
2. All circles, squares, equilateral triangles are similar.
3. BPT: $DE \parallel BC \Rightarrow \frac{AD}{DB} = \frac{AE}{EC}$.
4. Converse BPT: equal ratios $\Rightarrow$ parallel.
5. Criteria: AA, SSS, SAS. SSA is not one.
6. Shadow sums: height/shadow same for all objects at one time.
7. Perimeters, medians, altitudes follow the side ratio.
8. Trapezium: diagonals cut each other in the same ratio.

Apne notes / galtiyan yahan likho

Most asked: distance formula, section formula, 4th vertex, ratio

COORDINATE GEOMETRY

1. Basics of the Cartesian Plane

Two perpendicular number lines: **x-axis** (horizontal) and **y-axis** (vertical) meet at the **origin O(0, 0)**. A point is written $P(x, y)$.

- ✓ x = **abscissa** = distance from the y-axis; y = **ordinate** = distance from the x-axis.
- ✓ Point on the **x-axis**: $(x, 0)$. Point on the **y-axis**: $(0, y)$.
- ✓ Distance of $P(x, y)$ from the x-axis $= |y|$, from the y-axis $= |x|$.

Quadrant	Signs	Example
I	$(+, +)$	$(3, 2)$
II	$(-, +)$	$(-4, 1)$
III	$(-, -)$	$(-2, -5)$
IV	$(+, -)$	$(6, -3)$

Yaad karo: "Jis axis pe ho, doosra zero"

Point x-axis par hai to $y = 0$, y-axis par hai to $x = 0$. Equidistant point "on x-axis" maange to seedha $(x, 0)$ lo, ek hi unknown bachega.

2. Distance Formula

Distance between $A(x_1, y_1)$ and $B(x_2, y_2)$ (from Pythagoras in the right triangle with legs $x_2 - x_1$ and $y_2 - y_1$):

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

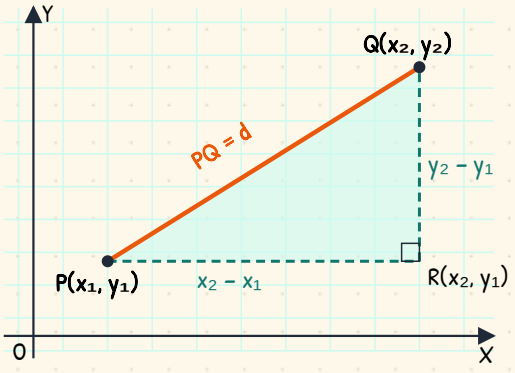


Fig: AB is the hypotenuse; legs are the x-difference and the y-difference

Distance of $P(x, y)$ from the origin: $OP = \sqrt{x^2 + y^2}$.

Order of points does not matter, because the differences are squared.

Quick fact The coordinate plane is also called the Cartesian plane, after the French mathematician Rene Descartes.

Solved Example 1

Find the distance between $A(-3, 5)$ and $B(5, -1)$.

1. $x_2 - x_1 = 5 - (-3) = 8$, $y_2 - y_1 = -1 - 5 = -6$
2. $AB = \sqrt{8^2 + (-6)^2} = \sqrt{64 + 36} = \sqrt{100}$

✓ $AB = 10$ units

Solved Example 2

Find x if the distance between $P(x, -1)$ and $Q(3, 2)$ is 5 units.

1. $(x - 3)^2 + (2 + 1)^2 = 25$
2. $(x - 3)^2 = 16 \Rightarrow x - 3 = \pm 4$

✓ $x = 7$ or $x = -1$ (both valid)

Uses of the distance formula

To show	Check
Collinear A, B, C	Sum of two distances = third ($AB + BC = AC$)
Isosceles triangle	Two sides equal
Equilateral triangle	All three sides equal
Right triangle	$(\text{longest})^2$ = sum of squares of other two
Parallelogram	Both pairs of opposite sides equal
Rectangle	Opposite sides equal + diagonals equal
Rhombus	All four sides equal (diagonals unequal)
Square	All four sides equal + diagonals equal

Solved Example 3

Show that $A(1, 2)$, $B(4, 3)$, $C(3, 6)$ and $D(0, 5)$ are the vertices of a square.

1. $AB^2 = 3^2 + 1^2 = 10$, $BC^2 = 1^2 + 3^2 = 10$, $CD^2 = 3^2 + 1^2 = 10$, $DA^2 = 1^2 + 3^2 = 10$
2. Diagonals: $AC^2 = 2^2 + 4^2 = 20$, $BD^2 = 4^2 + 2^2 = 20$
3. All sides equal and diagonals equal.

✓ ABCD is a square.

Quick fact Distance between $(a, 0)$ and $(0, b)$ is $\sqrt{a^2 + b^2}$.

Quick fact Distance between (x_1, y) and (x_2, y) on a horizontal line $= |x_2 - x_1|$.

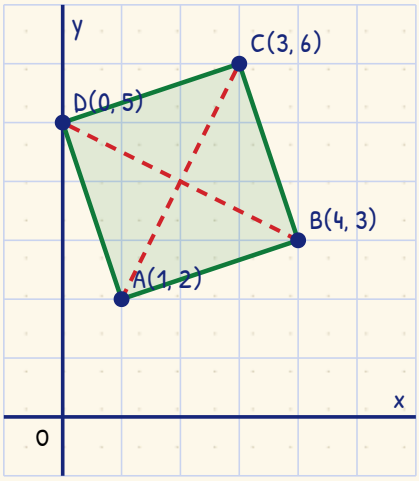


Fig: Ex 3. Square ABCD; diagonals (dashed) are equal

Solved Example 4

Find the point on the y-axis which is equidistant from $A(6,5)$ and $B(-4,3)$.

1. Let $P(0,y)$. $PA^2 = PB^2$
2. $36 + (y-5)^2 = 16 + (y-3)^2$
3. $61 - 10y = 25 - 6y \Rightarrow 4y = 36$

✔ $y = 9$, so $P(0,9)$

Meme samjho

Distance formula ko har shape wala question: **"Humare liye kaam karoge?"** Isosceles, right triangle, square, rhombus, collinear... sab isi se.

Asli baat: Sides aur diagonals ki lengths nikaalo, compare karo. Square = 4 equal sides + equal diagonals; rhombus = 4 equal sides only.

Common mistakes

- → Forgetting $\pm$ after taking the square root: $(x-3)^2 = 16$ gives **two** values.
- → Square vs rhombus: checking only the sides. Diagonals must be checked too.
- → Section formula: writing $mx_1 + nx_2$. Correct is $\dfrac{mx_2 + nx_1}{m+n}$.
- → Point on x-axis taken as $(0,y)$. It is $(x,0)$!
- → Sign slips: $(-1-5)^2 = 36$, not 16. Bracket every negative.

3. Section Formula

The point $P(x,y)$ dividing the join of $A(x_1,y_1)$ and $B(x_2,y_2)$ **internally** in the ratio $m:n$ (i.e. $AP:PB = m:n$):

$$P = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$$

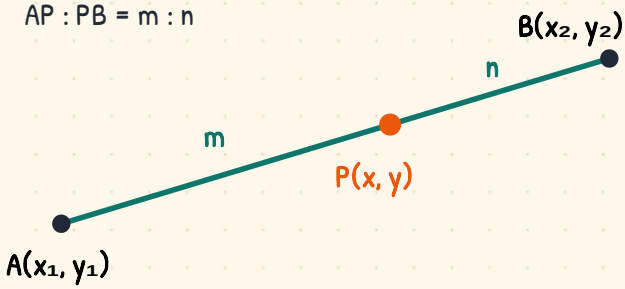


Fig: P divides AB internally in the ratio m : n

Mid-point $(m = n = 1)$: $M = \left(\dfrac{x_1+x_2}{2}, \dfrac{y_1+y_2}{2}\right)$.

Unknown ratio? Take it as $k:1$. Then $x = \dfrac{kx_2 + x_1}{k+1}$; solve for k from whichever coordinate is given.

Yaad karo: "m ka dost door wala"

Ratio $m:n$ mein m (A ke paas wala hissa) multiply hota hai door wale point B ke saath: mx_2; aur n multiply hota hai A ke saath: nx_1. Cross pairing!

Solved Example 5

Find the point which divides the join of $A(-2,5)$ and $B(8,-5)$ internally in the ratio $3:2$.

1. $x = \dfrac{3(8) + 2(-2)}{5} = \dfrac{20}{5} = 4$
2. $y = \dfrac{3(-5) + 2(5)}{5} = \dfrac{-5}{5} = -1$

✔ $P(4,-1)$

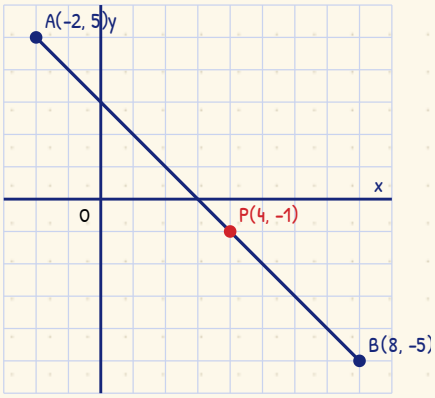


Fig: Solved Example 5. AP : PB = 3 : 2

Quick fact Points (x,y) and $(-x,-y)$ are equidistant from the origin; their mid-point is the origin.

Quick fact Centroid of the triangle with vertices $(0,0), (6,0), (0,9)$ is $(2,3)$.

Solved Example 6

In what ratio does the x-axis divide the join of $A(3,-4)$ and $B(-2,6)$? Also find the point.

1. Let ratio $k:1$; on x-axis $y=0$: $\frac{6k-4}{k+1}=0 \Rightarrow k=\frac{2}{3}$
2. Ratio $=2:3$. $x=\frac{2(-2)+3(3)}{5}=1$

✔ Ratio $2:3$; point $(1,0)$

Formula Box

- Distance: $AB=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$
- From origin: $OP=\sqrt{x^2+y^2}$
- Section $(m:n)$: $\left(\frac{mx_2+nx_1}{m+n},\frac{my_2+ny_1}{m+n}\right)$
- Mid-point: $\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}\right)$
- Ratio $k:1$: $k=\frac{x-x_1}{x_2-x}=\frac{y-y_1}{y_2-y}$
- x-axis divides in ratio $-y_1:y_2$; y-axis in ratio $-x_1:x_2$
- Trisection points: ratios $1:2$ and $2:1$
- Parallelogram ABCD: mid-point of AC = mid-point of BD, so $D=A+C-B$
- Centroid: $G=\left(\frac{x_1+x_2+x_3}{3},\frac{y_1+y_2+y_3}{3}\right)$

4. Trisection, Parallelogram, Centroid

Points of trisection divide AB into 3 equal parts: P in ratio $1:2$ and Q in ratio $2:1$ (Q is also the mid-point of PB).

Solved Example 7

Find the points of trisection of the join of $A(-3,4)$ and $B(6,-5)$.

1. P $(1:2)$: $\left(\frac{6+2(-3)}{3},\frac{-5+2(4)}{3}\right)=(0,1)$
2. Q $(2:1)$: $\left(\frac{2(6)-3}{3},\frac{2(-5)+4}{3}\right)=(3,-2)$

✔ $P(0,1)$ and $Q(3,-2)$

Fourth vertex of a parallelogram: diagonals bisect each other, so mid-point of AC = mid-point of BD.

Centroid (meeting point of medians) divides each median in $2:1$ from the vertex:

Solved Example 8

$A(1,2)$, $B(4,3)$ and $C(6,6)$ are three vertices of parallelogram ABCD. Find D.

1. Mid-point of AC $=\left(\frac{7}{2},4\right)$
2. Mid-point of BD $=\left(\frac{4+x}{2},\frac{3+y}{2}\right)$
3. $4+x=7, 3+y=8$

✔ $D(3,5)$ (shortcut $A+C-B$)

$$G=\left(\frac{x_1+x_2+x_3}{3},\frac{y_1+y_2+y_3}{3}\right)$$

Yaad karo: "Parallelogram = A + C - B"

Teen vertices A, B, C (order mein) diye hon to D = A + C minus B, coordinate-wise. Mid-point wali do lines ek second mein.

Shivam Bhaiya says

Coordinate geometry = **full marks wala chapter**. Bas har step mein formula pehle likho, phir values daalo. Negative numbers ko bracket mein rakho: $(-3)^2=9$. Aur answer ke saath "units" likhna mat bhoolo.

Meme samjho

Viva mein sir: "Point $(0,5)$ kis axis pe hai?" Main, x aur y dono soch ke:

Asli baat: $x=0$ hai to point y-axis par hai. $(0,5)$ is on the y-axis, 5 units above the origin.

5. Board Favourite Question Types

(a) Vertices from mid-points of sides

If D, E, F are mid-points of BC, CA, AB, then $B+C=2D$, $C+A=2E$, $A+B=2F$ (coordinate-wise). Add all: $A+B+C=D+E+F$. So $A=(D+E+F)-2D$, etc.

↳Example: mid-points $D(3,4)$, $E(4,1)$, $F(1,2)$. Sum $=(8,7)$. $A=(8,7)-(6,8)=(2,-1)$, $B=(8,7)-(8,2)=(0,5)$, $C=(8,7)-(2,4)=(6,3)$. Check: mid-point of BC $=(3,4)$ ✓

(b) Collinearity using the section formula

To find k so that A, P, B are collinear, assume P divides AB in $\lambda:1$, find λ from one coordinate, then use the other coordinate to get k.

↳Example: for which k are $(1,-1)$, $(3,k)$, $(7,5)$ collinear? x: $\frac{7\lambda+1}{\lambda+1}=3 \Rightarrow \lambda=\frac{1}{2}$. y:

↳ $k = \frac{\frac{1}{2}(5)+(-1)}{\frac{3}{2}} = 1$.

(c) Circle problems

→ Centre is the mid-point of any diameter.

→ Point P lies inside / on / outside the circle as $OP < r$, $OP = r$, $OP > r$ (O = centre).

→ Other end of a diameter: $B = 2O - A$.

↳ Example: centre $(2, -3)$, one end of diameter $(-1, 1)$: other end $= (4+1, -6-1) = (5, -7)$; radius $= \sqrt{9+16} = 5$.

Yaad karo: "Teen mid-points ka jod = teen vertices ka jod"

$A + B + C = D + E + F$. Har vertex = total minus 2 times opposite side ka mid-point. 3 lines mein answer!

Chapter at a glance

Topic	Key line	Marks
Distance formula	$\sqrt{\Delta x^2 + \Delta y^2}$; shapes, equidistant point	1 to 5
Section formula	$m : n$, cross pairing; unknown ratio $k : 1$	2 to 3
Mid-point	Average; parallelogram, diameter	1 to 3
Case study	Map / ground / classroom grid	4

Shivam Bhaiya says

Shape wale question mein pehle **figure ka rough sketch** grid pe bana lo, order samajh aa jaata hai (ABCD mein AC aur BD diagonals hain). Phir 4 sides + 2 diagonals ki list banao, compare karo, aur conclusion likho.

Practice yourself (answers in brackets)

1. Distance between $(2, 3)$ and $(-4, -5)$. [10]
2. Mid-point of $(-7, 4)$ and $(3, -8)$. [$(-2, -2)$]
3. Point dividing $(0, 0)$ and $(10, 5)$ in $2 : 3$. [$(4, 2)$]
4. Ratio in which x-axis divides $(2, -3)$ and $(5, 6)$. [$1 : 2$]
5. Fourth vertex D of parallelogram $A(0, 0)$, $B(5, 0)$, $C(7, 3)$. [$(2, 3)$]

True or False? (quick self-check)

1. $(0, -4)$ lies on the y-axis. True
2. Distance of (a, b) from the x-axis is a. False ($|b|$)
3. A quadrilateral with 4 equal sides is a square. False (could be a rhombus)
4. Mid-point of (a, b) and $(-a, -b)$ is the origin. True
5. $(-2, 3)$ is in quadrant II. True

Most Important Questions

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. Distance of the point $(-6, 8)$ from the origin is 1M Board fav

(a) 6 (b) 8 (c) 10 (d) 14

Ans. (c) $\sqrt{36+64} = 10$

Q2. Mid-point of the segment joining $(2, -5)$ and $(-6, 3)$ is 1M

(a) $(-4, -2)$ (b) $(-2, -1)$ (c) $(2, 1)$ (d) $(-2, 1)$

Ans. (b) $\left(\frac{2-6}{2}, \frac{-5+3}{2}\right) = (-2, -1)$

Q3. The point on the x-axis equidistant from $(1, 4)$ and $(5, 2)$ is 1M

(a) $(1.5, 0)$ (b) $(2, 0)$ (c) $(0, 1.5)$ (d) $(3, 0)$

Ans. (a) $(x-1)^2 + 16 = (x-5)^2 + 4 \Rightarrow 8x = 12 \Rightarrow x = \frac{3}{2}$

Q4. The ratio in which the y-axis divides the join of $(-4, 3)$ and $(6, -7)$ is 1M

(a) 3 : 2 (b) 2 : 3 (c) 1 : 1 (d) 4 : 3

Ans. (b) $-x_1 : x_2 = 4 : 6 = 2 : 3$

Q5. If $(3, a)$ is the mid-point of $(1, 2)$ and $(5, 8)$, then $a =$ 1M

(a) 3 (b) 4 (c) 5 (d) 10

Ans. (c) $a = \frac{2+8}{2} = 5$

Q6. The points $A(0, 6)$, $B(-5, 3)$ and $C(3, 1)$ are vertices of a triangle which is 1M

(a) equilateral (b) scalene (c) right isosceles (d) obtuse scalene

Ans. (c) $AB = \sqrt{25+9} = \sqrt{34}$, $AC = \sqrt{9+25} = \sqrt{34}$, $BC = \sqrt{64+4} = \sqrt{68}$; $34 + 34 = 68$: right isosceles.

Q7. **A:** The points $(0, 0)$, $(3, 4)$ and $(6, 8)$ are collinear. **R:** Three points are collinear if the sum of two of the distances between them equals the third. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (a) $5 + 5 = 10$: A true; R is the reason.

Q8. **A:** The point $(-3, 0)$ lies on the y-axis. **R:** Every point on the x-axis has ordinate 0. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (d) $(-3, 0)$ has $y = 0$, so it lies on the x-axis: A false, R true.

Q9. **A:** The point $(5, -7)$ is the other end of the diameter of the circle with centre $(2, -3)$ through $(-1, 1)$. **R:** The centre of a circle is the mid-point of every diameter. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (a) $\left(\frac{5-1}{2}, \frac{-7+1}{2}\right) = (2, -3)$ ✓; R explains A.

Very short (2 marks)

Q10. Find k if the distance between $(k, 3)$ and $(2, -1)$ is 5 units. 2M

> **Ans.** $(k-2)^2 + 16 = 25 \Rightarrow (k-2)^2 = 9 \Rightarrow k = 5$ or $k = -1$.

Q11. Find the ratio in which $P(-1, 6)$ divides the join of $A(-3, 10)$ and $B(6, -8)$. 2M Board fav

> **Ans.** $k : 1$: $\frac{6k-3}{k+1} = -1 \Rightarrow 7k = 2 \Rightarrow k = \frac{2}{7}$. Ratio $2 : 7$ (check y: $\frac{2(-8)+7(10)}{9} = 6$).

Q12. If $A(4, -2)$ and $B(-2, 6)$ are ends of a diameter of a circle, find the centre and radius. 2M

> **Ans.** Centre = mid-point $= (1, 2)$. $AB = \sqrt{36+64} = 10$, radius $= 5$ units.

Q13. Show that $A(5, 1)$, $B(1, -1)$ and $C(11, 4)$ are collinear. 2M

> **Ans.** $AB = \sqrt{16+4} = 2\sqrt{5}$, $AC = \sqrt{36+9} = 3\sqrt{5}$, $BC = \sqrt{100+25} = 5\sqrt{5}$. $AB + AC = BC$, so collinear.

Q14. Find k if $B(5, k)$ is equidistant from $A(2, 3)$ and $C(6, 7)$. 2M

> **Ans.** $9 + (k-3)^2 = 1 + (k-7)^2 \Rightarrow 9 - 6k + 9 = 1 - 14k + 49 \Rightarrow 8k = 32 \Rightarrow k = 4$.

Q15. Find the perimeter of the triangle with vertices $(0, 4)$, $(0, 0)$ and $(3, 0)$. 2M

> **Ans.** Sides: 4, 3 and $\sqrt{9+16} = 5$. Perimeter $= 12$ units.

Q16. Find the distance between $A(a\cos\theta, 0)$ and $B(0, a\sin\theta)$. 2M

> **Ans.** $AB = \sqrt{a^2\cos^2\theta + a^2\sin^2\theta} = \sqrt{a^2(\cos^2\theta + \sin^2\theta)} = a$ units.

Q17. If the distance of $P(x, y)$ from $A(5, 1)$ and $B(-1, 5)$ is equal, prove that $3x = 2y$. 2M

> **Ans.** $(x-5)^2 + (y-1)^2 = (x+1)^2 + (y-5)^2 \Rightarrow -10x - 2y + 26 = 2x - 10y + 26 \Rightarrow 8y = 12x \Rightarrow 3x = 2y$.

Q18. Find y if the distance between $P(2, -3)$ and $Q(10, y)$ is 10 units. 2M

> **Ans.** $64 + (y+3)^2 = 100 \Rightarrow (y+3)^2 = 36 \Rightarrow y = 3$ or $y = -9$.

Q19. Find the ratio in which the y-axis divides the join of $P(-3, -4)$ and $Q(5, 2)$, and the point of division. 2M

> **Ans.** Ratio $= -x_1 : x_2 = 3 : 5$. Point $= \left(0, \frac{3(2)+5(-4)}{8}\right) = \left(0, -\frac{7}{4}\right)$.

Short answer (3 marks)

Q20. Show that $A(2, 1)$, $B(5, 5)$ and $C(-2, 4)$ form a right-angled isosceles triangle. 3M Board fav

> (contd.) $AB^2 = 9 + 16 = 25$, $AC^2 = 16 + 9 = 25$, $BC^2 = 49 + 1 = 50$. $AB = AC$ (isosceles) and $AB^2 + AC^2 = 50 = BC^2$ (right angle at A).

Q21. Find a relation between x and y such that $P(x, y)$ is equidistant from $A(3, 6)$ and $B(-3, 4)$. 3M

> **Ans.** $(x-3)^2 + (y-6)^2 = (x+3)^2 + (y-4)^2 \Rightarrow -6x - 12y + 45 = 6x - 8y + 25 \Rightarrow 12x + 4y = 20$, i.e. $3x + y = 5$.

Q22. If $P(2, -1)$ is the mid-point of the join of $A(a, 3)$ and $B(4, b)$, find a, b and the length AB. 3M

> **Ans.** $\frac{a+4}{2} = 2 \Rightarrow a = 0$; $\frac{3+b}{2} = -1 \Rightarrow b = -5$. $A(0, 3)$, $B(4, -5)$: $AB = \sqrt{16+64} = \sqrt{80} = 4\sqrt{5}$ units.

Q23. Find the points that divide the join of $A(-4, 0)$ and $B(0, 8)$ into four equal parts. 3M

> **Ans.** Mid-point M of AB $= (-2, 4)$. Mid-point of AM $= (-3, 2)$, of MB $= (-1, 6)$. Points: $(-3, 2)$, $(-2, 4)$, $(-1, 6)$.

Q24. $A(-2, 1)$, $B(a, 0)$, $C(4, b)$ and $D(1, 2)$ are vertices of parallelogram ABCD. Find a and b, and the length of AB. 3M

> **Ans.** Mid-point AC = mid-point BD: $\frac{-2+4}{2} = \frac{a+1}{2} \Rightarrow a = 1$; $\frac{1+b}{2} = \frac{0+2}{2} \Rightarrow b = 1$. $AB = \sqrt{9+1} = \sqrt{10}$ units.

Q25. The mid-points of the sides of a triangle are $(2, 1)$, $(-1, -3)$ and $(4, 5)$. Find its vertices. 3M

> **Ans.** Sum of mid-points $= (5, 3)$. Vertices: $(5, 3) - 2(2, 1) = (1, 1)$; $(5, 3) - 2(-1, -3) = (7, 9)$; $(5, 3) - 2(4, 5) = (-3, -7)$. Check: mid-point of $(7, 9)$ and $(-3, -7)$ is $(2, 1)$ ✓

Q26. Find the value of p for which $(-1, 3)$, $(2, p)$ and $(5, -1)$ are collinear. 3M

> **Ans.** $(2, p)$ divides in $\lambda : 1$: $\frac{5\lambda-1}{\lambda+1} = 2 \Rightarrow \lambda = 1$ (mid-point). $p = \frac{3+(-1)}{2} = 1$.

Q27. Name the type of quadrilateral formed by $(-1, -2)$, $(1, 0)$, $(-1, 2)$, $(-3, 0)$ and justify. 3M Board fav

> **Ans.** Each side $= \sqrt{4+4} = 2\sqrt{2}$; diagonals: $(-1, -2)$ to $(-1, 2) = 4$, $(1, 0)$ to $(-3, 0) = 4$. Equal sides and equal diagonals: **square**.

Q28. Find the coordinates of the point on the join of $A(2, -5)$ and $B(-3, 5)$ which is twice as far from A as from B. 3M

> **Ans.** $AP = 2PB \Rightarrow AP : PB = 2 : 1$. $P = \left(\frac{2(-3)+2}{3}, \frac{2(5)+(-5)}{3}\right) = \left(-\frac{4}{3}, \frac{5}{3}\right)$.

Q29. If $Q(0, 1)$ is equidistant from $P(5, -3)$ and $R(x, 6)$, find x. Also find QR and PR. 3M Board fav

> **Ans.** $QP^2 = 25 + 16 = 41$; $QR^2 = x^2 + 25 = 41 \Rightarrow x = \pm 4$. $QR = \sqrt{41}$. For $x = 4$: $PR = \sqrt{1+81} = \sqrt{82}$; for $x = -4$: $PR = \sqrt{81+81} = 9\sqrt{2}$.

Q30. Find the ratio in which $P(4, m)$ divides the join of $A(2, 3)$ and $B(6, -3)$. Hence find m. 3M

(contd.) $k:1:\frac{6k+2}{k+1}=4\Rightarrow 2k=2\Rightarrow k=1$. Ratio $1:1$ (P is the mid-point), so $m=\frac{3-3}{2}=0$.

Q31. Find the coordinates of the point P on AB such that $AP=\frac{3}{7}AB$, where $A(-2,-2)$ and $B(2,-4)$. 3M Board fav

Ans. $AP:PB=3:4$. $P=\left(\frac{3(2)+4(-2)}{7},\frac{3(-4)+4(-2)}{7}\right)=\left(-\frac{2}{7},-\frac{20}{7}\right)$.

Q32. Find the point equidistant from $A(8,6)$, $B(8,-2)$ and $C(2,-2)$ (the circumcentre). Also find the radius. 3M

Ans. AB is vertical, BC horizontal, so $\angle B=90°$; circumcentre = mid-point of hypotenuse $AC=(5,2)$. Check: distance to A, B, C $=\sqrt{9+16}=5$ each. Radius $=5$ units.

Q33. Show that (a,a), $(-a,-a)$ and $(-\sqrt{3}a,\sqrt{3}a)$ are vertices of an equilateral triangle. 3M

Ans. Side 1: $\sqrt{(2a)^2+(2a)^2}=2\sqrt{2}a$. Side 2: $\sqrt{(a+\sqrt{3}a)^2+(a-\sqrt{3}a)^2}=\sqrt{8a^2}=2\sqrt{2}a$. Side 3: $\sqrt{(\sqrt{3}a-a)^2+(\sqrt{3}a+a)^2}=\sqrt{8a^2}=2\sqrt{2}a$. All equal: equilateral.

Q34. Find the ratio in which $P(-3,k)$ divides the join of $A(-5,-4)$ and $B(-2,3)$. Hence find k. 3M

Ans. $m:1:\frac{-2m-5}{m+1}=-3\Rightarrow -2m-5=-3m-3\Rightarrow m=2$. Ratio $2:1$. $k=\frac{2(3)+1(-4)}{3}=\frac{2}{3}$.

Long answer (5 marks)

Q35. Show that $A(-2,-1)$, $B(4,0)$, $C(3,3)$ and $D(-3,2)$ are the vertices of a parallelogram. Is it a rectangle? 5M

Ans. $AB=\sqrt{36+1}=\sqrt{37}$, $CD=\sqrt{36+1}=\sqrt{37}$; $BC=\sqrt{1+9}=\sqrt{10}$, $DA=\sqrt{1+9}=\sqrt{10}$: opposite sides equal, so parallelogram. Diagonals: $AC=\sqrt{25+16}=\sqrt{41}$, $BD=\sqrt{49+4}=\sqrt{53}$; unequal, so **not** a rectangle.

Q36. If $A(-1,3)$, $B(1,-1)$ and $C(5,1)$ are vertices of $\triangle ABC$, find the length of the median through A, and the centroid. 5M

Ans. Mid-point of BC: $D=(3,0)$. $AD=\sqrt{16+9}=5$ units. Centroid $G=\left(\frac{-1+1+5}{3},\frac{3-1+1}{3}\right)=\left(\frac{5}{3},1\right)$. (G lies on AD with $AG:GD=2:1$: $\left(\frac{2(3)+1(-1)}{3},\frac{0+3}{3}\right)=\left(\frac{5}{3},1\right)$ ✓)

Q37. The line segment joining $A(-2,9)$ and $B(6,3)$ is a diameter of a circle with centre C. Find C, the radius, and check whether $P(1,2)$ lies inside, on or outside the circle. 5M

Ans. $C=(2,6)$. $AB=\sqrt{64+36}=10$, $r=5$. $CP=\sqrt{1+16}=\sqrt{17}\approx 4.12<5$, so P lies **inside** the circle.

Q38. Show that $A(4,0)$, $B(5,5)$, $C(0,4)$ and $D(-1,-1)$ form a rhombus, not a square. Find its area using $\frac{1}{2}\times d_1\times d_2$. 5M Board fav

(contd.) $AB=\sqrt{1+25}=\sqrt{26}$, $BC=\sqrt{25+1}=\sqrt{26}$, $CD=\sqrt{1+25}=\sqrt{26}$, $DA=\sqrt{25+1}=\sqrt{26}$: all sides equal. $AC=\sqrt{16+16}=4\sqrt{2}$, $BD=\sqrt{36+36}=6\sqrt{2}$: diagonals unequal, so rhombus, not square. Area $=\frac{1}{2}\times 4\sqrt{2}\times 6\sqrt{2}=24$ sq units.

Case based (4 marks)

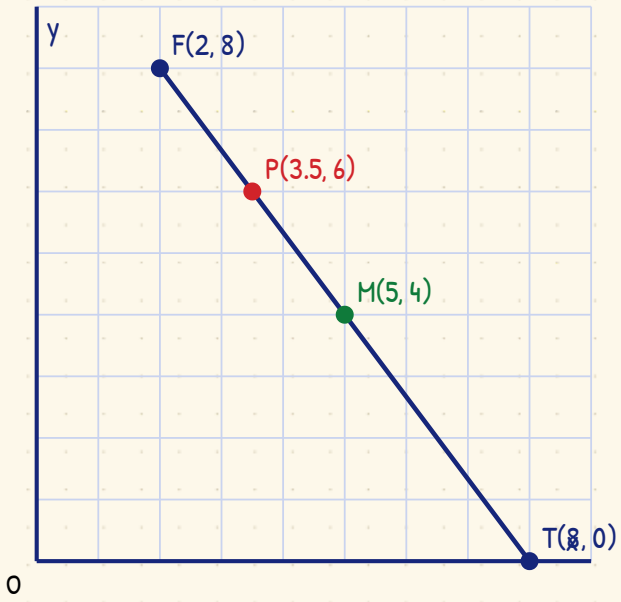


Fig: Q39. Flag F, tap T, mid-point M, cone P (FP : PT = 1 : 3)

Q39. In a school sports ground (grid in metres, origin at a corner), the flag post F is at $(2,8)$, the drinking-water tap T is at $(8,0)$. (i) Find the distance FT. (ii) Find the mid-point of FT, where a first-aid desk is placed. (iii) A cone is placed on FT at P such that $FP:PT=1:3$. Find P. 4M

Ans. (i) $FT=\sqrt{36+64}=10$ m. (ii) $(5,4)$. (iii) $P=\left(\frac{1(8)+3(2)}{4},\frac{1(0)+3(8)}{4}\right)=\left(\frac{7}{2},6\right)$.

Last minute: 8 one-liners

1. On x-axis $y=0$; on y-axis $x=0$.
2. $OP=\sqrt{x^2+y^2}$.
3. Distance formula: squares kill the order and the signs.
4. Section: $\frac{mx_2+nx_1}{m+n}$ (cross pairing).
5. Unknown ratio: take $k:1$.
6. Mid-point = average of coordinates.
7. Square: 4 sides + 2 diagonals equal. Rhombus: 4 sides only.
8. Parallelogram 4th vertex: $D=A+C-B$.

Apne notes / galtiyan yahan likho

Introduction to Trigonometry

1. Sides w.r.t. an Angle

Trigonometry = study of relations between the sides and angles of a triangle (Greek: tri = three, gon = sides, metron = measure). In Class 10 we work only with **right triangles** and **acute angles**.

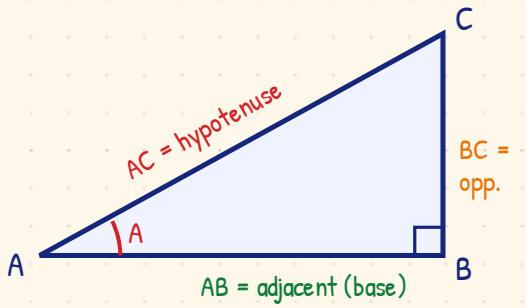


Fig 8.1: Right $\triangle ABC$, $\angle B = 90°$; sides named w.r.t. $\angle A$

- → **Hypotenuse**: side opposite the right angle; always the longest side.
- → **Opposite side** (perpendicular): the side facing the angle we are looking at.
- → **Adjacent side** (base): the side touching the angle, other than the hypotenuse.
- → Look from $\angle C$ instead: opposite and adjacent swap; the hypotenuse stays the same.

↳ Pehle angle pe ungli rakho, phir dekho uske saamne kaun si side hai.

2. The Six Trigonometric Ratios

Definition

For acute $\angle A$ in right $\triangle ABC$ ($\angle B = 90°$):

$$\sin A = \frac{\text{opp}}{\text{hyp}} = \frac{BC}{AC}, \quad \cos A = \frac{\text{adj}}{\text{hyp}} = \frac{AB}{AC}$$

$$\tan A = \frac{\text{opp}}{\text{adj}} = \frac{BC}{AB}$$

$$\operatorname{cosec} A = \frac{AC}{BC}, \quad \sec A = \frac{AC}{AB}, \quad \cot A = \frac{AB}{BC}$$

Ratio	Sides	Reciprocal
$\sin A$	P / H	$\operatorname{cosec} A = \frac{1}{\sin A}$
$\cos A$	B / H	$\sec A = \frac{1}{\cos A}$
$\tan A$	P / B	$\cot A = \frac{1}{\tan A}$

Quotient relations: $\tan A = \frac{\sin A}{\cos A}$, $\cot A = \frac{\cos A}{\sin A}$.

Yaad karo: "Pandit Badri Prasad Har Har Bole"

Upar wale: Pandit, Badri, Prasad. Neeche wale: Har, Har, Bole. So sin = P/H, cos = B/H, tan = P/B (P = perpendicular, B = base, H = hypotenuse). Baaki teen (cosec, sec, cot) = bas ulta kar do.

Important facts

- ✓ Ratios depend only on the angle, not on the size of the triangle (similar triangles have equal side ratios).
- ✓ For acute A: $0 < \sin A < 1$ and $0 < \cos A < 1$, so $\sec A > 1$ and $\operatorname{cosec} A > 1$. $\tan A$ can be any positive number.
- ✓ $\sin A$ is one symbol, not "sin × A". $\sin^2 A$ means $(\sin A)^2$, not $\sin(A^2)$.

3. All Ratios from One Ratio

Given ratio = two sides ($3k$, $5k$) → Third side by Pythagoras → Read all 6 ratios (k cancels)

Solved Example 1

If $\sin A = \frac{5}{13}$, find all the other trigonometric ratios of A.

1. Let $BC = 5k$ (opp) and $AC = 13k$ (hyp).
2. $AB = \sqrt{169k^2 - 25k^2} = \sqrt{144k^2} = 12k$.
3. $\cos A = \frac{AB}{AC} = \frac{12}{13}$, $\tan A = \frac{5}{12}$; reciprocals: $\operatorname{cosec} A = \frac{13}{5}$, $\sec A = \frac{13}{12}$, $\cot A = \frac{12}{5}$.

✓ $\cos A = \frac{12}{13}$, $\tan A = \frac{5}{12}$, $\operatorname{cosec} A = \frac{13}{5}$, $\sec A = \frac{13}{12}$, $\cot A = \frac{12}{5}$

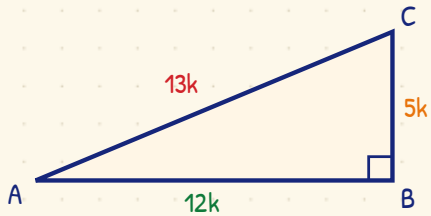


Fig 8.2: $\sin A = \frac{5}{13}$ gives $AB = 12k$

Solved Example 2

If $8 \tan A = 15$, find $\sin A$ and $\cos A$.

1. $\tan A = \frac{15}{8}$: opp $= 15k$, adj $= 8k$.
2. hyp $= \sqrt{225k^2 + 64k^2} = \sqrt{289k^2} = 17k$.

✓ $\sin A = \frac{15}{17}$, $\cos A = \frac{8}{17}$

Solved Example 3

If $3\tan\theta = 4$, evaluate $\dfrac{3\sin\theta + 2\cos\theta}{3\sin\theta - 2\cos\theta}$.

1. Divide num. and den. by $\cos\theta$: $\dfrac{3\tan\theta + 2}{3\tan\theta - 2}$.

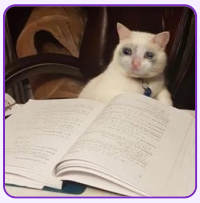
2. $= \dfrac{4+2}{4-2} = \dfrac{6}{2} = 3.$

✔ 3

Yaad karo: sin-cos wala fraction?

Agar $\tan\theta$ diya hai aur expression mein sirf $\sin\theta$, $\cos\theta$ hain, toh **upar-neeche $\cos\theta$ se divide** karo. Triangle banane ki zaroorat hi nahi.

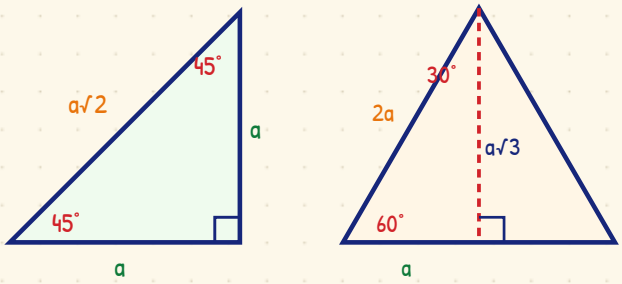
Meme samjho



Jab finally samajh aaya ki $\tan A$ diya ho toh baaki 5 ratios ek triangle se nikal jaate hain: **"Padhai ho rahi hai, disturb mat karo."**

Asli baat: Ek ratio = do sides. Teesri side Pythagoras se, phir saare 6 ratios.

4. Ratios of Specific Angles



Isosceles right △ | Half of equilateral △

45 degrees

In right △ with $\angle A = 45°$, the other angle is also $45°$, so the legs are equal: a, a. Hypotenuse $= a\sqrt{2}$. So $\sin 45° = \cos 45° = \dfrac{1}{\sqrt{2}}$, $\tan 45° = 1$.

30 and 60 degrees

Take equilateral △ of side $2a$; the altitude bisects the base and the top angle. Half: sides a, $2a$, altitude $\sqrt{4a^2 - a^2} = a\sqrt{3}$. So $\sin 30° = \dfrac{a}{2a} = \dfrac{1}{2}$, $\cos 30° = \dfrac{\sqrt{3}}{2}$, $\sin 60° = \dfrac{\sqrt{3}}{2}$, $\cos 60° = \dfrac{1}{2}$.

0 and 90 degrees: as $A \to 0°$, opp $\to 0$: $\sin 0° = 0$, $\cos 0° = 1$. As $A \to 90°$: $\sin 90° = 1$, $\cos 90° = 0$. $\tan 90°$, $\sec 90°$, $\operatorname{cosec} 0°$, $\cot 0°$ are **not defined**.

Quick fact $\tan 45° = 1$, so at $45°$ opposite side = adjacent side.

$\angle A$	$0°$	$30°$	$45°$	$60°$	$90°$
$\sin A$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos A$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan A$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	ND
$\operatorname{cosec} A$	ND	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
$\sec A$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	ND
$\cot A$	ND	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

Yaad karo: 0, 1, 2, 3, 4 trick

sin row: likho $0, 1, 2, 3, 4$ for $0°, 30°, 45°, 60°, 90°$. Har ek ko 4 se divide karo, phir root lo: $\sqrt{\frac{0}{4}}, \sqrt{\frac{1}{4}}, \sqrt{\frac{2}{4}}, \sqrt{\frac{3}{4}}, \sqrt{\frac{4}{4}} = 0, \frac{1}{2}, \frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{2}, 1$. **cos row = sin row ulta.** tan = sin / cos.

As A increases from $0°$ to $90°$: $\sin A$ increases $(0 \to 1)$, $\cos A$ decreases $(1 \to 0)$, $\tan A$ increases.

Solved Example 4

Evaluate:

$$\frac{\tan^2 60° + 4\cos^2 45° + 3\sec^2 30° + 5\cos^2 90°}{\operatorname{cosec} 30° + \sec 60° - \cot^2 30°}$$

1. Numerator $= 3 + 4 \cdot \frac{1}{2} + 3 \cdot \frac{4}{3} + 0 = 3 + 2 + 4 = 9.$

2. Denominator $= 2 + 2 - 3 = 1.$

✔ 9

Solved Example 5

If $\tan(A+B) = \sqrt{3}$ and $\tan(A-B) = \dfrac{1}{\sqrt{3}}$, $0° < A + B \le 90°$, $A > B$, find A and B.

1. $\tan(A+B) = \sqrt{3} = \tan 60° \Rightarrow A + B = 60°.$

2. $\tan(A-B) = \frac{1}{\sqrt{3}} = \tan 30° \Rightarrow A - B = 30°.$

3. Add: $2A = 90°$, so $A = 45°$; then $B = 15°$.

✔ $A = 45°, B = 15°$

Shivam Bhaiya says

Standard angle table ek baar khud bana ke likho, roz 2 minute. Exam mein ise rough sheet pe sabse pehle likh lo, phir galti ka chance zero.

Quick fact $\sin^2\theta + \cos^2\theta = 1$ for every θ: $\sin^2 30° + \cos^2 30° = \frac{1}{4} + \frac{3}{4} = 1.$

Common mistakes

- $\sin(A+B) \neq \sin A + \sin B$. Check: $A = B = 30°$ gives $\sin 60° = 0.866$ but $\sin 30° + \sin 30° = 1$.
- Taking the same "opposite side" for both acute angles. Opp of $\angle A$ is adj of $\angle C$.
- Writing $\tan 90° = \infty$ or 0: it is **not defined**.
- $\sin^2 A$ written as $\sin A^2$; or $\cos^2 30° = \frac{\sqrt{3}}{2}$ (forgetting to square: it is $\frac{3}{4}$).
- Using a value like $\sin\theta = \frac{4}{3}$: impossible, since hyp is the longest side.

5. Trigonometric Identities

An **identity** is an equation true for **all values of the angle** for which both sides are defined.

Theorem: $\sin^2 A + \cos^2 A = 1$

In right $\triangle ABC$ ($\angle B = 90°$): $AB^2 + BC^2 = AC^2$ (Pythagoras).

Divide by AC^2: $\left(\frac{AB}{AC}\right)^2 + \left(\frac{BC}{AC}\right)^2 = 1$, i.e. $\cos^2 A + \sin^2 A = 1$. True for $0° \leq A \leq 90°$.

Divide by AB^2: $1 + \tan^2 A = \sec^2 A$ ($0° \leq A < 90°$).

Divide by BC^2: $\cot^2 A + 1 = \operatorname{cosec}^2 A$ ($0° < A \leq 90°$).

Formula Box

- $\sin A = \frac{P}{H}, \cos A = \frac{B}{H}, \tan A = \frac{P}{B}$
- $\operatorname{cosec} A = \frac{1}{\sin A}, \sec A = \frac{1}{\cos A}, \cot A = \frac{1}{\tan A}$
- $\tan A = \frac{\sin A}{\cos A}, \cot A = \frac{\cos A}{\sin A}$
- $\sin^2 A + \cos^2 A = 1$
- $\sec^2 A - \tan^2 A = 1$
- $\operatorname{cosec}^2 A - \cot^2 A = 1$
- $\sin 30° = \cos 60° = \frac{1}{2}, \sin 60° = \cos 30° = \frac{\sqrt{3}}{2}$
- $\sin 45° = \cos 45° = \frac{1}{\sqrt{2}}; \tan 30° = \frac{1}{\sqrt{3}}, \tan 60° = \sqrt{3}$

Useful forms

- $\sin^2 A = 1 - \cos^2 A, \quad \cos^2 A = 1 - \sin^2 A$
- $(\sec A - \tan A)(\sec A + \tan A) = 1$
- $(\operatorname{cosec} A - \cot A)(\operatorname{cosec} A + \cot A) = 1$

→ Express every ratio in terms of one: $\sin A$ in terms of $\cot A$: $\sin A = \frac{1}{\sqrt{1 + \cot^2 A}}$.

Yaad karo: "1 + tan = sec, 1 + cot = cosec"

Jodi yaad rakho: **tan ki jodi sec, cot ki jodi cosec** (dono "t" wale ke saath "s/c" wala). $1 + \tan^2 = \sec^2$ aur $1 + \cot^2 = \operatorname{cosec}^2$.

How to prove an identity (board steps)

1. Start from the **more complicated side** (usually LHS).
2. Convert everything to sin and cos if stuck.
3. Take LCM, use $\sin^2 + \cos^2 = 1$, or multiply by the conjugate ($1 + \sin\theta$ with $1 - \sin\theta$).
4. Reach the other side; write "= RHS. Hence proved."

Solved Example 6

Prove that $\frac{1 - \sin\theta}{1 + \sin\theta} = (\sec\theta - \tan\theta)^2$.

1. RHS $= \left(\frac{1}{\cos\theta} - \frac{\sin\theta}{\cos\theta}\right)^2 = \frac{(1 - \sin\theta)^2}{\cos^2\theta}$
2. $= \frac{(1 - \sin\theta)^2}{1 - \sin^2\theta} = \frac{(1 - \sin\theta)^2}{(1 - \sin\theta)(1 + \sin\theta)}$
3. $= \frac{1 - \sin\theta}{1 + \sin\theta} =$ LHS.

✔ Hence proved.

Solved Example 7

Prove that $\sqrt{\frac{1 + \sin A}{1 - \sin A}} = \sec A + \tan A$.

1. Multiply num. and den. inside the root by $(1 + \sin A)$:
$$\sqrt{\frac{(1 + \sin A)^2}{1 - \sin^2 A}} = \sqrt{\frac{(1 + \sin A)^2}{\cos^2 A}}$$
2. $= \frac{1 + \sin A}{\cos A} = \frac{1}{\cos A} + \frac{\sin A}{\cos A} = \sec A + \tan A$.

✔ Hence proved.

Meme samjho

Identity ka sawaal aaya, sab kuch sin aur cos mein badla, LCM liya... aur LHS = RHS ho gaya. **Mood:**

Asli baat: Stuck ho? Sab sin, cos mein likho, LCM lo, $\sin^2 + \cos^2 = 1$ lagao. 90% proofs aise hi bante hain.

Quick fact Max value of $\sin\theta$ and $\cos\theta$ is 1; $\sec\theta, \operatorname{cosec}\theta \geq 1$.

Shivam Bhaiya says
Proof mein dono side ek saath mat chhedo. Ek side se shuru karo, line by line chalo, aur last mein "Hence proved" zaroor likho. Steps ke marks milte hain.

6. Ratios in Terms of One Ratio

Using the identities, every ratio can be written in terms of any one ratio. Board asks: "**Express** $\sin A$, $\sec A$ **and** $\tan A$ **in terms of** $\cot A$."

Solved Example 8

Express the ratios $\sin A$, $\sec A$ and $\tan A$ in terms of $\cot A$.

1. $\operatorname{cosec}^2 A = 1 + \cot^2 A \Rightarrow \sin A = \dfrac{1}{\sqrt{1+\cot^2 A}}$.
2. $\tan A = \dfrac{1}{\cot A}$.
3. $\sec^2 A = 1 + \tan^2 A = 1 + \dfrac{1}{\cot^2 A} = \dfrac{\cot^2 A + 1}{\cot^2 A}$, so $\sec A = \dfrac{\sqrt{1+\cot^2 A}}{\cot A}$.

✔ All three found in steps 1, 2, 3 above.

In terms of	$\sin A$	$\cos A$	$\tan A$
$\sin A$	$\sin A$	$\sqrt{1-\sin^2 A}$	$\dfrac{\sin A}{\sqrt{1-\sin^2 A}}$
$\cos A$	$\sqrt{1-\cos^2 A}$	$\cos A$	$\dfrac{\sqrt{1-\cos^2 A}}{\cos A}$
$\tan A$	$\dfrac{\tan A}{\sqrt{1+\tan^2 A}}$	$\dfrac{1}{\sqrt{1+\tan^2 A}}$	$\tan A$

↳Shortcut: triangle banao! $\tan A = t$ means opp t, adj 1, hyp $\sqrt{1+t^2}$. Baaki seedha padh lo.

7. Five Tricks for Identity Proofs

Trick	When to use
1. sin-cos mein badlo	tan, cot, sec, cosec mixed up
2. LCM lo	Two fractions added / subtracted
3. Conjugate se multiply	$1 \pm \sin\theta$, $1 \pm \cos\theta$, $\sec\theta \pm \tan\theta$ in denominator or under root
4. "1" ko identity se badlo	$1 = \sec^2\theta - \tan^2\theta$ or $\operatorname{cosec}^2\theta - \cot^2\theta$
5. $a^3 \pm b^3$, $a^2 - b^2$	Cubes or squares of sin, cos appear

Solved Example 9 (5 marks fav)

Prove that $\dfrac{\sin\theta - \cos\theta + 1}{\sin\theta + \cos\theta - 1} = \dfrac{1}{\sec\theta - \tan\theta}$.

1. Divide num. and den. by $\cos\theta$: LHS $= \dfrac{\tan\theta - 1 + \sec\theta}{\tan\theta + 1 - \sec\theta}$.
2. Replace 1 in the numerator by $\sec^2\theta - \tan^2\theta$: num $= (\tan\theta + \sec\theta) - (\sec^2\theta - \tan^2\theta)$
3. $= (\sec\theta + \tan\theta)\,[1 - (\sec\theta - \tan\theta)] = (\sec\theta + \tan\theta)(1 - \sec\theta + \tan\theta)$.
4. The bracket equals the denominator, so LHS $= \sec\theta + \tan\theta$.
5. Multiply and divide by $(\sec\theta - \tan\theta)$ and use $\sec^2\theta - \tan^2\theta = 1$: $\sec\theta + \tan\theta = \dfrac{1}{\sec\theta - \tan\theta}$ = RHS.

✔ Hence proved.

Solved Example 10

Prove that $\dfrac{\cos A}{1+\sin A} + \dfrac{1+\sin A}{\cos A} = 2\sec A$.

1. LCM: LHS $= \dfrac{\cos^2 A + (1+\sin A)^2}{(1+\sin A)\cos A}$.
2. Num $= \cos^2 A + 1 + 2\sin A + \sin^2 A = 2 + 2\sin A = 2(1+\sin A)$.
3. LHS $= \dfrac{2(1+\sin A)}{(1+\sin A)\cos A} = \dfrac{2}{\cos A} = 2\sec A$.

✔ Hence proved.

Solved Example 11

If $2\sin 2\theta = \sqrt{3}$, find θ (with 2θ acute). Also find $\cos 3\theta$.

1. $\sin 2\theta = \frac{\sqrt{3}}{2} = \sin 60^\circ \Rightarrow 2\theta = 60^\circ, \theta = 30^\circ$.
2. $\cos 3\theta = \cos 90^\circ = 0$.

✔ $\theta = 30^\circ$, $\cos 3\theta = 0$

Yaad karo: conjugate jodi

$(1-\sin)(1+\sin) = \cos^2$, $(1-\cos)(1+\cos) = \sin^2$, $(\sec-\tan)(\sec+\tan) = 1$, $(\operatorname{cosec}-\cot)(\operatorname{cosec}+\cot) = 1$. Root ya denominator mein dikhe, jodi wala multiply karo.

Chapter at a glance

6 ratios → reciprocal + quotient relations → standard angles table (0° to 90°) → 3 identities. Har sawaal in chaar cheezon se hi banta hai.

Board scan: what comes from this chapter

- 1M: value of an expression, one identity like $\sec^2 - \tan^2$, AR on range of sin.
- 2M: find A, B from $\tan(A+B)$ type; all ratios from one ratio.
- 3M: identity proof (one-step conjugate / LCM).
- 5M: longer identity (Example 9 type) or "if $\sec + \tan = p$" type.

8. Finding the Angle

If a ratio equals a standard value, read the angle from the table (only acute angles here).

If	then	If	then
$\sin\theta = \frac{1}{2}$	$\theta = 30^\circ$	$\tan\theta = 1$	$\theta = 45^\circ$
$\cos\theta = \frac{1}{2}$	$\theta = 60^\circ$	$\tan\theta = \sqrt{3}$	$\theta = 60^\circ$
$\sin\theta = \frac{\sqrt{3}}{2}$	$\theta = 60^\circ$	$\tan\theta = \frac{1}{\sqrt{3}}$	$\theta = 30^\circ$
$\cos\theta = \frac{\sqrt{3}}{2}$	$\theta = 30^\circ$	$\sec\theta = 2$	$\theta = 60^\circ$

Solved Example 12

In $\triangle ABC$, right-angled at C, $\tan A = \dfrac{1}{\sqrt{3}}$. Find $\sin A\cos B + \cos A \sin B$.

1. $\tan A = \frac{1}{\sqrt{3}} \Rightarrow A = 30^\circ$, so $B = 180^\circ - 90^\circ - 30^\circ = 60^\circ$.
2. $\sin 30^\circ \cos 60^\circ + \cos 30^\circ \sin 60^\circ = \frac{1}{2}\cdot\frac{1}{2} + \frac{\sqrt{3}}{2}\cdot\frac{\sqrt{3}}{2} = \frac{1}{4} + \frac{3}{4}$.

✔ $= 1$

Solved Example 13

If $\sin(A-B) = \dfrac{1}{2}$ and $\cos(A+B) = 0$, $0^\circ < A+B \le 90^\circ$, $A > B$, find A and B.

1. $\sin(A-B) = \frac{1}{2} \Rightarrow A - B = 30^\circ$; $\cos(A+B) = 0 = \cos 90^\circ \Rightarrow A + B = 90^\circ$.
2. Add: $2A = 120^\circ \Rightarrow A = 60^\circ$; subtract: $2B = 60^\circ \Rightarrow B = 30^\circ$.

✔ $A = 60^\circ, B = 30^\circ$

Solved Example 16

Find acute θ if $\sqrt{3}\tan 2\theta - 3 = 0$. Hence find $\sin\theta + \cos 2\theta$.

1. $\tan 2\theta = \frac{3}{\sqrt{3}} = \sqrt{3} = \tan 60^\circ \Rightarrow 2\theta = 60^\circ$, $\theta = 30^\circ$.
2. $\sin 30^\circ + \cos 60^\circ = \frac{1}{2} + \frac{1}{2} = 1$.

✔ $\theta = 30^\circ$; value $= 1$

True or False? (NCERT type, with reason)

- $\tan A$ is always less than 1. **False**: $\tan 60^\circ = \sqrt{3} > 1$.
- $\sec A = \frac{12}{5}$ for some angle A. **True**: $\sec A > 1$ is allowed (hyp 12, adj 5).
- $\cos A$ is short for cosecant of A. **False**: it is cosine of A.
- $\cot A$ is the product of cot and A. **False**: it is one symbol.
- $\sin\theta = \frac{4}{3}$ for some angle θ. **False**: $\sin\theta \le 1$.
- $\sin\theta$ increases as θ increases (0° to 90°). **True**.

Quick check: values (answers in green)

- $\sin^2 45^\circ + \cos^2 45^\circ = 1$
- $\sec^2 60^\circ - \tan^2 60^\circ = 4 - 3 = 1$
- $\sin 30^\circ + \cos 60^\circ + \tan 45^\circ = 2$
- $\operatorname{cosec}^2 30^\circ \cdot \sin^2 45^\circ - \sec^2 60^\circ = 4\cdot\frac{1}{2} - 4 = -2$
- $\cos 0^\circ + \sin 90^\circ + \sqrt{2}\sin 45^\circ = 3$

Solved Example 14

Prove that $\dfrac{1+\tan^2 A}{1+\cot^2 A} = \left(\dfrac{1-\tan A}{1-\cot A}\right)^2 = \tan^2 A$.

1. First: $\dfrac{1+\tan^2 A}{1+\cot^2 A} = \dfrac{\sec^2 A}{\operatorname{cosec}^2 A} = \dfrac{\sin^2 A}{\cos^2 A} = \tan^2 A$.
2. Second: $\dfrac{1-\tan A}{1-\cot A} = \dfrac{1-\tan A}{1-\frac{1}{\tan A}} = \dfrac{(1-\tan A)\tan A}{\tan A - 1} = -\tan A$.
3. Square: $(-\tan A)^2 = \tan^2 A$. All three are equal.

✔ Hence proved.

Solved Example 15

If $\sin\theta = \dfrac{3}{5}$, evaluate $\dfrac{\cos\theta - \frac{1}{\tan\theta}}{2\cot\theta}$.

1. opp 3, hyp 5, adj 4: $\cos\theta = \frac{4}{5}$, $\tan\theta = \frac{3}{4}$, $\cot\theta = \frac{4}{3}$.
2. Num $= \frac{4}{5} - \frac{4}{3} = \frac{12-20}{15} = -\frac{8}{15}$; Den $= \frac{8}{3}$.
3. Value $= -\frac{8}{15} \times \frac{3}{8} = -\frac{1}{5}$.

✔ $-\frac{1}{5}$

Quick fact $\sec^2\theta - \tan^2\theta = 1$, so $\sec\theta + \tan\theta = \dfrac{1}{\sec\theta - \tan\theta}$.

Last-minute revision

- → Opp / adj / hyp always w.r.t. the given angle.
- → sin, cos lie between 0 and 1 for acute angles; tan can exceed 1.
- → Table: sin row $\sqrt{\frac{0}{4}} \dots \sqrt{\frac{4}{4}}$; cos = sin reversed.
- → 3 identities: $\sin^2 + \cos^2 = 1$, $1 + \tan^2 = \sec^2$, $1 + \cot^2 = \operatorname{cosec}^2$.
- → Proofs: one side se shuru, sin-cos, LCM, conjugate.

MOST IMPORTANT QUESTIONS

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. If $\sin A = \frac{8}{17}$, then $\cos A$ is: 1M Board fav

(a) $\frac{15}{17}$ (b) $\frac{8}{15}$ (c) $\frac{17}{15}$ (d) $\frac{15}{8}$

Ans. (a) $\cos A = \sqrt{1 - \frac{64}{289}} = \sqrt{\frac{225}{289}} = \frac{15}{17}$.

Q2. $\frac{2\tan 30^\circ}{1 + \tan^2 30^\circ}$ is equal to: 1M Board fav

(a) $\sin 60^\circ$ (b) $\cos 60^\circ$ (c) $\tan 60^\circ$ (d) $\sin 30^\circ$

Ans. (a) $\frac{2/\sqrt{3}}{1 + 1/3} = \frac{2}{\sqrt{3}} \cdot \frac{3}{4} = \frac{\sqrt{3}}{2} = \sin 60^\circ$.

Q3. $9\sec^2 A - 9\tan^2 A$ is equal to: 1M

(a) 1 (b) 9 (c) 8 (d) 0

Ans. (b) $9(\sec^2 A - \tan^2 A) = 9 \times 1 = 9$.

Q4. $\frac{1 + \tan^2 A}{1 + \cot^2 A}$ is equal to: 1M

(a) $\sec^2 A$ (b) -1 (c) $\cot^2 A$ (d) $\tan^2 A$

Ans. (d) $\frac{\sec^2 A}{\operatorname{cosec}^2 A} = \frac{\sin^2 A}{\cos^2 A} = \tan^2 A$.

Q5. If $\sin\theta = \cos\theta$ for an acute angle θ, then θ is: 1M

(a) 30° (b) 45° (c) 60° (d) 90°

Ans. (b) $\tan\theta = 1 \Rightarrow \theta = 45^\circ$.

Q6. **A:** For an acute angle θ, $\sin\theta = \frac{4}{3}$ is not possible. **R:** In a right triangle the hypotenuse is the longest side, so $\sin\theta < 1$ for acute θ. 1M

(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true

Ans. (a) Both true; R explains A.

Q7. **A:** $\tan 45^\circ + \cot 45^\circ = 2$. **R:** $\tan 45^\circ = \cot 45^\circ = 1$. 1M

(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true

Ans. (a) Both true; R explains A ($1 + 1 = 2$).

Q8. If $\tan\theta = \frac{a}{b}$, then $\frac{a\sin\theta - b\cos\theta}{a\sin\theta + b\cos\theta}$ equals: 1M

(a) $\frac{a+b}{a-b}$ (b) 1 (c) $\frac{a^2-b^2}{a^2+b^2}$ (d) $\frac{a^2+b^2}{a^2-b^2}$

(contd.) (c) Divide by $\cos\theta$: $\frac{a\tan\theta - b}{a\tan\theta + b} = \frac{\frac{a^2}{b} - b}{\frac{a^2}{b} + b} = \frac{a^2 - b^2}{a^2 + b^2}$.

Q9. The value of $(\sin 45^\circ + \cos 45^\circ)$ is: 1M

(a) 1 (b) $\sqrt{2}$ (c) $\frac{1}{\sqrt{2}}$ (d) 2

Ans. (b) $\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2}$.

Q10. If $\cos A = \frac{4}{5}$, then $\tan A$ is: 1M

(a) $\frac{3}{5}$ (b) $\frac{3}{4}$ (c) $\frac{4}{3}$ (d) $\frac{5}{3}$

Ans. (b) opp $= \sqrt{25 - 16} = 3$, so $\tan A = \frac{3}{4}$.

Q11. **A:** $\sin^2 67^\circ + \cos^2 67^\circ = 1$. **R:** $\sin^2\theta + \cos^2\theta = 1$ for every angle θ (0° to 90°). 1M

(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true

Ans. (a) Both true; R explains A (an identity holds for every angle).

Very short (2 marks)

Q12. Evaluate: $2\tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ$. 2M Board fav

Ans. $= 2(1)^2 + \frac{3}{4} - \frac{3}{4} = 2$.

Q13. If $\sin(A + B) = 1$ and $\cos(A - B) = \frac{1}{2}$, $0^\circ < A + B \le 90^\circ$, $A > B$, find A and B. 2M

Ans. $A + B = 90^\circ$, $A - B = 60^\circ$. Adding: $2A = 150^\circ \Rightarrow A = 75^\circ$, $B = 15^\circ$.

Q14. If $3\cot A = 4$, check whether $\frac{1 - \tan^2 A}{1 + \tan^2 A} = \cos^2 A - \sin^2 A$. 2M

Ans. $\tan A = \frac{3}{4}$; opp $3k$, adj $4k$, hyp $5k$. LHS $= \frac{1 - \frac{9}{16}}{1 + \frac{9}{16}} = \frac{7}{25}$. RHS $= \frac{16}{25} - \frac{9}{25} = \frac{7}{25}$. Yes, equal.

Q15. Prove that $(\sin\theta + \cos\theta)^2 + (\sin\theta - \cos\theta)^2 = 2$. 2M

Ans. LHS $= (\sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta) + (\sin^2\theta + \cos^2\theta - 2\sin\theta\cos\theta) = 1 + 1 = 2 =$ RHS.

Q16. If $\cot\theta = \frac{7}{8}$, evaluate $\frac{(1 + \sin\theta)(1 - \sin\theta)}{(1 + \cos\theta)(1 - \cos\theta)}$. 2M Board fav

Ans. $= \frac{1 - \sin^2\theta}{1 - \cos^2\theta} = \frac{\cos^2\theta}{\sin^2\theta} = \cot^2\theta = \frac{49}{64}$.

Q17. If $\sin A = \frac{3}{4}$, find $\cos A$ and $\tan A$. 2M

Ans. $\cos A = \sqrt{1 - \frac{9}{16}} = \frac{\sqrt{7}}{4}$; $\tan A = \frac{3/4}{\sqrt{7}/4} = \frac{3}{\sqrt{7}}$.

Short answer (3 marks)

Q19. Evaluate: $\frac{4\cos^2 60^\circ + 3\sec^2 30^\circ - 2\tan^2 45^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$. 3M

(contd.) Num $= 4 \cdot \frac{1}{4} + 3 \cdot \frac{4}{3} - 2 \cdot 1 = 1 + 4 - 2 = 3$. Den $= \frac{1}{4} + \frac{3}{4} = 1$. Value $= 3$.

Q18. In $\triangle ABC$, right-angled at B, $AB = 24$ cm and $BC = 7$ cm. Find $\sin A, \cos A, \sin C, \cos C$. 3M Board fav

Ans. $AC = \sqrt{24^2 + 7^2} = \sqrt{625} = 25$ cm. $\sin A = \frac{BC}{AC} = \frac{7}{25}, \cos A = \frac{24}{25}; \sin C = \frac{AB}{AC} = \frac{24}{25}, \cos C = \frac{7}{25}$.

Q20. Prove that $\dfrac{1 + \sec A}{\sec A} = \dfrac{\sin^2 A}{1 - \cos A}$. 3M Board fav

Ans. LHS $= \dfrac{1 + \frac{1}{\cos A}}{\frac{1}{\cos A}} = \cos A + 1$. RHS $= \dfrac{1 - \cos^2 A}{1 - \cos A} = \dfrac{(1 - \cos A)(1 + \cos A)}{1 - \cos A} = 1 + \cos A$. LHS = RHS.

Q21. Prove that $\dfrac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta} = \tan\theta$. 3M

Ans. LHS $= \dfrac{\sin\theta(1 - 2\sin^2\theta)}{\cos\theta(2\cos^2\theta - 1)}$. Now $1 - 2\sin^2\theta = 1 - 2(1 - \cos^2\theta) = 2\cos^2\theta - 1$. So LHS $= \dfrac{\sin\theta}{\cos\theta} = \tan\theta$.

Q22. If $\sin\theta + \cos\theta = \sqrt{2}$, prove that $\tan\theta + \cot\theta = 2$. 3M

Ans. Square: $1 + 2\sin\theta\cos\theta = 2 \Rightarrow \sin\theta\cos\theta = \frac{1}{2}$. $\tan\theta + \cot\theta = \dfrac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta} = \dfrac{1}{1/2} = 2$.

Q23. If $\tan A = \dfrac{1}{\sqrt{3}}$ (A acute), find $7\sin^2 A + 3\cos^2 A$. Also verify $\sin^2 A + \cos^2 A = 1$. 3M

Ans. $\tan A = \frac{1}{\sqrt{3}} \Rightarrow A = 30°$. $7 \cdot \frac{1}{4} + 3 \cdot \frac{3}{4} = \frac{7}{4} + \frac{9}{4} = 4$. Check: $\frac{1}{4} + \frac{3}{4} = 1$. Value $= 4$.

Q24. In $\triangle PQR$, right-angled at Q, $PR + QR = 49$ cm and $PQ = 7$ cm. Find $\sin P, \cos P$ and $\tan P$. 3M Board fav

Ans. $PR^2 - QR^2 = PQ^2 = 49 \Rightarrow (PR - QR)(PR + QR) = 49 \Rightarrow PR - QR = 1$. So $PR = 25$ cm, $QR = 24$ cm. $\sin P = \frac{QR}{PR} = \frac{24}{25}, \cos P = \frac{PQ}{PR} = \frac{7}{25}, \tan P = \frac{24}{7}$.

Q25. Prove that $\dfrac{\sec A - 1}{\sec A + 1} = \dfrac{1 - \cos A}{1 + \cos A}$. 3M

Ans. LHS $= \dfrac{\frac{1}{\cos A} - 1}{\frac{1}{\cos A} + 1} = \dfrac{\frac{1 - \cos A}{\cos A}}{\frac{1 + \cos A}{\cos A}} = \dfrac{1 - \cos A}{1 + \cos A}$ = RHS.

Long answer (5 marks)

Q26. Prove that $\dfrac{\tan\theta}{1 - \cot\theta} + \dfrac{\cot\theta}{1 - \tan\theta} = 1 + \sec\theta\,\mathrm{cosec}\,\theta$. 5M Board fav

Ans. Put $t = \tan\theta$: LHS $= \dfrac{t}{1 - \frac{1}{t}} + \dfrac{\frac{1}{t}}{1 - t} = \dfrac{t^2}{t - 1} - \dfrac{1}{t(t-1)} = \dfrac{t^3 - 1}{t(t-1)}$

(contd.) $= \dfrac{(t-1)(t^2 + t + 1)}{t(t-1)} = \dfrac{t^2 + t + 1}{t} = t + 1 + \frac{1}{t}$. Now $\tan\theta + \cot\theta = \dfrac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta} = \sec\theta\,\mathrm{cosec}\,\theta$. So LHS $= 1 + \sec\theta\,\mathrm{cosec}\,\theta$ = RHS.

Q27. Prove that $(\mathrm{cosec}\,\theta - \cot\theta)^2 = \dfrac{1 - \cos\theta}{1 + \cos\theta}$. 5M

Ans. LHS $= \left(\dfrac{1}{\sin\theta} - \dfrac{\cos\theta}{\sin\theta}\right)^2 = \dfrac{(1 - \cos\theta)^2}{\sin^2\theta} = \dfrac{(1 - \cos\theta)^2}{1 - \cos^2\theta} = \dfrac{(1 - \cos\theta)^2}{(1 - \cos\theta)(1 + \cos\theta)} = \dfrac{1 - \cos\theta}{1 + \cos\theta}$ = RHS.

Q28. If $\sec\theta + \tan\theta = p$, prove that $\sin\theta = \dfrac{p^2 - 1}{p^2 + 1}$. 5M

Ans. $p^2 = \dfrac{(1 + \sin\theta)^2}{\cos^2\theta} = \dfrac{(1 + \sin\theta)^2}{(1 - \sin\theta)(1 + \sin\theta)} = \dfrac{1 + \sin\theta}{1 - \sin\theta}$. Then $\dfrac{p^2 - 1}{p^2 + 1} = \dfrac{(1 + \sin\theta) - (1 - \sin\theta)}{(1 + \sin\theta) + (1 - \sin\theta)} = \dfrac{2\sin\theta}{2} = \sin\theta$. Hence proved.

Q29. Prove that $\dfrac{\cos A - \sin A + 1}{\cos A + \sin A - 1} = \mathrm{cosec}\,A + \cot A$, using $\mathrm{cosec}^2 A = 1 + \cot^2 A$. 5M Board fav

Ans. Divide by $\sin A$: LHS $= \dfrac{\cot A - 1 + \mathrm{cosec}\,A}{\cot A + 1 - \mathrm{cosec}\,A}$. Write $1 = \mathrm{cosec}^2 A - \cot^2 A$ in the numerator: num $= (\mathrm{cosec}\,A + \cot A) - (\mathrm{cosec}\,A - \cot A)(\mathrm{cosec}\,A + \cot A) = (\mathrm{cosec}\,A + \cot A)(1 - \mathrm{cosec}\,A + \cot A)$. The bracket = denominator, so LHS $= \mathrm{cosec}\,A + \cot A$ = RHS.

Case based (4 marks)

Q30. A carpenter makes a right-angled triangular frame PQR with $\angle Q = 90°$. Side $PQ = 9$ cm, and $PR - QR = 1$ cm. (i) Find QR and PR. (1M) (ii) Find $\sin P$ and $\cos P$. (1M) (iii) Find $\sec P + \tan P$. (2M) 4M

Ans. (i) $PR^2 - QR^2 = 81 \Rightarrow (PR - QR)(PR + QR) = 81 \Rightarrow PR + QR = 81$. With $PR - QR = 1$: $PR = 41$ cm, $QR = 40$ cm. (ii) $\sin P = \frac{QR}{PR} = \frac{40}{41}$, $\cos P = \frac{PQ}{PR} = \frac{9}{41}$. (iii) $\sec P + \tan P = \frac{41}{9} + \frac{40}{9} = \frac{81}{9} = 9$.

Quick fact 3-4-5, 5-12-13, 8-15-17, 7-24-25: Pythagorean triplets save time.

Apne notes / galtiyan yahan likho

Most asked: two-angle tower sums, building + tower, depression

APPLICATIONS OF TRIGONOMETRY

1. Heights and Distances: Key Terms

Trigonometry helps find heights and distances that cannot be measured directly: height of a tower, width of a river, distance of a ship. CBSE uses only angles $30°, 45°, 60°$.

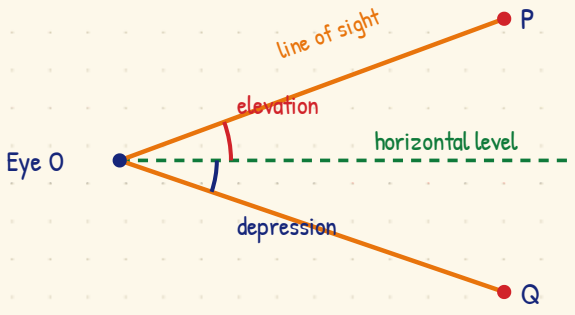


Fig 9.1: P above eye level: angle of elevation; Q below: angle of depression

Key terms

Line of sight: the line from the eye of the observer to the point (object) being viewed.

Horizontal level: the horizontal line through the eye of the observer.

Angle of elevation: angle between the line of sight and the horizontal, when the object is above the horizontal level (we raise our head).

Angle of depression: angle between the line of sight and the horizontal, when the object is below the horizontal level (we lower our head).

- ✓ Both angles are always measured from the horizontal, never from the vertical.
- ✓ Angle of elevation of P from Q = angle of depression of Q from P (**alternate angles**, horizontals are parallel).
- ✓ Heights (towers, poles, buildings) are taken **vertical**, i.e. at $90°$ to the ground.
- ✓ As you walk towards a tower, its angle of elevation **increases**; walking away, it **decreases**.
- ✓ The observer is treated as a point unless his height is given.

Yaad karo: "Z wala angle"

Upar wale ki **depression** = neeche wale ki **elevation**. Dono horizontal lines parallel hain, beech mein line of sight: Z shape banta hai, alternate angles barabar.

Quick fact Higher angle of elevation $\Rightarrow$ observer is nearer to the object.

2. Which Ratio to Use?

Known + wanted	Use
height and ground distance	$\tan\theta = \dfrac{\text{height}}{\text{distance}}$
height and slant (string, ladder, rope)	$\sin\theta = \dfrac{\text{height}}{\text{slant}}$
ground distance and slant	$\cos\theta = \dfrac{\text{distance}}{\text{slant}}$

θ	$30°$	$45°$	$60°$
$\tan\theta$	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$
$\sin\theta$	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$
$\cos\theta$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$

↳ Use $\sqrt{3} = 1.732$ and $\sqrt{2} = 1.414$ only at the last step (if the question says so).

Yaad karo: "Dori, seedhi, rassi = sin"

Jahan **slant length** (kite ki dori, seedhi, rassi, tar) di ho ya maangi ho, wahan **sin**. Zameen pe doori aur height ho toh **tan**. 80% sawaal tan se hi bante hain.

3. Method (5 steps)

Read + draw rough figure → Mark right angle, angles, knowns → Pick the ratio (mostly tan) → Solve, rationalise, put $\sqrt{3}$ value

1. Draw the figure: vertical object, horizontal ground, line of sight.
2. Mark the angle at the correct point (elevation at the observer on the ground).
3. In each right triangle, write one equation using the known side.
4. Two triangles? Find the common side (usually the height h or a distance x) and equate.
5. Write the answer with units; approximate only if asked.

Quick fact $\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3} \approx 0.577$; multiply by $\sqrt{3}$ to rationalise.

Quick fact km/h $=$ m/s $\times \frac{18}{5}$; m/s $=$ km/h $\times \frac{5}{18}$.

Quick fact Angle of depression is measured at the observer's eye, below the horizontal.

4. Type 1: One Right Triangle

Solved Example 1

The angle of elevation of the top of a tower from a point 30 m away from its foot is 30°. Find the height of the tower.

1. $\tan 30^\circ = \dfrac{h}{30} \Rightarrow \dfrac{1}{\sqrt{3}} = \dfrac{h}{30}$.
2. $h = \dfrac{30}{\sqrt{3}} = \dfrac{30\sqrt{3}}{3} = 10\sqrt{3}$ m.

✔ $h = 10\sqrt{3} \approx 17.32$ m

Solved Example 2

A kite is flying with a string of length 80 m. The string makes an angle of 60° with the ground. Find the height of the kite (string taut).

1. String = slant (hypotenuse), so use sine: $\sin 60^\circ = \dfrac{h}{80}$.
2. $h = 80 \times \dfrac{\sqrt{3}}{2} = 40\sqrt{3}$ m.

✔ $h = 40\sqrt{3} \approx 69.28$ m

Solved Example 3

From the top of a building 60 m high, the angle of depression of a car on the road is 30°. How far is the car from the foot of the building?

1. Angle of depression 30° = angle of elevation of the top from the car.
2. $\tan 30^\circ = \dfrac{60}{d} \Rightarrow d = 60\sqrt{3}$ m.

✔ $d = 60\sqrt{3} \approx 103.92$ m

Meme samjho

Sawaal mein angle 45° aaya: height = distance, koi calculation nahi. **Main, full sukoon mein:**

Asli baat: $\tan 45^\circ = 1$, so at 45° the height of the object equals its distance from the observer.

5. Type 2: Two Angles, Same Side

Observer moves towards (or away from) the tower: two right triangles share the same height h.

Quick fact Shadow problems: Sun's altitude = angle of elevation of the Sun.

Quick fact $\tan 60^\circ = \sqrt{3} \approx 1.732$: at 60° height is about $1.7\times$ the distance.

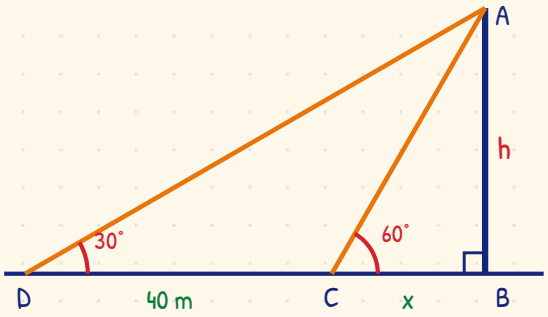


Fig 9.2: Solved Example 4 ($DC = 40$ m, $CB = x$)

Solved Example 4 (board fav)

The angle of elevation of the top of a tower changes from 30° to 60° as an observer walks 40 m towards it. Find the height of the tower.

1. In $\triangle ABC$: $\tan 60^\circ = \dfrac{h}{x} \Rightarrow h = \sqrt{3}\,x$.
2. In $\triangle ABD$: $\tan 30^\circ = \dfrac{h}{x+40} \Rightarrow h = \dfrac{x+40}{\sqrt{3}}$.
3. Equate: $3x = x + 40 \Rightarrow x = 20$ m.
4. $h = 20\sqrt{3}$ m.

✔ $h = 20\sqrt{3} \approx 34.64$ m

Yaad karo: 30-60 jodi = 1 : 3 rule

Same tower, angles 60° aur 30°: agar 60° wali doori x hai, toh 30° wali doori $3x$, aur chali hui doori $2x$. Height $= x\sqrt{3}$. Example 4: $2x = 40 \Rightarrow x = 20$, $h = 20\sqrt{3}$. (Answer check karne ke liye; exam mein steps likho.)

Shivam Bhaiya says

Figure bina sawaal mat shuru karo. Figure sahi bana, toh aadha sawaal wahi solve ho gaya. Aur angle hamesha zameen (horizontal) se naapo.

6. Type 3: Building and Tower

From the top of a building, the top of a taller tower is seen at an elevation and its foot at a depression. Draw a horizontal line from the building top to the tower: it splits the tower into two parts.

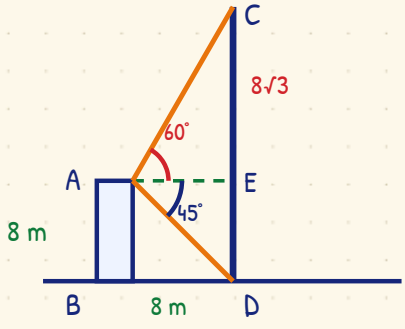


Fig 9.3: Building AB, tower CD (Solved Example 5)

Solved Example 5

From the top of an 8 m high building, the angle of elevation of the top of a tower is 60° and the angle of depression of its foot is 45°. Find the height of the tower.

1. $AE \parallel BD$, $DE = AB = 8$ m. In $\triangle AED$: $\tan 45^\circ = \dfrac{DE}{AE} \Rightarrow AE = 8$ m.
2. In $\triangle AEC$: $\tan 60^\circ = \dfrac{CE}{AE} \Rightarrow CE = 8\sqrt{3}$ m.
3. $CD = CE + ED = 8\sqrt{3} + 8 = 8(\sqrt{3}+1)$ m.

✔ $8(\sqrt{3}+1) \approx 21.86$ m

7. Type 4: Opposite Sides

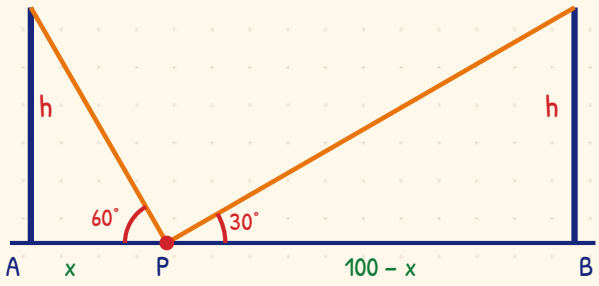


Fig 9.4: Two equal poles on opposite sides of a 100 m road

Solved Example 6

Two poles of equal height stand on either side of a road 100 m wide. From a point between them on the road, the angles of elevation of their tops are 60° and 30°. Find the height of the poles and the distances of the point from the poles.

1. Let $AP = x$, $PB = 100 - x$. $\tan 60^\circ = \dfrac{h}{x} \Rightarrow h = \sqrt{3}\,x$.
2. $\tan 30^\circ = \dfrac{h}{100-x} \Rightarrow h = \dfrac{100-x}{\sqrt{3}}$.
3. $3x = 100 - x \Rightarrow x = 25$ m, so $PB = 75$ m.
4. $h = 25\sqrt{3}$ m.

✔ $h = 25\sqrt{3} \approx 43.30$ m; distances 25 m and 75 m

8. Type 5: Observer's Height

If the observer's eye is at height a above the ground, the line of sight starts at height a. Then height of object = a + (part found by trig).

Quick fact Complementary angles from distances a, b: height $= \sqrt{ab}$.

Solved Example 7

A boy 1.6 m tall stands 32.4 m away from a tower and sees its top at an angle of elevation of 45°. Find the height of the tower.

1. Above eye level: $\tan 45^\circ = \dfrac{h'}{32.4} \Rightarrow h' = 32.4$ m.
2. Height $= 32.4 + 1.6 = 34$ m.

✔ 34 m

Formula Box

- → height $=$ distance $\times \tan\theta$; distance $= \dfrac{\text{height}}{\tan\theta}$
- → height $=$ slant $\times \sin\theta$; distance $=$ slant $\times \cos\theta$
- → At 45°: height = distance
- → At 60°: $h = \sqrt{3}\,d$; at 30°: $h = \dfrac{d}{\sqrt{3}}$
- → Elevation of P from Q = depression of Q from P
- → Two positions (same side, distance d apart, angles $\alpha < \beta$):

$$h = \frac{d\tan\alpha\tan\beta}{\tan\beta - \tan\alpha}$$

- → Observer of height a: object height $= a +$ trig part
- → $\sqrt{3} \approx 1.732$, $\sqrt{2} \approx 1.414$

Common mistakes

- → Measuring the angle from the vertical (tower) instead of from the horizontal.
- → Putting the angle of depression at the bottom point of the figure the wrong way: it is at the **top**, between the horizontal and the line of sight.
- → Using tan for a kite string or ladder: slant given means sin or cos.
- → Forgetting to add the observer's height (Type 5).
- → Writing $\frac{30}{\sqrt{3}} = 10$: it is $10\sqrt{3}$. Rationalise carefully.

Meme samjho

Shaam ko Sun, elevation ghatate ghatate: **"Samay aa gaya hai aapse vida lene ka."** Shadow: aur lambi ho gayi.

Asli baat: For a fixed pole, $\tan\theta = \dfrac{h}{\text{shadow}}$. Lower angle of elevation $\Rightarrow$ longer shadow.

Quick fact Always write units (m) with the final answer.

Shivam Bhaiya says
Is chapter se 5 marks ka LA lagbhag pakka aata hai. Upar ke 5 types ka ek-ek sawaal khud figure bana ke karo, board mein koi naya type nahi milega.

9. Type 6: Broken Tree

The broken part, the standing part and the ground form a right triangle. Original height = standing part + broken part.

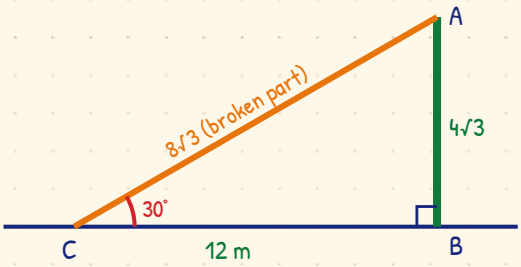


Fig 9.5: Broken tree (Solved Example 8)

Solved Example 8

A tree breaks due to a storm and the broken part bends so that its top touches the ground at 12 m from the foot, making an angle of 30° with the ground. Find the original height of the tree.

1. Standing part $AB = 12\tan 30^\circ = \dfrac{12}{\sqrt{3}} = 4\sqrt{3}$ m.
2. Broken part $AC = \dfrac{12}{\cos 30^\circ} = \dfrac{24}{\sqrt{3}} = 8\sqrt{3}$ m.
3. Height $= AB + AC = 4\sqrt{3} + 8\sqrt{3} = 12\sqrt{3}$ m.

✔ $12\sqrt{3} \approx 20.78$ m

Yaad karo: broken tree at 30°

30° pe toota hua ped: broken part = **2** × **standing part** (kyunki $\sin 30^\circ = \frac{1}{2}$). Toh total = **3** × **standing part**. Example 8: $3 \times 4\sqrt{3} = 12\sqrt{3}$.

10. Type 7: Opposite-side Depressions

From the top of a tower, two objects on **opposite sides** are seen: total distance = sum of the two distances. On the **same side**: distance between = difference.

Quick fact $\sin 30^\circ = \frac{1}{2}$: at 30° the height is half the slant length.

Quick fact $\sqrt{3} \approx 1.732$, so $10\sqrt{3} \approx 17.32$ and $20\sqrt{3} \approx 34.64$.

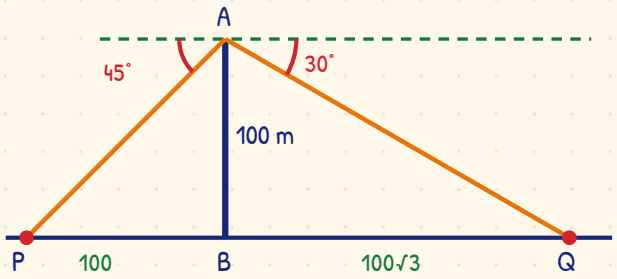


Fig 9.6: Cars on opposite sides (Solved Example 9)

Solved Example 9

From the top of a 100 m high tower, the angles of depression of two cars on opposite sides of it are 45° and 30°. Find the distance between the cars.

1. $BP = \dfrac{100}{\tan 45^\circ} = 100$ m.
2. $BQ = \dfrac{100}{\tan 30^\circ} = 100\sqrt{3}$ m.
3. $PQ = 100 + 100\sqrt{3} = 100(1+\sqrt{3})$ m.

✔ $100(1+\sqrt{3}) \approx 273.2$ m

Solved Example 10

The angles of elevation of the top of a tower from two points at distances 4 m and 16 m from its base, on the same side and in a straight line with it, are complementary (add up to 90°). Prove that the height of the tower is 8 m.

1. Let angles be θ (at 16 m) and $90^\circ - \theta$ (at 4 m). $\tan\theta = \dfrac{h}{16}$.
2. In the other triangle, the angle at the top is θ, so $\tan\theta = \dfrac{4}{h}$.
3. $\dfrac{h}{16} = \dfrac{4}{h} \Rightarrow h^2 = 64 \Rightarrow h = 8$ m.

✔ $h = 8$ m (in general $h = \sqrt{ab}$)

Rationalise fast

→ $\dfrac{1}{\sqrt{3}} = \dfrac{\sqrt{3}}{3}$, $\dfrac{30}{\sqrt{3}} = 10\sqrt{3}$, $\dfrac{90}{\sqrt{3}} = 30\sqrt{3}$

→ $\dfrac{2}{\sqrt{3}-1} = \dfrac{2(\sqrt{3}+1)}{3-1} = \sqrt{3}+1$

→ $\dfrac{1}{\sqrt{3}+1} = \dfrac{\sqrt{3}-1}{2}$; $\sqrt{3} - \dfrac{1}{\sqrt{3}} = \dfrac{2}{\sqrt{3}}$

Quick fact Line of sight is the hypotenuse of the right triangle.

Board scan: what comes from this chapter

- → **1M:** Sun's elevation from pole and shadow; ladder/kite height; AR on shadow length.
- → **2M / 3M:** one triangle sums, broken tree, statue on pedestal, tower on building.
- → **5M:** two-angle sums: moving observer, aeroplane speed, two ships, car approaching tower.
- → **Case study:** river width, lighthouse, kite festival.

Last-minute revision

- → Angles from the horizontal; elevation = depression (alternate angles).
- → tan for height-distance; sin/cos when slant is involved.
- → 30-60 pair: distances x and $3x$, moved $2x$, $h = x\sqrt{3}$.
- → Same side: subtract distances; opposite sides: add.
- → Add the observer's height at the end.

MOST IMPORTANT QUESTIONS

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. A pole 6 m high casts a shadow $2\sqrt{3}$ m long on the ground. The Sun's elevation is: 1M Board fav

(a) $30°$ (b) $45°$ (c) $60°$ (d) $90°$

Ans. (c) $\tan\theta = \dfrac{6}{2\sqrt{3}} = \sqrt{3} \Rightarrow \theta = 60°$.

Q2. A ladder 10 m long leans against a wall making $60°$ with the ground. The height it reaches on the wall is: 1M

(a) 5 m (b) $5\sqrt{3}$ m (c) $10\sqrt{3}$ m (d) $\frac{10}{\sqrt{3}}$ m

Ans. (b) $10\sin 60° = 5\sqrt{3}$ m.

Q3. If the height of a tower and the distance of a point from its foot are equal, the angle of elevation of its top from that point is: 1M

(a) $30°$ (b) $45°$ (c) $60°$ (d) $90°$

Ans. (b) $\tan\theta = 1 \Rightarrow \theta = 45°$.

Q4. The angle of elevation of the top of a tower from a point 100 m from its foot is $30°$. Its height is: 1M

(a) $100\sqrt{3}$ m (b) 50 m (c) $50\sqrt{3}$ m (d) $\frac{100\sqrt{3}}{3}$ m

Ans. (d) $h = 100\tan 30° = \dfrac{100}{\sqrt{3}} = \dfrac{100\sqrt{3}}{3}$ m.

Q5. The length of the shadow of a 15 m pole when the Sun's altitude is $45°$ is: 1M

(a) 15 m (b) $15\sqrt{3}$ m (c) $\frac{15}{\sqrt{3}}$ m (d) 30 m

Ans. (a) At $45°$, shadow = height $= 15$ m.

Q6. **A:** As the shadow of a tower becomes longer, the angle of elevation of the Sun decreases. **R:** $\tan\theta = \dfrac{h}{s}$; for fixed h, $\tan\theta$ decreases as s increases. 1M Board fav

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (a) Both true; R explains A.

Q7. **A:** The angle of elevation of the top of a tower from a point P equals the angle of depression of P from the top. **R:** Alternate angles between two parallel lines are equal. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (a) Both true; R explains A (both horizontals are parallel).

Q8. From the top of a 50 m high tower, the angle of depression of a car is $45°$. The distance of the car from the foot of the tower is: 1M

(a) $50\sqrt{3}$ m (b) 25 m (c) 50 m (d) $\frac{50}{\sqrt{3}}$ m

Ans. (c) At $45°$, distance = height $= 50$ m.

Q9. A wheelchair ramp rises 1.5 m and makes an angle of $30°$ with the ground. The length of the ramp is: 1M

(a) $1.5\sqrt{3}$ m (b) 3 m (c) $\sqrt{3}$ m (d) 0.75 m

Ans. (b) $\sin 30° = \dfrac{1.5}{l} \Rightarrow l = 3$ m.

Very short (2 marks)

Q10. A kite is flying at a height of 45 m. Its string makes an angle of $60°$ with the ground. Find the length of the string (no slack). 2M Board fav

Ans. $\sin 60° = \dfrac{45}{l} \Rightarrow l = \dfrac{45\times 2}{\sqrt{3}} = \dfrac{90}{\sqrt{3}} = 30\sqrt{3} \approx 51.96$ m.

Q11. From the top of a cliff 50 m high, the angle of depression of a boat is $30°$. Find the distance of the boat from the foot of the cliff. 2M

Ans. Elevation of cliff top from boat $= 30°$. $\tan 30° = \dfrac{50}{d} \Rightarrow d = 50\sqrt{3} \approx 86.6$ m.

Q12. A ladder reaches the top of a $6\sqrt{3}$ m high wall making an angle of $60°$ with the ground. Find the length of the ladder and the distance of its foot from the wall. 2M

Ans. $\sin 60° = \dfrac{6\sqrt{3}}{l} \Rightarrow l = 12$ m. Distance $= 12\cos 60° = 6$ m.

Q13. The angle of elevation of the top of a 30 m high tower from a point on the ground is $60°$. How far is the point from the foot of the tower? 2M

Ans. $\tan 60° = \dfrac{30}{d} \Rightarrow d = \dfrac{30}{\sqrt{3}} = 10\sqrt{3} \approx 17.32$ m.

Q14. The ratio of the height of a pole to the length of its shadow is $1 : \sqrt{3}$. Find the angle of elevation of the Sun. 2M

Ans. $\tan\theta = \dfrac{1}{\sqrt{3}} \Rightarrow \theta = 30°$.

Short answer (3 marks)

Q15. A 2 m tall boy is standing at some distance from a 32 m tall building. The angle of elevation from his eyes to the top of the building increases from 30° to 60° as he walks towards the building. Find the distance he walked. 3M Board fav

> **Ans.** Height above eye $= 30$ m. Distances: $30\sqrt{3}$ (at 30°) and $\frac{30}{\sqrt{3}} = 10\sqrt{3}$ (at 60°). Walked $= 30\sqrt{3} - 10\sqrt{3} = 20\sqrt{3} \approx 34.64$ m.

Q16. The shadow of a tower is 60 m longer when the Sun's altitude is 30° than when it is 60°. Find the height of the tower. 3M

> **Ans.** Shadows: $h\cot 30^\circ = h\sqrt{3}$ and $h\cot 60^\circ = \frac{h}{\sqrt{3}}$. $h\sqrt{3} - \frac{h}{\sqrt{3}} = 60 \Rightarrow \frac{2h}{\sqrt{3}} = 60 \Rightarrow h = 30\sqrt{3} \approx 51.96$ m.

Q17. From a point on the ground, the angles of elevation of the bottom and the top of a transmission tower fixed on top of a 25 m high building are 45° and 60°. Find the height of the tower. 3M

> **Ans.** Distance from building $= 25$ m ($\tan 45^\circ = 1$). Total height $= 25\tan 60^\circ = 25\sqrt{3}$. Tower $= 25\sqrt{3} - 25 = 25(\sqrt{3} - 1) \approx 18.30$ m.

Q18. A statue 2 m tall stands on a pedestal. From a point on the ground, the angle of elevation of the top of the statue is 60° and of the top of the pedestal is 45°. Find the height of the pedestal. 3M

> **Ans.** Let pedestal $= h$; distance $= h$ (45°). $\tan 60^\circ = \frac{h+2}{h} \Rightarrow \sqrt{3}h = h + 2 \Rightarrow h = \frac{2}{\sqrt{3}-1} = \sqrt{3} + 1 \approx 2.73$ m.

Q19. An observer 1.5 m tall is 20.5 m away from a tower 22 m high. Find the angle of elevation of the top of the tower from his eye. 3M

> **Ans.** Height above eye level $= 22 - 1.5 = 20.5$ m. $\tan\theta = \frac{20.5}{20.5} = 1 \Rightarrow \theta = 45^\circ$.

Q20. The angle of elevation of the top of a hill from the foot of a tower is 60° and the angle of elevation of the top of the tower from the foot of the hill is 30°. If the tower is 50 m high, find the height of the hill. 3M Board fav

> **Ans.** Let distance $= d$. Tower: $\tan 30^\circ = \frac{50}{d} \Rightarrow d = 50\sqrt{3}$. Hill: $H = d\tan 60^\circ = 50\sqrt{3} \times \sqrt{3} = 150$ m.

Q21. A tower stands on level ground and has a flagstaff 6 m high on its top. From a point on the ground, the angles of elevation of the bottom and the top of the flagstaff are 30° and 45°. Find the height of the tower. 3M

> **Ans.** Let tower $= h$, distance $= d$. $\tan 30^\circ = \frac{h}{d} \Rightarrow d = \sqrt{3}h$.
>
> (contd.) $\tan 45^\circ = \frac{h+6}{d} \Rightarrow h + 6 = \sqrt{3}h \Rightarrow h = \frac{6}{\sqrt{3}-1} = 3(\sqrt{3}+1) \approx 8.20$ m.

Long answer (5 marks)

Q22. An aeroplane flying horizontally at a height of $1200\sqrt{3}$ m is observed from a point on the ground. The angle of elevation changes from 60° to 30° in 12 seconds. Find the speed of the plane in km/h. 5M Board fav

> **Ans.** At 60°: distance $= \frac{1200\sqrt{3}}{\sqrt{3}} = 1200$ m. At 30°: distance $= 1200\sqrt{3} \times \sqrt{3} = 3600$ m. Distance flown $= 2400$ m in 12 s $= 200$ m/s $= 200 \times \frac{18}{5} = 720$ km/h.

Q23. From the top of a 60 m high lighthouse, the angles of depression of two ships on the same side of it are 30° and 45°. Find the distance between the ships. 5M

> **Ans.** Nearer ship (45°): $d_1 = 60$ m. Farther ship (30°): $d_2 = 60\sqrt{3}$ m. Distance between $= 60\sqrt{3} - 60 = 60(\sqrt{3} - 1) \approx 43.92$ m.

Q24. A straight highway leads to the foot of a tower. A man at the top of the tower sees a car at an angle of depression of 30°, approaching at uniform speed. After 10 minutes the angle of depression is 60°. Find the further time taken by the car to reach the tower. 5M Board fav

> **Ans.** Let tower $= h$. Distances: $h\sqrt{3}$ (at 30°) and $\frac{h}{\sqrt{3}}$ (at 60°). Covered in 10 min: $h\sqrt{3} - \frac{h}{\sqrt{3}} = \frac{2h}{\sqrt{3}}$. Remaining $\frac{h}{\sqrt{3}}$ is half of that, so time $= 5$ minutes.

Q25. From a point on a bridge across a river, the angles of depression of the banks on opposite sides of the river are 30° and 45°. If the bridge is at a height of 5 m from the banks, find the width of the river. 5M Board fav

> **Ans.** Distances of the banks from the foot of the point: $\frac{5}{\tan 45^\circ} = 5$ m and $\frac{5}{\tan 30^\circ} = 5\sqrt{3}$ m. Opposite sides, so width $= 5 + 5\sqrt{3} = 5(1 + \sqrt{3}) \approx 13.66$ m.

Case based (4 marks)

Q26. A student stands at point P on one bank of a river and sees the top of a tree on the opposite bank, exactly opposite to him, at an angle of elevation of 60°. He walks 20 m away from the bank to Q, and the angle becomes 30°. (i) Write two equations for height h and river width x. (1M) (ii) Find the width of the river. (1M) (iii) Find the height of the tree. (2M) 4M

> **Ans.** (i) $h = x\tan 60^\circ = \sqrt{3}x$ and $h = (x + 20)\tan 30^\circ = \frac{x+20}{\sqrt{3}}$. (ii) $3x = x + 20 \Rightarrow x = 10$ m. (iii) $h = 10\sqrt{3} \approx 17.32$ m.

CIRCLES

1. Circle, Secant and Tangent

A **circle** is the set of all points in a plane at a fixed distance (**radius**) from a fixed point (**centre**). For a line and a circle there are three possibilities:

Case	Common points	Name
Line misses the circle	0	Non-intersecting line
Line cuts the circle	2	Secant
Line touches the circle	1	Tangent

Key terms

Tangent: a line that intersects the circle at exactly one point. The common point is the **point of contact**.

Length of tangent: length of the segment from an external point to the point of contact.

A tangent is a special case of a secant when the two end points of the chord coincide.

- ✓ At any point of a circle there is **one and only one** tangent.
- ✓ A circle can have **infinitely many** tangents. A secant can be drawn in infinitely many ways too.
- ✓ Tangents at the two ends of a diameter are parallel.

Number of tangents from a point P

Position of P	Tangents
Inside the circle	0 (none)
On the circle	exactly 1
Outside the circle	exactly 2

Yaad karo: "Andar 0, upar 1, bahar 2"

Point circle ke andar: koi tangent nahi. Circle ke upar: sirf ek. Bahar: pure do, aur dono ki length barabar.

2. Theorem: Tangent and Radius

Statement: The tangent at any point of a circle is perpendicular to the radius through the point of contact.

Quick fact The word "tangent" comes from the Latin "tangere", meaning "to touch".

Quick fact A circle has infinitely many tangents but only one tangent at each point.

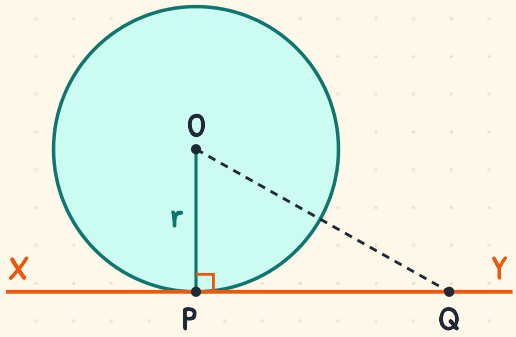


Fig: XY is the tangent at P; Q is any other point on XY, so OQ is greater than OP

Theorem: Proof (3 marks, board fav)

Given: Circle with centre O, tangent XY at point P. **To prove:** $OP \perp XY$.

1. Take any point Q on XY other than P, and join OQ.
2. Q lies outside the circle (if Q were inside, XY would be a secant, not a tangent).
3. So $OQ > OP$ (OP = radius). This is true for every point Q on XY except P.
4. So OP is the **shortest distance** from O to the line XY.
5. The shortest distance from a point to a line is the perpendicular. Hence $OP \perp XY$.

Converse: A line drawn through the end of a radius and perpendicular to it is a tangent. Also, the perpendicular to a tangent at its point of contact passes through the centre.

Length of a tangent

If $OP = d$ (distance of external point from centre) and r = radius, then $\triangle OTP$ is right-angled at T, so

$$PT = \sqrt{d^2 - r^2}$$

Solved Example 1

From a point P, 17 cm away from the centre O of a circle, the length of the tangent PT is to be found. The radius is 8 cm.

1. $OT \perp PT$ (tangent $\perp$ radius), so $\triangle OTP$ is right-angled at T.
2. $PT^2 = OP^2 - OT^2 = 17^2 - 8^2 = 289 - 64 = 225$

✓ $PT = 15$ cm

Quick fact Tangents at the ends of a diameter are parallel; their distance $= 2r$.

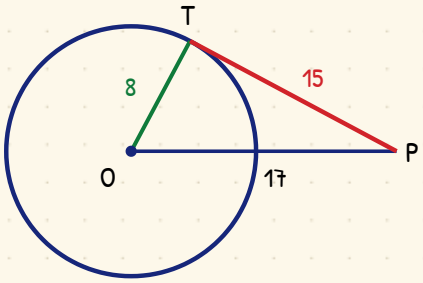


Fig: Solved Example 1. Right angle at T

Solved Example 2

The length of the tangent from a point A at distance 25 cm from the centre is 24 cm. Find the radius.

1. $r^2 = OA^2 - AT^2 = 625 - 576 = 49$

✔ $r = 7$ cm

3. Theorem: Equal Tangents

Statement: The lengths of the two tangents drawn from an external point to a circle are equal.

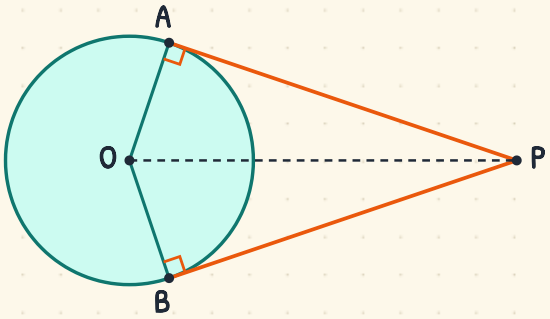


Fig: PA and PB are tangents from P; OA and OB are radii

Theorem: Proof (3 marks, board fav)

Given: PA, PB are tangents from external point P to a circle with centre O. **To prove:** $PA = PB$.

1. Join OP, OA, OB. $\angle OAP = \angle OBP = 90^\circ$ (tangent $\perp$ radius).
2. In right $\triangle OAP$ and $\triangle OBP$: $OA = OB$ (radii), OP common (hypotenuse).
3. So $\triangle OAP \cong \triangle OBP$ (RHS).
4. Hence $PA = PB$ (CPCT).

Results from the same congruence

- ✓ $\angle OPA = \angle OPB$: centre lies on the **bisector** of the angle between the two tangents.
- ✓ $\angle POA = \angle POB$: OP bisects $\angle AOB$.
- ✓ In quadrilateral OAPB, two angles are 90°, so $\angle APB + \angle AOB = 180^\circ$.
- ✓ OP is the perpendicular bisector of the chord of contact AB.

Yaad karo: "Tangent-Radius = 90, do tangent = jodwa bhai"

Contact point pe hamesha 90° mark karo. Bahar ke point se do tangents jodwa bhai jaise: length same, angle bhi OP barabar baantta hai.

Formula Box

- → Tangent $\perp$ radius: $\angle OTP = 90^\circ$
- → Length of tangent: $PT = \sqrt{d^2 - r^2}$, $d = OP$
- → Equal tangents: $PA = PB$
- → Angle between tangents: $\angle APB = 180^\circ - \angle AOB$
- → $\angle OPA = \angle OPB = \frac{1}{2}\angle APB$
- → Circumscribed quadrilateral: $AB + CD = AD + BC$
- → Concentric circles (radii $R > r$): chord of bigger circle touching smaller $= 2\sqrt{R^2 - r^2}$
- → Incircle of triangle: tangent lengths $x = s - a$, $y = s - b$, $z = s - c$, $s = \frac{a+b+c}{2}$

Solved Example 3

PA and PB are tangents from P to a circle with centre O. If $\angle APB = 70^\circ$, find $\angle AOB$ and $\angle OAB$.

1. $\angle OAP = \angle OBP = 90^\circ$, so $\angle AOB = 180^\circ - 70^\circ = 110^\circ$.
2. $OA = OB$ (radii), so $\triangle OAB$ is isosceles.
3. $\angle OAB = \dfrac{180^\circ - 110^\circ}{2}$

✔ $\angle AOB = 110^\circ$, $\angle OAB = 35^\circ$

Solved Example 4

Prove that $\angle PTQ = 2\angle OPQ$, where TP and TQ are tangents from T to a circle with centre O touching it at P and Q.

1. Let $\angle PTQ = \theta$. $TP = TQ$, so $\angle TPQ = \angle TQP = \dfrac{180^\circ - \theta}{2} = 90^\circ - \dfrac{\theta}{2}$.
2. $\angle OPT = 90^\circ$, so $\angle OPQ = 90^\circ - \angle TPQ = \dfrac{\theta}{2}$.

✔ $\angle PTQ = \theta = 2\angle OPQ$. **Hence proved.**

Meme samjho

Question: "Two tangents PA, PB from P... find PB." Mujhe pata hai $PA = PB$ hota hai:

Asli baat: Tangents from an external point are equal (RHS congruence). PA diya hai to PB wahi hai, koi calculation nahi!

4. Applications of Tangents

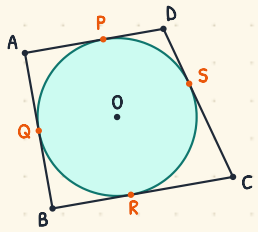


Fig: Quadrilateral ABCD circumscribing a circle; equal tangents from each vertex

Result: Quadrilateral circumscribing a circle

Sides AB, BC, CD, DA touch the circle at Q, R, S, P (see figure). Equal tangents: $AQ = AP$, $BQ = BR$, $CS = CR$, $DS = DP$. Adding:

$$AB + CD = AD + BC$$

So a **parallelogram** circumscribing a circle is a **rhombus** (opposite sides equal $\Rightarrow 2AB = 2BC$).

Yaad karo: "Aamne-saamne ka jod barabar"

Circle ke bahar bani quadrilateral mein opposite sides (aamne-saamne) ka sum equal: AB + CD = AD + BC. Adjacent mat jodna!

Solved Example 5

A quadrilateral ABCD is drawn to circumscribe a circle. $AB = 6$ cm, $BC = 7$ cm, $CD = 4$ cm. Find AD.

1. $AB + CD = AD + BC$
2. $6 + 4 = AD + 7$

✔ $AD = 3$ cm

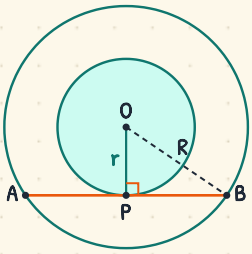


Fig: Concentric circles; chord AB of the bigger circle touches the smaller circle at P

Concentric circles: a chord AB of the bigger circle that touches the smaller circle at P is bisected at P (OP $\perp$ AB, and the perpendicular from the centre bisects a chord). So $AB = 2\sqrt{R^2 - r^2}$.

Incircle of a triangle: if the incircle of $\triangle ABC$ touches BC, CA, AB at D, E, F then $AF = AE = x$, $BF = BD = y$, $CD = CE = z$; so $x + y = AB$, $y + z = BC$, $z + x = CA$.

Solved Example 6

Two concentric circles have radii 10 cm and 6 cm. Find the length of the chord of the larger circle which touches the smaller circle.

1. $OP = 6$, $OA = 10$, $OP \perp AB$.
2. $AP = \sqrt{10^2 - 6^2} = \sqrt{64} = 8$
3. $AB = 2AP$

✔ $AB = 16$ cm

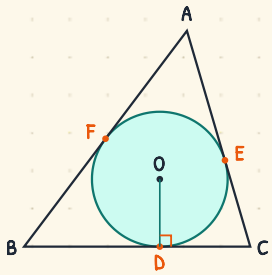


Fig: Incircle of triangle ABC; equal tangents from each vertex

Shivam Bhaiya says

Circles chapter chhota hai par **2 proofs pakke aate hain**: tangent $\perp$ radius, aur equal tangents. Dono ke figure + Given + To prove + steps yaad karo. Numerical mein pehla step hamesha: "$\angle OTP = 90°$ (tangent $\perp$ radius)" likho, isi ke marks hain.

Common mistakes

- → Drawing a tangent from a point **inside** the circle (not possible).
- → Forgetting to write the reason "tangent $\perp$ radius" for the $90°$.
- → Mixing $\angle APB$ (between tangents) and $\angle AOB$ (at centre): they add to $180°$, not equal.
- → Quadrilateral result: adding adjacent sides instead of opposite sides.
- → Concentric circles: giving half chord (AP) as the answer instead of full chord AB.

True or False? (quick self-check)

1. A tangent can touch a circle at two points. False
2. From a point on the circle exactly one tangent can be drawn. True
3. A rectangle can circumscribe a circle only if it is a square. True ($AB + CD = AD + BC$)
4. The angle between two tangents can be $180°$. False
5. The perpendicular at the point of contact passes through the centre. True

Quick fact The centre lies on the bisector of the angle between two tangents from a point.

Practice yourself (answers in brackets)

1. $d = 10$ cm, $r = 6$ cm: tangent length? [8 cm]
2. Tangents at $\angle APB = 110°$: find $\angle AOB$. [$70°$]
3. Concentric circles 13 cm and 5 cm: chord touching inner circle? [24 cm]
4. Circumscribed ABCD: $AB = 5, BC = 6, CD = 8$; AD? [7]
5. Incircle: sides 7, 8, 9; tangent lengths? [3, 4, 5]

MOST IMPORTANT QUESTIONS

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. The number of tangents that can be drawn to a circle from a point inside it is 1M
(a) 0 (b) 1 (c) 2 (d) infinitely many
Ans. (a) 0

Q2. From a point 13 cm from the centre of a circle of radius 5 cm, the length of the tangent is 1M Board fav
(a) 8 cm (b) 10 cm (c) 12 cm (d) 18 cm
Ans. (c) $\sqrt{169 - 25} = 12$ cm

Q3. PA and PB are tangents to a circle with centre O. If $\angle APB = 50°$, then $\angle AOB =$ 1M
(a) $50°$ (b) $90°$ (c) $100°$ (d) $130°$
Ans. (d) $180° - 50° = 130°$

Q4. The distance between two parallel tangents of a circle of radius 4.5 cm is 1M
(a) 4.5 cm (b) 6 cm (c) 9 cm (d) 13.5 cm
Ans. (c) diameter $= 9$ cm

Q5. PA, PB are tangents from P to a circle with centre O and $\angle APB = 80°$. Then $\angle POA =$ 1M
(a) $40°$ (b) $50°$ (c) $60°$ (d) $80°$
Ans. (b) $\angle APO = 40°$, $\angle OAP = 90°$, so $\angle POA = 50°$

Q6. Quadrilateral ABCD circumscribes a circle. AB = 9 cm, BC = 8 cm, CD = 7 cm. Then AD = 1M
(a) 6 cm (b) 8 cm (c) 10 cm (d) 24 cm
Ans. (b) $AD = AB + CD - BC = 9 + 7 - 8 = 8$ cm

Q7. **A:** Tangents drawn at the ends of a diameter of a circle are parallel. **R:** The tangent at any point of a circle is perpendicular to the radius through the point of contact. 1M
(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true
Ans. (a) Both tangents make 90 degrees with the same diameter, so they are parallel; R explains A.

Q8. **A:** Two tangents can be drawn to a circle from a point inside it. **R:** Tangents drawn from an external point are equal in length. 1M
(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true
(contd.) (d) No tangent from an inside point: A false; R true.

Very short (2 marks)

Q9. Prove that the tangents drawn at the ends of a diameter of a circle are parallel. 2M Board fav
Ans. Let AB be a diameter, tangents l at A and m at B. $OA \perp l$ and $OB \perp m$, so both make $90°$ with line AB. Alternate angles are equal ($90°$ each), so $l \parallel m$.

Q10. Two concentric circles have radii 5 cm and 3 cm. Find the length of the chord of the larger circle that touches the smaller circle. 2M
Ans. Half chord $= \sqrt{25 - 9} = 4$ cm, so chord $= 8$ cm.

Q11. The length of the tangent from a point Q, 26 cm from the centre, is 24 cm. Find the radius. 2M
Ans. $r = \sqrt{26^2 - 24^2} = \sqrt{676 - 576} = 10$ cm.

Q12. The incircle of $\triangle ABC$ touches AB, BC, CA at F, D, E. If $AB = 12$ cm, $BC = 8$ cm and $CA = 10$ cm, find AF, BD and CE. 2M
Ans. $x + y = 12, y + z = 8, z + x = 10 \Rightarrow x + y + z = 15$. $AF = x = 7$ cm, $BD = y = 5$ cm, $CE = z = 3$ cm.

Q13. TA and TB are tangents to a circle with centre O. If $\angle OAB = 25°$, find $\angle ATB$. 2M
Ans. $\angle OAT = 90° \Rightarrow \angle TAB = 65°$. $TA = TB \Rightarrow \angle TBA = 65°$. $\angle ATB = 180° - 130° = 50°$.

Q14. Prove that the perpendicular at the point of contact to the tangent to a circle passes through the centre. 2M
Ans. Let tangent touch at P. The radius OP is $\perp$ tangent (Theorem 1). Through P only one line can be perpendicular to the tangent, so the perpendicular at P is the line OP, which passes through O.

Short answer (3 marks)

Q15. Prove that a parallelogram circumscribing a circle is a rhombus. 3M Board fav
Ans. For circumscribed ABCD: $AB + CD = AD + BC$. In a parallelogram $AB = CD$, $AD = BC$, so $2AB = 2BC \Rightarrow AB = BC$. All sides equal: rhombus.

Q16. Prove that the angle between the two tangents drawn from an external point to a circle is supplementary to the angle subtended at the centre by the segment joining the points of contact. 3M
Ans. PA, PB tangents, O centre. $\angle OAP = \angle OBP = 90°$. In quadrilateral OAPB, angle sum $= 360°$: $\angle APB + \angle AOB = 360° - 180° = 180°$. Hence supplementary.

Quick fact Length of tangent from $(d = 10, r = 6)$ is $\sqrt{100 - 36} = 8$.

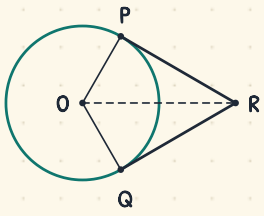


Fig: Tangents RP and RQ with angle PRQ = 120 degrees

Q17. Two tangents RP and RQ are drawn from an external point R to a circle with centre O. If $\angle PRQ = 120°$, prove that $OR = PR + RQ$. 3M

> **Ans.** OR bisects $\angle PRQ$, so $\angle ORP = 60°$. In right $\triangle OPR$ ($\angle OPR = 90°$): $\cos 60° = \frac{PR}{OR} \Rightarrow OR = 2PR$. Since $PR = RQ$, $OR = PR + RQ$.

Q18. From P, two tangents PA and PB are drawn to a circle of radius 3 cm with centre O, where OP = 5 cm. Find the length of the chord of contact AB. 3M

> **Ans.** $PA = \sqrt{25 - 9} = 4$ cm. OP $\perp$ AB at M and AM is the altitude of right $\triangle OAP$: $AM = \frac{OA \times PA}{OP} = \frac{3 \times 4}{5} = 2.4$ cm. $AB = 2AM = 4.8$ cm.

Q19. A circle touches all four sides of quadrilateral ABCD with $AB = 6$ cm, $BC = x + 3$, $CD = 8$ cm and $AD = 2x$ cm. Find x. 3M

> **Ans.** $AB + CD = AD + BC \Rightarrow 14 = 2x + x + 3 \Rightarrow 3x = 11 \Rightarrow x = \frac{11}{3}$ cm.

Q20. A circle is inscribed in $\triangle ABC$ with $AB = 10$ cm, $BC = 8$ cm and $AC = 12$ cm, touching AB, BC, CA at F, D, E. Find AF, BD and CE. 3M

> **Ans.** $x + y = 10, y + z = 8, z + x = 12 \Rightarrow x + y + z = 15$. $AF = 15 - 8 = 7$ cm, $BD = 15 - 12 = 3$ cm, $CE = 15 - 10 = 5$ cm.

Q21. A quadrilateral ABCD is drawn to circumscribe a circle. Prove that AB + CD = AD + BC. 3M Board fav

> **Ans.** Let AB, BC, CD, DA touch the circle at Q, R, S, P. Equal tangents: $AQ = AP, BQ = BR, CS = CR, DS = DP$. Add: $(AQ + BQ) + (CS + DS) = (AP + DP) + (BR + CR)$, i.e. $AB + CD = AD + BC$.

Long answer (5 marks)

Q22. Prove that the lengths of tangents drawn from an external point to a circle are equal. Hence, if a circle touches side BC of triangle ABC at P and AB, AC produced at Q and R, show that AQ is half the perimeter of triangle ABC. 5M Board fav

> **Ans.** Proof as in Theorem 2 (RHS congruence). Then $BQ = BP, CR = CP, AQ = AR$. Perimeter $= AB + BC + CA = AB + BP + PC + CA = AB + BQ + CR + CA = AQ + AR = 2AQ$. So $AQ = \frac{1}{2}$ (perimeter).

Q23. XY and X'Y' are two parallel tangents to a circle with centre O, and another tangent AB with point of contact C meets XY at A and X'Y' at B. Prove that $\angle AOB = 90°$. 5M

> **Ans.** Let XY touch at P and X'Y' at Q. Tangents from A: $\triangle OPA \cong \triangle OCA$ (RHS) so $\angle OAP = \angle OAC$, i.e. $\angle OAB = \frac{1}{2}\angle PAB$. Similarly $\angle OBA = \frac{1}{2}\angle QBA$. $XY \parallel X'Y'$ so $\angle PAB + \angle QBA = 180°$ (co-interior). Hence $\angle OAB + \angle OBA = 90°$, and in $\triangle AOB$, $\angle AOB = 180° - 90° = 90°$.

Q24. PQ is a chord of length 24 cm of a circle of radius 13 cm. The tangents at P and Q meet at T. Find the length TP. 5M

> **Ans.** OT $\perp$ PQ at R, $PR = 12, OR = \sqrt{169 - 144} = 5$. Let $TR = y$. $TP^2 = y^2 + 144$ and $TP^2 = OT^2 - OP^2 = (y + 5)^2 - 169$. So $y^2 + 144 = y^2 + 10y - 144 \Rightarrow y = 28.8$. $TP^2 = 829.44 + 144 = 973.44 \Rightarrow TP = 31.2$ cm.

Case based (4 marks)

Q25. A circular park has radius 10 m. Riya stands at point P, 26 m from the centre O, and walks along a straight path PA that just touches the park boundary at A. Another straight path PB touches it at B. (i) Find PA. (ii) Find the total length of the two paths PA + PB. (iii) For another point Q, the two tangent paths make an angle of $60°$. Find the angle they subtend at the centre. 4M

> **Ans.** (i) $PA = \sqrt{26^2 - 10^2} = \sqrt{576} = 24$ m. (ii) $PB = PA$, total $= 48$ m. (iii) $180° - 60° = 120°$.

Last minute: 8 one-liners

1. Tangent touches at exactly one point.
2. Inside 0, on 1, outside 2 tangents.
3. Tangent $\perp$ radius at the point of contact.
4. Tangent length $= \sqrt{d^2 - r^2}$.
5. Tangents from an external point are equal (RHS).
6. $\angle APB + \angle AOB = 180°$.
7. Circumscribed quadrilateral: $AB + CD = AD + BC$.
8. Concentric circles: chord $= 2\sqrt{R^2 - r^2}$, bisected at contact.

Apne notes / galtiyan yahan likho

Most asked: sector + arc length, segment (60, 90, 120), clock

AREAS RELATED TO CIRCLES

1. Circle Basics (recap)

Quantity	Formula
Circumference	$2\pi r = \pi d$
Area of circle	πr^2
Area of semicircle	$\frac{\pi r^2}{2}$
Perimeter of semicircle	$\pi r + 2r$
Area of quadrant	$\frac{\pi r^2}{4}$

✓ π is irrational; take $\pi = \frac{22}{7}$ (best when r is a multiple of 7) or 3.14, as the question says.

✓ **Distance covered in one revolution** of a wheel = its circumference. Revolutions $= \frac{\text{distance}}{2\pi r}$.

✓ Always change diameter to radius first: $r = \frac{d}{2}$.

↳ Units: length in cm/m, area in cm^2/m^2. Answer mein unit zaroor likho.

2. Sector and Arc

Definition

Sector: the region enclosed by two radii and the arc between them (pizza slice). The smaller one is the **minor sector**, the rest is the **major sector**.

Arc: a piece of the circumference. **Angle of the sector** θ = angle the arc makes at the centre.

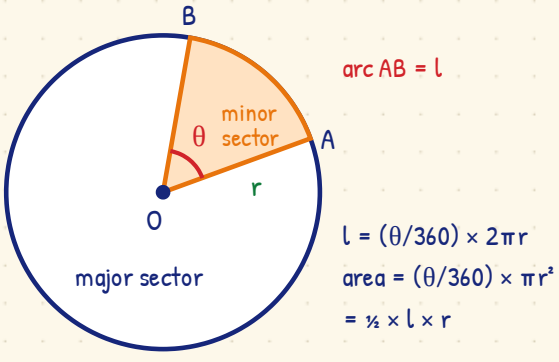


Fig 11.1: Sector OAB with angle θ; the rest is the major sector

Why the formula works: the whole circle is 360° with area πr^2. A sector of angle θ is the fraction $\frac{\theta}{360}$ of the circle (unitary method).

$$\text{Area of sector} = \frac{\theta}{360^\circ} \times \pi r^2$$

$$\text{Length of arc } l = \frac{\theta}{360^\circ} \times 2\pi r$$

→ Area of sector $= \frac{1}{2} l\, r$ (like a triangle with base l, height r).

→ Major sector area $= \pi r^2 -$ minor sector area; its angle is $360^\circ - \theta$.

→ Perimeter of sector $= l + 2r$.

Yaad karo: "θ by 360 = circle ka hissa"

Sector ho ya arc, bas poore circle wala formula likho aur aage $\frac{\theta}{360}$ laga do. Area ke liye πr^2, length ke liye $2\pi r$. $90^\circ = \frac{1}{4}$, $60^\circ = \frac{1}{6}$, $120^\circ = \frac{1}{3}$, $30^\circ = \frac{1}{12}$.

Solved Example 1

Find the length of the arc and the area of the sector of a circle of radius 21 cm with central angle 60°. Also find the area of the major sector. ($\pi = \frac{22}{7}$)

1. $l = \frac{60}{360} \times 2 \times \frac{22}{7} \times 21 = \frac{1}{6} \times 132 = 22$ cm.
2. Sector $= \frac{60}{360} \times \frac{22}{7} \times 21 \times 21 = \frac{1}{6} \times 1386 = 231\ \text{cm}^2$.
3. Major sector $= 1386 - 231 = 1155\ \text{cm}^2$.

✓ $l = 22$ cm; sector $= 231\ \text{cm}^2$; major sector $= 1155\ \text{cm}^2$

Solved Example 2

Find the perimeter of a sector of radius 10.5 cm and angle 60°. ($\pi = \frac{22}{7}$)

1. $l = \frac{60}{360} \times 2 \times \frac{22}{7} \times 10.5 = \frac{1}{6} \times 66 = 11$ cm.
2. Perimeter $= l + 2r = 11 + 21 = 32$ cm.

✓ 32 cm

3. Clock, Wiper, Umbrella, Lighthouse

Situation	Angle to use
Minute hand, t minutes	6° per minute ($\frac{360}{60}$)
Hour hand, t hours	30° per hour
Umbrella with n equal ribs	$\frac{360^\circ}{n}$ between two ribs
Wiper / lighthouse beam	angle swept (given)

Quick fact Semicircle = sector of 180°; quadrant = sector of 90°.

Solved Example 3

The minute hand of a clock is 14 cm long. Find the area swept by it in 5 minutes. ($\pi = \frac{22}{7}$)

1. Angle in 5 min $= 5 \times 6^\circ = 30^\circ$.
2. Area $= \frac{30}{360} \times \frac{22}{7} \times 14 \times 14 = \frac{1}{12} \times 616 = \frac{154}{3}$ cm^2.

✔ $\frac{154}{3} \approx 51.33$ cm^2

Solved Example 4

A car has two wipers which do not overlap. Each blade is 21 cm long and sweeps an angle of 120°. Find the total area cleaned at each sweep. ($\pi = \frac{22}{7}$)

1. One wiper $= \frac{120}{360} \times \frac{22}{7} \times 21 \times 21 = \frac{1}{3} \times 1386 = 462$ cm^2.
2. Two wipers $= 2 \times 462 = 924$ cm^2.

✔ 924 cm^2

Yaad karo: clock ki speed

Minute hand: **60 min mein 360**, toh **1 min** = 6°. Hour hand: **12 ghante mein 360**, toh 1 ghanta $= 30^\circ$ (1 min $= \frac{1}{2}^\circ$). Minute hand ki length = radius.

Shivam Bhaiya says

Radius 7 ka multiple ho toh $\pi = \frac{22}{7}$ lo, cancel ho jayega. Calculation ke time pehle cancel karo, phir multiply. Time bachega aur galti nahi hogi.

4. Segment of a Circle

Definition

Segment: the region between a chord and its arc. The smaller is the **minor segment**, the larger the **major segment**.

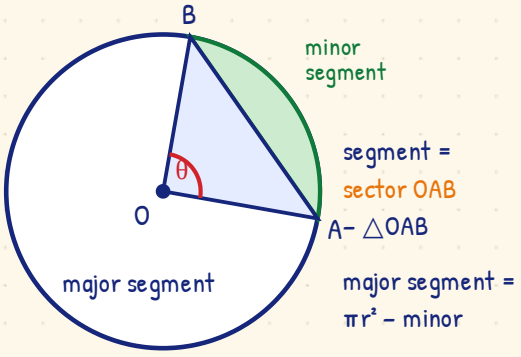


Fig 11.2: Chord AB cuts the circle into a minor (green) and a major segment

Minor segment = sector $OAB - \triangle OAB$

Major segment = πr^2 − minor segment

Area of $\triangle OAB$ ($OA = OB = r$)

$\angle AOB$	Triangle	Area
60°	equilateral, side r	$\frac{\sqrt{3}}{4}r^2$
90°	right isosceles	$\frac{1}{2}r^2$
120°	isosceles, base $\sqrt{3}r$	$\frac{\sqrt{3}}{4}r^2$

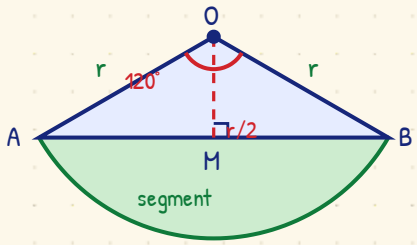


Fig 11.3: 120° case: OM = r/2, AB = √3 r, area of △OAB = (√3/4) r²

For 120°: draw $OM \perp AB$. It bisects $\angle AOB$, so $\angle AOM = 60^\circ$. In right $\triangle OMA$: $OM = r\cos 60^\circ = \frac{r}{2}$ and $AM = r\sin 60^\circ = \frac{\sqrt{3}}{2}r$, so $AB = \sqrt{3}\,r$. Area $= \frac{1}{2} \times \sqrt{3}r \times \frac{r}{2} = \frac{\sqrt{3}}{4}r^2$.

Yaad karo: 60 aur 120 ka triangle same area

60° aur 120° dono mein $\triangle OAB = \frac{\sqrt{3}}{4}r^2$, aur 90° mein $\frac{r^2}{2}$. Segment = **pizza slice** − **triangle**. Bas itna yaad rakho.

Solved Example 5

A chord of a circle of radius 20 cm subtends a right angle at the centre. Find the areas of the minor and major segments. ($\pi = 3.14$)

1. Sector $= \frac{90}{360} \times 3.14 \times 400 = 314$ cm^2.
2. $\triangle OAB = \frac{1}{2} \times 20 \times 20 = 200$ cm^2.
3. Minor segment $= 314 - 200 = 114$ cm^2.
4. Major segment $= 3.14 \times 400 - 114 = 1256 - 114 = 1142$ cm^2.

✔ **Minor** $= 114$ cm^2, **major** $= 1142$ cm^2

Quick fact $\pi \times 7^2 = 154$, $\pi \times 14^2 = 616$, $\pi \times 21^2 = 1386$ (with $\pi = \frac{22}{7}$).

Solved Example 6

A chord of a circle of radius 12 cm subtends an angle of 60° at the centre. Find the area of the minor segment. ($\pi = 3.14$, $\sqrt{3} = 1.73$)

1. Sector $= \frac{60}{360} \times 3.14 \times 144 = 75.36$ cm^2.
2. $\triangle OAB$ (equilateral) $= \frac{1.73}{4} \times 144 = 62.28$ cm^2.
3. Segment $= 75.36 - 62.28 = 13.08$ cm^2.

✔ 13.08 cm^2

Solved Example 7 (board fav)

A chord of a circle of radius 21 cm subtends an angle of 120° at the centre. Find the area of the minor segment. ($\pi = \frac{22}{7}$, $\sqrt{3} = 1.73$)

1. Sector $= \frac{120}{360} \times \frac{22}{7} \times 441 = 462$ cm^2.
2. $\triangle OAB = \frac{\sqrt{3}}{4} r^2 = \frac{1.73 \times 441}{4} = 190.7325$ cm^2.
3. Segment $= 462 - 190.7325 = 271.2675$ cm^2.

✔ ≈ 271.27 cm^2

Meme samjho

Segment ka area? Sector minus triangle. **Bas, presentation khatam.**

Asli baat: Minor segment $= \frac{\theta}{360}\pi r^2 -$ area of $\triangle OAB$. Major segment $= \pi r^2 -$ minor segment.

Formula Box

- Circumference $= 2\pi r$; area $= \pi r^2$
- Arc length $l = \frac{\theta}{360} \times 2\pi r$
- Sector area $= \frac{\theta}{360} \times \pi r^2 = \frac{1}{2} lr$
- Perimeter of sector $= l + 2r$
- Major sector $= \pi r^2 -$ minor sector
- Minor segment $=$ sector $- \triangle OAB$
- $\triangle OAB$: $\frac{\sqrt{3}}{4} r^2$ ($60^\circ, 120^\circ$); $\frac{1}{2} r^2$ (90°)
- Major segment $= \pi r^2 -$ minor segment
- Minute hand: 6°/min; revolutions $= \frac{\text{distance}}{2\pi r}$

Quick fact Circumference with $r = 7$: 44; $r = 14$: 88; $r = 21$: 132.

Common mistakes

- Using the diameter in place of the radius.
- Giving the sector area when the **segment** is asked (forgot to subtract the triangle).
- Perimeter of a sector written as only l (forgot to add $2r$).
- Writing area in cm (should be cm^2).
- Taking the 120° triangle as equilateral with side r: its base is $\sqrt{3}\,r$ (area formula is still $\frac{\sqrt{3}}{4} r^2$).

Shivam Bhaiya says

Segment ke sawaal mein 3 line likho: sector, triangle, minus. Har line ke marks hain. Value of $\sqrt{3}$ jo di ho wahi lagao, apni marzi se 1.732 mat lo.

5. Reverse Questions and Wheels

Sometimes the area or arc is given and r or θ is asked. Put the values in the same formula and solve.

Solved Example 8

The area of a sector of angle 60° is 231 cm^2. Find the radius of the circle. ($\pi = \frac{22}{7}$)

1. $\frac{60}{360} \times \frac{22}{7} \times r^2 = 231 \Rightarrow \frac{11}{21} r^2 = 231$.
2. $r^2 = 231 \times \frac{21}{11} = 441 \Rightarrow r = 21$ cm.

✔ $r = 21$ cm

Solved Example 9

The wheels of a cycle have a diameter of 70 cm. How many revolutions will each wheel make to cover 1.1 km? ($\pi = \frac{22}{7}$)

1. Circumference $= \pi d = \frac{22}{7} \times 70 = 220$ cm.
2. 1.1 km $= 110000$ cm. Revolutions $= \frac{110000}{220} = 500$.

✔ 500 revolutions

Yaad karo: 7 ka pahada

$\pi = \frac{22}{7}$ ke saath: $r = 7 \to$ area 154, $r = 14 \to 616$, $r = 21 \to 1386$, $r = 35 \to 3850$. Ye 4 values ratt lo, aadhe sawaal mental maths ban jaate hain.

Quick fact Sector area $= \frac{1}{2} \times$ arc $\times$ radius.

Quick fact 1 km $= 1000$ m $= 100000$ cm.

Board scan: what comes from this chapter

- → **1M:** arc length, quadrant area, minute-hand angle, AR on sector area.
- → **2M:** sector area / perimeter, angle from arc, wheel revolutions.
- → **3M:** minor + major segment ($60°$, $90°$), umbrella, horse grazing, lighthouse.
- → **5M / case:** $120°$ segment, clock hand area + distance, park or brooch design.

Last-minute revision

- → Sector $= \frac{\theta}{360}\pi r^2$; arc $= \frac{\theta}{360}2\pi r$; perimeter $= l + 2r$.
- → Segment $=$ sector $-$ triangle; triangle $\frac{\sqrt{3}}{4}r^2$ for $60°/120°$, $\frac{1}{2}r^2$ for $90°$.
- → Major $=$ whole circle $-$ minor.
- → Minute hand $6°$/min; ribs: $\frac{360}{n}$.
- → Radius, not diameter; units cm^2 for area.

Solved Example 10

Find the perimeter of the minor segment cut off by a chord that subtends $60°$ at the centre of a circle of radius 21 cm. ($\pi = \frac{22}{7}$)

1. Arc $= \frac{1}{6} \times 2 \times \frac{22}{7} \times 21 = 22$ cm.
2. For $60°$, $\triangle OAB$ is equilateral, so chord $AB = r = 21$ cm.
3. Perimeter of segment $=$ arc $+$ chord $= 22 + 21 = 43$ cm.

✔ 43 cm

Grazing problems: an animal tied at a corner of a square field with a rope of length r grazes a **quadrant** ($90°$ sector) of radius r, if r is less than the side.

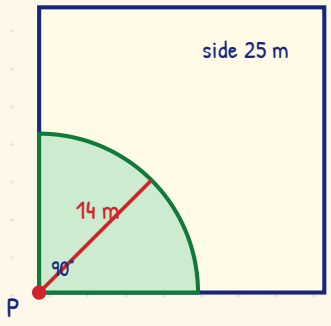
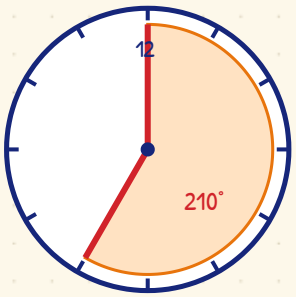


Fig 11.4: Horse tied at corner P grazes a quadrant

Quick fact Hour hand moves $\frac{1}{2}°$ every minute.



8:00 to 8:35
35 min × 6°
= 210°
r = 10.5 cm

Fig 11.5: Minute hand from 8:00 to 8:35 sweeps a $210°$ sector

MOST IMPORTANT QUESTIONS

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. If the circumference and the area of a circle are numerically equal, the radius is: 1M Board fav
(a) 1 (b) 2 (c) 4 (d) π
Ans. (b) $2\pi r = \pi r^2 \Rightarrow r = 2$ units.

Q2. The length of the arc of a sector of angle $90°$ in a circle of radius 14 cm is: ($\pi = \frac{22}{7}$) 1M
(a) 22 cm (b) 44 cm (c) 11 cm (d) 154 cm
Ans. (a) $\frac{1}{4} \times 2 \times \frac{22}{7} \times 14 = 22$ cm.

Q3. The area of a quadrant of a circle whose circumference is 22 cm is: 1M
(a) $\frac{77}{2}$ cm^2 (b) $\frac{77}{4}$ cm^2 (c) $\frac{77}{8}$ cm^2 (d) 154 cm^2
Ans. (c) $2\pi r = 22 \Rightarrow r = 3.5$ cm. Quadrant $= \frac{1}{4} \times \frac{22}{7} \times 3.5^2 = \frac{77}{8}$ cm^2.

Q4. The angle described by the minute hand of a clock in 20 minutes is: 1M
(a) $60°$ (b) $90°$ (c) $100°$ (d) $120°$
Ans. (d) $20 \times 6° = 120°$.

Q5. The area of a sector is one-sixth of the area of the circle. Its central angle is: 1M
(a) $30°$ (b) $60°$ (c) $90°$ (d) $120°$
Ans. (b) $\frac{\theta}{360} = \frac{1}{6} \Rightarrow \theta = 60°$.

Q6. The perimeter of a semicircular protractor of diameter 14 cm is: ($\pi = \frac{22}{7}$) 1M
(a) 22 cm (b) 29 cm (c) 36 cm (d) 44 cm
Ans. (c) $\pi r + 2r = 22 + 14 = 36$ cm.

Q7. **A:** The area of a sector of angle $90°$ of a circle of radius 7 cm is 38.5 cm^2. **R:** Area of a sector of angle $\theta = \frac{\theta}{360} \times \pi r^2$. 1M
(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true
Ans. (a) Both true; R explains A: $\frac{1}{4} \times \frac{22}{7} \times 49 = 38.5$.

Q8. **A:** If the radius of a circle is doubled, the area of a sector with the same angle becomes 4 times. **R:** Area of a sector is proportional to r^2 for a fixed angle. 1M
(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true

(contd.) (a) Both true; R explains A.

Q9. The area swept by the minute hand of length 7 cm in 15 minutes is: ($\pi = \frac{22}{7}$) 1M

(a) 38.5 cm^2 (b) 77 cm^2 (c) 11 cm^2 (d) 154 cm^2

Ans. (a) $15 \times 6^\circ = 90^\circ$; $\frac{1}{4} \times 154 = 38.5$ cm^2.

Very short (2 marks)

Q10. Find the area of a sector of a circle of radius 6 cm with angle 60°. ($\pi = 3.14$) 2M

Ans. $\frac{60}{360} \times 3.14 \times 36 = 18.84$ cm^2.

Q11. Find the perimeter of a quadrant of a circle of radius 7 cm. ($\pi = \frac{22}{7}$) 2M

Ans. Arc $= \frac{1}{4} \times 2 \times \frac{22}{7} \times 7 = 11$ cm. Perimeter $= 11 + 2 \times 7 = 25$ cm.

Q12. The arc of a sector of a circle of radius 21 cm is 44 cm long. Find the angle of the sector. 2M Board fav

Ans. $\frac{\theta}{360} \times 2 \times \frac{22}{7} \times 21 = 44 \Rightarrow \frac{\theta}{360} \times 132 = 44 \Rightarrow \theta = 120^\circ$.

Q13. A wheel has radius 35 cm. How many complete revolutions does it make in covering 11 km? ($\pi = \frac{22}{7}$) 2M

Ans. Circumference $= 2 \times \frac{22}{7} \times 35 = 220$ cm. 11 km $= 1100000$ cm. Revolutions $= \frac{1100000}{220} = 5000$.

Q14. Find the area of a sector whose arc length is 22 cm and radius is 10 cm. 2M

Ans. Area $= \frac{1}{2}lr = \frac{1}{2} \times 22 \times 10 = 110$ cm^2.

Short answer (3 marks)

Q15. A chord of a circle of radius 20 cm subtends an angle of 60° at the centre. Find the areas of the minor and major segments. ($\pi = 3.14$, $\sqrt{3} = 1.73$) 3M Board fav

Ans. Sector $= \frac{1}{6} \times 3.14 \times 400 = 209.33$ cm^2. $\triangle OAB = \frac{1.73}{4} \times 400 = 173$ cm^2. Minor $= 209.33 - 173 = 36.33$ cm^2. Major $= 1256 - 36.33 = 1219.67$ cm^2.

Q16. An umbrella has 8 ribs which are equally spaced. Taking it as a flat circle of radius 42 cm, find the area between two consecutive ribs. ($\pi = \frac{22}{7}$) 3M

Ans. Angle $= \frac{360}{8} = 45^\circ$. Area $= \frac{45}{360} \times \frac{22}{7} \times 42 \times 42 = \frac{1}{8} \times 5544 = 693$ cm^2.

Q17. A horse is tied to a peg at one corner of a square grass field of side 25 m by a 14 m rope (Fig 11.4). Find (i) the area of the field it can graze (ii) the increase in grazing area if the rope were 21 m long. ($\pi = \frac{22}{7}$) 3M Board fav

Ans. Quadrant (90°). (i) $\frac{1}{4} \times \frac{22}{7} \times 196 = 154$ m^2. (ii) $\frac{1}{4} \times \frac{22}{7} \times 441 = 346.5$ m^2; increase $= 346.5 - 154 = 192.5$ m^2.

Q18. A lighthouse spreads a red light over a sector of angle 60° to a distance of 21 km. Find the area of the sea over which ships are warned. ($\pi = \frac{22}{7}$) 3M

(contd.) $\frac{60}{360} \times \frac{22}{7} \times 21 \times 21 = \frac{1}{6} \times 1386 = 231$ km^2.

Q19. The area of a sector of a circle of radius 14 cm is 154 cm^2. Find the central angle and the length of the arc. ($\pi = \frac{22}{7}$) 3M

Ans. $\frac{\theta}{360} \times 616 = 154 \Rightarrow \theta = 90^\circ$. $l = \frac{1}{4} \times 2 \times \frac{22}{7} \times 14 = 22$ cm. (Check: $\frac{1}{2}lr = \frac{1}{2} \times 22 \times 14 = 154$.)

Q20. Find the area of the major sector of a circle of radius 7 cm whose minor sector has angle 40°. ($\pi = \frac{22}{7}$) 3M

Ans. Major angle $= 320^\circ$. Area $= \frac{320}{360} \times \frac{22}{7} \times 49 = \frac{8}{9} \times 154 = \frac{1232}{9} \approx 136.89$ cm^2.

Long answer (5 marks)

Q21. A chord of a circle of radius 14 cm subtends an angle of 120° at the centre. Find the areas of the minor and major segments. ($\pi = \frac{22}{7}$, $\sqrt{3} = 1.73$) 5M Board fav

Ans. Sector $= \frac{1}{3} \times \frac{22}{7} \times 196 = \frac{616}{3} = 205.33$ cm^2. $\triangle OAB = \frac{\sqrt{3}}{4}r^2 = \frac{1.73 \times 196}{4} = 84.77$ cm^2. Minor $= 205.33 - 84.77 = 120.56$ cm^2. Circle $= 616$ cm^2; major $= 616 - 120.56 = 495.44$ cm^2.

Q22. The minute hand of a clock is 10.5 cm long. Find the area swept and the distance moved by its tip from 8:00 am to 8:35 am. ($\pi = \frac{22}{7}$) 5M

Ans. 35 min $\Rightarrow \theta = 35 \times 6^\circ = 210^\circ$. Area $= \frac{210}{360} \times \frac{22}{7} \times 10.5 \times 10.5 = \frac{7}{12} \times 346.5 = 202.125$ cm^2. Arc $= \frac{7}{12} \times 2 \times \frac{22}{7} \times 10.5 = \frac{7}{12} \times 66 = 38.5$ cm.

Q23. A chord of a circle of radius 14 cm subtends a right angle at the centre. Find (i) the area of the minor segment (ii) the area of the major sector (iii) the area of the major segment. ($\pi = \frac{22}{7}$) 5M

Ans. Sector $= \frac{1}{4} \times 616 = 154$ cm^2; $\triangle OAB = \frac{1}{2} \times 14 \times 14 = 98$ cm^2. (i) Minor segment $= 154 - 98 = 56$ cm^2. (ii) Major sector $= 616 - 154 = 462$ cm^2. (iii) Major segment $= 616 - 56 = 560$ cm^2.

Case based (4 marks)

Q24. A circular park of radius 21 m is divided into 6 equal flower beds by straight paths from the centre to the boundary. (i) Find the angle of each bed at the centre. (1M) (ii) Find the area of one bed. (1M) (iii) A straight fence joins the two ends of one bed's arc. Find the area between this fence and the arc. ($\pi = \frac{22}{7}$, $\sqrt{3} = 1.73$) (2M) 4M

Ans. (i) $\frac{360}{6} = 60^\circ$. (ii) $\frac{1}{6} \times \frac{22}{7} \times 441 = 231$ m^2. (iii) Segment $= 231 - \frac{1.73}{4} \times 441 = 231 - 190.7325 = 40.2675 \approx 40.27$ m^2.

Apne notes / galtiyan yahan likho

Most asked: toy, tent, capsule, cavity + case-based glass

Surface Areas and Volumes

1. The 6 Basic Solids

Every board question is a **combination of any two** of: cuboid, cube, cylinder, cone, sphere, hemisphere. Know the CSA, TSA and volume of each and the chapter is done.

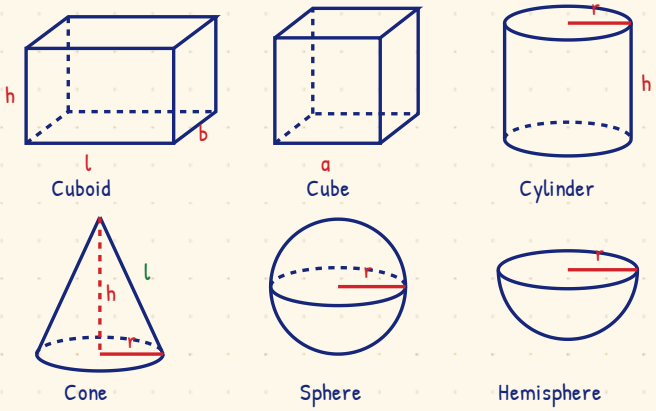


Fig: The 6 basic solids with their measurements

Key terms

CSA / LSA (curved or lateral surface area): area of the side surface only, no top or bottom.

TSA (total surface area): CSA + area of all flat faces (bases).

Volume: space occupied by the solid, in cubic units (cm^3, m^3). **Capacity** = volume a hollow vessel can hold.

Slant height of a cone: $l = \sqrt{r^2 + h^2}$ (Pythagoras on the side).

Solid	CSA	TSA	Volume
Cuboid l, b, h	$2h(l+b)$	$2(lb+bh+hl)$	lbh
Cube a	$4a^2$	$6a^2$	a^3
Cylinder r, h	$2\pi rh$	$2\pi r(r+h)$	$\pi r^2 h$
Cone r, h, l	πrl	$\pi r(l+r)$	$\frac{1}{3}\pi r^2 h$
Sphere r	$4\pi r^2$	$4\pi r^2$	$\frac{4}{3}\pi r^3$
Hemisphere r	$2\pi r^2$	$3\pi r^2$	$\frac{2}{3}\pi r^3$

↳ Use $\pi = \frac{22}{7}$ unless the question says 3.14. Radius 7, 14, 3.5, 21 dikhe toh $\frac{22}{7}$ hi lena.

Where do the formulas come from?

→ **Cylinder CSA:** cut the side open and it becomes a rectangle of length $2\pi r$ (circumference) and breadth h, so CSA $= 2\pi rh$.

→ **Cone CSA:** the side opens into a sector of radius l and arc $2\pi r$: area $= \frac{1}{2} \times 2\pi r \times l = \pi rl$.

Quick fact Hemisphere volume $= \frac{1}{2}$ sphere $= \frac{2}{3}\pi r^3$.

→ **Cone volume:** a cone filled with water 3 times fills a cylinder of the same r, h: $V = \frac{1}{3}\pi r^2 h$.

→ **Hemisphere:** exactly half a sphere, so CSA $= 2\pi r^2$, volume $= \frac{2}{3}\pi r^3$; its flat top adds πr^2.

Cylinder side = rectangle → $2\pi r \times h$ → CSA $= 2\pi rh$

Formula Box

→ Cone slant height: $l = \sqrt{r^2 + h^2}$

→ Cuboid diagonal $= \sqrt{l^2 + b^2 + h^2}$; cube diagonal $= \sqrt{3}\,a$

→ Hollow cylinder volume $= \pi(R^2 - r^2)h$

→ Combined SA = sum of **visible** surfaces only

→ Combined volume = sum of volumes (cavity: subtract)

→ Cube + hemisphere on top: SA $= 6a^2 + \pi r^2$

→ Capsule (cylinder + 2 hemispheres, total length L): SA $= 2\pi rL$

→ Cylinder with cone scooped out: SA $= 2\pi rh + \pi r^2 + \pi rl$

→ $1\ m^3 = 1000$ L; 1 L $= 1000\ cm^3$

Yaad karo: "C se Curved, T se Total"

TSA = CSA + jitne flat circles/faces. Hemisphere: $2\pi r^2$ (dome) + πr^2 (base) = $3\pi r^2$. Cylinder: $2\pi rh$ + 2 circles = $2\pi r(r+h)$.

2. Surface Area of Combined Solids

A **combined solid** is made by joining two basic solids, e.g. a cone on a hemisphere (toy), a cone on a cylinder (tent), a cylinder with hemispherical ends (capsule).

Golden rule

SA of combined solid = area that you can touch from outside. The faces where two solids are **glued** are hidden: never add TSA of each part.

1. Draw the solid and name each part.
2. For each part write only the visible surface (usually CSA, sometimes one base).
3. Add. If a part is **scooped out**, its curved surface is added and the circle removed is subtracted.

Quick fact Sphere ka CSA = TSA $= 4\pi r^2$ (koi flat face nahi).

Board question types at a glance

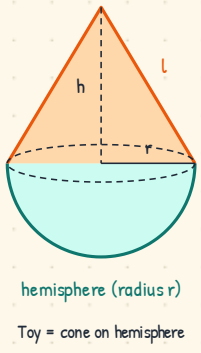


Fig: Toy = cone on hemisphere. SA $= \pi rl + 2\pi r^2$

Object	Made of	Surface area
Toy / top	cone + hemisphere	$\pi rl + 2\pi r^2$
Tent	cone + cylinder	$\pi rl + 2\pi rh$
Capsule, gulab jamun	cylinder + 2 hemispheres	$2\pi rh + 4\pi r^2$
Ice-cream cone	cone + hemisphere	(volume asked)
Vessel / bowl	hemisphere + cylinder	$2\pi rh + 2\pi r^2$
Cube with dome	cube + hemisphere	$6a^2 + \pi r^2$
Cavity / pen stand	solid − cone	add cavity CSA

Solved Example 1 (toy)

A toy is a cone of radius 6 cm mounted on a hemisphere of the same radius. Height of the cone is 8 cm. Find the total surface area and the volume of the toy. ($\pi = 3.14$)

1. $l = \sqrt{6^2 + 8^2} = \sqrt{100} = 10$ cm
2. SA $= \pi rl + 2\pi r^2 = \pi(60 + 72) = 132\pi$
3. $= 132 \times 3.14 = 414.48$ cm^2
4. Volume $= \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3 = \pi(96 + 144) = 240\pi$

✔ **SA** $= 414.48$ cm^2, **Volume** $= 753.6$ cm^3

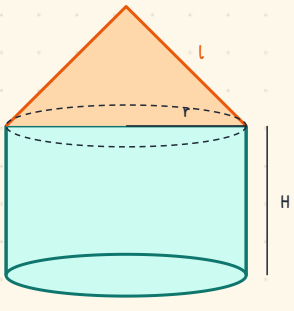


Fig: Tent = cone on cylinder (no floor)

Quick fact 1 L $= 1000$ cm^3, so 1 m^3 $= 1000$ L.

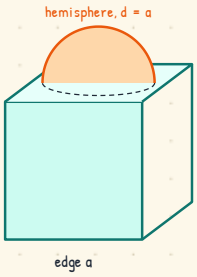


Fig: Cube with a hemispherical dome, $d = a$

Solved Example 2 (tent canvas)

A tent is a cylinder of diameter 14 m and height 4 m, surmounted by a cone of the same base with slant height 9 m. Find the canvas used (no floor) and its cost at Rs 120 per m^2.

1. $r = 7$ m. CSA of cylinder $= 2 \times \frac{22}{7} \times 7 \times 4 = 176$ m^2
2. CSA of cone $= \frac{22}{7} \times 7 \times 9 = 198$ m^2
3. Canvas $= 176 + 198 = 374$ m^2; cost $= 374 \times 120$

✔ **Canvas** $= 374$ m^2, **cost = Rs 44,880**

Solved Example 3 (cube with dome)

A cube of edge 10 cm is surmounted by a hemisphere of the largest possible diameter. Find the surface area of the solid. ($\pi = 3.14$)

1. Largest diameter = edge, so $r = 5$ cm
2. SA $= 6a^2 - \pi r^2 + 2\pi r^2 = 6a^2 + \pi r^2$
3. $= 600 + 3.14 \times 25 = 600 + 78.5$

✔ **SA** $= 678.5$ cm^2

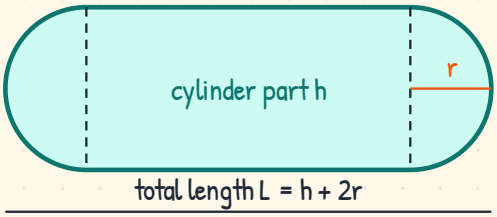


Fig: Capsule, total length $L = h + 2r$

Yaad karo: 10-second shortcuts

Cube + dome $= 6a^2 + \pi r^2$ (ek circle gaya, dome aaya, net $+\pi r^2$).

Capsule $= 2\pi rL$, L = poori length. Kyunki $2\pi r(L - 2r) + 4\pi r^2 = 2\pi rL$.

Quick fact Cube diagonal $= \sqrt{3}\,a$; edge 5 cm: $5\sqrt{3}$ cm.

Meme samjho

TSA + TSA jodne ki galti kabhi nahi hogi agar **pehle diagram bana lo** aur hidden face kaat do.

Asli baat: Toy ka SA $= \pi rl + 2\pi r^2$, not $\pi r(l + r) + 3\pi r^2$. Joined circle dikhta hi nahi.

3. Volume of Combined Solids

Volume has no hidden-face problem: volumes simply add. If a part is hollowed out (cavity), subtract its volume.

Joined

$V = V_1 + V_2$

toy, tent, capsule, ice-cream

Scooped out

$V = V_{\text{solid}} - V_{\text{cavity}}$

pen stand, cavity in block

Yaad karo: "Teen cone = ek cylinder"

Same r and h: cone $= \frac{1}{3}$ cylinder. So cylinder : cone : hemisphere (with $h = r$) volume $= 3 : 1 : 2$.

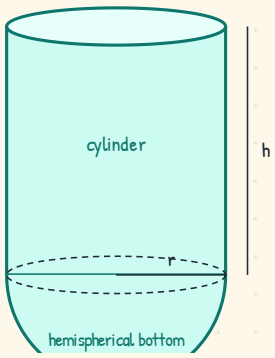


Fig: Vessel = hemispherical bowl + cylinder on top

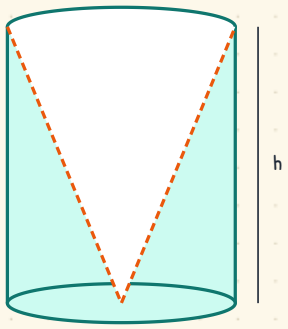
Solved Example 4 (vessel)

A vessel is a hollow hemisphere mounted by a hollow cylinder. Diameter of the hemisphere is 14 cm and total height of the vessel is 13 cm. Find its inner surface area and capacity.

1. $r = 7$ cm, cylinder height $h = 13 - 7 = 6$ cm
2. Inner SA $= 2\pi rh + 2\pi r^2 = 264 + 308 = 572\ \text{cm}^2$
3. Capacity $= \pi r^2 h + \frac{2}{3}\pi r^3 = 924 + \frac{2156}{3}$

✔ SA $= 572\ \text{cm}^2$, capacity $= \frac{4928}{3} \approx 1642.67\ \text{cm}^3$

Quick fact Cone CSA $= \pi rl$ uses slant l; volume uses height h.



conical cavity in a cylinder

Fig: Conical cavity in a solid cylinder

Solved Example 5 (cavity)

From a solid cylinder of height 12 cm and diameter 7 cm, a conical cavity of the same height and diameter is hollowed out. Find the volume and the total surface area of the remaining solid.

1. $r = 3.5$, $h = 12$. Volume left $= \pi r^2 h - \frac{1}{3}\pi r^2 h = \frac{2}{3}\pi r^2 h$
2. $= \frac{2}{3} \times \frac{22}{7} \times 12.25 \times 12 = 308\ \text{cm}^3$
3. $l = \sqrt{3.5^2 + 12^2} = \sqrt{156.25} = 12.5$ cm
4. SA $= 2\pi rh + \pi r^2 + \pi rl = 264 + 38.5 + 137.5$

✔ Volume $= 308\ \text{cm}^3$, SA $= 440\ \text{cm}^2$

Solved Example 6 (pencil shape)

A solid is a cylinder of radius 3.5 cm and length 10 cm with a cone of height 12 cm on one end (same radius). Find its total surface area and volume.

1. Cone: $l = \sqrt{3.5^2 + 12^2} = 12.5$ cm
2. Visible: flat base $\pi r^2 = 38.5$, cylinder CSA $2\pi rh = 220$, cone CSA $\pi rl = 137.5$
3. SA $= 38.5 + 220 + 137.5 = 396\ \text{cm}^2$
4. Volume $= \pi r^2 \left(10 + \frac{12}{3}\right) = 38.5 \times 14$

✔ SA $= 396\ \text{cm}^2$, Volume $= 539\ \text{cm}^3$

Solved Example 7 (hollow pipe)

An iron pipe is 7 m long. Its outer diameter is 10 cm and inner diameter is 8 cm. Find the volume of iron used.

1. $R = 5$ cm, $r = 4$ cm, $h = 700$ cm
2. $V = \pi(R^2 - r^2)h = \frac{22}{7} \times (25 - 16) \times 700$

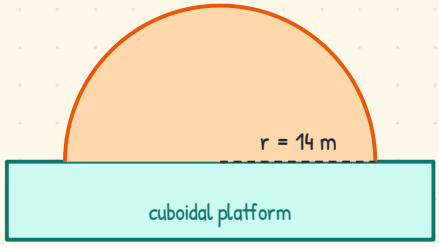
✔ Volume of iron $= 19800\ \text{cm}^3$

Read + draw → Write r, h, l → Visible faces / add volumes → Answer in π, then put value → Unit!

4. More Board-Type Combinations

Same golden rule, different shapes. Board loves these three:

stupa/monument (hemisphere on cuboid), **cube with hemisphere scooped**, and **gilli** (cylinder with cones at both ends).



Stupa: hemisphere on a cuboid

Fig: Monument = hemisphere on a cuboidal platform

Solved Example 8 (monument)

A monument is a hemisphere of radius 14 m standing on a cuboidal platform $30 \times 30 \times 2$ m. Find the area to be painted (all outer surfaces except the bottom) and the cost at Rs 20 per m^2.

1. Platform top visible $= 900 - \pi r^2 = 900 - 616 = 284\ \text{m}^2$
2. 4 side faces $= 4 \times 30 \times 2 = 240\ \text{m}^2$
3. Dome $= 2\pi r^2 = 2 \times 616 = 1232\ \text{m}^2$
4. Total $= 284 + 240 + 1232 = 1756\ \text{m}^2$; cost $= 1756 \times 20$

✔ **Area** $= 1756\ \text{m}^2$, **cost = Rs 35,120**

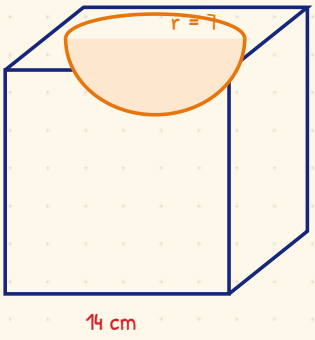


Fig (Ex 9): hemisphere scooped from the top face

Solved Example 9 (scooped cube)

A hemisphere of diameter 14 cm is scooped out from one face of a cube of edge 14 cm. Find the surface area and volume of the remaining solid.

1. $r = 7$. SA $= 6a^2 - \pi r^2 + 2\pi r^2 = 6a^2 + \pi r^2$
2. $= 6 \times 196 + 154 = 1176 + 154 = 1330\ \text{cm}^2$
3. Volume $= a^3 - \frac{2}{3}\pi r^3 = 2744 - \frac{2156}{3}$

✔ **SA** $= 1330\ \text{cm}^2$, **Volume** $= \frac{6076}{3} \approx 2025.33\ \text{cm}^3$

↳Notice: dome on top or bowl scooped inside, SA formula same: $6a^2 + \pi r^2$! Volume alag: dome mein add, scoop mein subtract.

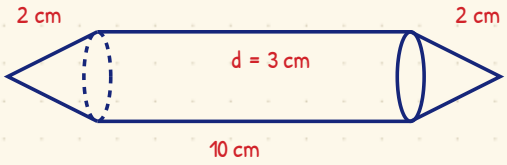


Fig (Ex 10): gilli = cylinder + 2 cones

Solved Example 10 (gilli)

A gilli is a cylinder of length 10 cm and diameter 3 cm with a cone of height 2 cm at each end. Find its volume. ($\pi = 3.14$)

1. $r = 1.5$, $r^2 = 2.25$
2. $V = \pi r^2 h + 2 \times \frac{1}{3}\pi r^2 h_c = \pi r^2 \left(10 + \frac{4}{3}\right)$
3. $= \pi \times 2.25 \times \frac{34}{3} = 25.5\pi$

✔ **Volume** $= 25.5 \times 3.14 = 80.07\ \text{cm}^3$

Meme samjho

Cube pe dome lagao ya cube mein bowl khodo: SA wahi $6a^2 + \pi r^2$. **Perfectly balanced**, jaise har cheez honi chahiye. **Asli baat:** One circle πr^2 goes, curved $2\pi r^2$ comes, both cases. But volume: dome adds $\frac{2}{3}\pi r^3$, scoop subtracts it.

Exam Alert

Melting/recasting (conversion of solids) and frustum are **out of syllabus** now. Board only asks combination of any two of the 6 basic solids. Case-based mein glass, tent, toy sabse common.

5. Units and Conversions

Convert	Rule	Example
m to cm	$\times 100$	1.2 m $= 120$ cm
m^2 to cm^2	$\times 10^4$	3.41 $\text{m}^2 = 34100\ \text{cm}^2$
m^3 to cm^3	$\times 10^6$	1 $\text{m}^3 = 10^6\ \text{cm}^3$
cm^3 to litre	$\div 1000$	2425.5 $\text{cm}^3 = 2.4255$ L
m^3 to litre	$\times 1000$	2 $\text{m}^3 = 2000$ L

✓ All lengths in **one unit** before using any formula (cm with m is the classic trap).

✓ Keep the answer in terms of π till the last step: 93π is easier than multiplying three times.

✓ $\pi = \frac{22}{7}$ when r is a multiple of 7 or 3.5; else 3.14 (or as told).

Yaad karo: "CC rule"

Do solids jude hain? To Curved + Curved (CSA + CSA). Flat base sirf tab jodo jab woh bahar dikh raha ho (jaise pencil ka bottom, tent ka floor nahi).

Solved Example 11 (bird bath)

A bird bath is a solid cylinder of height 1.2 m and radius 35 cm with a hemispherical depression at the top of the same radius. Find its total surface area (excluding the bottom).

1. Same unit: $h = 120$ cm, $r = 35$ cm
2. SA = outer CSA + bowl $= 2\pi rh + 2\pi r^2 = 2\pi r(h + r)$
3. $= 2 \times \frac{22}{7} \times 35 \times 155 = 220 \times 155$

✔ **SA** $= 34100\ \text{cm}^2 = 3.41\ \text{m}^2$

Solved Example 12 (solid in a tub)

A cylindrical tub of radius 5 cm and height 10 cm is full of water. A solid (cone of height 4 cm on a hemisphere, both of radius 3 cm) is put in completely. Find the water left in the tub. ($\pi = 3.14$)

1. Tub $= \pi(25)(10) = 250\pi$
2. Solid $= \frac{1}{3}\pi(9)(4) + \frac{2}{3}\pi(27) = 12\pi + 18\pi = 30\pi$ (this much water spills)
3. Left $= 250\pi - 30\pi = 220\pi = 220 \times 3.14$

✔ **Water left** $= 690.8\ \text{cm}^3$

↳ Solid dooba = utna paani bahar. Volume ka khel, SA ka nahi.

Shivam Bhaiya says

Har question mein pehle 3 cheezein likh lo: **r, h, l**. Diameter diya ho toh turant half karo. Aur answer ki unit check karo: area = cm^2, volume = cm^3. Itne se hi aadhe galti khatam!

Common mistakes

- → Adding TSA of each part (joined faces counted twice).
- → Using diameter as radius; forgetting $l = \sqrt{r^2 + h^2}$ for the cone.
- → Total height given: cone/cylinder height = total − radius of hemisphere.
- → Cavity: volume is subtracted but its curved surface is **added** to SA.
- → Litres: $1\ \text{m}^3 = 1000$ L, not 100 L.

Yaad karo: unit power check

Area mein r ki power **2** (πr^2, πrl), volume mein power **3** ($\pi r^2 h$, $\frac{4}{3}\pi r^3$). Formula ki power galat = formula galat.

Solved Example 13 (painting a bowl)

A hollow hemispherical bowl has inner radius 7 cm and outer radius 10.5 cm. Find the total area to be painted (inside, outside and the rim).

1. Inside $= 2\pi r^2 = 2 \times \frac{22}{7} \times 49 = 308\ \text{cm}^2$
2. Outside $= 2\pi R^2 = 2 \times \frac{22}{7} \times 110.25 = 693\ \text{cm}^2$
3. Rim (ring) $= \pi(R^2 - r^2) = \frac{22}{7} \times 61.25 = 192.5\ \text{cm}^2$

✔ **Total** $= 308 + 693 + 192.5 = 1193.5\ \text{cm}^2$

Solved Example 14 (domed building)

A building is a cylinder of radius 7 m and height 10 m with a hemispherical dome on top. Find the volume of air inside and the outer area to be painted (walls + dome).

1. Volume $= \pi r^2 h + \frac{2}{3}\pi r^3 = 1540 + \frac{2156}{3}$
2. Paint $= 2\pi rh + 2\pi r^2 = 440 + 308$

✔ **Volume** $= \frac{6776}{3} \approx 2258.67\ \text{m}^3$, **area** $= 748\ \text{m}^2$

6. One-Page Recap

Surface area

Only **visible** faces. Joined = CSA + CSA. Scoop: add curved, subtract circle.

Volume

Joined = add. Cavity / depression = subtract. Hollow pipe $= \pi(R^2 - r^2)h$.

If you see...	Do this first
Total height of toy	cone h = total − r
Cone + "slant" not given	$l = \sqrt{r^2 + h^2}$
"Largest" sphere/cone in cube	$d = a, h = a$
Canvas / painting cost	area × rate
Capacity in litres	$\text{cm}^3 \div 1000$
Inner surface of vessel	$2\pi rh + 2\pi r^2$

MOST IMPORTANT QUESTIONS

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. A cone and a cylinder have the same radius and height. Ratio of their volumes is: 1M Board fav

(a) 1 : 2 (b) 1 : 3 (c) 3 : 1 (d) 2 : 3

Ans. (b) 1 : 3, since $\frac{1}{3}\pi r^2 h : \pi r^2 h$.

Q2. The total surface area of a solid hemisphere of radius 7 cm is: 1M

(a) 308 cm^2 (b) 462 cm^2 (c) 616 cm^2 (d) 154 cm^2

Ans. (b) $3\pi r^2 = 3 \times \frac{22}{7} \times 49 = 462\ \text{cm}^2$.

Q3. The volume of the largest sphere that can be carved out of a cube of edge 6 cm is: 1M

(a) 36π cm^3 (b) 288π cm^3 (c) 72π cm^3 (d) 27π cm^3

Ans. (a) $r = 3$: $\frac{4}{3}\pi(27) = 36\pi$ cm^3.

Q4. A cone has radius 5 cm and height 12 cm. Its curved surface area is: 1M

(a) 60π (b) 65π (c) 90π (d) 25π (in cm^2)

Ans. (b) $l = \sqrt{25 + 144} = 13$, CSA $= \pi \times 5 \times 13 = 65\pi$ cm^2.

Q5. **A:** SA of a toy (cone on a hemisphere) = CSA of cone + CSA of hemisphere. **R:** The circular faces where the two solids are joined are hidden. 1M Board fav

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (a) Both true and R explains A.

Q6. **A:** If the radius of a sphere is doubled, its volume becomes 8 times. **R:** Volume of a sphere is proportional to r^3. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (a) $(2r)^3 = 8r^3$; R explains A.

Q7. Two cubes each of volume 64 cm^3 are joined end to end. The surface area of the resulting cuboid is: 1M

(a) 192 cm^2 (b) 128 cm^2 (c) 160 cm^2 (d) 176 cm^2

Ans. (c) Edge 4; cuboid $8 \times 4 \times 4$: $2(32 + 16 + 32) = 160$ cm^2.

Q8. If both the radius and the height of a cylinder are doubled, its volume becomes: 1M

(a) 2 times (b) 4 times (c) 6 times (d) 8 times

Ans. (d) $\pi(2r)^2(2h) = 8\pi r^2 h$.

Q9. A cone and a hemisphere have equal bases and equal volumes. The ratio of their heights (cone : hemisphere) is: 1M

(a) 1 : 2 (b) 2 : 1 (c) 3 : 1 (d) 1 : 1

Ans. (b) $\frac{1}{3}\pi r^2 h = \frac{2}{3}\pi r^3 \Rightarrow h = 2r$; hemisphere height $= r$. Ratio $2 : 1$.

Q10. The curved surface area of a hemisphere of radius 14 cm is: 1M

(a) 1232 cm^2 (b) 616 cm^2 (c) 1848 cm^2 (d) 2464 cm^2

Ans. (a) $2\pi r^2 = 2 \times \frac{22}{7} \times 196 = 1232$ cm^2.

Q11. The total surface area of a cube is 216 cm^2. Its volume is: 1M

(a) 36 cm^3 (b) 216 cm^3 (c) 196 cm^3 (d) 512 cm^3

Ans. (b) $6a^2 = 216 \Rightarrow a = 6$; $a^3 = 216$ cm^3.

Very short (2 marks)

Q12. Find the length of the longest rod that can be placed in a box of size $12 \times 4 \times 3$ cm. 2M

Ans. Diagonal $= \sqrt{144 + 16 + 9} = \sqrt{169} = 13$ cm.

Q13. Find the volume of a solid hemisphere of radius 21 cm. 2M

(contd.) $\frac{2}{3} \times \frac{22}{7} \times 21^3 = \frac{2}{3} \times \frac{22}{7} \times 9261 = 19404$ cm^3.

Q14. A cylinder and a cone have base radius 3 cm and height 4 cm each. Find the ratio of their curved surface areas. 2M

Ans. $l = 5$. $2\pi rh : \pi rl = 2h : l = 8 : 5$.

Q15. A medicine capsule is a cylinder with two hemispheres stuck to its ends. Its total length is 20 mm and diameter 7 mm. Find its surface area. 2M Board fav

Ans. SA $= 2\pi rL = 2 \times \frac{22}{7} \times 3.5 \times 20 = 440$ mm^2.

Q16. A solid is a cone standing on a hemisphere, both of radius 2 cm, height of cone equal to its radius. Find its volume in terms of π. 2M

Ans. $\frac{1}{3}\pi(4)(2) + \frac{2}{3}\pi(8) = \frac{8\pi}{3} + \frac{16\pi}{3} = 8\pi$ cm^3.

Q17. Three cubes of edge 4 cm are joined end to end. Find the surface area of the cuboid formed. 2M

Ans. Cuboid $12 \times 4 \times 4$: $2(48 + 16 + 48) = 224$ cm^2.

Q18. A cone has height 24 cm and base diameter 14 cm. Find its slant height and curved surface area. 2M

Ans. $l = \sqrt{7^2 + 24^2} = 25$ cm; CSA $= \frac{22}{7} \times 7 \times 25 = 550$ cm^2.

Short answer (3 marks)

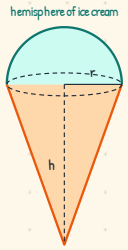


Fig: Ice-cream = cone + hemisphere on top

Q19. An ice-cream cone of radius 3 cm and height 8 cm is filled with ice-cream and has a hemisphere of ice-cream on top. Find the volume of ice-cream. ($\pi = 3.14$) 3M

Ans. Cone $= \frac{1}{3}\pi(9)(8) = 24\pi$; hemisphere $= \frac{2}{3}\pi(27) = 18\pi$. Total $= 42\pi = 42 \times 3.14 = 131.88$ cm^3.

Q20. A wooden article is made by scooping out a hemisphere from each end of a solid cylinder of height 12 cm and radius 4.2 cm. Find its total surface area. 3M Board fav

Ans. SA $= 2\pi rh + 2(2\pi r^2)$. $2\pi rh = 2 \times \frac{22}{7} \times 4.2 \times 12 = 316.8$; $4\pi r^2 = 4 \times \frac{22}{7} \times 17.64 = 221.76$. Total $= 538.56$ cm^2.

Q21. A pen stand is a cuboid $20 \times 12 \times 5$ cm with three conical depressions, each of radius 0.7 cm and depth 1.5 cm. Find the volume of wood in it. 3M

Ans. One cone $= \frac{1}{3} \times \frac{22}{7} \times 0.49 \times 1.5 = 0.77$ cm^3; three $= 2.31$. Cuboid $= 1200$. Wood $= 1200 - 2.31 = 1197.69$ cm^3.

Q22. A gulab jamun is a cylinder with two hemispherical ends, total length 5 cm and diameter 2.8 cm. It contains sugar syrup up to 30% of its volume. Find the syrup in 50 gulab jamuns. 3M Board fav

Ans. $r = 1.4$, cylinder length $= 5 - 2.8 = 2.2$. One $= \pi r^2(2.2) + \frac{4}{3}\pi r^3 = 13.552 + 11.499 \approx 25.05\ \text{cm}^3$. 50 of them ≈ 1252.5; syrup $= 30\% \approx 375.76\ \text{cm}^3$.

Long answer (5 marks)

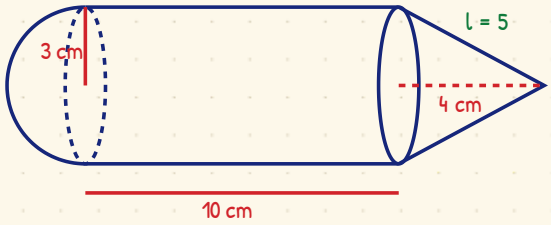


Fig (Q23): hemisphere + cylinder + cone, $r = 3$ cm

Q23. A solid toy is a cylinder with a hemisphere at one end and a cone at the other. Common radius 3 cm, cylinder height 10 cm, cone height 4 cm. Find its surface area and volume. ($\pi = 3.14$) 5M

Ans. $l = \sqrt{9+16} = 5$. SA $= 2\pi r^2 + 2\pi rh + \pi rl = \pi(18 + 60 + 15) = 93\pi = 292.02\ \text{cm}^2$. Volume $= \frac{2}{3}\pi(27) + \pi(9)(10) + \frac{1}{3}\pi(9)(4) = \pi(18 + 90 + 12) = 120\pi = 376.8\ \text{cm}^3$.

Q24. A tent is a cylinder of height 3 m and diameter 4.2 m, surmounted by a cone of the same base and slant height 3.5 m. Find (i) the canvas needed and its cost at Rs 600 per m^2, (ii) the volume of air inside. 5M Board fav

Ans. (i) $r = 2.1$. Cylinder CSA $= 2 \times \frac{22}{7} \times 2.1 \times 3 = 39.6$; cone CSA $= \frac{22}{7} \times 2.1 \times 3.5 = 23.1$. Canvas $= 62.7\ \text{m}^2$, cost = Rs 37,620. (ii) Cone height $= \sqrt{3.5^2 - 2.1^2} = \sqrt{7.84} = 2.8$ m. Volume $= \pi r^2\left(3 + \frac{2.8}{3}\right) = 13.86 \times 3.933 \approx 54.52\ \text{m}^3$.

Q25. A solid made of a cone of height 12 cm standing on a hemisphere, both of radius 6 cm, is placed upright in a cylinder full of water, touching the bottom. The cylinder has radius 6 cm and height 18 cm. Find the volume of water left in the cylinder. ($\pi = 3.14$) 5M

Ans. Cylinder $= \pi(36)(18) = 648\pi$. Solid $= \frac{1}{3}\pi(36)(12) + \frac{2}{3}\pi(216) = 144\pi + 144\pi = 288\pi$. Water left $= 648\pi - 288\pi = 360\pi = 360 \times 3.14 = 1130.4\ \text{cm}^3$.

Apne notes / galtiyan yahan likho

Shivam Bhaiya says
5 marks wale sawal mein formula pehle likho, phir values. Step marks milte hain! Answer π mein rakh ke end mein value daalo, calculation aadhi ho jayegi.

Case based (4 marks)

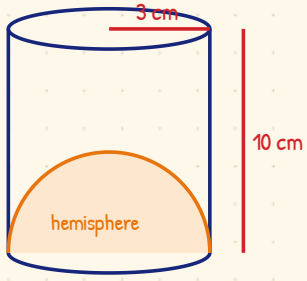


Fig (Q26): glass with raised hemispherical bottom

Q26. A juice seller uses cylindrical glasses of inner diameter 6 cm and height 10 cm, but the bottom has a raised hemispherical portion of the same radius. ($\pi = 3.14$) (i) Find the apparent capacity. (ii) Find the volume of the hemisphere. (iii) Find the actual capacity. 4M Board fav

Ans. $r = 3$. (i) $\pi r^2 h = 90\pi = 282.6\ \text{cm}^3$ (ii) $\frac{2}{3}\pi(27) = 18\pi = 56.52\ \text{cm}^3$ (iii) $72\pi = 226.08\ \text{cm}^3$.

Practice yourself (answers below)

1. Surface area of a sphere of radius 7 cm.
2. Volume, slant height and CSA of a cone with $r = 7$ cm, $h = 24$ cm.
3. TSA of a closed cylinder with $r = 7$ cm, $h = 10$ cm.
4. Capacity of a hemispherical bowl of inner radius 10.5 cm (in litres).
5. TSA and volume of a cuboid $10 \times 8 \times 6$ cm.
6. SA and volume of a toy: cone of height 12 cm on a hemisphere of radius 3.5 cm.
7. Volume of a sphere of radius 21 cm.
8. Volume of the largest cone that fits in a cube of edge 7 cm.
9. Volume of a cone of radius 6 cm and height 7 cm.
10. LSA and volume of a cube of edge 6 cm.
11. CSA and volume of a cylinder with $r = 14$ cm, $h = 20$ cm.

↳Ans: (1) 616 cm^2 (2) 1232 cm^3, 25 cm, 550 cm^2 (3) 748 cm^2 (4) 2425.5 cm^3 = 2.4255 L (5) 376 cm^2, 480 cm^3 (6) 214.5 cm^2, ≈243.83 cm^3 (7) 38808 cm^3 (8) ≈ 89.83 cm^3 (9) 264 cm^3 (10) 144 cm^2, 216 cm^3 (11) 1760 cm^2, 12320 cm^3

Apne notes / galtiyan yahan likho

Most asked: mean by step deviation, median + mode, missing f

STATISTICS

1. Grouped Data: the Basics

Large data is put in **class intervals** with frequencies. Class 10 asks three averages of grouped data: mean, median and mode. Every sum is 3 or 5 marks and fully scoring.

Key terms

Class interval: a group like 20-30. In 20-30, the value 30 goes to the **next** class 30-40 (upper limit not included).

Class size h: upper limit $-$ lower limit (here 10).

Class mark x_i: mid-point $= \frac{\text{lower}+\text{upper}}{2}$; it represents the whole class.

Frequency f_i: number of observations in the class. $\sum f_i = n$.

Cumulative frequency (cf): running total of frequencies up to that class.

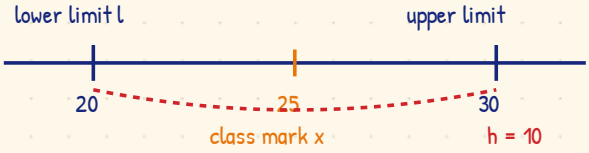


Fig: Class 20-30: $l = 20, h = 10, x_i = \frac{20+30}{2} = 25$

Quick recap: ungrouped data (Class 9)

→ **Mean** $= \frac{\text{sum of observations}}{\text{number of observations}}$; with frequencies $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$.

→ **Median**: arrange in order. n odd: $\left(\frac{n+1}{2}\right)$th value; n even: average of $\frac{n}{2}$th and $\left(\frac{n}{2}+1\right)$th values.

→ **Mode**: the value with the highest frequency.

Solved Example 0 (ungrouped)

Find the mean, median and mode of 4, 7, 7, 9, 10, 12, 7, 8.

1. Mean $= \frac{64}{8} = 8$
2. Order: 4, 7, 7, 7, 8, 9, 10, 12; $n = 8$ even: median $= \frac{7+8}{2} = 7.5$
3. 7 occurs 3 times: mode $= 7$

✔ **Mean** $= 8$, **Median** $= 7.5$, **Mode** $= 7$

Inclusive classes? First make them continuous

If classes are 10-19, 20-29, 30-39 (gap of 1), subtract **0.5** from every lower limit and add **0.5** to every upper limit: 9.5-19.5, 19.5-29.5, 29.5-39.5. Needed for median and mode (the value of l changes). Mean is unaffected because class marks stay the same.

Yaad karo: class mark

Class mark = **beech ka number** $= \frac{L+U}{2}$. 0-10 ka 5, 10-20 ka 15: har baar h jodte jao. Pehla sahi, baaki apne aap.

Formula Box

→ Class mark $x_i = \frac{\text{upper}+\text{lower}}{2}$

→ Direct: $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$

→ Assumed mean: $\bar{x} = a + \frac{\sum f_i d_i}{\sum f_i}, d_i = x_i - a$

→ Step deviation: $\bar{x} = a + \frac{\sum f_i u_i}{\sum f_i} \times h, u_i = \frac{x_i - a}{h}$

→ Mode $= l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$

→ Median $= l + \left(\frac{\frac{n}{2} - cf}{f}\right) \times h$

→ Empirical: $3\,\text{Median} = \text{Mode} + 2\,\text{Mean}$

↳ Mode: l = lower limit of modal class, f_1 its frequency, f_0 = class before, f_2 = class after. Median: l = lower limit of median class, cf = cf of class **before** it, f = its frequency.

2. Mean of Grouped Data

Three methods, same answer. Choose by the numbers:

Method	Use when	Extra column
Direct	small x_i and f_i	$f_i x_i$
Assumed mean	x_i large	$d_i = x_i - a$
Step deviation	d_i have common factor h	$u_i = \frac{x_i - a}{h}$

All three examples below use one data set: marks of 40 students.

Quick fact $\sum f_i(x_i - \bar{x}) = 0$: deviations from the mean cancel out.

Quick fact Mode $= 3$ Median $- 2$ Mean (empirical).

Quick fact Median class: first class with cf $\geq \frac{n}{2}$.

Solved Example 1 (direct method)

Find the mean marks: classes 0–10, 10–20, 20–30, 30–40, 40–50 with frequencies 6, 9, 15, 6, 4.

Class	f_i	x_i	$f_i x_i$
0–10	6	5	30
10–20	9	15	135
20–30	15	25	375
30–40	6	35	210
40–50	4	45	180
Total	**40**		**930**

1. $\bar{x} = \dfrac{\sum f_i x_i}{\sum f_i} = \dfrac{930}{40}$

✔ Mean $= 23.25$ marks

Solved Example 2 (assumed mean + step deviation)

Find the mean of the same data by the assumed mean and step-deviation methods.

x_i	f_i	$d_i = x_i - 25$	$u_i = \frac{d_i}{10}$
5	6	-20	-2
15	9	-10	-1
25	15	0	0
35	6	10	1
45	4	20	2

1. $\sum f_i d_i = -120 - 90 + 0 + 60 + 80 = -70$; $\bar{x} = 25 + \frac{-70}{40} = 23.25$
2. $\sum f_i u_i = -12 - 9 + 0 + 6 + 8 = -7$
3. $\bar{x} = 25 + \frac{-7}{40} \times 10 = 25 - 1.75$

✔ Mean $= 23.25$ (same as direct method)

Solved Example 2b (daily wages)

Find the mean daily wage of 50 workers: Rs 500–520: 12, 520–540: 14, 540–560: 8, 560–580: 6, 580–600: 10.

x_i	f_i	$u_i = \frac{x_i - 550}{20}$	$f_i u_i$
510	12	-2	-24
530	14	-1	-14
550	8	0	0
570	6	1	6
590	10	2	20
Total	**50**		-12

1. $\bar{x} = 550 + \frac{-12}{50} \times 20 = 550 - 4.8$

✔ Mean daily wage = Rs 545.20

Solved Example 2c (unequal classes)

Find the mean: classes 0–10: 4, 10–30: 10, 30–60: 6.

1. Class marks: 5, 20, 45 (widths differ, so only direct or assumed mean)
2. $\sum f_i x_i = 20 + 200 + 270 = 490$, $\sum f_i = 20$

✔ Mean $= \frac{490}{20} = 24.5$

Exam answer template (step deviation)

1. Table: Class, f_i, x_i, u_i, $f_i u_i$ (split in two tables if needed). 2. Write $a =$, $h =$, $\sum f_i =$, $\sum f_i u_i =$. 3. Formula. 4. Substitute. 5. Final answer with unit. Each line = step marks.

Are x_i small (1–2 digits)?

↓

Yes: **direct**. No: are all d multiples of h?

↓

Yes: **step deviation**. No: **assumed mean**

Yaad karo: "a beech ka, h common"

Assumed mean a = beech wala x_i (jiski f badi ho). Phir u_i bas $-2, -1, 0, 1, 2$ jaise chhote numbers. Last mein $\times h$ bhoolna mat!

Meme samjho

Bade bade x_i (110, 130, 150...) dekh ke dost ghabra gaya, humne step deviation se 2 minute mein mean nikala. **Jalwa hai hamara!**

Asli baat: $u_i = \frac{x_i - a}{h}$ makes numbers tiny; mean $= a + \frac{\sum f_i u_i}{\sum f_i} \times h$ gives the exact same answer.

Missing frequency (mean given)

Write $\sum f_i$ and $\sum f_i x_i$ in terms of the unknowns, then use $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$ and the total n: two linear equations, solve.

Missing f trick

Two unknowns need two equations: (1) $\sum f_i = n$ (2) $\sum f_i x_i = \bar{x} \times n$. One unknown? Equation (2) alone is enough.

Quick fact Class mark of 30–40 $= 35$; class size $= 10$.

Quick fact Ogive is out of syllabus now; median by formula only.

Solved Example 3 (missing frequencies)

The mean of the data below is 46 and $n = 70$. Find p and q. Classes 0-20, 20-40, 40-60, 60-80, 80-100 with frequencies 10, p, 25, q, 5.

1. x_i: 10, 30, 50, 70, 90. $40 + p + q = 70 \Rightarrow p + q = 30$
2. $\sum f_i x_i = 100 + 30p + 1250 + 70q + 450 = 1800 + 30p + 70q$
3. $1800 + 30p + 70q = 46 \times 70 = 3220 \Rightarrow 30p + 70q = 1420$
4. Put $p = 30 - q$: $900 + 40q = 1420 \Rightarrow q = 13$

✔ $p = 17, q = 13$

3. Mode of Grouped Data

Mode = the value that occurs most often. In grouped data, the class with the highest frequency is the **modal class**; the formula locates the mode inside it.

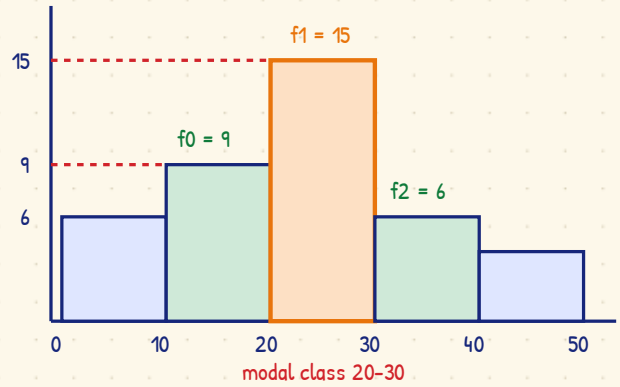


Fig: Histogram of the Example data. Tallest bar = modal class; its neighbours give f_0 and f_2

Solved Example 4 (mode)

Find the mode of the marks data (classes 0-10 to 40-50, frequencies 6, 9, 15, 6, 4).

1. Highest frequency 15, so modal class = 20-30
2. $l = 20, h = 10, f_1 = 15, f_0 = 9, f_2 = 6$
3. Mode $= 20 + \frac{15-9}{30-9-6} \times 10 = 20 + \frac{6}{15} \times 10$

✔ Mode $= 24$ marks

Solved Example 4c (big numbers)

Runs scored by batsmen in ODIs: 3000-4000: 4, 4000-5000: 18, 5000-6000: 9, 6000-7000: 7, 7000-8000: 6, 8000-9000: 3. Find the mode.

1. Modal class 4000-5000: $l = 4000, h = 1000, f_1 = 18, f_0 = 4, f_2 = 9$
2. Mode $= 4000 + \frac{18-4}{36-4-9} \times 1000 = 4000 + \frac{14000}{23}$

✔ Mode ≈ 4608.7 runs

↳ Modal class first or last ho toh missing neighbour ki frequency 0 lo.

Yaad karo: mode formula

"**Ek minus Zero** upar, **Do-Ek minus Zero minus Do** neeche": $\frac{f_1 - f_0}{2f_1 - f_0 - f_2}$, phir $\times h$ aur l jodo. f_0 = pehle wali, f_2 = baad wali class.

Solved Example 4b (modal class first)

Find the mode: classes 0-10, 10-20, 20-30, 30-40 with frequencies 20, 15, 10, 5.

1. Modal class 0-10 (first class): $f_0 = 0, f_1 = 20, f_2 = 15, l = 0, h = 10$
2. Mode $= 0 + \frac{20-0}{40-0-15} \times 10 = \frac{200}{25}$

✔ Mode $= 8$

4. Median of Grouped Data

Median = middle value. Steps:

Make cf column → Find $\frac{n}{2}$ → Median class: first cf $\geq \frac{n}{2}$ → Put in formula

Solved Example 5 (median)

Find the median of the marks data (frequencies 6, 9, 15, 6, 4).

Class	f	cf
0-10	6	6
10-20	9	15
20-30	15	30
30-40	6	36
40-50	4	40

1. $\frac{n}{2} = 20$; first cf ≥ 20 is 30, so median class = 20-30
2. $l = 20, cf = 15$ (class before), $f = 15, h = 10$
3. Median $= 20 + \frac{20-15}{15} \times 10 = 20 + 3.33$

✔ Median $= \frac{70}{3} \approx 23.33$ marks

↳ Check: $3(23.33) = 70$ and Mode + 2 Mean $= 24 + 46.5 = 70.5$. Empirical relation holds approximately.

Yaad karo: median formula

"**Aadha minus pichla cf**, apni f se bhaag, h se guna, l jodo." Pichla = median class se **pehle** wali class ka cf.

"Less than" data? Convert to frequencies first

Sometimes data comes as cumulative frequencies (less than 10: 5, less than 20: 12 ...). Subtract consecutive values to get each class frequency, then use the median formula.

Solved Example 5b (less than type)

Marks less than 10: 5, less than 20: 12, less than 30: 25, less than 40: 34, less than 50: 40. Find the median.

Class	f	cf
0–10	5	5
10–20	7	12
20–30	13	25
30–40	9	34
40–50	6	40

1. $\frac{n}{2} = 20$; median class 20–30; $l = 20$, $cf = 12$, $f = 13$
2. Median $= 20 + \frac{20-12}{13} \times 10 = 20 + \frac{80}{13}$

✔ Median $= \frac{340}{13} \approx 26.15$ marks

"More than" data

"More than or equal to" cumulative data: frequency of a class = its value − next value.

Solved Example 5c (more than type)

Marks of 50 students: ≥ 0: 50, ≥ 10: 46, ≥ 20: 38, ≥ 30: 24, ≥ 40: 10, ≥ 50: 0. Find the median.

Class	f	cf
0–10	4	4
10–20	8	12
20–30	14	26
30–40	14	40
40–50	10	50

1. $\frac{n}{2} = 25$; median class 20–30; $l = 20$, $cf = 12$, $f = 14$
2. Median $= 20 + \frac{25-12}{14} \times 10 = 20 + \frac{130}{14}$

✔ Median $= \frac{205}{7} \approx 29.29$ marks

Solved Example 5d (class size 5)

Weights (kg) of 30 students: 40–45: 2, 45–50: 3, 50–55: 8, 55–60: 6, 60–65: 6, 65–70: 3, 70–75: 2. Find the median.

1. cf: 2, 5, 13, 19, 25, 28, 30; $\frac{n}{2} = 15$
2. Median class 55–60: $l = 55$, $cf = 13$, $f = 6$, $h = 5$
3. Median $= 55 + \frac{15-13}{6} \times 5 = 55 + 1.67$

✔ Median $= \frac{170}{3} \approx 56.67$ kg

↳Median value diya ho toh pehle dekho woh kis class mein aata hai: wahi median class hai.

Solved Example 6 (missing f, median given)

The median of the data is 32 and $n = 50$. Find x and y. Classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60 with frequencies 4, x, 12, 10, y, 5.

1. $31 + x + y = 50 \Rightarrow x + y = 19$
2. Median 32 lies in 30–40: $l = 30$, $f = 10$, $cf = 4 + x + 12 = 16 + x$
3. $32 = 30 + \frac{25-(16+x)}{10} \times 10 \Rightarrow 2 = 9 - x$

✔ $x = 7, y = 12$

Shivam Bhaiya says

Statistics ke sawal mein table hi sab kuch hai. Saaf table banao: Class, f, x, fx ya cf. Table sahi = answer sahi, aur examiner ko step marks bhi dikhenge.

5. Relation between Mean, Median, Mode

For moderately skewed data:

$$3\,\text{Median} = \text{Mode} + 2\,\text{Mean}$$

$$\text{Mode} = 3\,\text{Median} - 2\,\text{Mean}$$

Yaad karo: "3-1-2"

3 Median = **1** Mode + **2** Mean. Bolo: "Teen Median barabar Ek Mode plus Do Mean", yaani **Me-Mo-Me = 3-1-2**.

Meme samjho

Median formula mein median class ka cf nahi, **pichli class ka cf** lagaya. Bilkul perfect!

Asli baat: cf in the median formula = cumulative frequency of the class just **before** the median class.

Common mistakes

- → Median: taking cf of the median class itself instead of the class before.
- → Mode: swapping f_0 and f_2, or using l of the wrong class.
- → Step deviation: forgetting to multiply by h at the end.
- → Inclusive classes (10–19) not converted before median/mode.
- → Dividing $\sum f_i x_i$ by the number of classes instead of $\sum f_i$.

Measure	What it tells	Class to find
Mean	average of all values	none (all classes)
Median	middle value	first cf $\geq \frac{n}{2}$
Mode	most frequent value	highest f

6. Which Average When?

Measure	Best for	Example
Mean	all values matter, no extreme values	average marks of a class
Median	a few very large or small values	salaries in a company, house prices
Mode	most popular / common item	shoe size to stock, favourite colour

Solved Example 7 (using the relation)

For a data set, the median is 42.5 and the mode is 45. Estimate the mean.

1. $3\,\text{Median} = \text{Mode} + 2\,\text{Mean}$
2. $127.5 = 45 + 2\,\text{Mean} \Rightarrow 2\,\text{Mean} = 82.5$

✔ Mean ≈ 41.25

↳Actual mean of that data (Q in MIQ) is 41.8; empirical relation sirf estimate deta hai, exact nahi.

7. What the Board Asks

Type	Marks	How often
Mean by step deviation	3	almost every year
Mode of grouped data	2-3	very often
Median, missing f	3-5	often
Empirical relation	1	MCQ / AR
Mean + median + mode together	5	LA / case

Practice yourself (answers below)

1. Mean of x: 3, 5, 7, 9, 11 with f: 2, 4, 6, 4, 2.
2. Class mark of 47.5-52.5.
3. Mode if $l = 60, h = 20, f_1 = 30, f_0 = 18, f_2 = 24$.
4. Median if $l = 40, n = 80, cf = 34, f = 12, h = 10$.
5. Mean $= 35$, median $= 33$. Find the mode.
6. $\sum f_i u_i = -6, \sum f_i = 30, a = 45, h = 10$. Find the mean.
7. Median class of 0-20: 6, 20-40: 10, 40-60: 14, 60-80: 12, 80-100: 8.
8. Mean of the first 10 natural numbers.
9. Mean of 0-10: 3, 10-20: 5, 20-30: 2.
10. Mode of 0-5: 4, 5-10: 10, 10-15: 6.
11. Mode $= 40$, mean $= 43$. Find the median.
12. Median of 0-10: 5, 10-20: 10, 20-30: 15 ($n = 30$).

↳Ans: (1) 7 (2) 50 (3) ≈ 73.33 (4) 45 (5) 29 (6) 43 (7) 40-60 (8) 5.5 (9) 14 (10) 8 (11) 42 (12) 20

Most Important Questions

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. The class mark of the class 15-25 is: 1M
(a) 10 (b) 15 (c) 20 (d) 25
Ans. (c) $\frac{15+25}{2} = 20$.

Q2. If mode $= 24$ and mean $= 30$, then median is: 1M Board fav
(a) 26 (b) 27 (c) 28 (d) 29
Ans. (c) $3\,\text{Md} = 24 + 60 = 84 \Rightarrow \text{Median} = 28$.

Q3. For classes 0-5, 5-10, 10-15, 15-20, 20-25 with frequencies 3, 7, 12, 8, 5, the upper limit of the median class is: 1M
(a) 10 (b) 15 (c) 20 (d) 25
Ans. (b) $n = 35$, $\frac{n}{2} = 17.5$; cf: 3, 10, 22, so median class 10-15; upper limit 15.

Q4. In step deviation, $\sum f_i u_i = 15, \sum f_i = 50, a = 35, h = 10$. The mean is: 1M
(a) 36 (b) 38 (c) 40 (d) 35.3
Ans. (b) $35 + \frac{15}{50} \times 10 = 38$.

Q5. If each observation of a data is increased by 5, the mean: 1M
(a) stays same (b) increases by 5 (c) becomes 5 times (d) decreases by 5
Ans. (b) increases by 5.

Q6. A: If mean $= 20$ and median $= 22$, then mode $= 26$. R: Mode $= 3$ Median $- 2$ Mean. 1M Board fav
(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true
Ans. (a) $66 - 40 = 26$; R explains A.

Q7. A: The modal class is the class with the highest cumulative frequency. R: Mode is the value that occurs most often. 1M
(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true
Ans. (d) Modal class has the highest **frequency**, not cf. A false, R true.

Q8. The mean of 10 observations is 15. If one observation 24 is replaced by 14, the new mean is: 1M
(a) 14 (b) 15 (c) 16 (d) 13.5
Ans. (a) New sum $= 150 - 24 + 14 = 140$; mean $= 14$.

Q9. If Mode $-$ Mean $= k$(Median $-$ Mean), then k is: 1M
(a) 1 (b) 2 (c) 3 (d) $\frac{1}{3}$
Ans. (c) Mode $= 3$ Median $- 2$ Mean, so Mode $-$ Mean $= 3$ (Median $-$ Mean).

Q10. The class mark of a class is 10 and its class size is 6. The class is: 1M

(a) 7-13 (b) 4-16 (c) 10-16 (d) 8-12

Ans. (a) $10-3=7$ to $10+3=13$.

Very short (2 marks)

Q11. Find the mean: x: 2, 4, 6, 8, 10 with f: 3, 5, 8, 6, 3. 2M

Ans. $\sum f=25, \sum fx=6+20+48+48+30=152$. Mean $=\frac{152}{25}=6.08$.

Q12. The mean of $x, x+2, x+4, x+6, x+8$ is 11. Find x. 2M

Ans. $\frac{5x+20}{5}=11 \Rightarrow 5x=35 \Rightarrow x=7$.

Q13. Find the mode if $l=40, h=10, f_1=20, f_0=12, f_2=14$. 2M

Ans. $40+\frac{20-12}{40-12-14}\times 10=40+\frac{80}{14}\approx 45.71$.

Q14. Classes are 10-19, 20-29, 30-39. Write the true class limits of 20-29 and its class mark. 2M

Ans. 19.5-29.5; class mark $=\frac{19.5+29.5}{2}=24.5$ (same as $\frac{20+29}{2}$).

Q15. Find the median and mode of 2, 3, 4, 3, 5, 3, 6. 2M

Ans. Order: 2, 3, 3, 3, 4, 5, 6; median = 4th value $=3$; mode $=3$ (occurs 3 times).

Q16. Find the mean of the first five multiples of 3. 2M

Ans. $\frac{3+6+9+12+15}{5}=\frac{45}{5}=9$.

Q17. For classes 0-10, 10-20, 20-30, 30-40, 40-50 with frequencies 5, 12, 18, 10, 5, write the modal class and the median class. 2M

Ans. Modal class: 20-30 (highest $f=18$). $n=50$, $\frac{n}{2}=25$; cf: 5, 17, 35, so median class is also 20-30.

Q18. The mean of 20 observations is 15. Later it was found that one observation 28 was wrongly read as 8. Find the correct mean. 2M

Ans. Wrong sum $=300$; correct sum $=300-8+28=320$; mean $=\frac{320}{20}=16$.

Short answer (3 marks)

Q19. Find the mean by step-deviation: classes 100-120, 120-140, 140-160, 160-180, 180-200 with frequencies 10, 14, 8, 12, 6. 3M Board fav

Ans. x_i: 110, 130, 150, 170, 190; $a=150, h=20$; u_i: $-2,-1,0,1,2$. $\sum f_iu_i=-20-14+0+12+12=-10$, $\sum f_i=50$. Mean $=150+\frac{-10}{50}\times 20=146$.

Q20. Find the mode: classes 0-20, 20-40, 40-60, 60-80, 80-100 with frequencies 8, 14, 22, 16, 10. 3M

Ans. Modal class 40-60; $l=40, f_1=22, f_0=14, f_2=16, h=20$. Mode $=40+\frac{8}{44-30}\times 20=40+\frac{80}{7}\approx 51.43$.

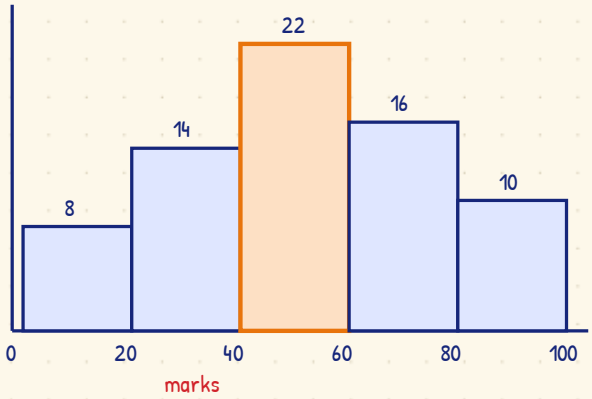


Fig (mode question above): modal class 40-60, $f_0=14, f_1=22, f_2=16$

Q21. Find the median: classes 0-10, 10-20, 20-30, 30-40, 40-50 with frequencies 8, 12, 20, 6, 4. 3M

Ans. $n=50$, $\frac{n}{2}=25$; cf: 8, 20, 40, so median class 20-30; $l=20, cf=20, f=20$. Median $=20+\frac{25-20}{20}\times 10=22.5$.

Q22. The mean of the data is 27. Find p: classes 0-10, 10-20, 20-30, 30-40, 40-50 with frequencies 5, 7, p, 12, 6. 3M

Ans. $\sum f=30+p; \sum fx=25+105+25p+420+270=820+25p$. $820+25p=27(30+p)=810+27p \Rightarrow p=5$.

Q23. Ages of 80 patients: 5-15: 5, 15-25: 10, 25-35: 20, 35-45: 25, 45-55: 15, 55-65: 5. Find the mean (assumed mean method) and the mode. 3M Board fav

Ans. x_i: 10, 20, ..., 60; $a=40$; d_i: $-30,-20,-10,0,10,20$; $\sum f_id_i=-150-200-200+0+150+100=-300$. Mean $=40-\frac{300}{80}=36.25$ years. Mode: modal class 35-45: $35+\frac{25-20}{50-20-15}\times 10=35+3.33\approx 38.33$ years.

Q24. Find the mean weight of the 30 students in Solved Example 5d by the step-deviation method. 3M

Ans. x_i: 42.5, 47.5, ..., 72.5; $a=57.5, h=5$; u_i: -3 to 3. $\sum f_iu_i=-6-6-8+0+6+6+6=-2$. Mean $=57.5+\frac{-2}{30}\times 5=57.5-0.33\approx 57.17$ kg.

Q25. Find the mean by the direct method: classes 0-4, 4-8, 8-12, 12-16 with frequencies 2, 6, 8, 4. 3M

Ans. x_i: 2, 6, 10, 14; f_ix_i: 4, 36, 80, 56; $\sum f_ix_i=176$, $\sum f_i=20$. Mean $=\frac{176}{20}=8.8$.

Long answer (5 marks)

Q26. Find the mean, median and mode: classes 10-20, 20-30, 30-40, 40-50, 50-60, 60-70 with frequencies 4, 8, 10, 12, 10, 6. 5M Board fav

Ans. $n=50$. Mean: $\sum fx=60+200+350+540+550+390=2090$, mean $=41.8$. Median: $\frac{n}{2}=25$, cf: 4, 12, 22, 34; class 40-50, $cf=22, f=12$: $40+\frac{3}{12}\times 10=42.5$. Mode: modal class 40-50, $f_1=12, f_0=10, f_2=10$: $40+\frac{2}{4}\times 10=45$.

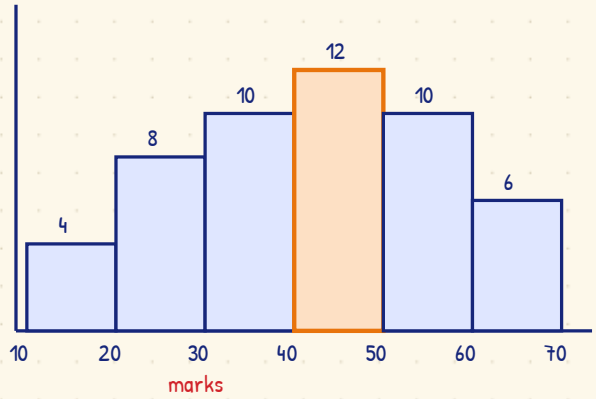


Fig (Q26 data): histogram; tallest bar 40–50 is the modal class

Q27. The median of the distribution is 35 and $n = 70$. Find x and y: classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60 with frequencies 6, x, 18, 16, y, 5. 5M

Ans. $45 + x + y = 70 \Rightarrow x + y = 25$. Median class 30–40: $l = 30, f = 16, cf = 24 + x$. $35 = 30 + \frac{35-24-x}{16} \times 10 \Rightarrow 8 = 11 - x \Rightarrow x = 3, y = 22$.

Q28. Find the mode and median of marks: 11–20, 21–30, 31–40, 41–50, 51–60 with frequencies 3, 7, 15, 10, 5. 5M

Ans. Make continuous: 10.5–20.5, ..., 50.5–60.5. Mode: modal class 30.5–40.5; $30.5 + \frac{15-7}{30-7-10} \times 10 = 30.5 + \frac{80}{13} \approx 36.65$. Median: $\frac{n}{2} = 20$, cf: 3, 10, 25; class 30.5–40.5, $cf = 10, f = 15$: $30.5 + \frac{10}{15} \times 10 \approx 37.17$.

Q29. The mean of the following distribution is 52 and $n = 100$. Find f_1 and f_2: classes 0–20, 20–40, 40–60, 60–80, 80–100 with frequencies 15, f_1, 30, f_2, 15. 5M Board fav

Ans. $60 + f_1 + f_2 = 100 \Rightarrow f_1 + f_2 = 40$. x_i: 10, 30, 50, 70, 90; $\sum f_i x_i = 150 + 30f_1 + 1500 + 70f_2 + 1350 = 3000 + 30f_1 + 70f_2$. $3000 + 30f_1 + 70f_2 = 5200 \Rightarrow 30f_1 + 70f_2 = 2200$. With $f_1 = 40 - f_2$: $1200 + 40f_2 = 2200 \Rightarrow f_2 = 25, f_1 = 15$.

Shivam Bhaiya says

Formula likhne ke baad har value alag line mein likho: l, f_1, f_0, f_2, h sabki value. Examiner ko clarity dikhegi aur tum khud galti pakad loge.

Apne notes / galtiyan yahan likho

Case based (4 marks)

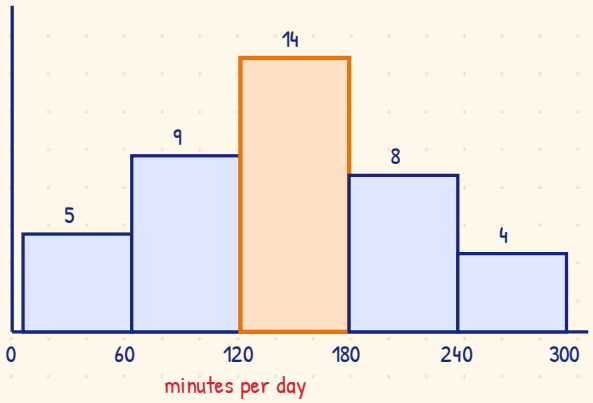


Fig (case study): screen time histogram; modal class 120–180

Q30. A school surveyed the daily screen time (minutes) of 40 students: 0–60: 5, 60–120: 9, 120–180: 14, 180–240: 8, 240–300: 4. (i) Write the modal class. (ii) Write the median class. (iii) Find the mean screen time by step deviation. 4M Board fav

Ans. (i) 120–180 (highest $f = 14$). (ii) $\frac{n}{2} = 20$, cf: 5, 14, 28, so 120–180. (iii) $a = 150, h = 60, u_i$: $-2, -1, 0, 1, 2$; $\sum f_i u_i = -10 - 9 + 0 + 8 + 8 = -3$; mean $= 150 + \frac{-3}{40} \times 60 = 145.5$ min.

Last-minute checklist

- ✓ Classes continuous? (convert 10–19 type first)
- ✓ x_i column right? (mid-points, gap $= h$)
- ✓ $\sum f_i = n$ checked?
- ✓ Median: cf of **previous** class; mode: f_0 before, f_2 after
- ✓ Step deviation: $\times h$ done? Unit written?

One-page recap

Symbol	Means
x_i	class mark (mid-point)
a	assumed mean (a middle x_i)
h	class size
l	lower limit of modal / median class
f_1, f_0, f_2	modal, before, after frequencies
cf	cf of class before the median class
f	frequency of the median class

Chapter in 4 lines

1. Mean: $a + \frac{\sum f_i u_i}{\sum f_i} \times h$ (step deviation is fastest). 2. Mode: modal class $=$ highest f, then $l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$. 3. Median: first cf $\geq \frac{n}{2}$, then $l + \frac{\frac{n}{2} - cf}{f} \times h$. 4. 3 Median $=$ Mode $+ 2$ Mean.

Apne notes / galtiyan yahan likho

Most asked: two dice, cards, find x, case-based game

PROBABILITY

1. Basic Terms

Key terms

Experiment: an action with well-defined outcomes (tossing a coin, throwing a die).

Outcome: a possible result. **Sample space:** set of all outcomes.

Event E: a collection of outcomes (e.g. "even number" $= \{2, 4, 6\}$).

Equally likely: no outcome is preferred over another (fair coin, unbiased die, well-shuffled cards).

Elementary event: an event with exactly one outcome.

Theoretical (classical) probability

$$P(E) = \frac{\text{number of outcomes favourable to } E}{\text{total number of outcomes}}$$

- ✓ $0 \le P(E) \le 1$ always. Never negative, never more than 1.
- ✓ **Sure event:** $P = 1$ (number < 7 on a die). **Impossible event:** $P = 0$ (8 on a die).
- ✓ **Complementary event** "not E" $= \bar{E}$: $P(E) + P(\bar{E}) = 1$.
- ✓ Sum of probabilities of all elementary events $= 1$.

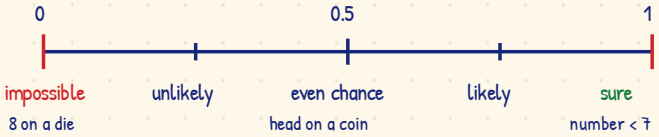


Fig: The probability scale from 0 to 1

Formula Box

- → $P(E) = \dfrac{\text{favourable outcomes}}{\text{total outcomes}}$
- → $0 \le P(E) \le 1$; sure event $= 1$, impossible $= 0$
- → $P(\bar{E}) = 1 - P(E)$
- → $P(\text{at least one}) = 1 - P(\text{none})$
- → n coins: 2^n outcomes; n dice: 6^n outcomes
- → Cards: 52; 4 suits $\times$ 13; face cards 12; aces 4
- → Leap year 53 Sundays: $\frac{2}{7}$; non-leap: $\frac{1}{7}$

Yaad karo: "Zero se Ek"

Probability hamesha **0 se 1** ke beech. Answer -0.2 ya $\frac{7}{5}$ aaya? Pakka galti hai, dobara gino. Percentage mein: 0% se 100%.

Quick fact 3 coins: P(all heads) $= \frac{1}{8}$.

Solved Example 1 (bag of balls)

A bag has 5 red, 8 white and 7 green balls. One ball is drawn at random. Find P(white) and P(not green).

1. Total $= 5 + 8 + 7 = 20$
2. P(white) $= \frac{8}{20} = \frac{2}{5}$
3. P(green) $= \frac{7}{20}$, so P(not green) $= 1 - \frac{7}{20}$

✓ P(white) $= \frac{2}{5}$, P(not green) $= \frac{13}{20}$

2. Coins

Coins	Sample space	Total
1	H, T	2
2	HH, HT, TH, TT	4
3	HHH, HHT, HTH, THH, HTT, THT, TTH, TTT	8

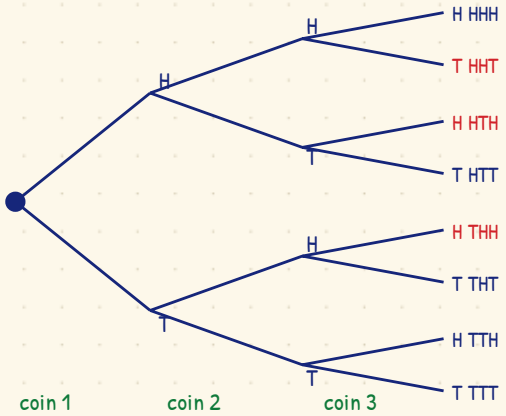


Fig: Tree for 3 coins: $2^3 = 8$ outcomes (red = exactly 2 heads)

↳HT aur TH alag outcomes hain! "Ek head" ke 2 tareeke, isliye P(exactly one head, 2 coins) $= \frac{2}{4}$, not $\frac{1}{3}$.

Solved Example 2 (two coins)

Two coins are tossed together. Find the probability of (i) at least one head (ii) exactly one head.

1. Outcomes: HH, HT, TH, TT (4)
2. At least one head: HH, HT, TH $= 3$ (or $1 - P(TT) = 1 - \frac{1}{4}$)
3. Exactly one head: HT, TH $= 2$

✓ (i) $\frac{3}{4}$ (ii) $\frac{1}{2}$

3. Dice

One die: 6 outcomes. Two dice: $6 \times 6 = 36$ outcomes written as ordered pairs (a, b); $(1, 2)$ and $(2, 1)$ are different.

Quick fact Two dice: P(sum 7) $= \frac{6}{36} = \frac{1}{6}$, the most likely sum.

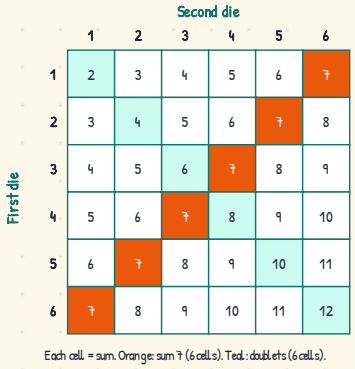


Fig: All 36 outcomes of two dice with their sums. Sum 7 appears most (6 times)

Sum	Ways	Sum	Ways
2	1	8	5
3	2	9	4
4	3	10	3
5	4	11	2
6	5	12	1
7	6	Total	36

Doublets $(1,1),(2,2),\ldots,(6,6)$: 6. Sum 1 or 13: impossible, $P = 0$.

Yaad karo: sum ke tareeke

Sum $s \leq 7$: tareeke $= s - 1$. Sum $s \geq 7$: tareeke $= 13 - s$. Jaise sum 9: $13 - 9 = 4$; sum 4: $4 - 1 = 3$. Bina table ke 5 second!

Solved Example 3 (two dice)

Two dice are thrown. Find the probability that (i) the sum is 8 (ii) a doublet (iii) the sum is at most 4 (iv) the product is 6.

1. Sum 8: (2,6), (3,5), (4,4), (5,3), (6,2) $= 5$
2. Doublets: 6; sum ≤ 4: sums 2, 3, 4 give $1 + 2 + 3 = 6$
3. Product 6: (1,6), (6,1), (2,3), (3,2) $= 4$

✔ (i) $\frac{5}{36}$ (ii) $\frac{1}{6}$ (iii) $\frac{1}{6}$ (iv) $\frac{1}{9}$

Meme samjho

Dost: "Do dice ke total 11 outcomes hain na, sum 2 se 12 tak?" Main, 36 wali table yaad karke: **muskurahat**.

Asli baat: Sums 2 to 12 are 11 values, but they are not equally likely. Outcomes are 36 ordered pairs; P(sum 7) $= \frac{6}{36}$, P(sum 2) $= \frac{1}{36}$.

4. Playing Cards

Each suit: A, 2 to 10, J, Q, K. **Face cards** = J, Q, K: $3 \times 4 = 12$ (6 red, 6 black). Aces: 4. Ace is not a face card.

Black: 26
Spades ♠ 13
Clubs ♣ 13

Red: 26
Hearts ♥ 13
Diamonds ♦ 13

Card type	Count	P
Red (or black)	26	$\frac{1}{2}$
One suit, e.g. spades	13	$\frac{1}{4}$
Face cards	12	$\frac{3}{13}$
Red face cards	6	$\frac{3}{26}$
Aces (or kings, queens...)	4	$\frac{1}{13}$
Black kings	2	$\frac{1}{26}$
Numbered cards 2 to 10	36	$\frac{9}{13}$
Jack of hearts	1	$\frac{1}{52}$

Yaad karo: "Dil aur Heera Laal"

Hearts (dil) aur Diamonds (heera) **red**; Spades, Clubs **black**. Face = **KQJ** = King, Queen, Jack: 12 cards. $52 = 4 \times 13$.

Solved Example 4 (cards)

One card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting (i) a red king (ii) a face card (iii) neither a red card nor a queen.

1. Red kings: K of hearts, K of diamonds $= 2$
2. Face cards $= 12$
3. Neither red nor queen = black non-queens $= 26 - 2 = 24$

✔ (i) $\frac{1}{26}$ (ii) $\frac{3}{13}$ (iii) $\frac{6}{13}$

5. Numbers, Letters, Calendar

Solved Example 5 (numbers 1 to 30)

A number is chosen at random from 1 to 30. Find P(prime) and P(multiple of 3 or 5).

1. Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29 $= 10$ (1 is not prime)
2. Multiples of 3: 10; of 5: 6; of both (15, 30): 2. So $10 + 6 - 2 = 14$

✔ P(prime) $= \frac{1}{3}$, P(3 or 5) $= \frac{14}{30} = \frac{7}{15}$

Solved Example 6 (53 Sundays)

Find the probability that a leap year has 53 Sundays.

1. 366 days $= 52$ weeks $+ 2$ days
2. The 2 extra days: (Sun, Mon), (Mon, Tue), ..., (Sat, Sun): 7 pairs
3. Pairs with Sunday: (Sat, Sun), (Sun, Mon) $= 2$

✔ P $= \frac{2}{7}$ (non-leap year: 1 extra day, P $= \frac{1}{7}$)

Solved Example 7 (find x)

A box has 12 balls of which x are black. If 6 more black balls are put in, the probability of drawing a black ball becomes double. Find x.

1. Before: $\frac{x}{12}$. After: $\frac{x+6}{18}$
2. $\frac{x+6}{18} = 2 \times \frac{x}{12} \Rightarrow 12(x+6) = 36x$
3. $12x + 72 = 36x \Rightarrow 24x = 72$

✔ $x = 3$

6. More Board Favourites

Solved Example 8 (spinning wheel)

An arrow on a wheel stops at one of the numbers 1 to 8 (equally likely). Find P(odd number), P(number greater than 2), P(number less than 9).

1. Odd: 1, 3, 5, 7 = 4: $\frac{4}{8}$
2. Greater than 2: 3 to 8 = 6: $\frac{6}{8}$
3. Less than 9: all 8 (sure event)

✔ $\frac{1}{2}$, $\frac{3}{4}$, 1

Solved Example 9 (defective items)

15 defective bulbs are mixed with 135 good ones. One bulb is taken at random. Find P(good bulb). If this bulb is defective and kept aside, find P(the next bulb is good).

1. Total = 150; P(good) = $\frac{135}{150} = \frac{9}{10}$
2. Defective kept aside: total 149, good still 135

✔ $\frac{9}{10}$, then $\frac{135}{149}$

Solved Example 10 (coin game)

Ravi tosses three coins. He wins if all three show the same face, else he loses. Find P(Ravi loses).

1. 8 outcomes; same face: HHH, TTT = 2; P(win) = $\frac{2}{8} = \frac{1}{4}$
2. P(lose) = $1 - \frac{1}{4}$

✔ P(lose) = $\frac{3}{4}$

↳Spinner, bulbs, games, sab mein ek hi formula: favourable ÷ total. Bas total sahi gino.

Solved Example 11 (at least one six)

Two dice are thrown. Find the probability of getting at least one six.

1. No six on either die: $5 \times 5 = 25$ outcomes
2. P(at least one six) = $1 - \frac{25}{36}$

✔ $\frac{11}{36}$

Quick fact Face cards 12, so P(face) = $\frac{3}{13}$.

Solved Example 12 (letters)

A letter is chosen at random from the word ASSASSINATION. Find P(vowel) and P(consonant).

1. Letters: 13. Vowels: A, A, I, A, I, O = 6
2. Consonants = $13 - 6 = 7$

✔ P(vowel) = $\frac{6}{13}$, P(consonant) = $\frac{7}{13}$

Yaad karo: "2 ki power, 6 ki power"

n coins $\Rightarrow 2^n$ outcomes (1: 2, 2: 4, 3: 8). n dice $\Rightarrow 6^n$ (1: 6, 2: 36). Repeat letters bhi alag gino: ASSASSINATION mein S chaar baar.

Solved Example 13 (between)

A die is thrown once. Find the probability of getting a number (i) between 2 and 6 (ii) at least 5.

1. "Between 2 and 6" excludes 2 and 6: 3, 4, 5 = 3
2. "At least 5" means 5 or 6 = 2

✔ (i) $\frac{1}{2}$ (ii) $\frac{1}{3}$

Shivam Bhaiya says

Case-based mein question lamba dikhta hai par sawal simple hota hai: favourable gino, total gino, divide karo. Ghabrao mat, bas list banao.

Exam answer template

1. Total outcomes = (write the sample space if small). 2. Favourable outcomes: list them. 3. $P(E) = \frac{\text{fav}}{\text{total}}$. 4. Simplify to lowest terms.

Shivam Bhaiya says

Probability mein sabse pehle **total outcomes** likho, phir favourable ki list banao. List likhoge toh galti nahi hogi, aur examiner ko bhi dikhega. Answer hamesha lowest form mein!

Meme samjho

"At least one" dekha? 10 cases mat gino, **time waste mat kar**: $1 - P(\text{none})$ lagao.

Asli baat: 3 coins: P(at least one head) $= 1 - P(TTT) = 1 - \frac{1}{8} = \frac{7}{8}$.

Yaad karo: "Not wala = 1 minus wala"

P(not E) $= 1 - P(E)$. "At least one" $= 1 -$ "none". Lamba counting bachao.

Common mistakes

- → Treating HT and TH (or (1,2) and (2,1)) as the same outcome.
- → Taking 1 as a prime number; taking Ace as a face card.
- → Two dice total taken as 12 or 11 instead of 36.
- → Answer more than 1 or negative: recount.
- → Not reducing the fraction to lowest terms.

Practice yourself (answers below)

1. P(red queen) from a deck of 52.
2. P(sum 13) with two dice.
3. P(not a face card).
4. P(at least one tail) when two coins are tossed.
5. P(multiple of 4) from numbers 1 to 20.
6. P(letter B) from the letters of PROBABILITY.
7. If $P(E) = 0.36$, find $P(\bar{E})$.
8. P(black face card) from 52 cards.
9. P(doublet) with two dice.
10. P(even number) from 1 to 25.

↳ Ans: (1) $\frac{1}{26}$ (2) 0 (3) $\frac{10}{13}$ (4) $\frac{3}{4}$ (5) $\frac{1}{4}$ (6) $\frac{2}{11}$ (7) 0.64 (8) $\frac{3}{26}$ (9) $\frac{1}{6}$ (10) $\frac{12}{25}$

↳ "Between 2 and 6" mein 2 aur 6 shaamil nahi; "at least 5" mein 5 shaamil hai.

MOST IMPORTANT QUESTIONS

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. Which of these cannot be the probability of an event? 1M Board fav

(a) $\frac{2}{3}$ (b) -1.5 (c) 15% (d) 0.7

Ans. (b) Probability is never negative.

Q2. If $P(E) = 0.05$, then $P(\text{not } E)$ is: 1M

(a) 0.05 (b) 0.5 (c) 0.95 (d) 0.9

Ans. (c) $1 - 0.05 = 0.95$.

Q3. A die is thrown once. The probability of a number greater than 4 is: 1M

(a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{2}{3}$ (d) $\frac{1}{6}$

Ans. (b) 5, 6: $\frac{2}{6} = \frac{1}{3}$.

Q4. One card is drawn from 52. The probability that it is an ace is: 1M

(a) $\frac{1}{13}$ (b) $\frac{1}{52}$ (c) $\frac{4}{13}$ (d) $\frac{1}{4}$

Ans. (a) $\frac{4}{52} = \frac{1}{13}$.

Q5. Two coins are tossed. The probability of getting no head is: 1M

(a) $\frac{1}{2}$ (b) $\frac{3}{4}$ (c) $\frac{1}{4}$ (d) 0

Ans. (c) Only TT: $\frac{1}{4}$.

Q6. A: The probability of getting a prime number on throwing a die is $\frac{1}{2}$. R: The prime numbers on a die are 2, 3 and 5. 1M

(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true

Ans. (a) $\frac{3}{6} = \frac{1}{2}$; R explains A.

Q7. A: The probability that a number chosen from 1 to 10 is prime is $\frac{1}{2}$. R: There are 4 prime numbers from 1 to 10. 1M Board fav

(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true

Ans. (d) Primes 2, 3, 5, 7: $P = \frac{4}{10} = \frac{2}{5}$. A false, R true.

Q8. A number is chosen from 1 to 25. The probability that it is a multiple of 6 is: 1M

(a) $\frac{4}{25}$ (b) $\frac{1}{5}$ (c) $\frac{6}{25}$ (d) $\frac{1}{6}$

Ans. (a) 6, 12, 18, 24: $\frac{4}{25}$.

Q9. The probability of a sure event is: 1M

(a) 0 (b) $\frac{1}{2}$ (c) 1 (d) more than 1

Ans. (c) 1.

Q10. Two dice are thrown. The probability that the sum is a multiple of 3 is: 1M

(a) $\frac{1}{6}$ (b) $\frac{1}{3}$ (c) $\frac{1}{4}$ (d) $\frac{5}{12}$

Ans. (b) Sums 3, 6, 9, 12: $2 + 5 + 4 + 1 = 12$; $\frac{12}{36} = \frac{1}{3}$.

Very short (2 marks)

Q11. A bag has 3 red and 5 black balls. Find the probability that a ball drawn is not red. 2M

Ans. P(red) $= \frac{3}{8}$, so P(not red) $= \frac{5}{8}$.

Q12. A letter is chosen from the word MATHEMATICS. Find the probability that it is a vowel. 2M

Ans. 11 letters; vowels A, E, A, I $= 4$. $P = \frac{4}{11}$.

Q13. A die is thrown. Find the probability of getting an odd prime number. 2M

Ans. Odd primes: 3, 5. $P = \frac{2}{6} = \frac{1}{3}$.

Q14. Find the probability that a non-leap year has 53 Mondays. 2M

Ans. 365 days $= 52$ weeks $+ 1$ day; the extra day is Monday in 1 of 7 cases: $\frac{1}{7}$.

Q15. A jar has 24 marbles, some green and the rest blue. P(green) $= \frac{2}{3}$. Find the number of blue marbles. 2M

Ans. Green $= \frac{2}{3} \times 24 = 16$; blue $= 24 - 16 = 8$.

Q16. A coin is tossed twice. Find the probability of getting at most one head. 2M

Ans. HT, TH, TT $= 3$ of 4: $\frac{3}{4}$ (or $1 - P(HH)$).

Q17. From a group of 3 boys and 2 girls, one child is chosen at random. Find the probability that a girl is chosen. 2M

Ans. Total 5; girls 2: $\frac{2}{5}$.

Q18. A bag has 12 balls, red and blue only. The probability of red is 3 times the probability of blue. Find the number of blue balls. 2M

(contd.) $r = 3b$ and $r + b = 12 \Rightarrow 4b = 12 \Rightarrow b = 3$ blue (9 red).

Short answer (3 marks)

Q19. Two dice are thrown. Find the probability that (i) the sum is 9 (ii) both show the same number (iii) the sum is 10 or more. 3M Board fav

Ans. (i) (3,6), (4,5), (5,4), (6,3): $\frac{4}{36} = \frac{1}{9}$ (ii) $\frac{6}{36} = \frac{1}{6}$ (iii) sums 10, 11, 12: $3 + 2 + 1 = 6$, P $= \frac{1}{6}$.

Q20. All red face cards are removed from a deck of 52. A card is drawn from the rest. Find P(black card), P(queen), P(face card). 3M

Ans. Removed 6; left 46. Black $= 26$: $\frac{26}{46} = \frac{13}{23}$. Queens left (black) $= 2$: $\frac{1}{23}$. Face cards left $= 6$: $\frac{6}{46} = \frac{3}{23}$.

Q21. Three coins are tossed. Find the probability of (i) exactly two heads (ii) at least two heads (iii) at most one head. 3M

Ans. 8 outcomes. (i) HHT, HTH, THH: $\frac{3}{8}$ (ii) add HHH: $\frac{4}{8} = \frac{1}{2}$ (iii) HTT, THT, TTH, TTT: $\frac{1}{2}$.

Q22. A box has discs numbered 1 to 80. One disc is drawn. Find P(two-digit number), P(perfect square), P(divisible by 5). 3M

Ans. Two-digit: 10 to 80 $= 71$: $\frac{71}{80}$. Squares: 1, 4, ..., 64 $= 8$: $\frac{1}{10}$. Divisible by 5: 16: $\frac{1}{5}$.

Q23. A card is drawn from cards numbered 1 to 50. Find the probability that the number is (i) prime (ii) a perfect square (iii) divisible by both 3 and 5. 3M

Ans. (i) Primes up to 50: 15: $\frac{15}{50} = \frac{3}{10}$ (ii) 1, 4, 9, 16, 25, 36, 49 $= 7$: $\frac{7}{50}$ (iii) 15, 30, 45 $= 3$: $\frac{3}{50}$.

Q24. A card is drawn from a deck of 52. Find the probability that it is (i) a black face card (ii) a spade or an ace (iii) not a heart. 3M

Ans. (i) 6: $\frac{6}{52} = \frac{3}{26}$ (ii) 13 spades + 3 other aces $= 16$: $\frac{16}{52} = \frac{4}{13}$ (iii) $52 - 13 = 39$: $\frac{3}{4}$.

Q25. Two dice are thrown. Find the probability that (i) the sum is even (ii) the sum is more than 8 (iii) the product is less than 6. 3M

Ans. (i) Both odd or both even: $9 + 9 = 18$: $\frac{1}{2}$. (ii) Sums 9, 10, 11, 12: $4 + 3 + 2 + 1 = 10$: $\frac{5}{18}$. (iii) (1,1)...(1,5), (2,1), (2,2), (3,1), (4,1), (5,1) $= 10$: $\frac{5}{18}$.

Long answer (5 marks)

Q26. A bag has 15 balls, of which x are white. (i) Write P(white). (ii) If 5 more white balls are added, P(white) becomes double the earlier value. Find x. (iii) Then find P(not white) in the new bag. 5M

Ans. (i) $\frac{x}{15}$. (ii) $\frac{x+5}{20} = \frac{2x}{15} \Rightarrow 15x + 75 = 40x \Rightarrow x = 3$. (iii) New bag: 20 balls, 8 white, 12 not white: $\frac{12}{20} = \frac{3}{5}$.

Apne notes / galtiyan yahan likho

Q27. Two dice are thrown. Find the probability that (i) the product is even (ii) the sum is a prime number (iii) the difference of the numbers is 2. 5M Board fav

(contd.) (i) Product odd only when both odd: $3 \times 3 = 9$; even $= 27$: $\frac{3}{4}$. (ii) Sums 2, 3, 5, 7, 11: $1 + 2 + 4 + 6 + 2 = 15$: $\frac{15}{36} = \frac{5}{12}$. (iii) (1,3), (3,1), (2,4), (4,2), (3,5), (5,3), (4,6), (6,4) $= 8$: $\frac{2}{9}$.

Case based (4 marks)

Q28. At a school fair, a box has 100 cards numbered 2 to 101. A student picks one card at random. Find the probability that the number is (i) even (ii) a perfect square (iii) divisible by 5. 4M Board fav

Ans. (i) Even: 2, 4, ..., 100 $= 50$: $\frac{1}{2}$. (ii) Squares: 4, 9, ..., 100 (2^2 to 10^2) $= 9$: $\frac{9}{100}$. (iii) 5, 10, ..., 100 $= 20$: $\frac{1}{5}$.

Total outcomes at a glance

Experiment	Total	Most asked
1 coin / 2 / 3 coins	2 / 4 / 8	at least one head
1 die / 2 dice	6 / 36	sum, doublet, product
Deck of cards	52	face, red, ace, king
Numbers 1 to n	n	prime, square, multiple
Days of week	7	53 Sundays

Last-minute checklist

- ✓ Total outcomes written? (36 for two dice, 52 for cards)
- ✓ Favourable listed, no repeats, no misses?
- ✓ 1 not prime, Ace not a face card
- ✓ Answer between 0 and 1, in lowest terms

Chapter in 4 lines

1. $P(E) = \frac{\text{fav}}{\text{total}}$, always between 0 and 1. 2. $P(E) + P(\bar{E}) = 1$. 3. Totals: coins 2^n, dice 6^n, cards 52. 4. List the favourable outcomes, never guess.

↳Board tip: Statistics + Probability unit = 11 marks. Probability ke sawal MCQ, 2-3 marks aur case-based mein aate hain; total aur favourable sahi gino, marks pakke.

Quick fact $P(E) + P(\bar{E}) = 1$ always.

Quick fact Primes 1 to 100: 25. And 1 is not prime.

Apne notes / galtiyan yahan likho

FORMULA SHEET All 14 chapters

Exam se pehle: har box ek baar padho, ek baar likh ke dekho

Saare Formulas, Ek Jagah

1. Real Numbers

- FTA: every composite = unique product of primes
- HCF = smallest powers of common primes; LCM = greatest powers of all primes
- $\text{HCF}(a,b) \times \text{LCM}(a,b) = a \times b$ (two numbers only)
- Greatest number dividing x, y with remainders r_1, r_2 = $\text{HCF}(x - r_1, y - r_2)$
- Smallest number leaving remainder r = LCM $+ r$
- p prime, $p \mid a^2 \Rightarrow p \mid a$; $\sqrt{2}, \sqrt{3}, \sqrt{5}$ irrational

2. Polynomials

- Quadratic $ax^2 + bx + c$: $\alpha + \beta = -\frac{b}{a}$, $\alpha\beta = \frac{c}{a}$
- Quadratic from zeroes: $k[x^2 - (\alpha+\beta)x + \alpha\beta]$
- Cubic: $\alpha + \beta + \gamma = -\frac{b}{a}$, $\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$, $\alpha\beta\gamma = -\frac{d}{a}$
- $\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta$; $(\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta$
- Zeroes = x-coordinates where $y = p(x)$ meets the x-axis; degree n: at most n zeroes

3. Pair of Linear Equations

- $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$
- $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$: intersecting lines, unique solution (consistent)
- $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$: coincident, infinitely many solutions
- $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$: parallel, no solution (inconsistent)
- Methods: graphical, substitution, elimination

4. Quadratic Equations

- Standard form $ax^2 + bx + c = 0$, $a \neq 0$
- $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, $D = b^2 - 4ac$
- $D > 0$: two distinct real roots; $D = 0$: two equal roots $-\frac{b}{2a}$; $D < 0$: no real roots
- Sum of roots $= -\frac{b}{a}$, product $= \frac{c}{a}$

5. Arithmetic Progressions

- $a_n = a + (n-1)d$; $d = a_2 - a_1$
- nth term from the end $= l - (n-1)d$
- $S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a + l)$
- $a_n = S_n - S_{n-1}$; three terms in AP: $a - d, a, a + d$
- a, b, c in AP $\Leftrightarrow 2b = a + c$; sum of first n natural numbers $= \frac{n(n+1)}{2}$

6. Triangles

- BPT: if $DE \parallel BC$ in $\triangle ABC$, then $\frac{AD}{DB} = \frac{AE}{EC}$ (converse also true)
- Similarity criteria: AA (AAA), SSS, SAS
- Similar triangles: corresponding angles equal, corresponding sides proportional
- All circles, squares, equilateral triangles are similar

7. Coordinate Geometry

- Distance: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$; from origin $\sqrt{x^2 + y^2}$
- Section formula $(m : n)$: $\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$
- Midpoint: $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$
- Ratio in which x-axis divides: use $y = 0$; y-axis: $x = 0$
- Collinear (3 points): $AB + BC = AC$

8. Introduction to Trigonometry

- $\sin A = \frac{P}{H}$, $\cos A = \frac{B}{H}$, $\tan A = \frac{P}{B}$; csc, sec, cot = reciprocals
- $\tan A = \frac{\sin A}{\cos A}$, $\cot A = \frac{\cos A}{\sin A}$
- $\sin^2 A + \cos^2 A = 1$; $1 + \tan^2 A = \sec^2 A$; $1 + \cot^2 A = \csc^2 A$

A	$0°$	$30°$	$45°$	$60°$	$90°$
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	not defined

Yaad karo sin row: $\frac{\sqrt{0}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{4}}{2}$; cos = same row ulta.

9. Some Applications of Trigonometry

- Angle of elevation: looking up from horizontal; depression: looking down
- Height = distance $\times \tan\theta$ (right triangle, θ at observer)
- $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 45^\circ = 1$, $\tan 60^\circ = \sqrt{3}$; $\sqrt{3} \approx 1.732$
- Angle of elevation from P to Q = angle of depression from Q to P (alternate angles)

10. Circles

- Tangent is perpendicular to the radius at the point of contact
- Lengths of tangents from an external point are equal
- Number of tangents: inside point 0, on circle 1, outside 2
- Tangent length $= \sqrt{d^2 - r^2}$ (d = distance of point from centre)

11. Areas Related to Circles

- Circumference $2\pi r$; area πr^2
- Arc length $= \frac{\theta}{360^\circ} \times 2\pi r$
- Area of sector $= \frac{\theta}{360^\circ} \times \pi r^2$
- Area of segment = area of sector − area of triangle
- Triangle area: $\theta = 60^\circ$: $\frac{\sqrt{3}}{4}r^2$; 90°: $\frac{1}{2}r^2$; 120°: $\frac{\sqrt{3}}{4}r^2$
- Minute hand: 6° per minute

Solid	CSA / LSA	TSA	Volume
Cuboid	$2h(l+b)$	$2(lb+bh+hl)$	lbh
Cube	$4a^2$	$6a^2$	a^3
Cylinder	$2\pi rh$	$2\pi r(r+h)$	$\pi r^2 h$
Cone	πrl	$\pi r(l+r)$	$\frac{1}{3}\pi r^2 h$
Sphere	$4\pi r^2$	$4\pi r^2$	$\frac{4}{3}\pi r^3$
Hemisphere	$2\pi r^2$	$3\pi r^2$	$\frac{2}{3}\pi r^3$

12. Surface Areas and Volumes

- Cone slant height $l = \sqrt{r^2 + h^2}$; cube diagonal $a\sqrt{3}$, cuboid diagonal $\sqrt{l^2 + b^2 + h^2}$
- Combined solid: add only the **outer visible** surfaces; volume = sum of volumes
- 1 litre $= 1000\ \text{cm}^3$; $1\ \text{m}^3 = 1000$ litres

13. Statistics

- Direct mean $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$
- Assumed mean $\bar{x} = a + \frac{\sum f_i d_i}{\sum f_i}$, $d_i = x_i - a$
- Step deviation $\bar{x} = a + \left(\frac{\sum f_i u_i}{\sum f_i}\right) h$, $u_i = \frac{x_i - a}{h}$
- Mode $= l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) h$
- Median $= l + \left(\frac{\frac{n}{2} - cf}{f}\right) h$
- Empirical: $3\,\text{Median} = \text{Mode} + 2\,\text{Mean}$

14. Probability

- $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$, $0 \le P(E) \le 1$
- $P(E) + P(\text{not } E) = 1$; sure event 1, impossible event 0
- Coin: 2 outcomes; die: 6; two dice: 36; pack of cards: 52 (26 red, 26 black, 12 face cards, 4 aces)
- Two coins: HH, HT, TH, TT (4 outcomes)

Algebra + values (sab jagah kaam aate hain)

- $(a+b)^2 = a^2 + 2ab + b^2$; $(a-b)^2 = a^2 - 2ab + b^2$
- $a^2 - b^2 = (a+b)(a-b)$
- $(a+b)^3 = a^3 + b^3 + 3ab(a+b)$; $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$
- $\sqrt{2} \approx 1.414$, $\sqrt{3} \approx 1.732$, $\sqrt{5} \approx 2.236$, $\pi = \frac{22}{7} \approx 3.14$
- Pythagoras (right triangle): $H^2 = P^2 + B^2$; triplets 3-4-5, 5-12-13, 8-15-17, 7-24-25

Units + conversions

- 1 m = 100 cm, 1 km = 1000 m, 1 hectare = 10000 m^2
- 1 cm^3 = 1 mL, 1000 cm^3 = 1 L, 1 m^3 = 1000 L
- Speed: 1 km/h $= \frac{5}{18}$ m/s; distance = speed $\times$ time
- Revolutions of a wheel = distance ÷ circumference

Kaunsa formula kab? (question pehchano)

Question mein dikhe	Formula / step
HCF / LCM word problem (tiles, rows, packs)	HCF = 'largest/maximum'; LCM = 'together again / least'
Zeroes given, polynomial banao	$x^2 - (S)x + P$
$\alpha^2 + \beta^2$, $\frac{1}{\alpha} + \frac{1}{\beta}$ type	$S^2 - 2P$, $\frac{S}{P}$

Question mein dikhe	Formula / step
Unique / no / infinite solutions	compare $\frac{a_1}{a_2}, \frac{b_1}{b_2}, \frac{c_1}{c_2}$
Equal roots / nature of roots	$D = b^2 - 4ac$
Which term is ...? / how many terms	$a_n = a + (n-1)d$, solve for n
Sum of first n terms / last term given	$S_n = \frac{n}{2}(2a + (n-1)d) = \frac{n}{2}(a + l)$
Line parallel to a side of triangle	BPT: $\frac{AD}{DB} = \frac{AE}{EC}$
Point dividing a segment / ratio	Section formula (ratio $k : 1$ if unknown)
Prove identity	convert to sin, cos; $\sin^2 + \cos^2 = 1$
Height / distance with angle	$\tan\theta = \frac{\text{height}}{\text{base}}$
Two tangents from a point	equal lengths; radius $\perp$ tangent
Segment area	sector $-$ triangle
Solid melted and recast	volume same: $V_1 = V_2$
Grouped data middle value	Median $= l + \frac{\frac{n}{2} - cf}{f} h$
At least one / not	$P(E) = 1 - P(\text{not } E)$
Angle in semicircle / right triangle check	$a^2 + b^2 = c^2$ (converse of Pythagoras)
Mean with big x_i	assumed mean or step deviation
Most frequent value (grouped)	Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} h$
Two-digit number / age problems	form linear or quadratic equation, check validity

Squares + cubes (calculation fast)

n	n^2	n	n^3
11	121	11	1331
12	144	12	1728
13	169	13	2197
14	196	14	2744
15	225	15	3375
16	256	16	4096
17	289	17	4913
18	324	18	5832
19	361	19	6859
20	400	20	8000

$\sqrt{2} \approx 1.414, \sqrt{3} \approx 1.732, \sqrt{5} \approx 2.236; \pi = \frac{22}{7} \approx 3.14$

Graph + lines at a glance

Graph cuts x-axis at	Zeroes
0 points	0
1 point (touches)	1
2 points	2
3 points (cubic)	3

Lines	Condition	Solutions
Intersecting	$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$	exactly 1
Parallel	$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$	none
Coincident	all three equal	infinitely many

Plane figures: area + perimeter

Figure	Area	Perimeter
Rectangle	lb	$2(l + b)$
Square	a^2	$4a$
Triangle	$\frac{1}{2} \times b \times h$	sum of sides
Equilateral	$\frac{\sqrt{3}}{4} a^2$	$3a$
Parallelogram	$b \times h$	$2(a + b)$
Rhombus	$\frac{1}{2} d_1 d_2$	$4a$
Trapezium	$\frac{1}{2}(a + b)h$	sum of sides
Circle	πr^2	$2\pi r$
Ring	$\pi(R^2 - r^2)$	$2\pi(R + r)$

Formula lagate waqt 6 traps

- → Diameter diya ho to pehle $r = \frac{d}{2}$.
- → Units same karo (cm aur m mix nahi).
- → Sum of zeroes mein minus: $-\frac{b}{a}$.
- → Section formula mein m_1 ke saath x_2.
- → Median mein cf pichli class ka.
- → Probability 1 se zyada aaye to galti pakki.

Theorems: answer mein naam likho

- → **FTA:** every composite number = unique product of primes.
- → **BPT (Thales)** and its converse; similarity by AA, SSS, SAS.
- → **Tangents:** tangent $\perp$ radius; tangents from an external point are equal.

Shivam Bhaiya says

Ye 3 pages roz subah 5 minute padho. Exam ke din formula yaad karne mein time nahi jaana chahiye, sirf lagane mein!

LAST-NIGHT REVISION

Real Numbers + Polynomials

1. HCF: smallest powers of common primes; LCM: greatest powers of all.
2. HCF x LCM = product only for 2 numbers.
3. Prove irrational: assume $\frac{a}{b}$ coprime, square, divide, contradiction.
4. $\alpha+\beta=-\frac{b}{a}$, $\alpha\beta=\frac{c}{a}$; polynomial $=k[x^2-Sx+P]$.
5. Zeroes on a graph = points on the x-axis (not y-axis).

Linear Equations + Quadratics

1. Compare $\frac{a_1}{a_2}$, $\frac{b_1}{b_2}$, $\frac{c_1}{c_2}$: unique / infinite / no solution.
2. Graph: intersecting (1), coincident (infinite), parallel (none).
3. Quadratic formula: $x=\frac{-b\pm\sqrt{D}}{2a}$, $D=b^2-4ac$.
4. Equal roots: $D=0$; real roots: $D\geq 0$; no real roots: $D<0$.
5. Word problems: define the variable, form the equation, reject negative/absurd roots.

AP

1. $a_n=a+(n-1)d$; $S_n=\frac{n}{2}[2a+(n-1)d]$.
2. $a_n=S_n-S_{n-1}$; d can be negative or a fraction.
3. Three terms in AP: $a-d, a, a+d$.
4. Is 0 / 301 a term? Solve a_n = value: n must be a natural number.

Triangles + Circles

1. BPT: line parallel to one side divides the other two in the same ratio.
2. Criteria: AA, SSS, SAS (no SSA!). Order of letters matters in $\triangle ABC\sim\triangle PQR$.
3. Tangent $\perp$ radius; tangents from an external point are equal.
4. Quadrilateral circumscribing a circle: $AB+CD=AD+BC$.
5. Angle between two tangents + angle at centre = 180°.

Yaad rakho $\tan\theta=\frac{\sin\theta}{\cos\theta}$; never divide by $\cos 90^\circ=0$.

Coordinate Geometry

1. Distance formula: $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$.
2. Section: $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$; midpoint $m=n=1$.
3. Point on x-axis: $(x,0)$; on y-axis: $(0,y)$.
4. Parallelogram: diagonals bisect each other (same midpoint).

Trigonometry + Heights

1. $\sin^2 A+\cos^2 A=1$, $\sec^2 A-\tan^2 A=1$, $\csc^2 A-\cot^2 A=1$.
2. Values: sin row $0, \frac{1}{2}, \frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{2}, 1$; cos is the reverse.
3. $\tan 90^\circ$ not defined; $\sin A, \cos A\leq 1$.
4. Heights: draw the figure, mark the right angle, use $\tan\theta=\frac{\text{height}}{\text{base}}$.
5. Elevation from A = depression from B (alternate angles).

Areas + Surface Areas, Volumes

1. Sector $=\frac{\theta}{360}\pi r^2$, arc $=\frac{\theta}{360}2\pi r$.
2. Segment = sector − triangle (angles 60, 90, 120 only).
3. Combined solid: TSA = only outer visible surfaces; volume = sum.
4. Cone: $l=\sqrt{r^2+h^2}$; hemisphere CSA $2\pi r^2$, TSA $3\pi r^2$.
5. Units: convert all lengths to the same unit before calculating.

Statistics + Probability

1. Modal class = class with the highest frequency.
2. Median class: the class where cumulative frequency first reaches $\frac{n}{2}$.
3. Empirical: Mode $=3$ Median -2 Mean.
4. $P(E)+P(\bar{E})=1$; $0\leq P(E)\leq 1$.
5. Two dice: 36 outcomes; cards: 52, face cards 12, aces 4.

Quick check Probability answers always lie between 0 and 1.

Units Area in square units, volume in cubic units.

Pakka aane wale sawal (chapter-wise)

Chapter	Most expected
Real Numbers	Prove $\sqrt{p}$ irrational; HCF/LCM word problem
Polynomials	Find zeroes and verify; form quadratic from S, P
Linear Equations	Word problem by elimination/substitution; consistency conditions
Quadratic Equations	Nature of roots (find k); speed/age/area word problem
AP	S_n and a_n word problem; find n for given sum
Triangles	Prove BPT; use similarity to find a length/shadow
Coordinate Geometry	Section formula ratio; point equidistant from two points
Trigonometry	Prove an identity; evaluate using standard values
Heights and Distances	Two angles of elevation (tower/building), 5 marks
Circles	Prove tangents from a point are equal; tangent length
Areas Related to Circles	Area of segment/sector; minute-hand area
Surface Areas and Volumes	Combined solid TSA/volume (toy, tent, capsule)
Statistics	Mean + mode or median from a grouped table
Probability	Cards / two dice / numbered discs

2-minute self test (answers in brackets)

1. $\text{HCF}(12, 18) \times \text{LCM}(12, 18)$ = ? (216)
2. Discriminant of $x^2 - 4x + 4$ = ? (0, equal roots)
3. 10th term of 3, 7, 11, ... = ? (39)
4. Distance of $(3, 4)$ from origin = ? (5)
5. $\sin 30^\circ + \cos 60^\circ$ = ? (1)
6. Volume of a cube of edge 3 cm = ? (27 cm^3)
7. Probability of a prime on a die = ? ($\frac{1}{2}$)
8. Mode of 2, 3, 3, 5, 3, 7 = ? (3)

Recall S_n of first n odd numbers $= n^2$.

Trap Diameter given? Halve it first: $r = \frac{d}{2}$.

Recall $\sin$ increases, $\cos$ decreases from 0° to 90°.

Top 10 board traps (inko mat bhoolna)

1. Sum of zeroes is $-\frac{b}{a}$, not $\frac{b}{a}$.
2. Writing units (cm^2, cm^3, m) in final answers.
3. Using diameter in place of radius.
4. Taking $\sqrt{3} = 1.73$ when the question says 1.732.
5. Not rejecting the negative root in quadratic word problems.
6. In mode formula, f_0 = class before and f_2 = class after the modal class.
7. Median: use cf of the class **before** the median class.
8. Similarity: matching the wrong vertices.
9. Probability greater than 1 (always recheck).
10. Forgetting "coprime" in irrationality proofs.

Theorems: statement word-to-word likho

1. **FTA:** every composite number can be expressed as a product of primes, uniquely apart from order.
2. **BPT:** if a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio.
3. **Converse of BPT:** if a line divides any two sides of a triangle in the same ratio, it is parallel to the third side.
4. **Tangent theorem:** the tangent at any point of a circle is perpendicular to the radius through the point of contact.
5. **Equal tangents:** the lengths of tangents drawn from an external point to a circle are equal.

Exam day strategy (3 hours, 80 marks)

- → First 15 min reading time: mark easy questions, plan the order.
- → Section A (MCQ + AR) in about 20-25 min; never leave an MCQ blank.
- → Do sure-shot 2 and 3 markers next, then case-based, then 5 markers.
- → Draw neat figures in pencil for geometry, trigonometry and mensuration.
- → Keep the last 10 minutes to check calculations and units.

Rapid Fire: 56 one-mark checks

Answer column dhak ke 10 min mein karo. 50+ sahi = ready!

Question	Answer
1. $\text{HCF}(18, 24)$	6
2. $\text{LCM}(18, 24)$	72
3. Is $\sqrt{9}$ irrational?	No, $= 3$

Question	Answer
4. Zeroes of $x^2 - 9$	± 3
5. Sum of zeroes of $2x^2 - 8x + 5$	4
6. Product of zeroes of $x^2 + x - 6$	-6
7. $x + y = 5, x - y = 1$: x, y	3, 2
8. Lines $x + 2y = 3, 2x + 4y = 6$ are	coincident
9. Pair with no solution: ratio test	$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$
10. Discriminant of $x^2 + 4x + 4$	0 (equal roots)
11. Roots of $x^2 - 5x + 6 = 0$	2, 3
12. Nature of roots of $x^2 + x + 1 = 0$	no real roots
13. 10th term of 2, 5, 8, ...	29
14. Sum of 1st 10 odd numbers	100

Question	Answer
15. d of $\sqrt{2}, \sqrt{8}, \sqrt{18}$	$\sqrt{2}$
16. Ratio of areas, similar triangles, sides 2 : 3	4 : 9
17. Sides 6, 8, 10: right-angled?	Yes, $36 + 64 = 100$
18. BPT also called	Thales theorem
19. Distance $(0, 0)$ to $(5, 12)$	13
20. Midpoint of $(2, 4), (6, 8)$	$(4, 6)$
21. Point on y-axis has	$x = 0$
22. $\sin 30^\circ + \cos 60^\circ$	1
23. $\tan 45^\circ \times \cot 45^\circ$	1
24. $\sec^2 \theta - \tan^2 \theta$	1
25. Shadow = height: sun's altitude	45°
26. $\tan 60^\circ$	$\sqrt{3} \approx 1.732$
27. Tangents from an external point	equal
28. Angle between tangent and radius	90°

Question	Answer
29. Tangent length, $d = 5, r = 3$	4
30. Circumference, $r = 7$ cm	44 cm
31. Area of quadrant, $r = 14$ cm	154 cm^2
32. Sector area, $\theta = 90^\circ, r = 7$	38.5 cm^2
33. Volume of cone, $r = 3, h = 7$	66 cm^3
34. CSA of cylinder, $r = 7, h = 10$	440 cm^2
35. Volume of hemisphere, $r = 21$	19404 cm^3
36. Mean of 2, 4, 6, 8, 10	6
37. Empirical relation	3 Median = Mode + 2 Mean
38. Class mark of 10–20	15

39. $P(E) + P(\text{not } E)$	1
40. P(king) from 52 cards	$\frac{1}{13}$
41. P(sum 7) with two dice	$\frac{1}{6}$

Question	Answer
42. P(impossible event)	0

Question	Answer
43. LCM $\times$ HCF of 12, 15	180
44. $\sqrt{5}$ is	irrational
45. Graph of quadratic is a	parabola
46. Max zeroes of a cubic	3
47. $2x + 3y = 5$ type eqn is	linear in two variables
48. Quadratic formula	$x = \frac{-b \pm \sqrt{D}}{2a}$
49. a_n of AP	$a + (n-1)d$
50. S_n of 3, 7, 11 for $n = 5$	55
51. Similarity criteria (3)	AA, SSS, SAS
52. Section formula x	$\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}$
53. $\cos 0^\circ$	1
54. $\sin^2 A + \cos^2 A$	1
55. Area of circle, $r = 7$ cm	154 cm^2
56. Mode formula $l + \ldots$	$\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$

Board ke pakke proofs (steps yaad karo)

1. **$\sqrt{2}$ irrational:** assume $\frac{a}{b}$ (coprime) $\Rightarrow 2b^2 = a^2 \Rightarrow 2$ divides $a \Rightarrow 2$ divides b: contradiction.
2. **BPT:** join BE, CD; use ar(ADE)/ar(BDE) $= \frac{AD}{DB}$ and ar(BDE) = ar(CED) (same base DE, between same parallels).
3. **Tangent $\perp$ radius:** every other point of the tangent lies outside, so OP is the shortest distance.
4. **Two tangents equal:** $\triangle OPA \cong \triangle OPB$ (RHS) $\Rightarrow PA = PB$.
5. **S_n of AP:** write S_n forwards + backwards and add: $2S_n = n(2a + (n-1)d)$.
6. **Identity proofs:** convert to sin, cos; take LCM; use $\sin^2 + \cos^2 = 1$; work on the heavier side.

Shivam Bhaiya says
Raat ko naya chapter mat kholo. Sirf ye revision pages, formula sheet aur apni galtiyon wali list dekho. Achhi neend lo, subah fresh dimaag se paper do. All the best!

Trap Volume units are cubed: cm^3, never cm^2.

PRACTICE PAPER 1 CBSE board pattern, with answers

Time: 3 hours | Max marks: 80
38 questions, 5 sections

Mock Paper 1

Instructions

- Section A: 20 MCQ/AR (1 mark each). Section B: 5 VSA (2 marks). Section C: 6 SA (3 marks). Section D: 4 LA (5 marks). Section E: 3 case-based (4 marks).
- Pehle khud solve karo with a timer, phir green answers se milao. Use $\pi = \frac{22}{7}$, $\sqrt{3} = 1.732$ where needed.

Section A: MCQ + Assertion-Reason (1 x 20)

Q1. LCM of $2^3 \times 3^2$ and $2^2 \times 3^3$ is 1M
(a) 72 (b) 108 (c) 216 (d) 36
Ans. (c) $2^3 \times 3^3 = 216$

Q2. If 2 is a zero of $x^2 + 3x + k$, then $k =$ 1M
(a) -10 (b) 10 (c) 5 (d) -5
Ans. (a) $4 + 6 + k = 0 \Rightarrow k = -10$

Q3. The pair $2x + 3y = 5$, $4x + 6y = 15$ has 1M
(a) one solution (b) two solutions (c) no solution (d) infinitely many
Ans. (c) $\frac{1}{2} = \frac{1}{2} \neq \frac{1}{3}$: parallel lines, no solution.

Q4. The equation $2x^2 - 4x + 3 = 0$ has 1M
(a) two distinct real roots (b) two equal roots (c) one root (d) no real roots
Ans. (d) $D = 16 - 24 = -8 < 0$

Q5. The 11th term of the AP $-3, -\frac{1}{2}, 2, \ldots$ is 1M
(a) 28 (b) 22 (c) -38 (d) $-48\frac{1}{2}$
Ans. (b) $d = \frac{5}{2}$, $a_{11} = -3 + 10 \times \frac{5}{2} = 22$

Q6. In $\triangle ABC$, $DE \parallel BC$ with $AD = 2$ cm, $DB = 3$ cm, $AE = 4$ cm. Then $EC =$ 1M
(a) 4 cm (b) 5 cm (c) 6 cm (d) 8 cm
Ans. (c) $\frac{2}{3} = \frac{4}{EC} \Rightarrow EC = 6$ cm

Q7. The distance between $(2, 3)$ and $(4, 1)$ is 1M
(a) 2 (b) $2\sqrt{2}$ (c) 4 (d) $\sqrt{10}$
Ans. (b) $\sqrt{4 + 4} = 2\sqrt{2}$

Q8. The midpoint of the segment joining $(-1, 7)$ and $(3, -3)$ is 1M
(a) $(1, 2)$ (b) $(2, 1)$ (c) $(1, 5)$ (d) $(2, 5)$
Ans. (a) $(1, 2)$

Q9. If $\sin A = \frac{3}{5}$ (A acute), then $\tan A =$ 1M
(a) $\frac{3}{4}$ (b) $\frac{4}{3}$ (c) $\frac{4}{5}$ (d) $\frac{5}{3}$
Ans. (a) $\cos A = \frac{4}{5}$, $\tan A = \frac{3}{4}$

Q10. $(\sec^2\theta - 1)(1 - \sin^2\theta) =$ 1M
(a) $\cos^2\theta$ (b) $\sin^2\theta$ (c) 1 (d) 0
Ans. (b) $\tan^2\theta \cos^2\theta = \sin^2\theta$

Q11. A ladder makes 60° with the ground and its foot is 2.5 m from the wall. Its length is 1M
(a) 2.5 m (b) 4 m (c) 5 m (d) $2.5\sqrt{3}$ m
Ans. (c) $\frac{2.5}{\cos 60^\circ} = 5$ m

Q12. PA, PB are tangents from P to a circle with centre O. If $\angle APB = 80^\circ$, then $\angle AOB =$ 1M
(a) 80° (b) 90° (c) 100° (d) 50°
Ans. (c) $180^\circ - 80^\circ = 100^\circ$

Q13. Area of a quadrant (sector of 90°) of a circle of radius 7 cm is 1M
(a) 154 cm^2 (b) 38.5 cm^2 (c) 77 cm^2 (d) 11 cm^2
Ans. (b) $\frac{1}{4} \times \frac{22}{7} \times 49 = 38.5$ cm^2

Q14. Volume of a hemisphere of radius 3 cm is 1M
(a) 18π cm^3 (b) 36π cm^3 (c) 9π cm^3 (d) 27π cm^3
Ans. (a) $\frac{2}{3}\pi(27) = 18\pi$ cm^3

Q15. A cylinder and a cone have the same radius and height. Ratio of their volumes is 1M
(a) 1 : 3 (b) 2 : 1 (c) 3 : 1 (d) 1 : 1
Ans. (c) $\pi r^2 h : \frac{1}{3}\pi r^2 h = 3 : 1$

Q16. If median = 22 and mean = 20, then mode (empirical relation) = 1M
(a) 24 (b) 21 (c) 18 (d) 26
Ans. (d) $3(22) - 2(20) = 26$

Q17. Two dice are thrown. P(sum = 7) is 1M
(a) $\frac{7}{36}$ (b) $\frac{1}{6}$ (c) $\frac{1}{12}$ (d) $\frac{5}{36}$
Ans. (b) 6 favourable out of 36: $\frac{1}{6}$

Q18. A card is drawn from a pack of 52. P(red king) = 1M
(a) $\frac{1}{26}$ (b) $\frac{1}{13}$ (c) $\frac{1}{52}$ (d) $\frac{2}{13}$
Ans. (a) $\frac{2}{52} = \frac{1}{26}$

Q19. **A:** $2 + \sqrt{3}$ is irrational. **R:** The sum of a rational and an irrational number is irrational. 1M
(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true
Ans. (a)

Q20. **A:** The point $(0, 4)$ lies on the y-axis. **R:** The x-coordinate of every point on the y-axis is 0. 1M
(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true
Ans. (a)

Section B: Very short answer (2 x 5)

Q21. Find the HCF and LCM of 96 and 404 by prime factorisation. 2M
Ans. $96 = 2^5 \times 3$, $404 = 2^2 \times 101$. HCF $= 4$, LCM $= 2^5 \times 3 \times 101 = 9696$.

Q22. Solve by elimination: $2x + 3y = 11$, $2x - 4y = -24$. 2M

(contd.) Subtract: $7y = 35 \Rightarrow y = 5; 2x = 11 - 15 \Rightarrow x = -2$.

Q23. Find the ratio in which the y-axis divides the segment joining $(5, -6)$ and $(-1, -4)$. Also find the point. 2M

Ans. Let ratio $k : 1$: $\frac{-k+5}{k+1} = 0 \Rightarrow k = 5$, ratio $5 : 1$. Point $\left(0, \frac{5(-4)+1(-6)}{6}\right) = \left(0, -\frac{13}{3}\right)$.

Q24. Evaluate $2\tan^2 45° + \cos^2 30° - \sin^2 60°$. 2M

Ans. $2(1) + \frac{3}{4} - \frac{3}{4} = 2$.

Q25. Two concentric circles have radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle. 2M

Ans. Radius to the point of contact $\perp$ chord and bisects it: half chord $= \sqrt{25 - 9} = 4$, chord $= 8$ cm.

Section C: Short answer (3 x 6)

Q26. Given that $\sqrt{5}$ is irrational, prove that $3 + 2\sqrt{5}$ is irrational. 3M

Ans. If $3 + 2\sqrt{5} = \frac{a}{b}$ (a, b integers, $b \neq 0$), then $\sqrt{5} = \frac{a-3b}{2b}$ is rational: contradiction. So $3 + 2\sqrt{5}$ is irrational.

Q27. Find the zeroes of $5x^2 - 8x - 4$ and verify the relationship between zeroes and coefficients. 3M

Ans. $5x^2 - 10x + 2x - 4 = (5x + 2)(x - 2)$: zeroes $2, -\frac{2}{5}$. Sum $= \frac{8}{5} = -\frac{-8}{5}$; product $= -\frac{4}{5} = \frac{-4}{5}$ ✔

Q28. A fraction becomes $\frac{9}{11}$ if 2 is added to both numerator and denominator, and $\frac{5}{6}$ if 3 is added to both. Find the fraction. 3M

Ans. $\frac{x+2}{y+2} = \frac{9}{11} \Rightarrow 11x - 9y = -4$; $\frac{x+3}{y+3} = \frac{5}{6} \Rightarrow 6x - 5y = -3$. Solving: $x = 7, y = 9$. Fraction $\frac{7}{9}$.

Q29. Prove that $\frac{1+\tan^2 A}{1+\cot^2 A} = \tan^2 A$. 3M

Ans. LHS $= \frac{\sec^2 A}{\csc^2 A} = \frac{1/\cos^2 A}{1/\sin^2 A} = \frac{\sin^2 A}{\cos^2 A} = \tan^2 A =$ RHS.

Q30. Prove that the lengths of tangents drawn from an external point to a circle are equal. 3M

Ans. Let PQ, PR be tangents from P to a circle with centre O. Join OQ, OR, OP. In $\triangle OQP$ and $\triangle ORP$: $OQ = OR$ (radii), OP common, $\angle OQP = \angle ORP = 90°$. So $\triangle OQP \cong \triangle ORP$ (RHS), hence $PQ = PR$.

Q31. Find the mean: classes 0-10, 10-20, 20-30, 30-40, 40-50 with frequencies 7, 8, 12, 13, 10. 3M

Ans. x_i: 5, 15, 25, 35, 45; $\sum f_i = 50$, $\sum f_i x_i = 35 + 120 + 300 + 455 + 450 = 1360$. Mean $= \frac{1360}{50} = 27.2$.

Section D: Long answer (5 x 4)

Q32. A train travels 360 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 1 hour less. Find the speed of the train. 5M

Ans. $\frac{360}{x} - \frac{360}{x+5} = 1 \Rightarrow 1800 = x(x + 5) \Rightarrow x^2 + 5x - 1800 = 0 \Rightarrow (x + 45)(x - 40) = 0$. Speed $= 40$ km/h (reject -45).

Q33. From the top of a 10 m high building, the angle of elevation of the top of a tower is $60°$ and the angle of depression of its foot is $45°$. Find the height of the tower. ($\sqrt{3} = 1.732$) 5M

Ans. Distance $= 10\cot 45° = 10$ m. Extra height above building $= 10\tan 60° = 10\sqrt{3}$. Tower $= 10 + 10\sqrt{3} = 10(1 + \sqrt{3}) = 27.32$ m.

Q34. The sum of the first 7 terms of an AP is 49 and that of the first 17 terms is 289. Find the sum of its first n terms. 5M

Ans. $\frac{7}{2}(2a + 6d) = 49 \Rightarrow a + 3d = 7$; $\frac{17}{2}(2a + 16d) = 289 \Rightarrow a + 8d = 17$. So $d = 2, a = 1$. $S_n = \frac{n}{2}[2 + 2(n - 1)] = n^2$.

Q35. A toy is a cone of radius 3.5 cm mounted on a hemisphere of the same radius. The total height of the toy is 15.5 cm. Find its total surface area. 5M

Ans. Cone height $= 15.5 - 3.5 = 12$, $l = \sqrt{144 + 12.25} = 12.5$ cm. CSA cone $= \frac{22}{7} \times 3.5 \times 12.5 = 137.5$; CSA hemisphere $= 2 \times \frac{22}{7} \times 12.25 = 77$. TSA $= 214.5$ cm^2.

Section E: Case based (4 x 3)

Q36. Piggy bank. Riya saves Rs 50 in the first week and increases her saving by Rs 20 every week. (i) How much does she save in the 10th week? (1) (ii) Total saved in 10 weeks? (2) (iii) In which week will she save Rs 410? (1) 4M

Ans. (i) $50 + 9(20) =$ Rs 230. (ii) $\frac{10}{2}(50 + 230) =$ Rs 1400. (iii) $50 + (n - 1)20 = 410 \Rightarrow n =$ 19th week.

Q37. Classroom desks. Desks are at A(3, 1), B(6, 4) and C(8, 6). (i) Find AB. (1) (ii) Show that A, B, C are collinear. (2) (iii) Find the midpoint of AC. (1) 4M

Ans. (i) $\sqrt{9 + 9} = 3\sqrt{2}$. (ii) $BC = \sqrt{4 + 4} = 2\sqrt{2}$, $AC = \sqrt{25 + 25} = 5\sqrt{2} = AB + BC$, so collinear. (iii) $(5.5, 3.5)$.

Q38. Lucky draw. A box has 20 discs numbered 1 to 20. One disc is drawn at random. Find P(i) an even number (1) (ii) a multiple of 3 (1) (iii) a prime number (2). 4M

Ans. (i) $\frac{10}{20} = \frac{1}{2}$ (ii) 3, 6, 9, 12, 15, 18: $\frac{6}{20} = \frac{3}{10}$ (iii) 2, 3, 5, 7, 11, 13, 17, 19: $\frac{8}{20} = \frac{2}{5}$.

Score yourself (Paper 1)

Section	Max	My marks	Time taken
A (MCQ + AR)	20		25 min
B (2 marks)	10		15 min
C (3 marks)	18		35 min
D (5 marks)	20		45 min
E (case based)	12		25 min
Total	**80**		2 h 25 min + checking

PRACTICE PAPER 2 CBSE board pattern, with answers

Time: 3 hours | Max marks: 80
timer laga ke solve karo

Mock Paper 2

Section A: MCQ + Assertion-Reason (1 x 20)

Q1. If HCF$(a,b)=12$ and $a\times b=1800$, then LCM $(a,b)=$ 1M
(a) 150 (b) 1800 (c) 900 (d) 120
Ans. (a) $\frac{1800}{12}=150$

Q2. A quadratic polynomial with zeroes 5 and -5 is 1M
(a) x^2+25 (b) x^2-10x (c) x^2-25 (d) x^2+5x
Ans. (c) sum 0, product -25

Q3. $kx+2y=5$ and $3x+y=1$ have a unique solution if 1M
(a) $k=6$ (b) $k\neq 6$ (c) $k=-6$ (d) $k\neq -6$
Ans. (b) $\frac{k}{3}\neq\frac{2}{1}\Rightarrow k\neq 6$

Q4. The roots of $x^2-3x=0$ are 1M
(a) 0, 3 (b) 3, -3 (c) 0, -3 (d) 1, 3
Ans. (a) $x(x-3)=0$

Q5. The common difference of the AP $\frac{1}{p},\frac{1-p}{p},\frac{1-2p}{p},\ldots$ is 1M
(a) p (b) $-p$ (c) -1 (d) 1
Ans. (c) $\frac{1-p}{p}-\frac{1}{p}=-1$

Q6. The sum of the first 20 natural numbers is 1M
(a) 200 (b) 210 (c) 220 (d) 420
Ans. (b) $\frac{20\times 21}{2}=210$

Q7. $\triangle ABC\sim\triangle DEF$ with $AB=3$ cm, $DE=6$ cm, $BC=4$ cm. Then $EF=$ 1M
(a) 2 cm (b) 4 cm (c) 6 cm (d) 8 cm
Ans. (d) $\frac{3}{6}=\frac{4}{EF}\Rightarrow EF=8$ cm

Q8. The distance of $(-6,8)$ from the origin is 1M
(a) 8 (b) 6 (c) 10 (d) 14
Ans. (c) $\sqrt{36+64}=10$

Q9. The point dividing the join of $(0,0)$ and $(6,9)$ in the ratio $1:2$ is 1M
(a) $(2,3)$ (b) $(4,6)$ (c) $(3,4.5)$ (d) $(2,6)$
Ans. (a) $\left(\frac{6}{3},\frac{9}{3}\right)=(2,3)$

Q10. $\sin 60^\circ\cos 30^\circ+\sin 30^\circ\cos 60^\circ=$ 1M
(a) 0 (b) $\frac{1}{2}$ (c) $\frac{\sqrt{3}}{2}$ (d) 1
Ans. (d) $\frac{3}{4}+\frac{1}{4}=1$

Q11. If $\tan\theta=1$ (θ acute), then $\sin^2\theta-\cos^2\theta=$ 1M
(a) 0 (b) 1 (c) $\frac{1}{2}$ (d) -1
Ans. (a) $\theta=45^\circ$: $\frac{1}{2}-\frac{1}{2}=0$

Q12. A 30 m tower casts a shadow $10\sqrt{3}$ m long. The sun's altitude is 1M
(a) 30° (b) 45° (c) 60° (d) 90°
Ans. (c) $\tan\theta=\frac{30}{10\sqrt{3}}=\sqrt{3},\theta=60^\circ$

Q13. The length of the tangent from a point 13 cm from the centre of a circle of radius 5 cm is 1M
(a) 8 cm (b) 12 cm (c) 18 cm (d) 10 cm
Ans. (b) $\sqrt{169-25}=12$ cm

Q14. The perimeter of a semicircular protractor of radius 7 cm is 1M
(a) 22 cm (b) 29 cm (c) 36 cm (d) 44 cm
Ans. (c) $\pi r+2r=22+14=36$ cm

Q15. Area of a sector of angle 60° in a circle of radius 6 cm is 1M
(a) 3π cm^2 (b) 6π cm^2 (c) 12π cm^2 (d) 36π cm^2
Ans. (b) $\frac{60}{360}\pi(36)=6\pi$ cm^2

Q16. Surface area of a sphere of radius 7 cm is 1M
(a) 154 cm^2 (b) 308 cm^2 (c) 462 cm^2 (d) 616 cm^2
Ans. (d) $4\times\frac{22}{7}\times 49=616$ cm^2

Q17. The mean of $x,x+2,x+4,x+6,x+8$ is 11. Then $x=$ 1M
(a) 7 (b) 9 (c) 11 (d) 5
Ans. (a) $\frac{5x+20}{5}=11\Rightarrow x=7$

Q18. The probability that a leap year has 53 Sundays is 1M
(a) $\frac{1}{7}$ (b) $\frac{2}{7}$ (c) $\frac{3}{7}$ (d) $\frac{5}{7}$
Ans. (b) 366 days = 52 weeks + 2 days; 2 of the 7 day-pairs contain a Sunday: $\frac{2}{7}$

Q19. **A:** The equation $x^2+x+1=0$ has no real roots. **R:** If $D=b^2-4ac<0$, a quadratic equation has no real roots. 1M
(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true
Ans. (a) $D=1-4=-3<0$

Q20. **A:** The probability of an event can be 1.2. **R:** For any event E, $0\le P(E)\le 1$. 1M
(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true
Ans. (d) A is false, R is true.

Section B: Very short answer (2 x 5)

Q21. Find the HCF and LCM of 26 and 91 and verify HCF x LCM = product. 2M
Ans. $26=2\times 13, 91=7\times 13$. HCF $=13$, LCM $=182$. $13\times 182=2366=26\times 91$ ✔

Q22. Find k so that $x^2+kx+9=0$ has equal roots. 2M
Ans. $D=k^2-36=0\Rightarrow k=\pm 6$.

Q23. Find the point on the y-axis equidistant from $(-5,-2)$ and $(3,2)$. 2M
Ans. Let $(0,y)$: $25+(y+2)^2=9+(y-2)^2\Rightarrow 8y=-16\Rightarrow y=-2$. Point $(0,-2)$.

Q24. If $\tan(A+B)=\sqrt{3}$ and $\tan(A-B)=\frac{1}{\sqrt{3}}$ ($0^\circ < A+B \le 90^\circ$, $A>B$), find A and B. 2M

> **Ans.** $A+B=60^\circ$, $A-B=30^\circ$, so $A=45^\circ$, $B=15^\circ$.

Q25. A wheel has diameter 70 cm. How many revolutions will it make to cover 1.1 km? 2M

> **Ans.** Circumference $=\frac{22}{7}\times 70=220$ cm. Revolutions $=\frac{110000}{220}=500$.

Section C: Short answer (3 x 6)

Q26. Find two consecutive positive integers whose product is 306. 3M

> **Ans.** $x(x+1)=306\Rightarrow x^2+x-306=0\Rightarrow (x+18)(x-17)=0$. $x=17$: integers 17 and 18.

Q27. Which term of the AP 3, 15, 27, 39, ... will be 132 more than its 54th term? 3M

> **Ans.** $d=12$, $a_{54}=3+53(12)=639$. $3+(n-1)12=639+132=771\Rightarrow n-1=64$, $n=65$th term.

Q28. In $\triangle ABC$, $DE \parallel BC$ with D on AB and E on AC. If $\frac{AD}{DB}=\frac{3}{4}$ and $AC=21$ cm, find AE. 3M

> **Ans.** By BPT $\frac{AE}{EC}=\frac{3}{4}$. So $AE=\frac{3}{7}\times 21=9$ cm.

Q29. Prove that $\sqrt{\frac{1+\sin A}{1-\sin A}}=\sec A+\tan A$. 3M

> **Ans.** Multiply inside by $\frac{1+\sin A}{1+\sin A}$: $\sqrt{\frac{(1+\sin A)^2}{1-\sin^2 A}}=\frac{1+\sin A}{\cos A}=\sec A+\tan A$.

Q30. Find the area of the minor segment of a circle of radius 14 cm when the central angle is 60°. ($\pi=\frac{22}{7}$, $\sqrt{3}=1.73$) 3M

> **Ans.** Sector $=\frac{60}{360}\times\frac{22}{7}\times 196=102.67$ cm^2. Triangle (equilateral) $=\frac{\sqrt{3}}{4}\times 196=49\sqrt{3}=84.77$. Segment ≈ 17.90 cm^2.

Q31. Find the median: classes 0-10, 10-20, 20-30, 30-40, 40-50, 50-60 with frequencies 5, 8, 20, 15, 7, 5. 3M

> **Ans.** $n=60$, $\frac{n}{2}=30$; cf: 5, 13, 33, ... so median class 20-30: $l=20, cf=13, f=20, h=10$. Median $=20+\frac{30-13}{20}\times 10=28.5$.

Section D: Long answer (5 x 4)

Q32. 5 pencils and 7 pens together cost Rs 50, and 7 pencils and 5 pens cost Rs 46. Find the cost of one pencil and one pen. 5M

> **Ans.** $5x+7y=50$, $7x+5y=46$. Add: $12(x+y)=96\Rightarrow x+y=8$. Subtract: $2y-2x=4\Rightarrow y-x=2$. So $y=5$, $x=3$: pencil Rs 3, pen Rs 5.

Apne notes / galtiyan yahan likho

Q33. The angle of elevation of the top of a tower from a point on the ground is 30°. On walking 40 m towards the tower, it becomes 60°. Find the height of the tower. 5M

> **Ans.** Let height h, near distance x. $h=x\sqrt{3}$ and $h=\frac{x+40}{\sqrt{3}}$. So $3x=x+40\Rightarrow x=20$, $h=20\sqrt{3}\approx 34.64$ m.

Q34. Prove that the tangent at any point of a circle is perpendicular to the radius through the point of contact. Hence, if a quadrilateral ABCD circumscribes a circle with $AB=6$, $BC=7$, $CD=4$ cm, find AD. 5M

> **Ans.** Let tangent XY touch the circle (centre O) at P. Every point Q on XY other than P lies outside the circle, so $OQ>OP$. Thus OP is the shortest distance from O to XY, so $OP\perp XY$. For ABCD (equal tangents): $AB+CD=AD+BC\Rightarrow AD=6+4-7=3$ cm.

Q35. A capsule is a cylinder with two hemispherical ends. Its total length is 14 mm and diameter 5 mm. Find its surface area. 5M

> **Ans.** $r=2.5$ mm, cylinder length $=14-5=9$ mm. CSA cylinder $=2\pi(2.5)(9)=45\pi$; two hemispheres $=4\pi(2.5)^2=25\pi$. Total $=70\pi=70\times\frac{22}{7}=220$ mm^2.

Section E: Case based (4 x 3)

Q36. Society park. A rectangular park is 3 m longer than it is broad and its area is 180 m^2. (i) Form the quadratic equation. (1) (ii) Find its length and breadth. (2) (iii) Find its perimeter. (1) 4M

> **Ans.** (i) $x(x+3)=180\Rightarrow x^2+3x-180=0$. (ii) $(x+15)(x-12)=0$, breadth 12 m, length 15 m. (iii) $2(15+12)=54$ m.

Q37. Daily wages. Wages (Rs) 100-120, 120-140, 140-160, 160-180, 180-200 with workers 12, 14, 8, 6, 10. (i) Modal class? (1) (ii) Find the mode. (2) (iii) Class mark of the modal class? (1) 4M

> **Ans.** (i) 120-140. (ii) $l=120, f_1=14, f_0=12, f_2=8, h=20$: mode $=120+\frac{14-12}{28-12-8}\times 20=125$. (iii) 130.

Q38. Ludo night. Two dice are thrown together. Find the probability of (i) a doublet (1) (ii) sum 10 (1) (iii) product 6 (2). 4M

> **Ans.** 36 outcomes. (i) $\frac{6}{36}=\frac{1}{6}$ (ii) (4,6), (5,5), (6,4): $\frac{3}{36}=\frac{1}{12}$ (iii) (1,6), (6,1), (2,3), (3,2): $\frac{4}{36}=\frac{1}{9}$.

Galti analysis (Paper 2 ke baad)

Galti ka type	Fix
Formula bhool gaya	Formula Sheet roz 5 min
Calculation mistake	Har step likho, last 10 min check
Sign error ($-\frac{b}{a}$ etc.)	Common mistakes boxes dobara padho
Figure galat	Pencil + ruler, labels match karo
Time kam pada	Section A max 25 min, 5-markers last

Maths One Shot

Mathematics | One Shot Handwritten Notes | Class 10

Is notebook mein kya hai

- ✓ All 14 chapters of CBSE Class 10 Maths, NCERT 2026-27 rationalised syllabus
- ✓ Handwritten-style notes, definitions, theorems and clean figures
- ✓ Formula Box in every chapter + a full Formula Sheet
- ✓ Solved examples with steps, "Yaad karo" tricks, common mistakes
- ✓ Most Important Questions (MCQ, A-R, short, long, case based) with solutions
- ✓ Last-night revision, 56 rapid-fire checks + 2 full mock papers with answers

Shivam Bhaiya says

Maths mein marks practice se aate hain. Is notebook ko 3 baar karo: pehli baar samjho, doosri baar saare examples khud solve karo, teesri baar sirf Formula Sheet, "Yaad karo" boxes aur questions. All the best!

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